Cardiac Diagnostics

Comprehensive Medical Knowledge Base

A Clinical Reference for ECG-Based Disease Recognition

57 Cardiac Conditions | Uniform Template | RAG-Optimized

Definition | Mechanism | ECG Findings | Diagnostic Criteria

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MEDICAL KNOWLEDGE BASE

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1. Introduction

This document transforms the original Cardiac Diagnostics RAG Book into a comprehensive, uniformly structured Medical Knowledge Base optimized for clinical decision support and Retrieval-Augmented Generation (RAG) systems.

1.1 Purpose

This knowledge base serves as a comprehensive clinical reference for ECG-based disease recognition, covering 57 cardiac conditions across the full spectrum of cardiovascular pathology. Each disease entry follows a uniform 12-section template ensuring consistent, complete, and clinically actionable information.

1.2 Scope

The knowledge base encompasses conditions organized by clinical priority:

- **Critical Priority (15 conditions)** - Life-threatening emergencies requiring immediate intervention
- **High Priority (25 conditions)** - Serious conditions requiring prompt diagnosis and management
- **Moderate Priority (17 conditions)** - Important conditions requiring structured evaluation

1.3 Uniform Disease Template

Every disease entry contains the following 12 standardized sections:

Standard Disease Template (12 Sections)

1. **Definition** - Precise clinical definition of the condition
2. **Mechanism** - Underlying pathophysiology and disease mechanism
3. **ECG Findings** - Complete electrocardiographic manifestations
4. **Required ECG Features** - Mandatory diagnostic ECG criteria
5. **Diagnostic Criteria** - Established clinical and investigational criteria
6. **Differential Diagnosis** - Key conditions to exclude with distinguishing features
7. **Symptoms** - Clinical presentation and patient-reported complaints
8. **Risk Factors** - Predisposing factors and patient populations
9. **Immediate Actions** - Emergency management and first-line interventions
10. **Treatment Overview** - Comprehensive management strategy
11. **Follow-up Questions** - Targeted clinical assessment questions
12. **References** - Supporting clinical guidelines and literature

1.4 Priority Classification

Priority	Definition	Response Time	Examples
CRITICAL	Life-threatening, immediate risk of death or permanent organ damage	Immediate (<10 min)	STEMI, Cardiac Arrest, Tamponade
HIGH	Serious condition requiring prompt diagnosis to prevent complications	Urgent (<60 min)	HCM, Myocarditis, Endocarditis
MODERATE	Important condition requiring structured evaluation and management	Priority (<4 hours)	Pericarditis, MVP, ASD

1.5 How to Use This Knowledge Base

This document is designed for dual use: (1) as a standalone clinical reference for healthcare providers interpreting ECGs and managing cardiac conditions, and (2) as a structured knowledge source for RAG-based clinical decision support systems. The uniform template ensures every disease entry contains all information needed for both clinical practice and AI-assisted retrieval.

2. Knowledge Base Architecture

This section provides architectural recommendations for structuring, storing, and retrieving this medical knowledge base in clinical information systems and RAG pipelines.

2.1 Recommended Document Structure

Hierarchical Organization

The knowledge base follows a 4-level hierarchy optimized for both human readability and machine retrieval:

Level	Type	Purpose	Example
L1	Document Root	Entire knowledge base collection	Cardiac_Diagnostics_Complete_KB
L2	Priority Category	Urgency-based grouping	Critical_Priority_Conditions
L3	Disease Entity	Individual disease (atomic unit)	Acute_Coronary_Syndrome
L4	Section Field	Specific knowledge component	ECG_Findings, Mechanism

Recommended JSON Schema for System Integration


```json
{
  "knowledge_base": {
    "version": "2.0",
    "total_diseases": 57,
    "last_updated": "2025-01-15",
    "categories": [
      {
        "priority_level": "CRITICAL",
        "count": 15,
        "diseases": [
          {
            "id": "ACS-001",
            "name": "Acute Coronary Syndrome",
            "category": "Coronary Artery Disease",
            "priority": "CRITICAL",
            "sections": {
              "definition": "...",
              "mechanism": "...",
              "ecg_findings": ["..."],
              "required_ecg_features": ["..."],
              "diagnostic_criteria": ["..."],
              "differential_diagnosis": ["..."],
              "symptoms": ["..."],
              "risk_factors": ["..."],
              "immediate_actions": ["..."],
              "treatment_overview": "...",
              "follow_up_questions": ["..."],
              "references": ["..."]
            }
          }
        ]
      }
    ]
  }
}
```

Recommended Flattened Schema for Vector Databases

```
{
  "chunk_id": "ACS-001-ECG-001",
  "disease_id": "ACS-001",
  "disease_name": "Acute Coronary Syndrome",
  "category": "Coronary Artery Disease",
  "priority": "CRITICAL",
  "section_type": "ECG_Findings",
  "content": "ST elevation >= 1mm in 2+ contiguous limb leads...",
  "metadata": {
    "source_version": "2.0",
    "clinical_domain": "emergency_cardiology",
    "ecg_relevance": "high",
    "life_threatening": true
  }
}
```

File Organization Strategy

Storage Model	Structure	Use Case
Single Document	One comprehensive PDF/JSON	Reference manual, printing
Disease-Per-File	57 individual JSON/Markdown files	Version control, selective updates
Section-Per-Chunk	~684 atomic chunks (57 x 12)	Vector database ingestion
Hybrid	Full document + pre-chunked JSON	Dual human-AI use (recommended)

RECOMMENDATION

Hybrid model is recommended: maintain the complete document for human reference while generating a companion JSON file with pre-chunked sections for RAG ingestion. This ensures consistency between human-readable and machine-retrievable formats.

2.2 Metadata Schema

Core Metadata Fields

Every disease entry and chunk should carry the following metadata for optimal retrieval:

Field	Type	Description	Example Values
`disease_id`	string	Unique identifier	"ACS-001", "HCM-001"
`disease_name`	string	Primary name	"Acute Coronary Syndrome"
`category`	string	Clinical category	"Coronary Artery Disease"
`priority`	enum	Urgency level	"CRITICAL", "HIGH", "MODERATE"
`section_type`	enum	Template section	"ECG_Findings", "Mechanism"
`ecg_relevance`	enum	ECG diagnostic value	"pathognomonic", "high", "supportive", "nonspecific"
`life_threatening`	boolean	Immediate mortality risk	true, false
`clinical_domain`	string	Specialty area	"emergency_cardiology", "electrophysiology"
`age_groups`	array	Commonly affected ages	["adult", "elderly"]
`version`	string	KB version	"2.0"
`source_type`	string	Content origin	"clinical_guideline", "expert_consensus"

Extended Metadata for RAG Optimization

Field	Type	RAG Purpose
`keywords`	array	Enable keyword-based pre-filtering before semantic search
`synonyms`	array	Map alternative disease names (e.g., "MI" = "Myocardial Infarction")
`icd10_codes`	array	Link to billing/EMR systems for integration
`snomed_ct`	array	Standardized clinical terminology mapping
`related_conditions`	array	Enable graph-based retrieval and recommendation
`ecg_patterns`	array	Specific pattern tags for ECG-image-based retrieval
`drug_interactions`	array	Medication safety alerts in CDS
`confidence_level`	enum	Evidence strength (A/B/C guideline levels)

Complete Metadata Example


```
{
  "metadata": {
    "disease_id": "BRUGADA-001",
    "disease_name": "Brugada Syndrome",
    "synonyms": ["BrS", "Sudden Unexplained Nocturnal Death Syndrome"],
    "category": "Channelopathy",
    "priority": "CRITICAL",
    "section_type": "ECG_Findings",
    "icd10_codes": ["I45.81"],
    "snomed_ct": ["418818005"],
    "ecg_relevance": "pathognomonic",
    "ecg_patterns": ["coved_st_elevation", "saddleback_st_elevation"],
    "life_threatening": true,
    "clinical_domain": "electrophysiology",
    "age_groups": ["adult", "young_adult"],
    "confidence_level": "A",
    "keywords": ["sodium_channel", "SCN5A", "sudden_cardiac_death", "ICD"],
    "related_conditions": ["Long_QT_Syndrome", "Early_Repolarization"],
    "version": "2.0",
    "source_type": "clinical_guideline"
  }
}
```

METADATA MAINTENANCE

Metadata should be version-controlled alongside content. When updating disease information (e.g., new ESC guidelines), increment the version and update the `last_updated` timestamp. Maintain a changelog mapping version differences.

2.3 Chunking Strategy

Chunking Principles for Medical Content

Effective chunking balances semantic coherence with retrieval precision. Medical content requires special attention to maintaining clinical context and avoiding dangerous information fragmentation.

Recommended Chunking Approaches

Strategy	Chunk Size	Method	Best For	Trade-off
Section-Based	200-800 tokens	Each template section = 1 chunk	Precision retrieval of specific clinical aspects	May lose cross-section context
Disease-Based	1500-3000 tokens	Full disease entry = 1 chunk	Comprehensive disease overview	Lower precision for specific queries
Hybrid (Recommended)	Variable	Section chunks + disease summary chunk	Both precision and comprehensive retrieval	Higher storage, more complex indexing
Sliding Window	300 tokens / 100 overlap	Overlapping text segments	Catching content at boundaries	Redundant storage, noisy retrieval

RECOMMENDED: HYBRID CHUNKING STRATEGY

Implement a **two-tier chunking system**:

- **Tier 1 - Atomic Chunks**: Each of the 12 template sections becomes an individual chunk (~684 total). Enables precise retrieval (e.g., "What are the ECG findings of Brugada?")
- **Tier 2 - Disease Summary**: A condensed 200-300 token summary per disease (~57 total). Enables overview retrieval (e.g., "Tell me about Brugada Syndrome")
- **Tier 3 - Priority Group**: Combined summaries per priority level (3 chunks). Enables comparative queries (e.g., "What are all the critical cardiac emergencies?")

Chunk Structure Example (Tier 1 - Section-Based)


```
{
  "chunk_id": "ACS-001-ECG-001",
  "content": "ECG Findings for Acute Coronary Syndrome: "
             "STEMI: ST elevation >=1mm in 2+ contiguous limb "
             "leads or >=2mm in precordial leads. NSTEMI: ST "
             "depression >=0.5mm horizontal/downsloping in 2+ "
             "contiguous leads. T wave inversion >=1mm deep in "
             "leads with dominant R wave. Pathological Q waves "
             ">=40ms wide or >=25% of subsequent R wave.",
  "metadata": {
    "disease_id": "ACS-001",
    "disease_name": "Acute Coronary Syndrome",
    "section_type": "ECG_Findings",
    "priority": "CRITICAL",
    "ecg_relevance": "pathognomonic",
    "life_threatening": true
  }
}
```

Chunk Structure Example (Tier 2 - Summary)

```
{
  "chunk_id": "ACS-001-SUMMARY",
  "content": "Acute Coronary Syndrome (ACS) is a medical emergency "
             "describing reduced blood flow to the heart, including "
             "STEMI, NSTEMI, and unstable angina. Mechanism: plaque "
             "rupture with thrombus formation causing coronary "
             "occlusion. Key ECG: ST elevation (STEMI), ST depression "
             "(NSTEMI), T wave inversion, pathological Q waves. "
             "Immediate actions: aspirin, P2Y12 inhibitor, anticoagulation, "
             "primary PCI within 90min for STEMI.",
  "metadata": {
    "disease_id": "ACS-001",
    "chunk_type": "disease_summary",
    "priority": "CRITICAL"
  }
}
```

Critical Chunking Rules for Medical Content

SAFETY-CRITICAL CHUNKING RULES

1. **Never split diagnostic criteria across chunks** - Complete criteria must reside in a single retrievable unit
2. **Never separate ECG findings from Required ECG Features** - These sections should be in the same chunk or closely linked
3. **Always include disease name in chunk content** - Self-contained chunks prevent context loss
4. **Immediate Actions must be standalone** - Emergency interventions must be retrievable without requiring other chunks
5. **Maintain differential diagnosis as complete units** - Splitting can lead to dangerous omission of alternative diagnoses

Optimal Chunk Size by Section Type

Section Type	Recommended Chunking	Rationale
Definition	Single chunk	Short, self-contained
Mechanism	Single chunk (max 500 tokens)	Needs completeness for understanding
ECG Findings	Single chunk with all findings	Never split - safety critical
Required ECG Features	Merge with ECG Findings	Closely related diagnostic data
Diagnostic Criteria	Single chunk	Must be complete for accuracy
Differential Diagnosis	Single chunk	Partial lists are dangerous
Symptoms	Single chunk	Clinical completeness required
Risk Factors	Single chunk	Moderate length, keep together
Immediate Actions	Single chunk, highest priority	Emergency content must be atomic
Treatment Overview	Single chunk or 2 if very long	Keep first-line treatments together
Follow-up Questions	Single chunk	Assessment checklist completeness
References	Single chunk	Bibliographic completeness

2.4 Embedding Recommendations

Embedding Model Selection

Model	Dimensions	Context	Strengths	Best Use Case
text-embedding-3-large (OpenAI)	3072	8191	Excellent medical retrieval; instruction-tuned	Production RAG systems
text-embedding-3-small (OpenAI)	1536	8191	Cost-effective; good medical performance	Large-scale deployment
gte-large (Alibaba)	1024	512	Strong biomedical performance; open source	On-premise deployment
BioClinicalBERT embeddings	768	512	Trained on MIMIC; understands clinical terminology	Specialized clinical systems
MedCPT (Microsoft)	768	512	Optimized for biomedical QA and retrieval	Medical QA systems
e5-large-v2 (Microsoft)	1024	512	Strong general performance with medical text	General medical KB

RECOMMENDED: TEXT-EMBEDDING-3-LARGE

For this cardiac diagnostics knowledge base, **OpenAI text-embedding-3-large** is recommended due to its superior performance on medical terminology, large context window (8191 tokens) that accommodates full disease sections, and excellent retrieval accuracy demonstrated in medical RAG benchmarks.

Embedding Strategy

Single vs. Multi-Vector Representations

Approach	Description	Advantage	Disadvantage
Single Vector	One embedding per chunk	Simple, fast retrieval	Loss of hierarchical information
Multi-Vector (Recommended)	Multiple embeddings per chunk (content + metadata)	Rich retrieval signals	Higher storage, more complex
Matryoshka	Truncate embeddings to variable dimensions	Flexible accuracy-speed tradeoff	Requires compatible model

Recommended Multi-Vector Approach

For each chunk, generate three embeddings:

```
{
  "chunk_id": "ACS-001-ECG-001",
  "embeddings": {
    "content_vector": "[0.023, -0.156, ...]",
    "semantic_vector": "[0.089, -0.042, ...]",
    "metadata_vector": "[0.034, -0.198, ...]"
  },
  "description": {
    "content_vector": "Embedding of the full section text",
    "semantic_vector": "Embedding of structured keywords/tags",
    "metadata_vector": "Embedding of disease_name + category + priority"
  }
}
```

Query Strategies

Query Type	Example	Retrieval Strategy
ECG Pattern Match	"ST elevation in V1-V2 with coved morphology"	Search content_vector with ECG_Finding chunks
Disease Lookup	"What is Brugada Syndrome?"	Search metadata_vector for disease name match
Emergency Protocol	"Patient with VF, what to do now?"	Filter life_threatening=true, search Immediate_Actions
Differential	"Chest pain with normal ECG"	Search content_vector across Symptom + ECG chunks
Treatment Query	"How to treat NSTEMI?"	Search content_vector, rank Treatment_Overview chunks

Retrieval Configuration

```
{
  "retrieval_config": {
    "vector_search": {
      "top_k": 10,
      "similarity_threshold": 0.75,
      "reranking": true
    },
    "metadata_filters": {
      "priority_boost": {
        "CRITICAL": 1.5,
        "HIGH": 1.2,
        "MODERATE": 1.0
      },
      "ecg_relevance_boost": {
        "pathognomonic": 1.4,
        "high": 1.2,
        "supportive": 1.0,
        "nonspecific": 0.9
      }
    },
    "hybrid_search": {
      "vector_weight": 0.7,
      "keyword_weight": 0.3
    },
    "reranking": {
      "model": "cross-encoder/ms-marco-MiniLM-L-6-v2",
      "top_n": 5
    }
  }
}
```

Vector Database Recommendations

Database	Best For	Medical KB Suitability
Pinecone	Managed, scalable	Excellent - metadata filtering, hybrid search
Weaviate	Graph + vector	Excellent - native hybrid search, modular
Chroma	Local/embedded	Good - simple, Python-native
Milvus/Zilliz	Enterprise scale	Excellent - GPU acceleration, billion-scale
pgvector	PostgreSQL integration	Good - if already using PostgreSQL

Indexing Best Practices

INDEXING RECOMMENDATIONS

1. **Create separate collections per priority level** - Enables fast filtering for emergency queries
2. **Index metadata fields** - priority, ecg_relevance, life_threatening, category should all be indexed
3. **Use HNSW index** - For approximate nearest neighbor search with good recall
4. **Set ef_construction to 200+** - Higher quality graph for medical accuracy requirements
5. **Include keyword inverted index** - Enable hybrid search for exact term matching
6. **Store full text alongside vectors** - Enable re-ranking and context assembly

Recommended System Architecture

```
Clinical Query (e.g., "ST elevation V1-V2, syncope")
       |
       v
[Query Understanding Module]
       |
       v
[Hybrid Retrieval Engine]
  |-- Vector Search (content_vector, top_k=20)
  |-- Keyword Search (exact ECG terms)
  |-- Metadata Filter (relevant categories)
       |
       v
[Re-ranking Module (Cross-encoder)]
       |
       v
[Context Assembly]
  |-- Prioritize: Immediate_Actions if emergency
  |-- Include: ECG_Findings + Required_Features
  |-- Add: Differential_Diagnosis for completeness
       |
       v
[Response Generation (LLM)]
       |
       v
[Clinical Decision Support Output]
```

3. Critical Priority Conditions

Life-threatening emergencies requiring immediate recognition and intervention

Acute Coronary Syndrome (ACS)

Category: Coronary Artery Disease **Priority: CRITICAL** **ICD-10:** I24.9

1. Definition

Acute coronary syndrome (ACS) is a spectrum of clinical presentations caused by acute myocardial ischemia ranging from unstable angina to non-ST-elevation myocardial infarction (NSTEMI) and ST-elevation myocardial infarction (STEMI). ACS represents a medical emergency requiring immediate evaluation and management to prevent myocardial necrosis, arrhythmias, and death.

2. Mechanism

The pathophysiology involves disruption of atherosclerotic plaque with subsequent platelet aggregation, thrombus formation, and coronary artery occlusion. STEMI results from complete coronary occlusion with transmural ischemia. NSTEMI involves subtotal occlusion or distal microembolization with detectable myocardial necrosis (elevated troponins). Unstable angina presents with acute ischemic symptoms without biomarker elevation. Type 2 MI results from supply-demand mismatch without plaque disruption.

3. ECG Findings

- ST elevation >= 1mm in 2+ contiguous limb leads or >= 2mm in precordial leads (STEMI)
- ST depression >= 0.5mm horizontal or downsloping in 2+ contiguous leads (NSTEMI/high-risk unstable angina)
- T wave inversion >= 1mm deep in leads with dominant R wave
- Pathological Q waves >= 40ms wide or >= 25% of subsequent R wave (established infarction)
- New left bundle branch block with ischemic symptoms (Sgarbossa criteria apply)
- Wellens syndrome: biphasic or deeply inverted T waves in V2-V3 indicating critical proximal LAD stenosis
- Hyperacute T waves: tall, broad-based T waves representing earliest sign of transmural ischemia
- Posterior MI: tall R waves with ST depression in V1-V2 (mirror pattern)

4. Required ECG Features

DIAGNOSTIC ECG CRITERIA

- **STEMI:** ST elevation in 2+ anatomically contiguous leads with >= 1mm (limb) or >= 2mm (precordial), in the absence of left ventricular hypertrophy or LBBB
- **NSTEMI:** ST depression >= 0.5mm or dynamic T wave changes with positive cardiac troponins
- **Wellens:** Characteristic T wave changes in V2-V3 (biphasic or deeply inverted) in pain-free state after anginal episodes

5. Diagnostic Criteria

- Universal Definition of MI (Fourth Universal Definition, 2018): detection of acute myocardial injury with clinical evidence of acute myocardial ischemia
- At least one of: symptoms of myocardial ischemia, new significant ST-T changes or new LBBB, pathological Q waves, imaging evidence of new loss of viable myocardium, or intracoronary thrombus
- Elevated cardiac troponin above 99th percentile upper reference limit with rising and/or falling pattern
- GRACE score for risk stratification in NSTEMI (in-hospital and 6-month mortality prediction)
- TIMI risk score for unstable angina/NSTEMI

6. Differential Diagnosis

Condition	Differentiating Features
Pericarditis	Diffuse ST elevation with PR depression; pain worse supine, better sitting forward; friction rub
Myocarditis	Younger patients, recent viral illness; diffuse ST changes; elevated troponin with non-obstructive coronaries
Takotsubo cardiomyopathy	Postmenopausal women, emotional trigger; apical ballooning on echo; no culprit lesion on angiography
PE with RV strain	S1Q3T3 pattern, tachycardia, risk factors for thrombosis; CT pulmonary angiography diagnostic
Aortic dissection	Tearing pain radiating to back; pulse deficit; widened mediastinum on CXR; usually normal ECG
Early repolarization	Concave ST elevation, notching at J-point, stable over time; no reciprocal changes

7. Symptoms

- **Classic:** Crushing, pressure-like, or squeezing substernal chest pain lasting >20 minutes
- **Radiation:** To left arm (ulnar distribution), jaw, neck, back, or epigastrium
- **Associated symptoms:** Diaphoresis (cold sweats), nausea/vomiting, dyspnea, syncope or presyncope
- **Non-classical presentations:** Isolated dyspnea, epigastric pain, fatigue (common in elderly, diabetics, women)
- **Sense of impending doom** (often reported in STEMI)
- **Relief pattern:** Unrelieved by rest or nitroglycerin suggests infarction vs. angina

8. Risk Factors

- Non-modifiable: Age (>45 men, >55 women), male sex, family history of premature CAD
- Modifiable: Hypertension, diabetes mellitus, dyslipidemia (LDL >100), smoking, obesity (BMI >30)
- Lifestyle: Sedentary behavior, high-fat diet, chronic stress
- Emerging: Lp(a) elevation, chronic kidney disease, inflammatory conditions (RA, psoriasis), HIV

9. Immediate Actions

EMERGENCY MANAGEMENT - FIRST 10 MINUTES

1. Aspirin 325 mg (if not already given) - chewable preferred for rapid absorption
2. IV access, continuous ECG monitoring (12-lead), pulse oximetry
3. Nitroglycerin 0.4 mg SL every 5 minutes x 3 for persistent pain (avoid if SBP <90, RV infarct, or PDE5 inhibitor use)
4. Morphine 2-4 mg IV for refractory pain (use sparingly, associated with worse outcomes)
5. Supplemental O2 (2-4 L NC) only if SpO2 <90%, respiratory distress, or high-risk features
6. **STEMI:** Emergent primary PCI target within 90 minutes of first medical contact (FMC-to-device); thrombolysis if PCI delay >120 minutes (door-to-needle <30 min)
7. **NSTEMI:** Dual antiplatelet + anticoagulation; early invasive strategy for high-risk (GRACE >140)

10. Treatment Overview

Acute phase: Dual antiplatelet therapy (aspirin + P2Y12 inhibitor - ticagrelor preferred, prasugrel if proceeding to PCI), parenteral anticoagulation (heparin, enoxaparin, or bivalirudin), high-intensity statin (atorvastatin 40-80 mg), beta-blocker within 24 hours if no contraindications (Killip Class < III), ACE inhibitor within 24 hours for anterior MI or EF <40%.

Definitive therapy: Primary PCI with stenting (drug-eluting preferred) for STEMI. Coronary angiography with revascularization (PCI or CABG) for high-risk NSTEMI. CABG preferred for left main disease, complex multivessel disease, or diabetes with multivessel involvement.

Secondary prevention: DAPT for 12 months post-PCI (balance bleeding risk), lifelong aspirin, high-intensity statin, ACE-I/ARB, beta-blocker, cardiac rehabilitation, aggressive risk factor modification.

11. Follow-up Questions

- When did the pain start? (Time is myocardium - document exact onset time)
- Characterize the pain: pressure, squeezing, burning, stabbing? Rate 1-10 severity.
- Does it radiate to arm, jaw, neck, or back?
- Is this the first episode or has the pattern changed? (New onset, crescendo, or rest pain?)
- Any relief with rest or nitroglycerin?
- Associated symptoms: sweating, nausea, shortness of breath, palpitations?

- Risk factor assessment: smoking, diabetes, hypertension, family history?
- Current medications, especially PDE5 inhibitors (sildenafil, tadalafil - contraindicate nitroglycerin)?
- History of CAD, PCI, CABG, or prior MI?
- Any bleeding history or contraindication to anticoagulation?

12. References

1. Thygesen K, et al. Fourth universal definition of myocardial infarction (2018). *J Am Coll Cardiol*. 2018;72(18):2231-2264.
2. Neumann FJ, et al. 2018 ESC Guidelines for myocardial revascularization. *Eur Heart J*. 2018;40(2):87-165.
3. Amsterdam EA, et al. 2014 AHA/ACC Guideline for the Management of Patients with NSTE-ACS. *Circulation*. 2014;130(25):e344-e426.
4. O'Gara PT, et al. 2013 ACCF/AHA Guideline for the Management of STEMI. *Circulation*. 2013;127(4):e362-e425.
5. Ibanez B, et al. 2017 ESC Guidelines for STEMI. *Eur Heart J*. 2018;39(2):119-177.

Aortic Dissection

Category: Aortic Emergency **Priority: CRITICAL** **ICD-10:** I71.0

1. Definition

Aortic dissection is a life-threatening condition involving an intimal tear in the aortic wall allowing blood to enter the media, creating a false lumen that propagates along the vessel. It is classified by location (DeBakey Type I-III or Stanford Type A/B) and represents the most common acute aortic emergency with mortality increasing 1-2% per hour in untreated Type A dissection.

2. Mechanism

Initiated by an intimal tear, typically in the right lateral wall of the ascending aorta (Type A) or distal to the left subclavian (Type B). Blood under pressure enters the media through the tear, creating a false lumen. Propagation is driven by hemodynamic forces and dP/dT. Complications include branch vessel occlusion (coronary, carotid, mesenteric, renal, spinal), aortic rupture into pericardium (tamponade), pleura (hemothorax), or mediastinum. Degeneration of the aortic media (cystic medial necrosis) predisposes to dissection.

3. ECG Findings

- Often normal or nonspecific - **ECG is NOT diagnostic of dissection itself**
- ST elevation if dissection extends into coronary ostia (RCA most common) - can mimic acute MI
- Diffuse ST elevation if rupture into pericardium causes pericarditis pattern
- Electrical alternans if large pericardial effusion/tamponade develops
- Inferior MI pattern if mesenteric ischemia correlation
- Limb lead voltage differences with branch vessel involvement
- Left ventricular hypertrophy pattern in chronic hypertension

4. Required ECG Features

> **IMPORTANT NOTE**
>
> No ECG finding is diagnostic or required for aortic dissection diagnosis. The ECG serves primarily to exclude concurrent MI and detect complications. The key clinical principle: **dissect before you thrombolyse** - always consider dissection before giving thrombolytics for STEMI, especially with atypical features.

5. Diagnostic Criteria

- IRAD criteria: Clinical suspicion based on pain characteristics plus risk factors
- Definitive imaging: CT angiography (first-line), TEE, or MRI demonstrating intimal flap with true and false lumens
- D-dimer <500 ng/mL has high negative predictive value for ruling out dissection in low-risk patients
- Stanford classification: Type A (involves ascending aorta, surgical emergency) vs. Type B (distal to left subclavian)
- DeBakey classification: Type I (ascending + descending), Type II (ascending only), Type III (descending only)

6. Differential Diagnosis

Condition	Differentiating Features
Acute MI	More gradual pain onset; ECG changes correspond to coronary territory; biomarker elevation; no pulse deficit
Pericarditis	Diffuse ST elevation, PR depression; positional pain; friction rub
Pulmonary embolism	Pleuritic pain, dyspnea, tachypnea; S1Q3T3 pattern; CT angiography differentiates
Pneumothorax	Sudden unilateral pleuritic pain, dyspnea; absent breath sounds; CXR diagnostic
Esophageal rupture	Boerhaave syndrome; severe vomiting history; subcutaneous emphysema; mediastinal air on CT
Penetrating atherosclerotic ulcer	Elderly, hypertensive; ulcer crater on imaging; no intimal flap or false lumen

7. Symptoms

- **Classic:** Sudden, severe tearing or ripping chest pain, maximal at onset (differentiates from MI's crescendo pain)
- **Pain migration:** From anterior chest to back, abdomen, or neck indicates propagation
- **Type A:** Anterior chest pain (ascending aorta involvement)
- **Type B:** Interscapular back pain (descending aorta)
- **Syncope:** May indicate tamponade, rupture, or cerebral involvement
- **Hoarseness:** Recurrent laryngeal nerve compression
- **Stroke symptoms:** Carotid artery involvement
- **Paraplegia:** Spinal cord ischemia from intercostal artery compromise
- **Abdominal pain:** Mesenteric ischemia
- **Oliguria:** Renal artery involvement

8. Risk Factors

- **Hypertension:** Present in 70-80% of patients (strongest risk factor)
- **Connective tissue disorders:** Marfan syndrome, Ehlers-Danlos (vascular type), Loeys-Dietz syndrome
- **Congenital:** Bicuspid aortic valve, coarctation of aorta
- **Inflammatory:** Giant cell arteritis, Takayasu arteritis, Behçet disease
- **Trauma:** Blunt chest trauma, iatrogenic (cardiac catheterization, surgery)
- **Cocaine use:** Acute hypertension and vasoconstriction
- **Pregnancy:** Third trimester and postpartum (hormonal + hemodynamic)
- **Male sex and age >60:** Higher incidence

9. Immediate Actions

EMERGENCY MANAGEMENT

1. Immediate IV beta-blockade: Esmolol (50-200 mcg/kg/min) or labetalol (20-80 mg IV q10min) to reduce dP/dT
2. Avoid isolated vasodilation (nitrates, hydralazine) - increases dP/dT and may worsen dissection
3. IV morphine for pain control (catecholamine reduction)
4. **Type A:** Immediate cardiothoracic surgery consultation - emergent surgical repair required
5. **Type B:** Medical management first-line; endovascular repair for complications (malperfusion, rupture, uncontrolled hypertension, expansion)
6. Arterial line for continuous BP monitoring
7. Type and crossmatch blood products

10. Treatment Overview

Type A: Emergent surgical repair with replacement of ascending aorta +/- aortic valve resuspension or replacement. Hypothermic circulatory arrest may be needed for arch involvement. Operative mortality 10-30% depending on complications.

Type B (uncomplicated): Medical management with IV beta-blockade (goal SBP <120, HR <60) and vasodilators once beta-blocked. Transition to oral antihypertensives. Serial imaging to monitor false lumen.

Type B (complicated): Thoracic endovascular aortic repair (TEVAR) is standard. Open surgery reserved for TEVAR failures or unsuitable anatomy.

Long-term: Aggressive BP control (goal <130/80), beta-blockers lifelong, statin therapy, avoidance of heavy lifting, surveillance imaging (CT/MR) at 1, 3, 6, 12 months then annually.

11. Follow-up Questions

- When did the pain start? (Maximal at onset is hallmark - differentiates from MI)
- How would you describe the pain? (Tearing/ripping quality highly suggestive)
- Did the pain migrate from chest to back, neck, or abdomen?
- Any blood pressure difference between arms? (>20 mmHg is significant)
- History of hypertension, connective tissue disorder, or bicuspid aortic valve?
- Family history of aortic dissection or sudden death?
- Any new neurologic symptoms - weakness, numbness, vision changes?

- Recent trauma, cardiac catheterization, or surgery?
- Cocaine use?
- Pregnancy or postpartum status?

12. References

1. Erbel R, et al. 2014 ESC Guidelines on the diagnosis and treatment of aortic diseases. *Eur Heart J*. 2014;35(41):2873-2926.
2. Hiratzka LF, et al. 2010 ACCF/AHA/AATS Guidelines for thoracic aortic disease. *Circulation*. 2010;121(13):e266-e369.
3. Evangelista A, Isselbacher EM, Bossone E, et al. Insights from the International Registry of Acute Aortic Dissection. *Circulation*. 2018;137(17):1846-1860.
4. Nienaber CA, Clough RE. Management of acute aortic dissection. *Lancet*. 2015;385(9970):800-811.

Arrhythmogenic Cardiomyopathy (ACM/ARVC)

Category: Cardiomyopathy **Priority: CRITICAL** **ICD-10:** I42.8

1. Definition

Arrhythmogenic cardiomyopathy (ACM), previously called arrhythmogenic right ventricular cardiomyopathy/dysplasia (ARVC/D), is a progressive inherited cardiomyopathy characterized by fibrofatty replacement of myocardium, primarily affecting the right ventricle, leading to ventricular arrhythmias and sudden cardiac death. ACM affects approximately 1 in 2,000 to 1 in 5,000 individuals worldwide.

2. Mechanism

Caused by mutations in desmosomal proteins (plakoglobin, desmoplakin, plakophilin-2, desmoglein-2, desmocollin-2) disrupting cell-to-cell adhesion. Mechanical stress during exercise causes myocyte detachment, inflammation, and progressive replacement by fibrofatty tissue. This creates a substrate for reentrant ventricular tachycardia. The "hot phase" involves active inflammation; the "cold phase" involves scar-related arrhythmias. Biventricular and left-dominant variants are increasingly recognized.

3. ECG Findings

- **Epsilon wave:** Small deflection at terminal portion of QRS complex in V1-V3 (pathognomonic but low sensitivity)
- **T wave inversion:** In right precordial leads V1-V3 (beyond V1 in adults >14 years; in V1-V2 alone may be normal variant)
- **Prolonged S wave upstroke:** Terminal activation delay >= 55 ms in V1-V3 (QRS duration >=110 ms in V1-V3 without RBBB)
- **Ventricular tachycardia:** LBBB morphology indicating RV origin, typically left inferior axis (RVOT tachycardia)
- **QRS dispersion:** >40 ms difference between maximal and minimal QRS duration across 12 leads
- **Paroxysmal complete heart block:** Due to septal involvement
- **Low ECG voltage:** In advanced disease with massive RV dilation

4. Required ECG Features

REVISED TASK FORCE CRITERIA (2010) - ECG MAJOR CRITERIA

- Repolarization abnormalities: T wave inversion in V1-V3 (in adults, excludes RBBB)
- Depolarization abnormalities: Epsilon wave in V1-V3 OR terminal QRS activation >= 55 ms in V1-V3
- Arrhythmias: NSVT or sustained VT with LBBB morphology and superior axis (RV inferolateral origin) or inferior axis (RVOT origin)

5. Diagnostic Criteria

- 2010 Revised Task Force Criteria: Major and minor criteria across 6 categories (global/regional RV dysfunction, tissue characterization, repolarization, depolarization, arrhythmias, family history/genetics)
- Definite ACM: 2 major, or 1 major + 2 minor, or 4 minor criteria from different categories
- Borderline: 1 major + 1 minor, or 3 minor from different categories
- Possible: 1 major, or 2 minor from different categories
- Genetic testing positive for known pathogenic desmosomal mutation counts as a major criterion
- CMR imaging for myocardial tissue characterization (LGE) increasingly important

6. Differential Diagnosis

Condition	Differentiating Features
RVOT tachycardia (idiopathic)	Normal echo/CMR; no structural RV abnormalities; negative genetic testing
Brugada syndrome	Coved ST elevation in V1-V2; no RV structural changes; different genetic basis (SCN5A)
Myocarditis	Acute presentation; viral prodrome; diffuse inflammation on CMR; resolves over time
Sarcoidosis	Extracardiac manifestations; non-coronary LGE pattern; granulomas on biopsy
Dilated cardiomyopathy	LV-dominant dysfunction; LBBB-VT less common; different genetic basis often
Athlete's heart	Physiologic RV enlargement with training; T wave inversion normalizes with deconditioning

7. Symptoms

- **Palpitations:** Especially with exertion, due to ventricular arrhythmias
- **Syncope or presyncope:** With exertion - major risk factor for sudden death, indicates hemodynamically significant VT
- **Chest discomfort:** Often atypical, not anginal
- **Dyspnea:** Right or biventricular heart failure in advanced disease
- **Peripheral edema, ascites:** Right heart failure signs
- **Many patients asymptomatic:** Diagnosed after family screening or SCD event

8. Risk Factors

- **Genetic:** Autosomal dominant inheritance with variable penetrance; PKP2 most common (60%), followed by DSP, DSG2, DSC2, JUP
- **Family history:** First-degree relative with ACM or sudden cardiac death <50 years
- **Exercise:** Competitive endurance exercise accelerates disease progression
- **Male sex:** Higher penetrance and worse prognosis in males
- **Age:** Presentation typically 20-40 years (young athletes)

9. Immediate Actions

EMERGENCY MANAGEMENT (VT/VF OR SYNCOPE)

1. Immediate ECG, continuous telemetry monitoring
2. If sustained VT with hemodynamic instability: synchronized cardioversion
3. If cardiac arrest: ACLS protocol with defibrillation for VF/pulseless VT
4. IV antiarrhythmics: amiodarone for recurrent VT storm
5. Emergent electrophysiology consultation
6. Athletes: immediate disqualification from competitive sports

10. Treatment Overview

Lifestyle modification: Complete avoidance of competitive and high-intensity endurance exercise (Class I recommendation); this is the only intervention proven to modify disease progression.

Arrhythmia management: Beta-blockers (sotalol preferred) as first-line antiarrhythmic. Flecainide combined with beta-blocker for refractory cases. Catheter ablation (endoepicardial approach) for recurrent VT despite medical therapy.

Sudden death prevention: ICD implantation for: survivors of cardiac arrest (Class I), sustained VT (Class I), syncope with VT inducible on EP study (Class IIa), major risk factors (severe RV dysfunction, LV involvement, genetic mutation + exercise, NSVT on monitoring).

Heart failure management: Standard HF therapy (ACE-I, beta-blockers, diuretics, aldosterone antagonists) for symptomatic HF. Cardiac transplantation for refractory biventricular failure.

11. Follow-up Questions

- Any episodes of syncope, especially during or after exercise?
- Do you experience palpitations? With what activities?
- Any family history of sudden death at young age (<50 years)?
- Any family member diagnosed with ACM, ARVC, or cardiomyopathy?
- What is your exercise history? Competitive sports? Endurance training?
- Any symptoms of right heart failure - leg swelling, abdominal bloating?
- Any previous ECG abnormalities detected incidentally?
- Genetic testing results for family members?

12. References

1. Marcus FI, et al. Diagnosis of arrhythmogenic right ventricular cardiomyopathy/dysplasia. *Circulation*. 2010;121(13):1533-1541.
2. Corrado D, et al. Treatment of arrhythmogenic right ventricular cardiomyopathy/dysplasia. *Circulation*. 2015;132(5):441-453.
3. Corrado D, et al. Cardiovascular pre-participation screening of young competitive athletes. *Eur Heart J*. 2021;42(1):1-19.
4. Te Riele ASJM, et al. Approach to family screening in ARVC/D. *Circulation*. 2014;130(21):1882-1892.

Brugada Syndrome

Category: Channelopathy **Priority: CRITICAL** **ICD-10:** I45.81

1. Definition

Brugada syndrome (BrS) is an inherited channelopathy characterized by characteristic ECG patterns (coved or saddleback ST elevation in V1-V2) and increased risk of malignant ventricular arrhythmias (polymorphic VT, VF) leading to sudden cardiac death. It demonstrates autosomal dominant inheritance with variable penetrance and is responsible for up to 20% of sudden deaths in patients with structurally normal hearts.

2. Mechanism

Loss-of-function mutations in the SCN5A gene encoding the cardiac sodium channel (Nav1.5) account for 20-30% of cases. Reduced sodium current (INa) creates a transmural voltage gradient between the right ventricular epicardium (predominantly in the outflow tract) and endocardium due to loss of the action potential dome in epicardial cells. This creates a substrate for phase 2 reentry initiating polymorphic VT/VF. Fever, vagotonic medications, alcohol, and large meals can unmask or accentuate the ECG pattern by further reducing INa.

3. ECG Findings

- **Type 1 (coved type):** >= 2mm ST elevation in V1-V2 with descending ST segment followed by negative T wave (diagnostic)
- **Type 2 (saddleback):** >= 2mm ST elevation with high take-off J point followed by ascending ST segment and positive or biphasic T wave
- **Type 3:** Coved or saddleback ST elevation < 1 mm (non-diagnostic, may convert to Type 1)
- ECG pattern may be concealed and unmasked by: fever, medications (sodium channel blockers, certain antiarrhythmics), alcohol, large meals, vagal stimulation
- Positioning V1-V2 in 2nd or 3rd intercostal space increases sensitivity
- PR prolongation, QRS widening may be present in SCN5A mutation carriers
- Atrial fibrillation may coexist (20% of patients)

4. Required ECG Features

DIAGNOSTIC CRITERIA (SHANGHAI SCORE SYSTEM 2018)

- **Type 1 ECG pattern** (spontaneous or drug-induced) in >= 1 right precordial lead (V1-V2), positioned in standard, 2nd, or 3rd intercostal space, in the absence of other causative factors
- Score >= 3.5 points indicates definite Brugada syndrome
- Type 2 or 3 pattern alone is NOT diagnostic - must convert to Type 1 with sodium channel blocker challenge (ajmaline 1 mg/kg or flecainide 2 mg/kg IV)

5. Diagnostic Criteria

- Shanghai Score System (2018): Points awarded for ECG pattern (spontaneous Type 1 = 3.5 pts, drug-induced Type 1 = 2.5 pts), clinical history (documented VF = 3 pts, unexplained syncope suggestive of arrhythmia = 2 pts), family history (SCD <45 years = 1 pt, Type 1 ECG in family = 0.5 pt), genetic testing (pathogenic SCN5A = 0.5 pt). Score >= 3.5 = definite, 2-3 = probable, <2 = possible.
- Previous consensus: Type 1 ECG + at least one of: documented VF/polymorphic VT, self-terminating polymorphic VT, family history of SCD <45 years with Type 1 Brugada, coved-type ECG in family members, syncope suggestive of arrhythmic origin, nocturnal agonal respiration

6. Differential Diagnosis

Condition	Differentiating Features
Early repolarization	Concave ST elevation, notching, most prominent in lateral leads; no arrhythmic risk
ARVC	Structural RV abnormalities on echo/CMR; epsilon wave; T wave inversion V1-V3; desmosomal mutations
Acute anterior STEMI	Dynamic changes over time; reciprocal changes; cardiac biomarker elevation; pain symptoms
Hyperkalemia	Peaked T waves, QRS widening; elevated potassium; reversible with treatment
Arrhythmogenic cardiomyopathy (left dominant)	LV structural changes; different ECG distribution
Pectus excavatum / RVOT abnormality	Structural chest wall or RVOT anomaly; no arrhythmic history

7. Symptoms

- **Syncope:** Often the presenting symptom, frequently nocturnal or at rest (circadian pattern with peak incidence at night/early morning)
- **Palpitations:** Due to ventricular arrhythmias or concomitant AF
- **Nocturnal agonal respiration:** Witnessed gasping during sleep, sign of aborted SCD
- **Seizures:** Cerebral hypoperfusion during VT/VF can mimic seizure disorder
- **Many patients asymptomatic:** ECG finding detected incidentally or through family screening

8. Risk Factors

- **Genetic:** Pathogenic SCN5A mutation (20-30%); autosomal dominant with variable penetrance
- **Family history:** SCD in first-degree relative <45 years, especially if nocturnal
- **Prior cardiac arrest:** Strongest predictor of recurrence
- **Spontaneous Type 1 pattern:** Higher risk than drug-induced pattern
- **Syncope:** Arrhythmic syncope indicates higher risk
- **Male sex:** 8-10:1 male predominance; testosterone may modulate risk
- **Fever:** Can trigger arrhythmias and unmask ECG pattern

- **Drugs:** Sodium channel blockers, certain antidepressants, antipsychotics

9. Immediate Actions

EMERGENCY MANAGEMENT (CARDIAC ARREST OR SUSTAINED VT)

1. ACLS protocol with immediate defibrillation for VF/pulseless VT
2. Treat fever aggressively - antipyretics and cooling (fever can precipitate arrhythmias)
3. IV isoproterenol (1-4 mcg/min) for electrical storm unresponsive to defibrillation
4. Quinidine (oral or IV) as adjunct for VT storm - restores epicardial action potential dome
5. Identify and discontinue any arrhythmogenic medications
6. Emergent electrophysiology consultation

10. Treatment Overview

ICD implantation: Only therapy proven to prevent SCD. Class I indication for survivors of cardiac arrest or documented spontaneous sustained VT. Class IIa for syncope suggestive of arrhythmic origin with spontaneous Type 1 pattern.

Pharmacotherapy: Quinidine (600-1500 mg/day) as adjunctive therapy to reduce VT burden and ICD shocks; effective for both suppression and acute treatment of electrical storm. Isoproterenol for acute electrical storm.

Catheter ablation: Epicardial ablation of RVOT electroanatomic substrate for recurrent VT/VF despite quinidine + ICD; reduces arrhythmia burden but does not replace ICD.

Lifestyle: Treat fever promptly, avoid large meals (especially carbohydrate-heavy), avoid alcohol, avoid hot baths/saunas, avoid drugs that block sodium channels or prolong QT (maintain updated BrugadaDrugs.org list).

11. Follow-up Questions

- Any episodes of syncope, fainting, or near-fainting, especially at night?
- Any witnessed gasping or seizure-like episodes during sleep?
- Family history of sudden death, especially at young age (<45) or during sleep?
- Family history of Brugada syndrome or similar ECG findings?
- Any personal history of ventricular fibrillation or cardiac arrest?
- Any history of fever-triggered events?
- Current medications - any antiarrhythmics, antidepressants, antipsychotics?
- Do you have the ECG pattern spontaneously or only with provocation testing?

12. References

1. Antzelevitch C, et al. Brugada syndrome: report of the second consensus conference. *Heart Rhythm*. 2005;2(4):429-440.
2. Priori SG, et al. Executive summary: HRS/EHRA/APHRS expert consensus statement on inherited primary arrhythmia syndromes. *Europace*. 2013;15(10):1389-1406.
3. Adler A, et al. An international, multicentered, evidence-based reappraisal of genes reported to cause congenital long QT syndrome. *Circulation*. 2020;141(18):418-428.
4. Bayés de Luna A, et al. Current electrocardiographic criteria for diagnosis of Brugada pattern. *J Am Coll Cardiol*. 2012;60(19):1940-1941.

Cardiac Tamponade

Category: Pericardial Emergency **Priority: CRITICAL** **ICD-10:** I31.9

1. Definition

Cardiac tamponade is a life-threatening clinical syndrome caused by accumulation of fluid in the pericardial space under pressure, resulting in impaired ventricular filling, reduced cardiac output, and hemodynamic compromise. It represents a continuum from asymptomatic pericardial effusion to pulseless electrical activity and death. Tamponade is a clinical diagnosis requiring emergent intervention.

2. Mechanism

Fluid accumulation in the fixed pericardial space raises intrapericardial pressure. When pericardial pressure exceeds ventricular filling pressures, diastolic collapse occurs. The right ventricle (lower pressure system) collapses first, followed by the right atrium. Equalization of pressures across all cardiac chambers occurs (RA, RV diastolic, PA diastolic, and PCWP become similar). Pulsus paradoxus (>10 mmHg inspiratory drop in SBP) results from exaggerated ventricular interdependence - increased RV filling during inspiration shifts the septum leftward, further reducing LV filling. Cardiac output falls, leading to hypotension and shock.

3. ECG Findings

- **Low QRS voltage:** <5 mm in limb leads, <10 mm in precordial leads (due to insulating effect of pericardial fluid)
- **Electrical alternans:** Beat-to-beat alternation of QRS amplitude (highly specific for large effusion with tamponade; due to swinging heart in fluid)
- **Sinus tachycardia:** Compensatory response to falling cardiac output
- **Diffuse ST elevation:** If concurrent acute pericarditis (concave, without reciprocal changes)
- **PR depression:** If concurrent pericarditis
- **Atrial arrhythmias:** Atrial fibrillation or ectopy may occur

4. Required ECG Features

> **IMPORTANT NOTE**
>
> No single ECG finding is diagnostic of tamponade. Electrical alternans is highly specific but insensitive (sensitivity ~20-30%). The clinical triad (Beck's triad) plus echocardiographic findings establishes the diagnosis. ECG is supportive, not definitive.

5. Diagnostic Criteria

- **Clinical:** Beck's triad (hypotension, muffled heart sounds, elevated JVP) - present in minority of cases
- **Echocardiographic (definitive):** RA systolic collapse, RV diastolic collapse, swinging heart in large effusion, IVC plethora without inspiratory collapse, exaggerated respiratory variation in mitral (>25%) and tricuspid (>40%) inflow velocities
- **Hemodynamic:** Equalization of diastolic pressures (RA = RV diastolic = PA diastolic = PCWP), pulsus paradoxus >10 mmHg
- **Blunt/traumatic:** Presence of effusion + hemodynamic instability in trauma patient

6. Differential Diagnosis

Condition	Differentiating Features
Constrictive pericarditis	Chronic presentation; pericardial knock; rapid x and y descents on JVP; no effusion on echo; calcification on CT
Cardiogenic shock	Large heart on CXR, pulmonary edema, echo shows poor contractility; no pericardial effusion
RV infarction	ST elevation in inferior leads with right-sided leads; hypotension with clear lungs; dilated RV on echo
Massive pulmonary embolism	Acute dyspnea, tachycardia, RV strain on echo; CT pulmonary angiography diagnostic
Tension pneumothorax	Unilateral absent breath sounds, tracheal deviation; CXR or ultrasound diagnostic

7. Symptoms

- **Dyspnea:** Most common symptom (80-90%), due to reduced cardiac output and pulmonary venous congestion
- **Chest discomfort or fullness:** Due to pericardial distension
- **Syncope or presyncope:** Reduced cerebral perfusion
- **Tachycardia, anxiety, restlessness:** Compensatory response to falling cardiac output
- **Weakness, fatigue:** Low-output state
- **Cold, clammy extremities:** Vasoconstriction from shock
- **Symptoms of underlying cause:** Fever (infectious), trauma, malignancy symptoms

8. Risk Factors / Etiologies

- **Malignancy:** Lung, breast, lymphoma, leukemia (most common in developed countries)
- **Viral pericarditis:** Most common overall cause; usually small-moderate effusions
- **Trauma:** Penetrating chest trauma, cardiac surgery (post-pericardiotomy syndrome), post-cardiac intervention (PCI, EP ablation, pacemaker lead perforation)
- **Autoimmune:** SLE, rheumatoid arthritis, scleroderma
- **Hypothyroidism:** Myxedema - slow accumulation, often large effusion without tamponade due to gradual adaptation
- **Infectious:** TB (endemic areas), bacterial purulent pericarditis (HIV patients)
- **Aortic dissection:** Rupture into pericardium - rapidly fatal tamponade
- **Anticoagulation:** Hemorrhagic tamponade (warfarin, DOACs, heparin)

9. Immediate Actions

EMERGENCY MANAGEMENT

1. Immediate IV volume resuscitation with crystalloids to maintain preload (500-1000 mL bolus) - temporary measure only
2. Avoid diuretics, vasodilators, and positive pressure ventilation if possible (reduce venous return)
3. Avoid nitrates (preload reduction worsens tamponade)
4. **Urgent pericardiocentesis** is definitive therapy; do not delay for additional imaging if tamponade clinically suspected
5. If due to aortic dissection: immediate surgical repair, not pericardiocentesis (risk of worsening hemorrhage)
6. Emergency department or bedside ultrasound to confirm effusion and guide drainage
7. Pericardial drain placement after drainage; send fluid for analysis

10. Treatment Overview

Immediate: Pericardiocentesis under echocardiographic or fluoroscopic guidance. Apical or subxiphoid approach. Leave pericardial catheter for 24-48 hours to prevent reaccumulation. If due to aortic dissection: OR for surgical repair.

Refractory/recurrent: Surgical pericardial window (subxiphoid pericardiostomy or pericardiectomy) for recurrent effusions, loculated effusions, or thick/clot-filled pericardium.

Specific therapy: Treat underlying cause - anticoagulation reversal if hemorrhagic, antibiotics for bacterial, antineoplastic therapy for malignant, dialysis for uremic, thyroid replacement for myxedema, TB treatment.

Malignant effusion: Intrapericardial chemotherapy, systemic therapy, or balloon pericardiotomy for recurrent cases.

11. Follow-up Questions

- Onset and progression of symptoms - hours, days, or weeks?
- Any recent cardiac procedure, surgery, or intervention?
- Recent chest trauma or penetrating injury?
- History of cancer, especially lung, breast, or lymphoma?
- Fever, night sweats, or recent infection?
- Any autoimmune disease (lupus, rheumatoid arthritis)?
- Current anticoagulant or antiplatelet medications?
- Signs of hypothyroidism (cold intolerance, weight gain, fatigue)?
- Recent travel to TB-endemic regions?
- Known aortic aneurysm or risk factors for dissection?

12. References

1. Adler Y, et al. 2015 ESC Guidelines for the diagnosis and management of pericardial diseases. *Eur Heart J*. 2015;36(42):2921-2964.
2. Hoit BD. Pathophysiology of cardiac tamponade. *Chest*. 2017;151(3):533-535.
3. Spodick DH. Acute cardiac tamponade. *N Engl J Med*. 2003;349(7):684-690.
4. LeWinter MM. Clinical practice: acute pericarditis. *N Engl J Med*. 2014;371(25):2410-2416.

Cardiogenic Shock

Category: Cardiac Emergency **Priority: CRITICAL** **ICD-10:** R57.0

1. Definition

Cardiogenic shock is a state of inadequate tissue perfusion due to cardiac dysfunction, characterized by persistent hypotension (SBP <90 mmHg or MAP <65 mmHg) despite adequate filling status, accompanied by clinical and biochemical evidence of tissue hypoperfusion (cool extremities, oliguria, altered mental status, elevated lactate). It complicates 5-10% of acute myocardial infarctions and carries 40-50% mortality despite modern interventions.

2. Mechanism

Primary mechanism: extensive myocardial necrosis (>40% of LV mass) causing pump failure, most commonly from acute STEMI. A vicious cycle ensues: decreased cardiac output reduces coronary perfusion pressure, worsening ischemia and further reducing contractility. Compensatory sympathetic activation increases afterload and myocardial oxygen demand, exacerbating the cycle. Mechanical complications of MI (papillary muscle rupture, ventricular septal defect, free wall rupture/tamponade) account for ~15% of cases. Non-MI causes include acute myocarditis, end-stage cardiomyopathy, severe valvular disease, and sustained arrhythmias.

3. ECG Findings

- **ST elevation MI (STEMI):** Most common cause (~80% of cases); extensive anterior STEMI (LAD occlusion) most likely to cause shock
- **New murmur + ECG changes:** Suggests mechanical complication (papillary muscle rupture, VSD)
- **Atrial/ventricular arrhythmias:** AF with rapid response, sustained VT, complete heart block
- **Diffuse ST-T changes:** Myocarditis pattern
- **Electrical alternans + low voltage:** Suggests tamponade (especially post-MI free wall rupture)
- **Right-sided leads:** RV infarction (inferior STEMI with right ventricular involvement) contributes to shock
- **Sinus tachycardia:** Compensatory; bradycardia ominous sign of failing compensatory mechanisms

4. Required ECG Features

ECG CLUES TO ETIOLOGY

- Extensive ST elevation (anterior, multi-territory) in hypotensive patient = cardiogenic shock from acute MI
- Inferior STEMI + right-sided ST elevation = RV infarct contributing to shock
- New LBBB or complete heart block in acute setting with shock = consider mechanical complications
- Diffuse ST elevation with PR depression in shock = pericardial tamponade

5. Diagnostic Criteria

- SBP <90 mmHg or MAP <65 mmHg for >30 minutes despite adequate volume resuscitation
- Cardiac index <2.2 L/min/m² (measured invasively via PAC)
- Pulmonary capillary wedge pressure (PCWP) >15 mmHg (elevated filling pressures)
- Elevated lactate (>2 mmol/L) indicating tissue hypoperfusion
- Clinical signs: cool clammy extremities, decreased urine output (<30 mL/hr), altered mental status
- SHOCK trial criteria validated the clinical definition

6. Differential Diagnosis

Condition	Differentiating Features
Hypovolemic shock	Low CVP/PCWP, clear lungs, flat neck veins; responds to volume resuscitation
Septic shock	Warm extremities (vasodilated), fever, infectious source; low SVR; pressor responsive
Obstructive shock (PE)	S1Q3T3, acute RV strain on echo; CT angiography confirms
Obstructive shock (tamponade)	Equalized pressures, echo shows pericardial effusion with chamber collapse
Opioid overdose	Pinpoint pupils, respiratory depression; responds to naloxone
Tension pneumothorax	Unilateral absent breath sounds, tracheal deviation; ultrasound/CXR diagnostic

7. Symptoms

- **Hypotension:** SBP <90 mmHg despite adequate filling
- **Cool, clammy extremities:** Cutaneous hypoperfusion from vasoconstriction
- **Altered mental status:** Confusion, agitation, or decreased consciousness from cerebral hypoperfusion
- **Oliguria:** Urine output <30 mL/hr from renal hypoperfusion
- **Dyspnea:** Pulmonary edema from elevated left-sided filling pressures
- **Chest pain:** Ongoing ischemia; may be absent if cardiogenic shock from non-ischemic cause
- **Tachycardia:** Compensatory; bradycardia is ominous

8. Risk Factors

- Extensive anterior STEMI (>40% LV infarction)
- Prior MI (reduced LVEF, limited compensatory reserve)
- Elderly age (>75 years)
- Diabetes mellitus
- Multivessel CAD
- Prior heart failure
- Anterior STEMI location
- Delayed reperfusion (>6 hours from symptom onset)
- Mechanical complications: papillary muscle rupture, VSD, free wall rupture

9. Immediate Actions

EMERGENCY MANAGEMENT

1. Emergent PCI for STEMI-related shock (Class I): door-to-balloon time critical; revascularize culprit lesion first, consider multivessel if refractory
2. IV vasopressors: norepinephrine first-line (0.05-1 mcg/kg/min) to maintain MAP >65
3. Add dobutamine (2-20 mcg/kg/min) if low cardiac output state persists despite norepinephrine
4. Invasive hemodynamic monitoring (arterial line, consider PAC)
5. Avoid beta-blockers, ACE inhibitors, and nitrates in acute phase (reduce contractility/afterload)
6. Mechanical circulatory support if refractory: Impella (preferred), intra-aortic balloon pump (IABP), VA-ECMO
7. Emergent echocardiogram to rule out mechanical complications
8. Intubation and mechanical ventilation if respiratory failure

10. Treatment Overview

Acute MI-related: Emergent PCI with stenting of culprit artery (SHOCK trial showed 13% absolute mortality reduction at 6 months with early revascularization). If PCI not feasible within 90 minutes, consider thrombolysis + rescue PCI if failed lysis.

Mechanical complications: Papillary muscle rupture (emergent valve repair/replacement), VSD (surgical repair or percutaneous closure), free wall rupture (emergent surgical repair + pericardiocentesis).

Non-MI causes: Treat underlying etiology - myocarditis (supportive, consider immunosuppression if giant cell), acute valvular regurgitation (emergent surgery), sustained arrhythmias (cardioversion/ablation).

Supportive: Mechanical ventilation if needed, renal replacement if AKI develops, targeted temperature management if cardiac arrest preceded shock.

Long-term: GDMT for HFrEF (ACE-I/ARB/ARNI, beta-blockers, MRA, SGLT2i), cardiac rehabilitation, ICD if EF <35% post-recovery.

11. Follow-up Questions

- When did symptoms begin? (Time-critical for STEMI-related shock)
- Any prior MI, heart failure, or known reduced ejection fraction?
- Any chest pain preceding the shock state?

- Current urine output? (Marker of end-organ perfusion)
- Any prior cardiac catheterization or known CAD?
- Any valvular disease history?
- Recent viral illness suggesting myocarditis?
- Current medications: beta-blockers, ACE inhibitors, nitrates?
- Code status and goals of care?

12. References

1. Hochman JS, et al. Early revascularization in acute myocardial infarction complicated by cardiogenic shock. *N Engl J Med*. 1999;341(9):625-634.
2. Thiele H, et al. Intraaortic balloon support for myocardial infarction with cardiogenic shock. *N Engl J Med*. 2012;367(14):1287-1296.
3. van Diepen S, et al. Contemporary management of cardiogenic shock: A scientific statement from the AHA. *Circulation*. 2017;136(16):e232-e268.
4. Thiele H, et al. Percutaneous short-term active mechanical support devices in cardiogenic shock. *Eur Heart J*. 2021;42(18):1734-1739.

Heart Attack (Myocardial Infarction)

Category: Coronary Artery Disease **Priority: CRITICAL** **ICD-10:** I21.9

1. Definition

Myocardial infarction (MI) is myocardial cell death due to prolonged ischemia from acute coronary artery occlusion. Diagnosed by the rise and/or fall of cardiac biomarkers (cardiac troponins) with at least one of: symptoms of acute myocardial ischemia, new significant ST-T changes or new LBBB, pathological Q waves, or imaging evidence of new loss of viable myocardium. Classified as STEMI (complete occlusion), NSTEMI (subtotal occlusion with biomarker rise), or Type 2 MI (supply-demand mismatch).

2. Mechanism

Type 1 MI (most common): Atherosclerotic plaque rupture, ulceration, fissuring, erosion, or dissection triggers platelet aggregation and thrombus formation causing acute coronary occlusion. STEMI results from complete occlusion; NSTEMI from subtotal occlusion or distal embolization. The "wavefront phenomenon" describes progressive necrosis from endocardium to epicardium over 3-6 hours. Reperfusion within 12 hours salvages viable myocardium. Reperfusion injury (stunning) and no-reflow phenomenon can limit recovery despite patent epicardial artery.

3. ECG Findings

- **Anterior/Septal STEMI:** ST elevation in V1-V4 (proximal LAD occlusion); proximal LAD causes more extensive changes including aVR elevation
- **Inferior STEMI:** ST elevation in II, III, aVF (RCA 80%, LCX 20%); right-sided leads for RV involvement
- **Lateral STEMI:** ST elevation in I, aVL, V5-V6 (LCX or diagonal branch)
- **Posterior MI:** Tall R waves with ST depression in V1-V2 (mirror pattern); confirm with V7-V9 ST elevation
- **Right ventricular MI:** ST elevation in V4R (proximal RCA occlusion); associated with inferior STEMI
- **NSTEMI:** ST depression >= 1mm, T wave inversion, or nondiagnostic ECG with elevated troponins
- **Reciprocal changes:** ST depression in leads opposite to STEMI territory increases specificity
- **Hyperacute T waves:** Earliest sign, tall, broad-based, present before ST elevation
- **Wellens syndrome:** Biphasic/deeply inverted T waves V2-V3 indicating critical LAD stenosis

4. Required ECG Features

> **STEMI DIAGNOSTIC CRITERIA**
>
> - New ST elevation at the J point in 2+ contiguous leads: >= 1 mm in all leads except V2-V3
> - V2-V3: >= 2.5 mm in men <40 years, >= 2 mm in men >40 years, >= 1.5 mm in women
> - New horizontal/downsloping ST depression >= 0.5 mm in V1-V3 suggests posterior STEMI
> - New ST elevation in aVR >= 1 mm suggests left main or multi-vessel disease

5. Diagnostic Criteria

- Fourth Universal Definition of MI (2018): Detection of rise and/or fall of cardiac troponin with at least one value above 99th percentile URL plus evidence of acute myocardial ischemia
- Clinical: Ischemic symptoms, new significant ST-T changes, new pathological Q waves, loss of viable myocardium on imaging, or intracoronary thrombus on angiography
- Type 1: Spontaneous atherosclerotic plaque rupture
- Type 2: Supply-demand mismatch without plaque disruption (tachyarrhythmia, anemia, hypotension)
- Type 3: Sudden cardiac death with symptoms suggestive of MI
- Type 4: PCI-related MI (troponin >5x 99th percentile with ischemic symptoms, ECG changes, imaging loss, or angiographic complication)
- Type 5: CABG-related MI (troponin >10x 99th percentile with supporting features)

6. Differential Diagnosis

Condition	Differentiating Features
Unstable angina	Same pathophysiology but no biomarker elevation; may have identical symptoms and ECG changes
Myocarditis	Younger, viral prodrome, diffuse ST changes, elevated troponin with non-obstructive coronaries
Takotsubo cardiomyopathy	Postmenopausal woman, emotional trigger, apical ballooning, no culprit lesion
Pericarditis	Diffuse concave ST elevation, PR depression, pain positional
Early repolarization	Stable pattern, concave ST elevation, no reciprocal changes, no symptoms
Pulmonary embolism	S1Q3T3, tachycardia, pleuritic pain, risk factors for DVT

7. Symptoms

- **Classic:** Crushing, pressure-like substernal pain >30 minutes, not relieved by rest or nitroglycerin
- **Atypical presentations:** Dyspnea, fatigue, nausea, epigastric pain, syncope (elderly, diabetics, women)
- **Radiation:** Left arm (ulnar), jaw, neck, interscapular region
- **Associated symptoms:** Diaphoresis, vomiting, pallor, sense of impending doom
- **Silent MI:** Especially in diabetics (autonomic neuropathy) and elderly
- **Women more likely:** Atypical symptoms including fatigue, sleep disturbance, anxiety, indigestion

8. Risk Factors

- Modifiable: Smoking, hypertension (>140/90), diabetes, dyslipidemia (LDL >100, HDL <40 men/<50 women), obesity (BMI >30), sedentary lifestyle
- Non-modifiable: Age >45 (men), >55 (women), male sex, family history of premature CAD
- Emerging: Lp(a) >50 mg/dL, CKD, inflammatory diseases, HIV, psychosocial stress/depression

9. Immediate Actions

EMERGENCY MANAGEMENT

1. Aspirin 325 mg chewable (unless true allergy) - immediate
2. P2Y12 inhibitor: ticagrelor 180 mg loading (or prasugrel 60 mg if proceeding to PCI)
3. Anticoagulation: heparin (UFH or enoxaparin) or bivalirudin
4. Nitroglycerin for pain (avoid if SBP <90, RV infarct, sildenafil use)
5. Morphine only if refractory (associated with worse outcomes)
6. Beta-blocker orally within 24 hours if no contraindications (avoid if acute decompensated HF, shock, bradycardia, heart block)
7. High-intensity statin (atorvastatin 40-80 mg)
8. Primary PCI target: FMC-to-device <90 minutes, door-to-balloon <90 minutes
9. If PCI delay >120 minutes: thrombolytics (door-to-needle <30 minutes)

10. Treatment Overview

Reperfusion: Primary PCI preferred for all STEMI within 12 hours of symptom onset; also consider 12-48 hours if ongoing symptoms, hemodynamic instability, or malignant arrhythmias. Facilitated PCI (lytics followed by immediate PCI) no longer recommended. Rescue PCI for failed lysis.

NSTEMI: Early invasive strategy (angiography within 24-72 hours) for high-risk (GRACE >140, dynamic ECG changes, elevated troponins). Ischemia-guided strategy acceptable for low-risk.

Medical therapy: DAPT for 12 months (aspirin + ticagrelor/prasugrel), high-intensity statin, ACE-I/ARB (anterior MI, EF <40%, diabetes), beta-blocker, aldosterone antagonist (EF <40% with HF or diabetes).

Secondary prevention: Cardiac rehabilitation, smoking cessation, glycemic control (HbA1c <7%), BP control (<130/80), LDL goal <70 for very high risk, weight management.

11. Follow-up Questions

- Exact time of symptom onset? (Critical for reperfusion decisions)
- Character of pain - pressure, crushing, burning, stabbing? (Pressure/crushing most typical)
- Any relief with rest or nitroglycerin? (MI typically unrelieved)
- Prior similar episodes? (New vs. progressive symptoms?)
- Risk factors: smoking, diabetes, hypertension, high cholesterol, family history?
- Prior CAD, PCI, or CABG?

- Any medications including PDE5 inhibitors, anticoagulants?
- Any bleeding history or conditions?
- Renal function? (Contrast risk for angiography)
- Code status and goals of care?

12. References

1. Thygesen K, et al. Fourth universal definition of myocardial infarction. *J Am Coll Cardiol*. 2018;72(18):2231-2264.
2. Lawton JS, et al. 2021 ACC/AHA/SCAI Guideline for Coronary Artery Revascularization. *J Am Coll Cardiol*. 2022;79(2):e21-e129.
3. Collet JP, et al. 2020 ESC Guidelines for the management of acute coronary syndromes in patients presenting without persistent ST-segment elevation. *Eur Heart J*. 2021;42(14):1289-1367.
4. Van de Werf F, et al. Management of acute myocardial infarction in patients presenting with persistent ST-segment elevation. *Eur Heart J*. 2008;29(23):2909-2945.

Long QT Syndrome

Category: Channelopathy **Priority: CRITICAL** **ICD-10:** I45.81

1. Definition

Long QT syndrome (LQTS) is a cardiac channelopathy characterized by prolonged ventricular repolarization (QTc >= 480 ms definite, 460-479 ms borderline in females; >= 460 ms borderline in males) with increased risk of polymorphic ventricular tachycardia (Torsades de Pointes), syncope, seizures, and sudden cardiac death. Congenital LQTS results from mutations in cardiac ion channel genes; acquired LQTS results from medications, electrolyte disturbances, or medical conditions.

2. Mechanism

Congenital: Loss-of-function mutations in potassium channels (KCNQ1 - LQT1, KCNH2 - LQT2) reducing repolarizing IKs and IKr currents, or gain-of-function mutations in sodium channel (SCN5A - LQT3) increasing late INa. Net effect: prolonged ventricular action potential duration. Acquired: Drug block of hERG potassium channel (Ikr), electrolyte disturbances (hypokalemia, hypomagnesemia, hypocalcemia), or bradycardia. Torsades de Pointes triggered by early afterdepolarizations during prolonged repolarization, often initiated by pause-dependent mechanisms.

3. ECG Findings

- **QTc intervals:** Male: >= 480 ms definite, 460-479 ms borderline; Female: >= 480 ms definite, 470-479 ms borderline
- **QTc >= 500 ms:** High risk for Torsades de Pointes
- **LQT1 (KCNQ1):** Broad-based T waves, exercise-triggered events
- **LQT2 (KCNH2):** Notched, low-amplitude, bifid T waves, auditory/emotion-triggered
- **LQT3 (SCN5A):** Late-onset peaked/biphasic T waves, sleep/rest-triggered
- **T wave alternans:** Beat-to-beat variation in T wave amplitude - marker of electrical instability and imminent TdP
- **Prominent U waves:** May merge with T wave, contributing to pseudo-long QT
- **Bradycardia:** May be present, especially LQT3; contributes to QT prolongation

4. Required ECG Features

DIAGNOSTIC QTC CRITERIA

- Schwartz Score: QTc >= 480 ms = 3 points, 460-479 ms = 2 points, 450-459 ms (male) = 1 point
- Score >= 3.5 = high probability, 2-3 = intermediate, <=1 = low probability
- Torsades de Pointes: Polymorphic VT with QRS axis twisting around baseline, typically rate 150-250 bpm, often pause-dependent, self-terminating but may degenerate to VF

5. Diagnostic Criteria

- Schwartz Score incorporating: QTc duration, congenital deafness, syncope with stress, family history, T wave alternans, notched T waves in 3 leads, bradycardia
- Genetic testing positive for pathogenic LQTS mutation confirms diagnosis regardless of QTc
- Definite LQTS: Schwartz score >= 3.5 in absence of secondary causes, or presence of confirmed LQTS mutation
- Provocative testing: Exercise testing (failure to shorten QT with increased HR in LQT1), epinephrine QT stress test

6. Differential Diagnosis

Condition	Differentiating Features
Drug-induced QT prolongation	Temporal relation to medication initiation; resolves with drug discontinuation; check CredibleMeds.org
Electrolyte abnormalities	Hypokalemia, hypomagnesemia, hypocalcemia; corrects with repletion
Vasovagal syncope	Prodrome, situational triggers, normal QTc, negative genetic testing
Brugada syndrome	Coved ST elevation V1-V2; no QT prolongation; different genetic basis
Andersen-Tawil syndrome (LQT7)	Triad: periodic paralysis, dysmorphic features, ventricular arrhythmias; KCNJ2 mutation
Timothy syndrome (LQT8)	Extreme QT prolongation (>500 ms), syndactyly, autism, CACNA1C mutation

7. Symptoms

- **Syncope:** Exercise-triggered (LQT1), emotion/auditory-triggered (LQT2 - alarm clock, telephone), sleep/rest-triggered (LQT3)

- **Seizures:** Cerebral hypoperfusion during Torsades can mimic epileptic seizures
- **Palpitations:** During Torsades episodes
- **Sudden cardiac death:** First manifestation in up to 10% of cases; aborted SCD increasingly recognized
- **Congenital deafness:** Jervell and Lange-Nielsen syndrome (homozygous KCNQ1, very rare)
- **Many patients asymptomatic:** Diagnosed through family screening or incidental ECG

8. Risk Factors

- **Genetic:** KCNQ1 (LQT1, 35%), KCNH2 (LQT2, 30%), SCN5A (LQT3, 10%); autosomal dominant (Romano-Ward) with variable penetrance
- **QTc duration:** >500 ms significantly increases risk; >550 ms very high risk
- **Prior syncope or aborted SCD:** Strongest predictors of future events
- **Gender:** LQT1: males higher risk; LQT2/LQT3: females higher risk (especially postpartum)
- **Acquired triggers:** QT-prolonging drugs (antiarrhythmics, macrolides, fluoroquinolones, antipsychotics, methadone), hypokalemia, hypomagnesemia, bradycardia
- **Swimming:** Specific trigger for LQT1 (diving activates sympathetic response)

9. Immediate Actions

EMERGENCY MANAGEMENT (TORSADES DE POINTES)

1. If pulseless: immediate unsynchronized defibrillation (treat as VF)
2. If hemodynamically stable: IV magnesium sulfate 2g push (even if serum Mg normal)
3. Correction of hypokalemia (target K > 4.5 mEq/L) and hypomagnesesia
4. Overdrive pacing (transcutaneous or transvenous) at 90-110 bpm to shorten QT
5. Isoproterenol if pacing unavailable (increase HR to shorten QT)
6. Identify and discontinue all QT-prolonging medications
7. ICD if recurrent Torsades despite medical therapy

10. Treatment Overview

Lifestyle: Avoid competitive sports (especially swimming for LQT1), avoid QT-prolonging medications (check CredibleMeds.org), treat fever promptly (fever prolongs QT - antipyretics), avoid sudden loud noises (LQT2).

Beta-blockers: First-line therapy for all LQTS types. Nadolol preferred (long-acting, non-selective); propranolol or metoprolol alternatives. Reduces cardiac events by ~70%. Lifelong therapy recommended.

ICD: Survivors of cardiac arrest (Class I), recurrent syncope despite beta-blocker (Class IIa), very high-risk profiles (QTc >550 ms with syncope). LQT3 may benefit from ICD earlier due to higher lethality.

LQT3 specific: Mexiletine (sodium channel blocker) shortens QT in LQT3; may be used as adjunct to beta-blocker.

Left cardiac sympathetic denervation (LCSD): For recurrent syncope despite beta-blocker, especially in young patients where ICD complications are high.

11. Follow-up Questions

- Any syncope episodes? What were you doing? (Exercise, emotion, sleep?)
- Any near-drowning episodes (especially while swimming)?
- Any seizures - consider if actually cardiac in origin?
- Family history of sudden death, drowning, or seizures at young age?
- Current medications - any new prescriptions? (Check QT-prolonging potential)
- Any electrolyte abnormalities, vomiting/diarrhea (loss of K, Mg)?
- Genetic testing results for LQTS mutations?
- Has QTc been measured before? What were the values?
- Any history of hearing loss? (Jervell-Lange-Nielsen)

12. References

1. Schwartz PJ, et al. Diagnostic criteria for the long QT syndrome: an update. *Circulation*. 1993;88(2):782-784.
2. Priori SG, et al. HRS/EHRA/APHRS expert consensus statement on inherited primary arrhythmia syndromes. *Europace*. 2013;15(10):1389-1406.
3. Schwartz PJ, et al. Left cardiac sympathetic denervation in the management of high-risk patients affected by the long-QT syndrome. *Circulation*. 2004;109(15):1826-1833.
4. Ackerman MJ, et al. HRS/EHRA expert consensus statement on the state of genetic testing. *Heart Rhythm*. 2011;8(8):1308-1339.

Pulmonary Embolism

Category: Vascular Emergency **Priority: CRITICAL** **ICD-10:** I26.9

1. Definition

Pulmonary embolism (PE) is the obstruction of pulmonary arterial vasculature by thrombus, most commonly originating from deep vein thrombosis (DVT) in the lower extremities (proximal ileofemoral veins in 90% of cases). PE represents the third most common cause of cardiovascular death worldwide and ranges from incidental subsegmental PE to massive PE causing hemodynamic collapse and death.

2. Mechanism

Virchow's triad drives thrombus formation: venous stasis (immobility, post-surgical), endothelial injury (trauma, surgery, central lines), and hypercoagulability (inherited: Factor V Leiden, prothrombin mutation; acquired: malignancy, pregnancy, OCP, APS). Thrombus embolizes to pulmonary circulation causing vascular obstruction, increased pulmonary vascular resistance, right ventricular afterload, and potential RV failure. Massive PE causes hemodynamic collapse through acute RV failure and reduced LV preload. Ventilation-perfusion mismatch causes hypoxemia.

3. ECG Findings

- **Sinus tachycardia:** Most common finding (present in 50%); nonspecific but expected
- **S1Q3T3 pattern:** S wave in lead I, Q wave in lead III, inverted T wave in lead III - classic but present in only 16%; specific when present
- **Right heart strain:** Right axis deviation, T wave inversions V1-V4 (anterior ischemia from RV dilation), incomplete or complete RBBB
- **P pulmonale:** Tall peaked P waves in lead II (right atrial enlargement)
- **Atrial fibrillation:** May occur due to right atrial strain
- **Low voltage QRS in limb leads:** In massive PE
- **Clockwise rotation:** Transition zone shifts to V5-V6
- **ST elevation:** Rarely aVR (right ventricular strain)

4. Required ECG Features

ECG IN PE

No ECG finding is diagnostic of PE. The ECG's primary role is excluding acute MI and assessing RV strain severity. New incomplete RBBB, right axis deviation, or T wave inversions V1-V4 in a patient with dyspnea and risk factors should raise suspicion for PE. The principal value of ECG is **risk stratification** - signs of RV strain indicate submassive or massive PE requiring escalated therapy.

5. Diagnostic Criteria

- CT pulmonary angiography (CTPA): First-line diagnostic test; directly visualizes filling defects
- V/Q scan: For pregnancy, CTPA contraindication, or contrast allergy; high probability scan diagnostic
- D-dimer: ELISA <500 ng/mL (high sensitivity, low specificity) excludes PE in low-intermediate pretest probability (Wells score <=4 or Geneva score <=3)
- Wells score: Clinical probability assessment (PE likely >4, unlikely <=4)
- Modified Geneva score: Alternative clinical probability tool
- Echocardiography: RV dilation, McConnell's sign (akinesia of mid-free wall with preserved apex), elevated PA pressures
- Lower extremity venous ultrasound: DVT evidence supports PE diagnosis

6. Differential Diagnosis

Condition	Differentiating Features
Acute MI	Typical chest pain, ST changes in coronary territory, elevated troponin, no risk factors for DVT
Pneumonia	Fever, productive cough, focal infiltrate on CXR; elevated WBC
Pneumothorax	Sudden unilateral pain, absent breath sounds; CXR diagnostic
Aortic dissection	Tearing pain radiating to back, pulse deficit, widened mediastinum on CXR
Pericarditis	Positional pain, friction rub, diffuse ST elevation, PR depression
Anxiety/hyperventilation	No hypoxemia, normal ECG, paresthesias, no DVT risk factors

7. Symptoms

- **Dyspnea:** Most common symptom; sudden onset, out of proportion to pain
- **Pleuritic chest pain:** Sharp, worse with inspiration (peripheral PE with lung infarction)
- **Tachypnea, tachycardia:** Nearly universal in significant PE
- **Syncope:** Indicates massive PE with hemodynamic compromise
- **Hemoptysis:** Indicates pulmonary infarction (peripheral PE)
- **Unilateral leg swelling/pain:** Signs of DVT (examined in 50% of PE patients)
- **Apprehension, diaphoresis:** Anxiety symptoms from hypoxia
- **Cough:** May be dry or with hemoptysis

8. Risk Factors

- **Strong:** Surgery (especially orthopedic, abdominal within 4 weeks), hospitalization/immobility >3 days, active cancer, prior VTE, lower limb trauma/fracture
- **Moderate:** Age >75, pregnancy/postpartum, estrogen therapy, obesity (BMI >30), central venous catheter
- **Weak:** Long-distance travel (>6 hours), smoking, HTN, COPD
- **Genetic:** Factor V Leiden, prothrombin G20210A, protein C/S deficiency, antithrombin deficiency, antiphospholipid syndrome
- **COVID-19:** Significantly increased risk of thromboembolic events

9. Immediate Actions

EMERGENCY MANAGEMENT

1. Supplemental O2 to maintain SpO2 >90%
2. IV access, continuous monitoring (telemetry, pulse oximetry)
3. Anticoagulation: unfractionated heparin (preferred if considering thrombolysis or embolectomy due to reversibility) or LMWH
4. **Massive PE (shock):** Systemic thrombolysis (tPA 100 mg IV over 2 hours or 0.6 mg/kg bolus) unless contraindicated
5. **Submassive PE (RV strain, normotensive):** Anticoagulation + consider thrombolysis if high-risk features; catheter-directed therapy if bleeding risk
6. **Surgical embolectomy:** For massive PE with contraindication to thrombolysis or failed thrombolysis
7. **Catheter-directed thrombolysis/embolectomy:** For intermediate-high risk with bleeding contraindications
8. Vasopressors (norepinephrine) for hemodynamic support
9. Avoid aggressive volume loading (may worsen RV distension)

10. Treatment Overview

Anticoagulation: Initial parenteral anticoagulation (UFH, enoxaparin, fondaparinux) overlapping with oral anticoagulation. DOACs preferred for most (rivaroxaban, apixaban can be started without parenteral bridge). Warfarin (target INR 2-3) for severe renal impairment, pregnancy, or mechanical valves.

Duration: Provoked PE (surgery, transient risk factor): 3 months. Unprovoked PE: extended therapy (indefinite) if bleeding risk low. Active cancer: LMWH or DOAC for 3-6 months minimum.

Thrombolysis: Massive PE (hemodynamic instability). Systemic tPA preferred. Absolute contraindications: active bleeding, recent hemorrhagic stroke, recent intracranial/spinal surgery, suspected aortic dissection.

Embolectomy: Surgical (bypass circuit) or catheter-based (AngioVac, FlowTriever) for massive PE with failed/contraindicated lysis.

IVC filter: Only if anticoagulation absolutely contraindicated; retrievable filter preferred.

11. Follow-up Questions

- When did symptoms start? Sudden or gradual onset?

- Any recent surgery, hospitalization, or immobilization >3 days?
- Any recent long-distance travel (>6 hours)?
- Any active cancer or cancer treatment?
- Pregnancy or postpartum status?
- Estrogen therapy (OCP, HRT)?
- Prior DVT or PE? Any family history of blood clots?
- Any leg swelling, pain, redness, or warmth?
- Hemoptysis?
- Any recent COVID-19 infection?
- Active bleeding history or bleeding disorder?

12. References

1. Konstantinides SV, et al. 2019 ESC Guidelines for the diagnosis and management of acute pulmonary embolism. *Eur Heart J*. 2020;41(4):543-603.
2. Stevens SM, et al. Guidance for the treatment of VTE and thrombophilia. *J Thromb Thrombolysis*. 2016;41(1):32-47.
3. Kearon C, et al. Antithrombotic therapy for VTE disease: CHEST Guideline. *Chest*. 2016;149(2):315-352.
4. Jaff MR, et al. Management of massive and submassive PE, iliofemoral DVT, and chronic thromboembolic pulmonary hypertension. *Circulation*. 2011;123(16):1788-1830.

Short QT Syndrome

Category: Channelopathy **Priority: CRITICAL** **ICD-10:** I45.81

1. Definition

Short QT syndrome (SQTS) is a rare inherited channelopathy characterized by a shortened QT interval on the ECG (QTc <340 ms diagnostic, 340-360 ms borderline) associated with increased risk of atrial fibrillation, ventricular fibrillation, and sudden cardiac death. It is one of the most lethal channelopathies if untreated but is also the rarest. Unlike long QT syndrome, there is no effective pharmacological therapy; ICD is the mainstay of treatment.

2. Mechanism

Gain-of-function mutations in potassium channels (KCNH2 - SQT1, KCNQ1 - SQT2, KCNJ2 - SQT3) increase repolarizing IKr, IKs, or IK1 currents. Alternatively, loss-of-function mutations in calcium channels (CACNA1C, CACNB2b, CACNA2D1) reduce depolarizing ICa,L. Net effect: abbreviated ventricular action potential duration and effective refractory period, creating a substrate for reentrant atrial and ventricular fibrillation. Shortened atrial refractoriness explains the high incidence of AF.

3. ECG Findings

- **Short QTc:** <340 ms definite, 340-360 ms borderline (use Hodges or Framingham correction for HR >60)
- **T waves:** Tall, peaked, narrow-based (symmetrical, high amplitude)
- **Short or absent ST segment:** T wave follows immediately after S wave
- **Atrial fibrillation:** May be presenting rhythm, especially in young patients with "lone AF"
- **Prominent U waves:** May be present
- **Little or no QT variation with heart rate:** Fails to shorten further with tachycardia
- **Hypercalcemia or hyperthermia:** Exclude acquired causes of short QT

4. Required ECG Features

DIAGNOSTIC CRITERIA (GOLLOB CRITERIA)

- QTc <370 ms: 1 point; <350 ms: 2 points; <330 ms: 3 points
  >Jpoint-Tpeak interval <120 ms: 1 point
- Tall, peaked T waves in precordial leads: 1 point
- Pathogenic mutation: 2 points; family history of SQTS: 2 points; survived cardiac arrest: 2 points
- Score >= 4 = high probability, 3 = intermediate, <3 = low probability

5. Diagnostic Criteria

- Gollob score >= 4: high probability; requires genetic confirmation or strong phenotype
- Genetic testing positive for pathogenic mutation in known SQTS gene confirms diagnosis
- Exclusion of acquired causes: hypercalcemia, hyperkalemia, digitalis effect, hyperthermia, acidosis
- Family history of sudden death or documented SQTS supports diagnosis
- Absence of structural heart disease

6. Differential Diagnosis

Condition	Differentiating Features
Hypercalcemia	Elevated serum calcium; tall, peaked T waves; corrects with calcium normalization
Digitalis effect	Sagging ST depression, not true short QT; therapeutic drug level; clinical context
Hyperkalemia	Peaked T waves, QRS widening; elevated potassium; correctable
Early repolarization	Concave ST elevation, notching; normal QTc; no arrhythmic risk
Normal variant (athlete)	QTc may be 340-360; no family history; no events; no mutation

7. Symptoms

- **Syncope:** Often first manifestation of ventricular arrhythmia
- **Palpitations:** Due to AF or ventricular arrhythmias
- **Resuscitated cardiac arrest:** VF is often the presenting event

- **Atrial fibrillation at young age:** May be the only symptom; SQTS should be considered in young patients with "lone AF"
- **Many patients asymptomatic:** Family screening identifies carriers

8. Risk Factors

- **Genetic:** Autosomal dominant; KCNH2 (SQT1), KCNQ1 (SQT2), KCNJ2 (SQT3), CACNA1C, CACNB2b
- **Family history:** Sudden cardiac death or documented SQTS
- **Prior cardiac arrest:** Highest risk for recurrence
- **QTc <330 ms:** Higher risk than 330-340 ms
- **Young age at presentation:** Earlier onset suggests more severe phenotype
- **Acquired:** Hypercalcemia, hyperthermia can transiently shorten QT

9. Immediate Actions

> **EMERGENCY MANAGEMENT**
>
> 1. ACLS protocol if cardiac arrest from VF
> 2. Immediate defibrillation for VF/pulseless VT
> 3. Quinidine (oral load 600-1000 mg) may prolong QT and prevent recurrence
> 4. Correction of any electrolyte abnormalities (hypercalcemia, hyperkalemia)
> 5. Emergent electrophysiology consultation

10. Treatment Overview

ICD implantation: Definitive therapy for all symptomatic patients (survivors of cardiac arrest, documented VF, syncope with VT). Consider for asymptomatic with QTc <330 ms and family history of SCD.

Quinidine: Prolongs QT by blocking IKr; used as adjunct to ICD to reduce shock frequency. May be used alone if ICD contraindicated or declined. Target QTc 360-380 ms.

Hydroquinidine: Alternative with similar mechanism; better tolerated by some patients.

Disopyramide: Alternative antiarrhythmic for QT prolongation.

Sotalol and amiodarone: Generally ineffective and may paradoxically shorten QT further.

Family screening: All first-degree relatives should undergo ECG and genetic testing.

11. Follow-up Questions

- Any syncope or fainting episodes?
- Any history of cardiac arrest or resuscitation?
- Any palpitations, especially irregular heartbeat?
- Family history of sudden death at young age?
- Has anyone in the family been diagnosed with short QT syndrome?
- Any history of atrial fibrillation, especially at young age?
- Any electrolyte abnormalities, especially high calcium?
- Any medication use, especially antiarrhythmics or digoxin?

12. References

1. Gollob MH, et al. Recommendations for the use of genetic testing in the clinical evaluation of inherited cardiac arrhythmias associated with sudden cardiac death. *Can J Cardiol*. 2011;27(2):215-220.
2. Giustetto C, et al. Long-term follow-up of patients with short QT syndrome. *J Am Coll Cardiol*. 2011;58(6):587-595.
3. Mazzanti A, et al. Quinidine prevents arrhythmic events in short QT syndrome. *J Am Coll Cardiol*. 2017;70(24):3010-3015.
4. Priori SG, et al. HRS/EHRA/APHRS expert consensus statement on inherited primary arrhythmia syndromes. *Europace*. 2013;15(10):1389-1406.

Spontaneous Coronary Artery Dissection (SCAD)

Category: Coronary Artery Disease **Priority: CRITICAL** **ICD-10:** I25.4

1. Definition

Spontaneous coronary artery dissection (SCAD) is a non-atherosclerotic, non-iatrogenic, non-traumatic cause of acute coronary syndrome caused by separation of coronary arterial wall layers creating an intramural hematoma that compresses the true lumen, leading to myocardial ischemia or infarction. SCAD is the most common cause of MI in women under 50 years and is increasingly recognized with advancing coronary imaging techniques. It accounts for up to 4% of all ACS cases and 35% of MIs in women <50.

2. Mechanism

Two proposed mechanisms: (1) Intimal tear allowing blood to enter media and create false lumen (classic dissection); (2) Rupture of vasa vasorum causing intramural hematoma without intimal tear (more common). The hemorrhage within the vessel wall compresses the true lumen, reducing distal perfusion. Associated with: coronary artery tortuosity, extracoronary vascular abnormalities (FMD, aneurysms), hormonal factors (estrogen/progesterone affect vessel wall integrity), and connective tissue disorders. Unlike atherosclerotic ACS, SCAD does not involve plaque rupture and typically has a benign natural history with spontaneous healing.

3. ECG Findings

- **ST elevation:** In territory of dissected vessel (STEMI pattern) - present in ~50%
- **ST depression, T wave inversion:** NSTEMI pattern - present in ~40%
- **ECG patterns same as conventional ACS:** Cannot distinguish SCAD from atherosclerotic ACS on ECG alone
- **May involve distal vessels:** More common in SCAD than atherosclerotic ACS
- **May involve less typical territories:** Inferior and distal LAD most common

4. Required ECG Features

ECG IN SCAD

No ECG feature is specific for SCAD. The ECG changes mirror those of conventional ACS. The key differentiator is **patient demographics and context** - SCAD should be suspected in: young/middle-aged women (<60 years), peripartum period, absence of traditional risk factors, and minimal/absent coronary disease in other territories on angiography.

5. Diagnostic Criteria

- Coronary angiography: Definitive diagnosis; classified by Saw angiographic classification
- Type 1: Pathognomonic - arterial wall staining with contrast hang-up, multiple radiolucent lumens
- Type 2: Most common (~70%) - diffuse, smooth stenosis (mild <50%, severe >50%) often extending beyond typical atherosclerotic distribution
- Type 3: Focal or tubular stenosis mimicking atherosclerosis (<20%); requires intracoronary imaging (OCT/IVUS) to confirm
- Type 4: Total occlusion, typically in distal vessels
- OCT (optical coherence tomography): Gold standard for confirmation - identifies intimal tear, intramural hematoma, double lumen
- IVUS: Alternative intracoronary imaging modality

6. Differential Diagnosis

Condition	Differentiating Features
Atherosclerotic ACS	Older patients, traditional risk factors, plaque rupture on angiography, proximal/mid vessel involvement
Vasospastic (Prinzmetal) angina	Transient ST elevation at rest; responds to nitrates; acetylcholine challenge positive; no dissection on angiography
Coronary embolism	AF, prosthetic valve, infective endocarditis as source; filling defect on angiography
Takotsubo cardiomyopathy	Postmenopausal woman, emotional trigger; apical ballooning; no coronary obstruction
Myocarditis	Younger, viral prodrome, diffuse ST changes, elevated troponin with non-obstructive coronaries
Coronary artery vasculitis	Systemic autoimmune disease, involvement of multiple vascular territories

7. Symptoms

- **ACS symptoms:** Chest pain (pressure, tightness, burning), dyspnea, diaphoresis, nausea
- **SCAD-triggering stressors:** Emotional stress, intense physical exercise, Valsalva maneuvers
- **Postpartum status:** Often occurs within first few weeks postpartum
- **Absence of exertional angina history:** Many patients have no prior cardiac symptoms

- **Associated symptoms:** Migraine history (associated with vascular abnormalities), anxiety/depression (common)

8. Risk Factors

- **Female sex:** 90% of SCAD cases occur in women
- **Age:** Predominantly 45-55 years (perimenopausal)
- **Pregnancy/postpartum:** Accounts for ~20% of SCAD cases; highest risk in third trimester and early postpartum
- **Emotional stress:** Major life stressors, grief, anxiety
- **Physical stress:** Heavy lifting, intense exercise, Valsalva (retching, coughing)
- **Fibromuscular dysplasia (FMD):** Present in up to 50-80% of SCAD patients; screen with CT/MR angiography
- **Connective tissue disorders:** Marfan, EDS, Loeys-Dietz (minority of cases)
- **Hormonal therapy:** Oral contraceptives, fertility treatment, hormone replacement
- **Systemic inflammation:** SLE, Crohn's disease
- **Absence of traditional CAD risk factors** is characteristic

9. Immediate Actions

EMERGENCY MANAGEMENT

1. Standard ACS protocol: aspirin, anticoagulation as per institutional guidelines
2. Clopidogrel (preferred over ticagrelor/prasugrel due to lower bleeding risk; avoid prasugrel)
3. Avoid aggressive anticoagulation if possible (theoretical risk of extending dissection)
4. **Conservative management preferred:** If hemodynamically stable, TIMI 2-3 flow - observe with medical management; most heal spontaneously
5. **PCI if:** Ongoing ischemia, hemodynamic instability, ventricular arrhythmias, total/progressive occlusion
6. **CABG:** Reserved for left main dissection, failed PCI, or multivessel involvement with ongoing ischemia
7. **Special considerations:** Avoid routine IABP (may extend dissection); stenting technically challenging due to vessel fragility

10. Treatment Overview

Conservative (preferred): Aspirin indefinitely, clopidogrel 12 months if stented or high-risk, beta-blockers (reduce vascular shear stress, may prevent recurrence), ACE-I if LV dysfunction, statin (controversial, benefit in non-atherosclerotic disease unclear). Avoid strenuous exercise and heavy lifting for 4-6 weeks.

Revascularization: Only if ongoing/recurrent ischemia, hemodynamic/electrical instability. PCI technically difficult due to fragile vessels, long lesion length, and involvement of distal segments. Stent malapposition common as hematoma resorbs. CABG for left main or failed PCI.

Secondary prevention: Screen for FMD (CT/MR angiography of head, neck, abdomen, pelvis), avoid hormonal therapy, cardiac rehabilitation (modified), beta-blockers indefinitely (evidence suggests reduced recurrence), treat anxiety/depression.

Recurrence: 10-20% recurrence rate over 10 years; highest risk in first year.

11. Follow-up Questions

- What were you doing when symptoms started? (Emotional stress, physical exertion?)
- Are you pregnant or recently postpartum?
- Any history of fibromuscular dysplasia, aneurysms, or arterial dissections?
- Any history of migraines?
- Any connective tissue disorder (Marfan, Ehlers-Danlos)?
- Any hormonal therapy - birth control, fertility treatment, HRT?
- Family history of arterial dissections or sudden death?
- Any recent major life stressors?
- What is your exercise pattern? Any heavy lifting or straining?

12. References

1. Saw J, et al. Spontaneous coronary artery dissection: clinical outcomes and risk of recurrence. *J Am Coll Cardiol*. 2017;70(9):1148-1158.

2. Hayes SN, et al. Spontaneous coronary artery dissection: current state of the science. *Circulation*. 2018;137(5):523-535.

3. Adlam D, et al. European Society of Cardiology, acute cardiovascular care association, SCAD study group. *Eur Heart J*. 2018;39(36):3354-3368.

4. Saw J, et al. Spontaneous coronary artery dissection: association with predisposing arteriopathies and precipitating stressors. *J Am Coll Cardiol*. 2015;67(14):1713-1722.

Stroke

Category: Cerebrovascular **Priority: CRITICAL** **ICD-10:** I64

1. Definition

Stroke is an acute neurological deficit caused by disruption of blood supply to the brain, classified as ischemic (85% - thrombotic or embolic arterial occlusion) or hemorrhagic (15% - intracerebral hemorrhage or subarachnoid hemorrhage). Time-critical diagnosis and treatment are essential; "Time is brain" - approximately 1.9 million neurons are lost per minute of untreated ischemic stroke.

2. Mechanism

Ischemic stroke: Large vessel atherosclerosis (internal carotid, vertebral, basilar), cardioembolism (AF most common, 20-25% of ischemic strokes), small vessel lacunar disease (hypertensive arteriolosclerosis), cryptogenic (embolic stroke of undetermined source - ESUS), dissection, hypercoagulable states. **Hemorrhagic stroke:** Hypertensive intraparenchymal hemorrhage (basal ganglia, thalamus, pons, cerebellum), cerebral amyloid angiopathy (lobar hemorrhage in elderly), anticoagulant-related bleeding, vascular malformations (AVM, cavernous malformation), aneurysmal subarachnoid hemorrhage.

3. ECG Findings

- **Atrial fibrillation:** Cause of 20-25% of ischemic strokes; irregularly irregular rhythm, absent P waves
- **Sick sinus syndrome:** With embolism risk
- **PFO/ASD:** Paradoxical embolism source; may have incomplete RBBB
- **Diffuse giant T wave inversion:** Acute CNS event (subarachnoid hemorrhage, ICH) causing autonomic-mediated repolarization changes
- **QT prolongation:** Autonomic changes from intracranial pathology
- **ST elevation or depression:** Neurogenic pattern from acute CNS insult (Takotsubo-like changes)
- **U waves:** May be prominent
- **Bradycardia or tachycardia:** Autonomic dysregulation

4. Required ECG Features

ECG ASSESSMENT IN STROKE

No specific ECG finding is required for stroke diagnosis (neuroimaging is definitive). However, ECG is mandatory to identify stroke etiology: AF detection guides anticoagulation, recent MI patterns suggest cardioembolic source, and acute ST-T changes may indicate neurogenic heart injury requiring cardiac evaluation.

5. Diagnostic Criteria

- Clinical: Sudden onset focal neurological deficit consistent with a vascular territory
- CT brain: Excludes hemorrhage (first-line), demonstrates early ischemic changes
- MRI with DWI: Gold standard for ischemic stroke detection; hyperintense within minutes
- NIH Stroke Scale (NIHSS): Quantifies stroke severity; guides treatment decisions
- TOAST classification: Large artery atherosclerosis, cardioembolism, small vessel occlusion, other determined cause, undetermined
- CT angiography or MRA: Identifies large vessel occlusion (LVO) amenable to thrombectomy

6. Differential Diagnosis

Condition	Differentiating Features
TIA	Same mechanism but symptoms resolve within 24 hours (usually <1 hour); tissue not infarcted
Seizure with Todd's paralysis	History of epilepsy; postictal state; symptoms resolve over hours; EEG changes
Hypoglycemia	Symptoms resolve with glucose correction; sweating, tremulousness; check bedside glucose
Complicated migraine	History of migraine; positive symptoms (scintillations) preceding deficit; gradual onset
Conversion disorder	Inconsistent examination; non-anatomical deficit; psychiatric history
Brain tumor	Subacute progressive course; seizures; mass effect on imaging

7. Symptoms

- **BEFAST mnemonic:** Balance loss, Eyes (vision change), Face droop, Arm weakness, Speech difficulty, Time to call emergency
- **Sudden onset:** Maximal at onset ("thunderclap" for SAH); unlike migraine's crescendo
- **Focal deficits:** Unilateral weakness/numbness, aphasia (expressive, receptive, global), dysarthria
- **Visual:** Hemianopia, diplopia, gaze deviation
- **Vestibular:** Vertigo, ataxia (posterior circulation)
- **Altered consciousness:** Large territory stroke, brainstem involvement, hemorrhage with mass effect
- **"Worst headache of life"** - subarachnoid hemorrhage
- **Nausea/vomiting, neck stiffness:** Increased ICP or meningeal irritation

8. Risk Factors

- **Non-modifiable:** Age (>55, doubles each decade), male sex, family history, race (Black > Hispanic > White), prior stroke/TIA
- **Vascular:** Hypertension (strongest modifiable risk), diabetes, dyslipidemia, smoking, AF, CAD, PAD
- **Lifestyle:** Obesity, physical inactivity, excessive alcohol, diet (high sodium, low potassium)
- **Cardiac:** AF, valvular disease (rheumatic mitral stenosis, prosthetic valves), cardiomyopathy with EF <30%, PFO
- **Other:** OSA, chronic kidney disease, hypercoagulable states, sickle cell disease

9. Immediate Actions

EMERGENCY MANAGEMENT

1. Immediate non-contrast CT brain to exclude hemorrhage (<20 minutes from arrival)
2. CT angiography (head/neck) to identify large vessel occlusion
3. **Ischemic stroke within 4.5 hours:** IV thrombolysis (tPA 0.9 mg/kg) if no contraindications
4. **Large vessel occlusion:** Mechanical thrombectomy (up to 24 hours in selected patients per DAWN/DEFUSE 3 criteria)
5. BP management: Allow permissive hypertension up to 220/120 pre-treatment; lower to <185/110 if thrombolysis planned
6. Aspirin 325 mg within 24-48 hours (if not thrombolysed; delay 24 hours post-tPA)
7. Airway protection if decreased consciousness, aspiration risk
8. Reverse anticoagulation if hemorrhagic stroke (PCC for warfarin, idarucizumab for dabigatran, andexanet for factor Xa inhibitors)

10. Treatment Overview

Acute ischemic: IV thrombolysis (alteplase, tenecteplase) within 4.5 hours; mechanical thrombectomy for LVO (internal carotid, M1, basilar) up to 24 hours. Aspirin 81-325 mg daily. Management in stroke unit. Prevent complications (DVT prophylaxis, aspiration precautions).

Secondary prevention: Antiplatelet (aspirin 81 mg + clopidogrel 75 mg for 21-90 days for minor stroke/TIA per CHANCE/POINT), statin (high-intensity), BP control (target <130/80), diabetes management (HbA1c <7%), smoking cessation, anticoagulation for AF (direct oral anticoagulants preferred over warfarin).

Carotid stenosis: CEA or stenting for symptomatic 70-99% stenosis within 2 weeks of event.

Hemorrhagic: BP control (target SBP <140 if safe), reverse anticoagulation, neurosurgical consultation for cerebellar hemorrhage (>3 cm), IVH, hydrocephalus, or deteriorating clinical status. Minimally invasive surgery for supratentorial ICH (MISTIE) in selected patients.

Rehabilitation: Early mobilization, PT/OT/speech therapy, dysphagia screening before oral intake.

11. Follow-up Questions

- When were you last seen normal? (Last known well time - critical for treatment window)
- Were you awake or asleep when symptoms started? (Wake-up stroke needs advanced imaging)

- Any prior stroke or TIA?
- History of atrial fibrillation, valvular disease, or heart failure?
- Blood pressure history? Diabetes? High cholesterol? Smoking?
- Current medications, especially anticoagulants, antiplatelets?
- Recent surgery, trauma, or bleeding?
- Any headache at onset? (Suggests hemorrhage or arterial dissection)
- Family history of stroke?
- Alcohol or drug use? (Cocaine associated with both hemorrhagic and ischemic stroke)

12. References

1. Powers WJ, et al. 2019 Update to the 2018 Guidelines for the Early Management of Acute Ischemic Stroke. *Stroke*. 2019;50(12):e344-e418.
2. Hemphill JC 3rd, et al. Guidelines for the Management of Spontaneous Intracerebral Hemorrhage. *Stroke*. 2015;46(7):2032-2060.
3. Nogueira RG, et al. Thrombectomy 6 to 24 hours after stroke with a mismatch between deficit and infarct. *N Engl J Med*. 2018;378(1):11-21.
4. Kleindorfer DO, et al. 2021 Guideline for the Prevention of Stroke in Patients With Stroke and TIA. *Stroke*. 2021;52(7):e364-e467.

Sudden Cardiac Arrest

Category: Cardiac Emergency **Priority: CRITICAL** **ICD-10:** I46.9

1. Definition

Sudden cardiac arrest (SCA) is the abrupt cessation of cardiac function with loss of effective systemic circulation, causing loss of consciousness and absence of palpable pulses. If not immediately reversed, it leads to sudden cardiac death (SCD). SCA is most commonly caused by ventricular fibrillation (VF) or pulseless ventricular tachycardia (VT), but may also result from pulseless electrical activity (PEA) or asystole. It accounts for approximately 15-20% of all natural deaths in developed countries.

2. Mechanism

Ventricular fibrillation: chaotic, disorganized ventricular electrical activity causing no meaningful cardiac output. Pulseless VT: organized ventricular rhythm at rapid rate with inadequate filling time. Both are "shockable" rhythms amenable to defibrillation. PEA: organized electrical activity on ECG without palpable pulse (mechanical failure); causes include hypovolemia, tamponade, tension pneumothorax, massive PE, MI, acidosis. Asystole: absence of ventricular electrical activity; flat line. Non-shockable rhythms carry much poorer prognosis. Underlying substrates include coronary artery disease (80% of SCD in adults >35 years), cardiomyopathies, channelopathies, and structural congenital abnormalities.

3. ECG Findings

- **Ventricular fibrillation:** Chaotic, disorganized electrical activity; no identifiable P waves, QRS complexes, or T waves; coarse VF (amplitude >3 mm) or fine VF (amplitude <3 mm - poorer prognosis)
- **Pulseless VT:** Wide QRS tachycardia (>120 ms), rate usually >150 bpm; monomorphic or polymorphic
- **PEA:** Organized electrical rhythm on ECG (sinus, atrial, junctional, or ventricular) with no palpable pulse
- **Asystole:** Flat line; must confirm in multiple leads (may fine VF masquerade as asystole)
- **Agonal rhythm:** Very slow, wide QRS complexes immediately preceding asystole

4. Required ECG Features

RHYTHM IDENTIFICATION

- **Shockable rhythms (treat immediately with defibrillation):** VF and pulseless VT
- **Non-shockable rhythms (CPR + epinephrine):** PEA and asystole
- Confirm rhythm in more than one lead to exclude artifact or fine VF
- Torsades de Pointes (polymorphic VT with twisting QRS axis) - treat with magnesium + defibrillation if unstable

5. Diagnostic Criteria

- Clinical: Sudden collapse, unresponsive, absent carotid/femoral pulse, agonal or absent breathing
- ECG: Confirms rhythm (VF, VT, PEA, asystole)
- Time-based definitions: SCD within 1 hour of symptom onset (witnessed) or within 24 hours of last being seen alive (unwitnessed)
- H's and T's: Systematic approach to reversible causes (Hypovolemia, Hypoxia, Hydrogen ion/acidosis, Hyper/Hypokalemia, Hypothermia, Tension pneumothorax, Tamponade, Toxins, Thrombosis - coronary and pulmonary)

6. Differential Diagnosis

Condition	Differentiating Features
Syncope	Brief loss of consciousness with spontaneous recovery; palpable pulse returns
Seizure	Tonic-clonic movements, postictal confusion; usually palpable pulse present
Stroke	Focal neurological deficit, usually alert if anterior circulation; vital signs typically present
Drug overdose	Pinpoint pupils (opioids), track marks, medication bottles nearby; may have respiratory arrest before cardiac
Hypoglycemia	Check bedside glucose; altered mental status, sweating; resolves with glucose

7. Symptoms (Preceding Events)

- **Witnessed:** Sudden collapse, often with gasping (agonal respirations)
- **Prodromal symptoms (hours to weeks before):** Chest pain, dyspnea, palpitations, syncope, fatigue

- **No warning:** Especially in channelopathies (Brugada, LQTS) and cardiomyopathies (HCM)
- **Seizure-like activity:** Cerebral hypoperfusion can cause myoclonic jerks

8. Risk Factors

- **Coronary artery disease:** 80% of SCD in adults; prior MI with reduced EF (<35%) highest risk
- **Cardiomyopathy:** Dilated (EF <35%), hypertrophic, arrhythmogenic right ventricular
- **Channelopathies:** Brugada, LQTS, SQTS, CPVT
- **Valvular disease:** Aortic stenosis (exertional syncope/SCD), mitral valve prolapse (rare)
- **Congenital:** Tetralogy of Fallot, transposition, coronary artery anomalies
- **Electrolyte:** Severe hypokalemia, hypomagnesemia
- **Drugs:** Proarrhythmic antiarrhythmics, QT-prolonging drugs, stimulants (cocaine)
- **Family history:** SCD <50 years suggests inherited condition

9. Immediate Actions

ACLS PROTOCOL

1. Activate emergency response; confirm unresponsiveness and absence of normal breathing/pulse (<10 seconds)
2. Immediate high-quality CPR: rate 100-120/min, depth 5-6 cm, full recoil, minimize interruptions (<10 seconds)
3. Defibrillate immediately if shockable rhythm (VF/pulseless VT): biphasic 200J, monophasic 360J
4. 2 minutes CPR between rhythm checks; defibrillate after each 2-minute cycle if still shockable
5. Advanced airway when feasible (do not interrupt CPR >10 seconds)
6. IV/IO epinephrine 1 mg every 3-5 minutes
7. Amiodarone 300 mg IV/IO for refractory VF/VT, then 150 mg if needed
8. Treat reversible causes (H's and T's)
9. Post-resuscitation care: targeted temperature management (32-36°C for 24 hours), coronary angiography if suspected cardiac etiology, hemodynamic optimization

10. Treatment Overview

Immediate: ACLS per AHA guidelines. Defibrillation is the single most important intervention for shockable rhythms; every minute without defibrillation reduces survival by 7-10%. Public access defibrillation programs improve outcomes.

Post-arrest care: Targeted temperature management for comatose survivors. Emergency coronary angiography for ST-elevation or suspected cardiac cause. Hemodynamic optimization (MAP >65-80). Neurological prognostication delayed until >72 hours after rewarming.

Secondary prevention: ICD implantation is Class I for survivors of SCA from reversible causes (unless completely reversible cause identified and eliminated, e.g., MI with complete revascularization and no recurrent VT on EP testing). Workup for underlying cause: coronary angiography, echocardiography, cardiac MRI, EP study, genetic testing.

Out-of-hospital SCA chain of survival: Immediate recognition + activation, early CPR, rapid defibrillation, effective ACLS, integrated post-arrest care.

11. Follow-up Questions

- Was the collapse witnessed? What time did it occur?
- How long until CPR started? How long until first defibrillation?
- What was the initial rhythm? (VF/VT vs. PEA/asystole)
- Any prodromal symptoms in hours/days before collapse?
- History of heart disease, prior MI, or reduced ejection fraction?
- Family history of sudden death, drowning, or seizures at young age?
- Current medications, especially antiarrhythmics or QT-prolonging drugs?
- Any recent substance use (cocaine, stimulants)?
- Electrolyte abnormalities?
- What is the neurologic status post-resuscitation?

12. References

1. Priori SG, et al. 2015 ESC Guidelines for the management of patients with ventricular arrhythmias. *Eur Heart J*. 2015;36(41):2793-2867.
2. Panchal AR, et al. Part 3: Adult Basic and Advanced Life Support: 2020 AHA Guidelines for CPR and ECC. *Circulation*. 2020;142(16_suppl_2):S366-S468.
3. Al-Khatib SM, et al. 2017 AHA/ACC/HRS Guideline for Management of Patients With Ventricular Arrhythmias. *Circulation*. 2018;138(13):e210-e271.
4. Stub D, et al. Refractory cardiac arrest treated with mechanical CPR, hypothermia, ECMO and early reperfusion (the CHEER trial). *Resuscitation*. 2015;86:88-94.

Ventricular Tachycardia

Category: Arrhythmia **Priority: CRITICAL** **ICD-10:** I47.2

1. Definition

Ventricular tachycardia (VT) is a wide-complex tachycardia (>120 ms) originating in the ventricles, defined as three or more consecutive ventricular beats at a rate >100 bpm (typically 150-200 bpm). Sustained VT lasts >30 seconds or causes hemodynamic compromise requiring intervention. Non-sustained VT (NSVT) is 3-30 consecutive beats. VT can be monomorphic (uniform QRS morphology) or polymorphic (varying QRS morphology). It is a life-threatening arrhythmia that may degenerate into ventricular fibrillation.

2. Mechanism

Reentry: Most common mechanism; scar tissue from prior MI creates anatomical or functional conduction block with slow conduction pathway allowing circus movement. Late potentials on signal-averaged ECG identify the substrate. **Enhanced automaticity:** Abnormal spontaneous depolarization of ventricular cells, often in ischemic tissue. **Triggered activity:** Early afterdepolarizations (phase 2 reentry in Brugada, long QT) or delayed afterdepolarizations (digoxin toxicity, catecholaminergic polymorphic VT). Monomorphic VT usually has a reentrant mechanism; polymorphic VT may be due to acute ischemia, channelopathies, or proarrhythmia.

3. ECG Findings

- **Rate:** 100-250 bpm (typical 150-200 bpm)
- **QRS duration:** >= 120 ms (typically >140 ms)
- **Bizarre QRS morphology:** Not typical of bundle branch block pattern
- **AV dissociation:** P waves marching through at their own rate, unrelated to QRS complexes
- **Fusion beats:** Hybrid QRS morphology (partial ventricular activation from supraventricular impulse) - diagnostic of VT
- **Capture beats:** Narrow QRS from conducted sinus beat within VT - diagnostic of VT
- **Negative precordial concordance:** All precordial leads show negative QRS - strongly suggests VT
- **Positive precordial concordance:** All precordial leads positive - VT (rarely WPW with antidromic tachycardia)
- **Brugada criteria:** VT vs. SVT with aberrancy algorithm (AV dissociation, QRS >140 ms, morphology criteria in aVR and precordial leads)

4. Required ECG Features

DIAGNOSTIC FEATURES (VT VS. SVT WITH ABERRANCY)

- AV dissociation = diagnostic of VT (do not need other criteria)
- Capture or fusion beats = diagnostic of VT
- Extreme axis deviation (northwest quadrant, positive aVR) favors VT
- No RS complex in any precordial lead favors VT
- Initial R wave in aVR > 40 ms, or notching on descending limb of S wave in aVR favors VT
- When in doubt: treat as VT (safer approach)

5. Diagnostic Criteria

- ECG: Wide complex tachycardia with QRS >120 ms at rate >100 bpm
- Clinical: Palpitations, syncope, hypotension, chest pain, or cardiac arrest
- Monomorphic VT: Uniform QRS morphology throughout; suggests reentry around fixed substrate
- Polymorphic VT: Changing QRS axis/amplitude; suggests acute ischemia, channelopathy, or proarrhythmia
- Torsades de Pointes: Polymorphic VT with QRS twisting around baseline, QT prolongation
- Bidirectional VT: Alternating QRS axis (frontal plane ±180°); diagnostic of CPVT or digoxin toxicity
- EP study: Confirms diagnosis, mechanism, and guides ablation

6. Differential Diagnosis

Condition	Differentiating Features
SVT with aberrancy	Typical RBBB or LBBB pattern; QRS <140 ms; rate <200; AV relationship; Valsalva/adenosine responsive
Antidromic AVRT (WPW)	Short RP tachycardia; delta wave visible in sinus rhythm; pre-excited AF can mimic polymorphic VT
Hyperkalemia	Very wide QRS, peaked T waves, sine wave pattern; history of renal failure; corrects with calcium/glucose/insulin
Pacing rhythm	Pacing spikes visible; history of pacemaker/ICD
Artifact	Not captured in all leads simultaneously; examine multiple leads; patient may be stable

7. Symptoms

- **Palpitations:** Rapid, regular heartbeat
- **Dyspnea, chest pain:** Due to decreased cardiac output and myocardial ischemia
- **Syncope or presyncope:** Especially with fast rates or onset of VT
- **Hypotension:** Hemodynamic instability when rate >170 or in reduced EF
- **Cardiac arrest:** If VT degenerates to VF or sustained at very rapid rate
- **Asymptomatic (NSVT):** Detected on monitoring or ICD interrogation

8. Risk Factors

- **Structural heart disease:** Prior MI with scar (most common), DCM, HCM, ARVC, sarcoidosis
- **Reduced EF:** <35% strongest predictor of sudden death from VT/VF
- **Channelopathies:** Brugada, LQTS, CPVT, SQTS
- **Electrolyte abnormalities:** Hypokalemia, hypomagnesemia, hypocalcemia
- **Drug toxicity:** Digoxin, antiarrhythmics (class IA, IC), QT-prolonging drugs
- **Acute ischemia:** Can trigger polymorphic VT
- **Family history:** Inherited arrhythmia syndromes

9. Immediate Actions

EMERGENCY MANAGEMENT

1. Assess hemodynamic stability: altered mental status, hypotension (SBP <90), signs of shock, ischemic chest pain, acute heart failure
2. **Unstable VT:** Immediate synchronized cardioversion (100-200J biphasic, 200-360J monophasic)
3. **Stable monomorphic VT:** IV antiarrhythmic - amiodarone 150 mg IV over 10 minutes (repeat as needed, max 2.2 g/24h); procainamide or sotalol alternatives
4. **Polymorphic VT (QT normal):** Cardioversion if unstable; IV magnesium, beta-blockers; treat ischemia
5. **Torsades de Pointes:** IV magnesium 2 g; correct K to >4.5, Mg >2.0; overdrive pacing; cardioversion if unstable
6. **Digoxin-induced bidirectional VT:** Digoxin-specific antibody fragments (Digibind)
7. Prepare for cardioversion if any deterioration

10. Treatment Overview

Acute termination: Unstable = cardioversion. Stable monomorphic VT = amiodarone, procainamide, or sotalol. Polymorphic without QT prolongation = magnesium + beta-blockers + treat ischemia. Torsades = magnesium + correct electrolytes + overdrive pacing.

Chronic therapy: ICD for secondary prevention (survived cardiac arrest, sustained VT with syncope). Primary prevention ICD for ischemic cardiomyopathy with EF <35% despite optimal medical therapy and >40 days post-MI.

Antiarrhythmics: Sotalol, amiodarone, or mexiletine as adjunct to ICD for recurrent VT reducing shock burden. Avoid class IA and IC in structural heart disease.

Catheter ablation: Endocardial (and epicardial if needed) ablation of VT substrate. Class I for recurrent monomorphic VT despite antiarrhythmics. Standalone therapy for idiopathic VT (RVOT, fascicular VT).

Revascularization: If VT due to acute ischemia; reduces recurrence risk.

11. Follow-up Questions

- Rate the palpitations - how fast? Regular or irregular?
- Any syncope or near-syncope during episodes?
- History of heart disease, prior MI, or known reduced ejection fraction?
- Family history of sudden death or cardiomyopathy?

- Current medications, especially antiarrhythmics, digoxin?
- Any electrolyte abnormalities, renal dysfunction?
- Any ICD? When was it last interrogated?
- Substance use - cocaine, amphetamines?
- Triggers - exercise, stress, rest?

12. References

1. Al-Khatib SM, et al. 2017 AHA/ACC/HRS Guideline for Management of Patients With Ventricular Arrhythmias. *Circulation*. 2018;138(13):e210-e271.
2. Priori SG, et al. 2015 ESC Guidelines for the management of patients with ventricular arrhythmias. *Eur Heart J*. 2015;36(41):2793-2867.
3. Brugada P, et al. A new approach to the differential diagnosis of a regular tachycardia with a wide QRS complex. *Circulation*. 1991;83(5):1649-1659.
4. Vereckei A, et al. New algorithm using only lead aVR for differential diagnosis of wide QRS complex tachycardia. *Heart Rhythm*. 2008;5(1):89-98.

4. High Priority Conditions

Serious conditions requiring prompt diagnosis and management to prevent complications

Angina

Category: Coronary Artery Disease **Priority: HIGH** **ICD-10:** I20.9

1. Definition

Angina pectoris is chest pain or discomfort caused by myocardial ischemia due to imbalance between myocardial oxygen supply and demand, most commonly from epicardial coronary artery stenosis. Stable angina is predictable with exertion and relieved by rest. Unstable angina is new-onset, crescendo, or rest pain without biomarker elevation - a medical emergency representing acute coronary syndrome. Variant (Prinzmetal) angina is caused by coronary artery spasm.

2. Mechanism

Stable angina: Fixed atherosclerotic stenosis (>70% of luminal diameter) limits coronary flow reserve. During exertion, increased myocardial oxygen demand cannot be met by increased supply through the stenotic vessel. Unstable angina: Plaque rupture or erosion with non-occlusive thrombus causes dynamic, unpredictable obstruction. Vasospastic (Prinzmetal) angina: Hyperreactivity of vascular smooth muscle causing focal coronary spasm, often at sites of minimal atherosclerosis; associated with smoking, cocaine, and altered autonomic tone.

3. ECG Findings

- **Stable angina:** Normal between episodes or nonspecific ST-T changes; ST depression >= 1 mm during chest pain or stress testing
- **Stress test positive:** Horizontal or downsloping ST depression >= 1 mm at 80 ms after J-point, persisting >0.08 seconds
- **Unstable angina:** ST depression, T wave inversion at rest; dynamic changes (worsening with pain, improving with rest)
- **Prinzmetal angina:** Transient ST elevation during rest pain (coronary spasm); resolves spontaneously or with nitrates within minutes
- **Microvascular angina:** ST depression during stress with non-obstructive coronaries (INOCA)
- **Persistent ST-T changes after pain resolution:** Suggests recent ischemia or non-ST-elevation ACS

4. Required ECG Features

ECG DIAGNOSTIC FEATURES

- Exercise-induced ST depression >= 1 mm (horizontal/downsloping) in 2+ contiguous leads = positive stress test
- Resting ST depression or T wave inversion that normalizes during ischemia ("pseudonormalization") is a high-risk finding
- Transient ST elevation during rest pain that resolves with nitrates = Prinzmetal angina (document with ambulatory ECG)

5. Diagnostic Criteria

- Clinical: Characteristic chest pain (retrosternal pressure/tightness, provoked by exertion, relieved by rest or nitrates within 5 minutes)
- CCS classification: I (only strenuous activity), II (moderate activity), III (minimal activity), IV (rest pain = unstable)
- Stress testing: Treadmill ECG, stress echo, or nuclear perfusion imaging demonstrates reversible ischemia
- Coronary CTA: Non-invasive anatomic assessment; CAC scoring for risk stratification
- Invasive coronary angiography: Gold standard for anatomic evaluation; defines stenosis severity and guides revascularization
- FFR (fractional flow reserve): <0.80 confirms hemodynamic significance of intermediate lesions (40-70%)

6. Differential Diagnosis

Condition	Differentiating Features
MI/NSTEMI	Similar pain but biomarker elevation; ECG changes at rest without exertion; unrelieved by rest/nitroglycerin
GERD/esophageal spasm	Burning pain, relation to meals, relief with antacids; may improve with nitrates (smooth muscle relaxation)
Costochondritis	Sharp, localized, reproducible chest wall tenderness
Pericarditis	Sharp pleuritic pain, diffuse ST elevation, pericardial friction rub, positional
Pulmonary embolism	Pleuritic pain, dyspnea, tachycardia, risk factors for DVT
Anxiety/panic attack	Chest tightness with hyperventilation, paresthesias; normal ECG; exclusion diagnosis

7. Symptoms

- **Stable angina:** Pressure, squeezing, heaviness, burning or tightness in substernal region
- **Provocation:** Exertion, emotional stress, cold exposure, heavy meals
- **Duration:** 2-15 minutes; <20 minutes in stable angina
- **Relief:** Rest or sublingual nitroglycerin within 5 minutes
- **Radiation:** Left arm (ulnar), jaw, neck, epigastrium, interscapular
- **Unstable angina:** New onset (within 2 months), crescendo (increasing frequency/severity), rest pain (>20 min)
- **Prinzmetal:** Pain at rest, typically midnight-early morning, cyclical, associated with arrhythmias

8. Risk Factors

- Traditional: Age, male sex, smoking, hypertension, diabetes, dyslipidemia, obesity, sedentary lifestyle, family history
- Emerging: Lp(a), CKD, inflammatory diseases, OSA, psychosocial stress
- Prinzmetal-specific: Smoking (most important), cocaine, amphetamines, magnesium deficiency

9. Immediate Actions

UNSTABLE ANGINA = MEDICAL EMERGENCY

1. Aspirin 325 mg if not taken; clopidogrel loading
2. Nitroglycerin 0.4 mg SL q5min x 3 for pain (avoid if SBP <90)
3. Anticoagulation: heparin or enoxaparin
4. Beta-blocker if no contraindications
5. High-intensity statin
6. Early invasive strategy if high-risk (GRACE >140, dynamic ECG changes, elevated troponins - then becomes NSTEMI)

10. Treatment Overview

Medical: Anti-ischemic (beta-blockers first-line, nitrates for symptom relief, CCB if contraindication to beta-blockers or vasospastic component), antiplatelet (aspirin lifelong, add P2Y12 if ACS), statin (high-intensity), ACE-I/ARB (especially diabetes or CKD). Antianginal add-ons: ranolazine, ivabradine (if sinus rhythm), nicorandil.

Revascularization: PCI or CABG for significant stenosis with symptoms despite OMT or high-risk anatomy (left main, proximal LAD, 3-vessel disease). CABG preferred for diabetes with multivessel disease, complex anatomy (SYNTAX score >22).

Prinzmetal: CCBs (amlodipine, diltiazem) first-line; nitrates for acute attacks; high-dose CCB often needed; smoking cessation essential.

11. Follow-up Questions

- What brings on the pain? How long does it last? What relieves it?
- CCS class I-IV? Has the pattern changed recently?
- Risk factors: smoking, diabetes, BP, cholesterol, family history?
- Prior stress test, angiography, or revascularization?
- Current antianginal medications and response?
- Rest pain or nocturnal symptoms? (suggests vasospastic component)

12. References

1. Fihn SD, et al. 2012 ACCF/AHA/ACP/AATS/PCNA/SCAI/STS Guideline for the diagnosis and management of stable ischemic heart disease. *Circulation*. 2012;126(25):e354-e471.
2. Knuuti J, et al. 2019 ESC Guidelines for the diagnosis and management of chronic coronary syndromes. *Eur Heart J*. 2020;41(3):407-477.

3. Levine GN, et al. 2015 ACC/AHA/SCAI Focused Update on Primary PCI for Patients With STEMI. *Circulation*. 2016;133(11):1135-1147.

Aortic Stenosis

Category: Valvular Heart Disease **Priority: HIGH** **ICD-10:** I35.0

1. Definition

Aortic stenosis (AS) is narrowing of the aortic valve orifice causing obstruction of left ventricular outflow, increased afterload, and pressure overload hypertrophy of the left ventricle. It is the most common valvular heart disease in developed countries. Severity graded by valve area and hemodynamics: mild (Vmax 2-2.9 m/s, mean gradient <20 mmHg, AVA >1.5 cm^2), moderate (Vmax 3-3.9 m/s, mean gradient 20-40 mmHg, AVA 1.0-1.5 cm^2), severe (Vmax >=4 m/s, mean gradient >=40 mmHg, AVA <1.0 cm^2 or indexed AVA <0.6 cm^2/m^2). Affects >6% of adults >75 years.

2. Mechanism

Calcific/degenerative AS (most common in adults >65): Progressive leaflet calcification similar to atherosclerosis - endothelial injury, lipid accumulation, inflammation, osteoblastic transformation of valvular interstitial cells. Rheumatic AS: commissural fusion with fusion of leaflets. Congenital bicuspid AS: abnormal turbulent flow causes accelerated calcification (presents 20-30 years earlier than tricuspid). LV pressure overload leads to concentric hypertrophy (wall thickening without dilation), which initially compensates but eventually leads to diastolic dysfunction, myocardial ischemia (supply-demand mismatch), and heart failure.

3. ECG Findings

- **LVH voltage:** Sokolow-Lyon (S V1 + R V5/V6 > 35 mm) or Cornell criteria (R aVL + S V3 > 28 mm men, > 20 mm women)
- **Strain pattern:** ST depression with T wave inversion in lateral leads (I, aVL, V5-V6) with downsloping ST segment - indicates severe AS with significant LVH
- **LBBB:** May develop due to calcification extending into conduction system or septal hypertrophy
- **Complete heart block:** If calcification involves the septum near the bundle of His
- **Left atrial enlargement:** Secondary to elevated LVEDP; P mitrale (broad, notched P in II), biphasic P in V1 with prominent negative terminal deflection
- **AF:** May precipitate acute decompensation in severe AS (loss of atrial kick critically reduces filling)

4. Required ECG Features

> **ECG IN AS**
>
> No single ECG finding is diagnostic of AS. LVH with strain pattern suggests severe AS but requires echocardiographic confirmation. New-onset LBBB or heart block in elderly patients with appropriate clinical context should prompt AS evaluation. AF in a patient with known AS is a warning sign for potential hemodynamic decompensation.

5. Diagnostic Criteria

- **Echocardiography (definitive):** Valve morphology, degree of calcification, transvalvular velocity (Vmax), mean gradient, valve area (continuity equation)
- **Severe AS criteria:** Vmax >= 4 m/s, mean gradient >= 40 mmHg, AVA < 1.0 cm^2 (all three do not need to be met simultaneously)
- **Low-flow low-gradient severe AS:** AVA < 1.0 cm^2 with low gradient (<40 mmHg) and low flow (SVI < 35 mL/m^2); dobutamine stress echo to distinguish true severe AS from pseudo-severe AS
- **CT calcium scoring:** Agatston score > 2000 (men) or > 1200 (women) suggests severe AS when echo is inconclusive
- **Cardiac catheterization:** Direct gradient measurement when non-invasive imaging is discordant with clinical findings

6. Differential Diagnosis

Condition	Differentiating Features
Aortic sclerosis	Valve thickening/calcification without significant obstruction; Vmax < 2.5 m/s; normal gradients
HOCM	Dynamic LVOT obstruction; systolic anterior motion of mitral valve; Vmax varies with maneuvers; family history
Supravalvular AS	Congenital; narrowing above valve; associated with Williams syndrome; different location on echo
Subvalvular AS	Membrane or tunnel below valve; continuous wave Doppler shows late-peaking dagger-shaped jet
Pulmonary hypertension	RV heave, loud P2, different murmur location; echo shows elevated PA pressures

7. Symptoms

- **Exertional dyspnea:** Earliest symptom; due to elevated LV filling pressures and reduced cardiac output
- **Exertional syncope:** Classic presentation; vasodepressor reflex from high LV pressure stimulating baroreceptors, or arrhythmia
- **Exertional angina:** Supply-demand mismatch due to LVH increasing demand and compression of intramural vessels reducing supply
- **Heart failure symptoms:** Orthopnea, PND, peripheral edema (late sign of decompensation)
- **Paradoxical low-flow, low-gradient AS:** Severe symptoms with lower-than-expected gradients in patients with reduced LVEF

8. Risk Factors

- **Age:** >65 years (degenerative calcification)
- **Bicuspid aortic valve:** Most common congenital heart defect (1-2% of population); presents 20 years earlier
- **Atherosclerotic risk factors:** Smoking, hypertension, hyperlipidemia, diabetes (similar risk profile but statins do not slow progression)
- **Male sex:** Higher incidence but women have worse outcomes after AVR
- **Chronic kidney disease:** Accelerated calcification
- **Radiation therapy:** To chest (breast cancer, Hodgkin lymphoma)
- **Rheumatic heart disease:** In developing countries

9. Immediate Actions

ACUTE DECOMPENSATION

1. If pulmonary edema: cautious diuresis (do not over-diurese - preload dependent)
2. If AF with RVR: rate control (avoid vasodilators); urgent cardioversion if unstable
3. Avoid vasodilators (nitrates, ACE-I) and inotropic agents (may cause hypotension)
4. Maintain sinus rhythm if possible (atrial kick essential for filling)
5. Emergent surgical or transcatheter AVR if decompensated severe AS

10. Treatment Overview

Asymptomatic severe AS: Watchful waiting with serial echocardiography (every 6 months). Consider early intervention if: very severe (Vmax > 5 m/s), rapid progression, markedly elevated BNP, or exercise-induced symptoms.

Symptomatic severe AS: Aortic valve replacement is indicated (Class I) - symptom onset without intervention carries 50% 2-year mortality. No medical therapy improves survival in symptomatic AS.

SAVR vs. TAVR: TAVR preferred for >75 years or high/prohibitive surgical risk. SAVR preferred for <65 years, low surgical risk, or bicuspid anatomy (evolving - select TAVR centers now performing bicuspid TAVR). Shared decision-making for 65-75 years.

Medical: No medications slow progression. Manage hypertension carefully (avoid excessive afterload reduction). Diuretics for volume overload. Statins have no proven benefit for AS progression.

11. Follow-up Questions

- Any chest discomfort with activity? Syncope? Shortness of breath?
- When did symptoms first appear? How have they progressed?
- Known bicuspid aortic valve? Prior echo results?
- Blood pressure history? Current medications?
- Exercise tolerance? How far can you walk? Climb stairs?
- Any episodes of rapid heartbeat?

12. References

1. Otto CM, et al. 2020 ACC/AHA Guideline for the Management of Patients With Valvular Heart Disease. *Circulation*. 2021;143(5):e72-e227.
2. Vahanian A, et al. 2021 ESC/EACTS Guidelines for the management of valvular heart disease. *Eur Heart J*. 2022;43(7):561-632.
3. Otto CM, Prendergast B. Aortic-valve stenosis - from patients at risk to severe valve obstruction. *N Engl J Med*. 2014;371(8):744-756.

Atrial Fibrillation

Category: Arrhythmia **Priority: HIGH** **ICD-10:** I48.2

1. Definition

Atrial fibrillation (AF) is a supraventricular tachyarrhythmia characterized by disorganized atrial electrical activity with loss of coordinated atrial contraction. The ECG shows irregularly irregular ventricular response with absent P waves, replaced by fine or coarse fibrillatory waves. AF is the most common sustained arrhythmia in adults, affecting approximately 3-6% of the population over 65 years and 10% over 80. It is associated with a 5-fold increased risk of stroke and 2-fold increased risk of all-cause mortality.

2. Mechanism

Multiple mechanisms: ectopic focal triggers (typically from pulmonary vein sleeves with enhanced automaticity and triggered activity) initiate AF, while structural and electrical atrial remodeling maintains it. "AF begets AF" - sustained AF causes progressive atrial dilation and fibrosis, promoting reentry circuits. Key mechanisms include: triggered activity from pulmonary vein myocardial sleeves, multiple wavelet reentry (maintaining AF), rotors (mother reentrant circuits), and autonomic ganglion plexus modulation. Risk factors promote atrial fibrosis and electrical instability.

3. ECG Findings

- **Irregularly irregular R-R intervals:** No pattern to ventricular response
- **Absent distinct P waves:** Baseline shows fine (low amplitude, <1 mm) or coarse (high amplitude, >1 mm) fibrillatory waves (f waves)
- **Atrial rate:** 350-600 bpm (chaotic, disorganized atrial activity - not visible as discrete waves)
- **Ventricular rate:** Controlled 60-100 bpm, rapid 100-180 bpm (with AV nodal conduction), slow <60 bpm (with AV nodal disease or drugs)
- **Narrow QRS:** <120 ms unless pre-existing bundle branch block or aberrant conduction
- **Ashman phenomenon:** Aberrantly conducted beats after long-short R-R cycle (wide QRS mimicking VT)
- **Regularization:** Suggests complete heart block with junctional escape, or digitalis toxicity

4. Required ECG Features

DIAGNOSTIC CRITERIA

- Irregularly irregular ventricular rhythm with no discernible P waves = diagnostic of AF
- Duration >30 seconds on rhythm strip or 12-lead ECG establishes diagnosis
- AF with RVR: ventricular rate >100-110 bpm requires rate control
- Coarse AF (f waves >1 mm amplitude): suggests mitral valve disease or recent-onset AF

5. Diagnostic Criteria

- ECG confirmation: Any ECG demonstrating AF establishes diagnosis
- Classification: First diagnosed (first detection regardless of duration or symptoms), Paroxysmal (self-terminating within 7 days), Persistent (>7 days or requiring cardioversion), Long-standing persistent (>12 months when rhythm control strategy adopted), Permanent (accepted as ongoing rhythm strategy)
- CHA_2DS_2-VASc score: Stroke risk stratification (Congestive heart failure 1, Hypertension 1, Age >=75 2, Diabetes 1, Stroke/TIA/thromboembolism 2, Vascular disease 1, Age 65-74 1, Sex category female 1)
- HADS-BLED score: Bleeding risk assessment on anticoagulation
- EHRA symptom score: I (none), IIa (mild), IIb (moderate), III (severe), IV (disabling)

6. Differential Diagnosis

Condition	Differentiating Features
Atrial flutter	Sawtooth flutter waves; regular atrial activity 250-350 bpm; usually regular ventricular response
Multifocal atrial tachycardia (MAT)	Irregular rhythm with >=3 distinct P wave morphologies; rate >100; usually in COPD
Sinus arrhythmia	Normal P waves before each QRS; phasic variation with respiration; normal finding
Frequent PACs	Irregularity but normal P waves between premature beats; isolated premature atrial contractions
Junctional tachycardia	Regular narrow complex tachycardia; retrograde P waves or no visible P waves; not irregularly irregular

7. Symptoms

- **Palpitations:** Irregular, chaotic heartbeat sensation; "fluttering" in chest
- **Dyspnea on exertion:** Reduced cardiac output (loss of atrial contribution, rapid rate reducing diastolic filling)
- **Fatigue, reduced exercise tolerance:** Most common symptom; often subtle
- **Chest discomfort:** Not anginal; may be due to rapid rate
- **Dizziness, lightheadedness:** From irregular rhythm or pauses
- **Syncope:** If rapid rate or prolonged pauses on conversion
- **Asymptomatic:** 10-40% of patients; detected incidentally; still carries stroke risk

8. Risk Factors

- **Age:** Prevalence doubles each decade after 50; 10% over 80
- **Hypertension:** Most common modifiable risk factor (causes atrial remodeling)
- **Structural heart disease:** Valvular disease (especially mitral), HF, CAD, cardiomyopathy
- **Obesity:** BMI >30 (increases risk 1.5-2 fold; weight loss reduces burden)
- **OSA:** Strong independent risk factor; CPAP treatment reduces recurrence
- **Alcohol:** Binge drinking ("holiday heart syndrome") and chronic heavy use
- **Hyperthyroidism:** Especially in elderly; check TSH in new-onset AF
- **Diabetes, CKD:** Metabolic risk factors
- **Genetic:** Family history increases risk 1.4-fold; GWAS identified multiple loci
- **Athletes:** Vagotonic AF in endurance athletes; resolves with deconditioning

9. Immediate Actions

ACUTE AF MANAGEMENT

1. Assess hemodynamic stability: If unstable (hypotension, ongoing ischemia, pulmonary edema) - urgent synchronized cardioversion
2. If stable and duration <48 hours: consider cardioversion (electrical or pharmacological - flecainide, propafenone, or amiodarone)
3. If stable and duration >48 hours (or unknown): Rate control + therapeutic anticoagulation for 3 weeks before cardioversion, or TEE-guided cardioversion if earlier rhythm restoration needed
4. Rate control: Beta-blockers (metoprolol, esmolol) or non-DHP CCBs (diltiazem, verapamil); digoxin if hypotensive or HFrEF
5. Anticoagulation: Initiate based on CHA_2DS_2-VASc score (>= 2 men, >= 3 women: oral anticoagulation indicated)
6. Identify and treat triggers: infection, electrolyte imbalance, thyrotoxicosis, alcohol

10. Treatment Overview

Rhythm control: Antiarrhythmics (flecainide, propafenone for structurally normal hearts; amiodarone, sotalol, dofetilide for structural heart disease; dronedarone for non-permanent AF without HFrEF). Catheter ablation (pulmonary vein isolation): first-line for symptomatic paroxysmal AF (Class I), recurrent symptomatic persistent AF, and selected patients with HFrEF (improves outcomes including mortality in CASTLE-AF).

Rate control: Beta-blockers or non-DHP CCBs first-line. Lenient rate control (<110 bpm at rest) acceptable for most; strict rate control (<80 bpm) if symptoms persist or LV dysfunction develops. AV nodal ablation + pacemaker for refractory cases.

Stroke prevention: DOACs preferred over warfarin for non-valvular AF (apixaban, rivaroxaban, dabigatran, edoxaban). Warfarin (INR 2-3) for mechanical valves and moderate-severe rheumatic mitral stenosis. LAA occlusion (Watchman) for contraindication to long-term anticoagulation.

Lifestyle: Weight loss (10% if BMI >27), BP control, alcohol moderation, OSA treatment, regular exercise, diabetes management (LEGACY study showed weight loss reduces AF burden).

11. Follow-up Questions

- When did you first notice the irregular heartbeat? Sudden or gradual onset?
- Any symptoms: palpitations, shortness of breath, fatigue, chest discomfort, dizziness?

- Does it start and stop on its own (paroxysmal) or is it constant (persistent)?
- Any history of hypertension, heart failure, valvular disease?
- Any prior stroke or TIA?
- Thyroid symptoms? Alcohol use? Sleep apnea symptoms?
- CHA_2DS_2-VASc score calculation: heart failure, hypertension, age, diabetes, stroke history, vascular disease?
- Bleeding risk factors: prior bleed, liver/kidney disease, alcohol, fall risk?
- Current medications and renal function (for DOAC dosing)?

12. References

1. Hindricks G, et al. 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation. *Eur Heart J*. 2021;42(5):373-498.
2. January CT, et al. 2019 AHA/ACC/HRS Focused Update of the 2014 Guideline for the Management of Patients With Atrial Fibrillation. *Circulation*. 2019;140(2):e125-e151.
3. Marrouche NF, et al. Catheter ablation for atrial fibrillation with heart failure. *N Engl J Med*. 2018;378(5):417-427.
4. Pathak RK, et al. Long-term effect of goal-directed weight management in an atrial fibrillation cohort. *J Am Coll Cardiol*. 2015;65(20):2159-2169.

Atrial Flutter

Category: Arrhythmia **Priority: HIGH** **ICD-10:** I48.3

1. Definition

Atrial flutter is a regular supraventricular tachycardia characterized by rapid, organized atrial depolarization at approximately 250-350 bpm (typically ~300 bpm), producing a distinctive sawtooth pattern of flutter (F) waves on the ECG. Unlike the chaotic atrial activity of atrial fibrillation, flutter represents a single reentrant macro-circuit, most commonly in the right atrium around the tricuspid annulus (cavotricuspid isthmus-dependent or "typical" flutter). It shares the same stroke risk as AF and often coexists or alternates with AF.

2. Mechanism

Typical atrial flutter: A macro-reentrant circuit rotating around the tricuspid annulus in a counterclockwise (caudad-to-cranial in the interatrial septum, cranial-to-caudad in the right atrial free wall) or clockwise direction. The cavotricuspid isthmus (CTI) - the region between the inferior vena cava and tricuspid annulus - is a critical conduction corridor and the target for ablation. Atypical flutters involve different circuits (left atrial, scar-related, post-surgical) and are not CTI-dependent. Atrial enlargement, fibrosis, and inflammation provide the substrate for reentry.

3. ECG Findings

- **Sawtooth flutter waves:** Regular, continuous atrial undulations at 250-350 bpm; best seen in II, III, aVF
- **Typical counterclockwise flutter:** Negative sawtooth flutter waves in II, III, aVF (inverted); positive in V1
- **Clockwise flutter:** Positive flutter waves in II, III, aVF; negative in V1 (less common)
- **Regular ventricular response:** With fixed AV block (2:1 most common = ventricular rate ~150 bpm; 4:1 = ~75 bpm)
- **Variable AV block:** Irregular ventricular response if AV conduction varies
- **Typical 2:1 block:** Atrial rate 300, ventricular 150 - every other flutter wave buried in QRS or T wave
- **4:1 block:** Suggests AV nodal disease, medication effect, or hypervagotonia
- **Narrow QRS:** <120 ms unless BBB or aberrancy

4. Required ECG Features

DIAGNOSTIC CRITERIA

- Regular atrial activity at 250-350 bpm with sawtooth morphology
- Negative flutter waves in inferior leads (II, III, aVF) with positive waves in V1 = typical counterclockwise CTI-dependent flutter
- Ventricular rate ~150 bpm with regular narrow complex rhythm should always prompt careful evaluation for 2:1 flutter
- Carotid sinus massage or adenosine can transiently increase AV block to reveal flutter waves

5. Diagnostic Criteria

- ECG: Characteristic sawtooth atrial activity at 250-350 bpm
- Typical (CTI-dependent): Counterclockwise or clockwise rotation around tricuspid annulus; ablatable via CTI line
- Atypical: Other macro-reentrant circuits (left atrial flutter, scar-related); requires electroanatomic mapping
- Adenosine or carotid massage: Transient AV block reveals underlying flutter waves
- Rate ~150 bpm with regular narrow QRS: Always consider 2:1 flutter until proven otherwise

6. Differential Diagnosis

Condition	Differentiating Features
Atrial fibrillation	Irregularly irregular rhythm; chaotic baseline without organized atrial activity; no sawtooth pattern
Sinus tachycardia	Normal P waves preceding each QRS; rate usually <160 in adults; varies with activity
AVNRT/AVRT	No visible atrial activity or retrograde P waves; abrupt onset/offset; rate 150-250
Atrial tachycardia	Discrete P waves with different morphology than sinus; rate usually 150-250; may have AV block

7. Symptoms

- **Palpitations:** Regular, rapid heartbeat (vs. irregular in AF)
- **Exercise intolerance, dyspnea:** Due to rapid ventricular rate and loss of atrial transport
- **Chest discomfort:** Rate-related, not anginal

- **Presyncope:** Especially with 1:1 conduction (very rapid rates) or upon conversion with pauses
- **Syncope:** If very rapid ventricular response or prolonged post-conversion pause
- **Asymptomatic:** Often in elderly or sedentary patients; detected incidentally

8. Risk Factors

- Same as AF: hypertension, structural heart disease, COPD (especially), obesity, OSA, alcohol
- Post-cardiac surgery: Atrial incisions create scar substrate (within 1-5 days post-op)
- Prior ablation for AF: Creates atrial scar enabling atypical flutter circuits
- Chronic lung disease: Right atrial dilation from pulmonary hypertension facilitates typical flutter

9. Immediate Actions

ACUTE MANAGEMENT

1. If unstable (hypotension, ischemia, pulmonary edema): Synchronized cardioversion (50-100 J typically effective)
2. If stable with rapid rate: Rate control with beta-blocker or non-DHP CCB; avoid AV nodal blockers if pre-excited (WPW)
3. If stable and desire rhythm control: Consider anticoagulation status; if >48 hours or unknown, need 3 weeks anticoagulation before cardioversion or TEE-guided approach
4. Adenosine: NOT for termination, but can aid diagnosis by unmasking flutter waves
5. Anticoagulation: Same criteria as AF (CHA_2DS_2-VASc)

10. Treatment Overview

Catheter ablation (first-line): CTI ablation for typical flutter has >95% success rate and very low recurrence; Class I recommendation. Should be offered to all symptomatic patients and considered even after first episode.

Rate control: Beta-blockers or non-DHP CCBs if ablation declined or atypical flutter not amenable to simple ablation. Digoxin as add-on.

Antiarrhythmics: Flecainide, propafenone, sotalol, amiodarone if ablation not available/declined; less effective than for AF; Class III agents preferred.

Stroke prevention: Same CHA_2DS_2-VASc criteria as AF; DOACs or warfarin based on risk score.

11. Follow-up Questions

- Regular or irregular heartbeat sensation?

- Onset: sudden or gradual? Duration of episodes?
- Any precipitating factors: exercise, alcohol, illness?
- History of cardiac surgery or prior ablation?
- COPD or other lung disease?
- CHA_2DS_2-VASc score components?

12. References

1. January CT, et al. 2019 AHA/ACC/HRS Focused Update for AF. *Circulation*. 2019;140(2):e125-e151.
2. Almendral J, et al. The electrocardiographic characteristics of atrial flutter. *Europace*. 2020;22(Supplement_3):iii13-iii22.

Bradycardia

Category: Arrhythmia **Priority: HIGH** **ICD-10:** R00.1

1. Definition

Bradycardia is defined as a heart rate <60 beats per minute in adults. It can be physiological (in trained athletes, during sleep, or with vagal tone) or pathological. Pathological bradycardia causes symptoms due to inadequate cardiac output (cerebral hypoperfusion, exercise intolerance) or pauses that predispose to escape rhythms or asystole. It encompasses sinus node dysfunction (sinus bradycardia, sinus arrest, sinoatrial exit block) and atrioventricular conduction disorders (first-degree AV block, second-degree AV block Mobitz I and II, third-degree AV block).

2. Mechanism

Sinus node dysfunction (sick sinus syndrome): Degeneration and fibrosis of sinus node and surrounding atrial tissue (age-related), autonomic imbalance (excessive vagal tone), or channelopathy (SCN5A, HCN4 mutations). Can cause inappropriate sinus bradycardia, sinus arrest, chronotropic incompetence (failure to increase HR with exertion), and bradycardia-tachycardia syndrome (alternating bradycardia with SVT/AF). **AV block:** First-degree (delayed conduction through AV node); Mobitz I (Wenckebach) - progressive PR prolongation until dropped beat, usually AV nodal; Mobitz II - intermittent non-conducted P waves without PR prolongation, usually infra-Hisian (below AV node); Third-degree - complete AV dissociation, escape rhythm (junctional 40-60 bpm or ventricular 20-40 bpm).

3. ECG Findings

- **Sinus bradycardia:** Normal P waves, regular rhythm, rate <60 bpm; severe <50 bpm, critical <40 bpm
- **Sinus arrest:** Absence of P waves for >3 seconds; escape beats may follow
- **Junctional rhythm:** Absent or retrograde P waves (inverted in II, III, aVF); rate 40-60 bpm
- **Idioventricular rhythm:** Wide QRS complexes, rate 20-40 bpm, AV dissociation
- **First-degree AV block:** PR interval >200 ms with 1:1 AV conduction
- **Mobitz I (Wenckebach):** Progressive PR prolongation until dropped QRS, then cycle repeats; grouped beating
- **Mobitz II:** Fixed PR interval with intermittent non-conducted P waves; more ominous
- **Third-degree AV block:** Complete AV dissociation; P waves march through at regular rate independent of QRS

- **Bradycardia-tachycardia syndrome:** Alternating episodes of AF/flutter with slow sinus rhythm or pauses

4. Required ECG Features

ECG CRITERIA BY TYPE

- **Symptomatic bradycardia:** HR <50 bpm with symptoms of hypoperfusion (syncope, fatigue, exercise intolerance) OR pauses >3 seconds
- **Chronotropic incompetence:** Failure to achieve 80% predicted HR with peak exercise
- **Mobitz II or high-grade AV block:** Any degree of second-degree AV block with infra-Hisian block (wide QRS) or multiple consecutive dropped beats
- **Third-degree AV block:** Complete AV dissociation regardless of rate or symptoms (requires pacing unless reversible)

5. Diagnostic Criteria

- ECG: Confirms rhythm and mechanism
- Symptomatic bradycardia: Correlation of symptoms with ECG-documented bradyarrhythmia (symptom-rhythm correlation via event monitor, loop recorder, or implantable monitor)
- Exercise testing: Chronotropic incompetence; failure to increase rate appropriately
- Atropine challenge (0.04 mg/kg IV): Failure to increase HR by >25% suggests intrinsic sinus node dysfunction
- Electrophysiology study: If non-invasive testing inconclusive; SNRT (sinus node recovery time) >1500 ms abnormal; HV interval >70 ms suggests infra-Hisian disease

6. Differential Diagnosis

Condition	Differentiating Features
Athlete's heart	Rate 40-50 normal; increases appropriately with exercise; asymptomatic; no pauses
Vagal syncope	Situational trigger, bradycardia only during episode, normal baseline rhythm
Medication effect	Beta-blockers, CCBs, digoxin, antiarrhythmics; resolves with dose reduction/discontinuation
Hypothyroidism	Elevated TSH; dry skin, fatigue, weight gain; improves with thyroid replacement
Obstructive sleep apnea	Nocturnal bradycardia; daytime somnolence; snoring; improves with CPAP
Hypothermia	Core temp <35°C; Osborn J waves; shivering; rewarmin

7. Symptoms

- **Fatigue, weakness:** Reduced cardiac output
- **Dizziness, lightheadedness:** Cerebral hypoperfusion
- **Syncope or near-syncope:** Especially with pauses >3 seconds
- **Exercise intolerance:** Chronotropic incompetence - cannot increase HR with exertion
- **Dyspnea:** Rate insufficient for metabolic demand
- **Symptom correlation with HR:** Symptoms worse when HR is low, improve when rate increases

8. Risk Factors

- Age >70 (degenerative fibrosis of conduction system - Lev's disease, Lenègre's disease)
- Medications: Beta-blockers, non-DHP CCBs, digoxin, amiodarone, sotalol
- Ischemic heart disease (especially inferior MI affecting AV node, anterior MI affecting bundle branches)
- Infiltrative diseases: Amyloidosis, sarcoidosis, hemochromatosis
- Collagen vascular diseases: Scleroderma, SLE, rheumatoid arthritis
- Genetic: SCN5A mutations (Lenègre-Lev disease), HCN4 mutations
- Electrolyte: Hyperkalemia, hypothyroidism
- Infectious: Lyme disease, endocarditis (abscess), Chagas disease
- Congenital: Congenital AV block (maternal lupus anti-Ro/SSA)

9. Immediate Actions

ACUTE MANAGEMENT

1. If unstable (hypotension, altered mental status, signs of shock, ischemic chest pain, acute heart failure): Atropine 0.5 mg IV q3-5 min (max 3 mg); if ineffective, transcutaneous pacing or transvenous pacing
2. Dopamine (5-20 mcg/kg/min) or epinephrine (2-10 mcg/min) if atropine ineffective
3. Transcutaneous pacing: Prepare immediately for any high-grade AV block or symptomatic bradycardia
4. Transvenous pacing if temporary pacing needed for > short period
5. Identify reversible causes: medications, electrolytes, ischemia, hypoxia, hypothyroidism
6. Stop offending medications if possible

10. Treatment Overview

Pacemaker indications (Class I): Symptomatic sinus bradycardia, chronotropic incompetence, sinus node dysfunction with symptomatic bradycardia, third-degree AV block (any level), Mobitz II second-degree AV block, alternating bundle branch block, bifascicular block with syncope and HV >70 ms on EP study, post-MI bilateral BBB or Mobitz II, carotid sinus hypersensitivity with pauses >3 seconds.

Pacemaker type: Sinus node dysfunction without AV block: AAIR. AV block: DDDR. Rate response for chronotropic incompetence. Leadless pacemakers available for AV block (single-chamber VVI).

Bradycardia-tachycardia syndrome: Pacemaker required before starting rate-slowing drugs for tachyarrhythmias; otherwise drugs worsen bradycardia.

Reversible causes: Inferior MI (often transient - observe), medication effect, Lyme disease (antibiotics), hyperkalemia, hypothyroidism. Treat underlying cause when identified.

11. Follow-up Questions

- Any fainting, near-fainting, or dizziness? Especially with position changes?
- Exercise tolerance - able to increase heart rate with activity?
- Current medications - beta-blockers, calcium channel blockers, digoxin?
- Any prior MI or known conduction disease?
- Symptoms of hypothyroidism?
- Any recent Lyme disease or tick exposure?

- History of collagen vascular disease?
- Syncope with carotid sinus stimulation (shaving, tight collars)?

12. References

1. Kusumoto FM, et al. 2018 ACC/AHA/HRS Guideline on the Evaluation and Management of Patients With Bradycardia and Cardiac Conduction Delay. *Circulation*. 2019;140(8):e382-e482.
2. Epstein AE, et al. 2012 ACCF/AHA/HRS Focused Update Incorporated Into the ACCF/AHA/HRS 2008 Guidelines for Device-Based Therapy of Cardiac Rhythm Abnormalities. *Circulation*. 2013;127(3):e283-e352.

Broken Heart Syndrome (Takotsubo Cardiomyopathy)

Category: Cardiomyopathy **Priority: HIGH** **ICD-10:** I51.81

1. Definition

Takotsubo cardiomyopathy (TCM), also called stress-induced cardiomyopathy or "broken heart syndrome," is an acute, transient cardiac syndrome characterized by transient regional systolic dysfunction of the left ventricle, mimicking acute MI, but with absence of obstructive coronary artery disease. Named for the Japanese octopus fishing pot (takotsubo) that resembles the apical ballooning shape seen on ventriculography. It predominantly affects postmenopausal women (90% of cases) and is typically triggered by intense emotional or physical stress.

2. Mechanism

The leading hypothesis is catecholamine-mediated myocardial stunning. Intense stress triggers massive sympathetic activation with circulating catecholamine surge (epinephrine, norepinephrine), causing direct myocyte toxicity, microvascular spasm, and abnormal calcium handling in cardiomyocytes. The apical LV is particularly vulnerable due to higher density of beta-adrenergic receptors and thinner wall stress. Multiple variants described: apical (most common, 80%), mid-ventricular, basal (inverted), focal, and biventricular. Transient nature distinguishes it from MI - typically recovers within days to weeks.

3. ECG Findings

- **ST elevation:** In anterior leads (V2-V5 most common), often more pronounced than typical anterior STEMI
- **Typically without reciprocal ST depression:** Unlike STEMI, reciprocal changes are usually absent
- **Deep, widespread T wave inversion:** Apical and mid-ventricular distribution; evolves over days 2-4
- **Markedly prolonged QT interval:** Often the most prominent finding in subacute phase; risk of Torsades
- **Transient small Q waves:** May appear in anterior leads but resolve with recovery
- **Evolution pattern:** ST elevation (acute) → T wave inversion → QT prolongation → complete resolution over 2-8 weeks
- **Lower voltage ST elevation** compared to STEMI in some cases

4. Required ECG Features

DIAGNOSTIC CLUES ON ECG

- Anterior ST elevation WITHOUT reciprocal changes in inferior leads strongly suggests Takotsubo over anterior STEMI
- ST elevation in aVR + widespread ST depression (diffuse ischemia pattern) can occur
- Prominent QT prolongation (>500 ms) in the subacute phase is characteristic
- Q waves if present are transient (unlike MI where they typically persist)

5. Diagnostic Criteria

- Mayo Clinic Criteria (2008, revised 2018): All four required: (1) Transient LV regional wall motion abnormality (akinesis/dyskinesis) extending beyond single epicardial vascular territory; (2) Absence of obstructive coronary disease or acute plaque rupture; (3) New ECG abnormalities (ST elevation and/or T wave inversion) or elevated troponins; (4) Absence of pheochromocytoma or myocarditis
- InterTAK Diagnostic Score: Age, sex, emotional trigger, absence of ST depression (except aVR), psychiatric/neurological disorder, QTc prolongation
- Echo/CMR: Apical ballooning with basal hypercontractility; no LGE on CMR (differentiates from MI scar)
- Elevated troponins (modest elevation relative to extent of wall motion abnormality - troponin modestly elevated despite extensive LV dysfunction)
- Elevated BNP/NT-proBNP disproportionate to troponin

6. Differential Diagnosis

Condition	Differentiating Features
Anterior STEMI	Obstructive culprit lesion on angiography; reciprocal changes; troponin rise matches wall motion extent
Myocarditis	Younger, viral prodrome, diffuse/global dysfunction; endomyocardial biopsy or CMR LGE patterns
Pheochromocytoma	Hypertension, episodic headaches, sweating; elevated catecholamines/metanephrines; excluded from TCM diagnosis
Seizure-related (neurogenic stunned myocardium)	Recent seizure; SAH; similar mechanism but different trigger
Drug-induced (chemotherapy)	5-FU, capecitabine; temporal relation to infusion

7. Symptoms

- **Recent severe emotional or physical stress:** Grief, death of loved one, financial loss, natural disaster, acute illness, surgery
- **Chest pain mimicking MI:** Pressure-like, substernal; may radiate
- **Dyspnea:** Due to acute LV dysfunction and pulmonary edema
- **Syncope:** From arrhythmia or reduced cardiac output
- **Symptoms of acute heart failure:** Orthopnea, PND (up to 20% present with pulmonary edema)
- **Cardiogenic shock:** In severe cases with extensive wall motion abnormality

8. Risk Factors

- **Female sex:** 90% of cases; postmenopausal state strongly predisposes (estrogen protective)
- **Age:** >50 years (mean age 65-70)
- **Stress trigger:** Emotional (50%) or physical (40%); 10% have no identifiable trigger
- **Psychiatric/neurological conditions:** Anxiety, depression, PTSD, subarachnoid hemorrhage, seizure disorders
- **Pheochromocytoma:** Must be excluded; produces identical catecholamine-mediated injury

9. Immediate Actions

ACUTE MANAGEMENT

1. Treat as ACS until coronary angiography excludes obstructive CAD
2. Aspirin, heparin, beta-blockers, statin (until diagnosis confirmed)
3. Urgent coronary angiography to exclude obstructive CAD (required for diagnosis)
4. Echocardiography: Apical ballooning pattern
5. Standard HF therapy if LV dysfunction: ACE-I (or ARNI), beta-blockers, diuretics for congestion
6. Avoid inotropes if possible (catecholamine-mediated mechanism); if cardiogenic shock, consider mechanical support (Impella)
7. Monitor QT interval - risk of Torsades de Pointes with QT prolongation
8. Anticoagulation if severe apical akinesis + risk of thrombus formation

10. Treatment Overview

Acute: Supportive care. Treat like acute HF with reduced EF. ACE-I/ARB (improve recovery), beta-blockers (counter sympathetic surge), diuretics for

congestion. Anticoagulation (apixaban or LMWH) if LV thrombus present or akinetic apex with EF <30% until recovery. Mechanical circulatory support if cardiogenic shock.

Arrhythmias: QT monitoring; avoid QT-prolonging drugs; treat torsades with magnesium if occurs.

Recovery: Complete in 2-8 weeks in most. Serial echo to document recovery. Continue ACE-I and beta-blocker until full recovery (typically 3-6 months).

Recurrence: 1-5% per year; higher with persistent psychiatric stressors.

Prevention: Stress management, beta-blocker continuation, treat underlying anxiety/depression.

11. Follow-up Questions

- What was happening in the hours/days before symptoms started? (Emotional or physical stressor?)
- Have you experienced a recent loss, trauma, or major life event?
- History of anxiety, depression, or PTSD?
- Any recent neurological event (stroke, seizure, SAH)?
- Postmenopausal? (Hormone history)
- Episodic headaches, sweating, hypertension? (pheochromocytoma screen)
- Any prior similar episode?

12. References

1. Ghadri JR, et al. International Expert Consensus Document on Takotsubo Syndrome (Part I). *Eur Heart J*. 2018;39(22):2032-2046.

2. Templin C, et al. Clinical Features and Outcomes of Takotsubo (Stress) Cardiomyopathy. *N Engl J Med*. 2015;373(10):929-938.

3. Prasad A, et al. U.S. Nationwide Assessment of Takotsubo Cardiomyopathy. *J Am Coll Cardiol*. 2018;71(15):1701-1703.

Cardiac Amyloidosis

Category: Infiltrative Cardiomyopathy **Priority: HIGH** **ICD-10:** E85.4

1. Definition

Cardiac amyloidosis is an infiltrative cardiomyopathy caused by extracellular deposition of insoluble amyloid fibrils in the myocardial interstitium, leading to restrictive cardiomyopathy, progressive heart failure, conduction disease, and arrhythmias. Two main types affect the heart: light-chain (AL) amyloidosis from plasma cell dyscrasia (most aggressive, requires chemotherapy), and transthyretin (ATTR) amyloidosis from wild-type (ATTRwt, formerly senile systemic) or hereditary mutated transthyretin (ATTRv, formerly familial amyloid cardiomyopathy). Cardiac involvement is the major determinant of prognosis.

2. Mechanism

Amyloid fibrils deposit in the myocardial interstitium between myocytes, causing extracellular expansion, myocyte compression, and restrictive physiology. In AL amyloidosis, monoclonal light chains (kappa or lambda) produced by plasma cell clone form amyloid fibrils. In ATTR, either age-related dissociation of TTR tetramers (wild-type) or destabilizing mutations (hereditary, most commonly Val122Ile in African Americans, Val30Met in early-onset familial) causes misfolding and deposition. Amyloid deposition increases ventricular wall thickness without cavity dilation, impairs diastolic filling, and eventually affects systolic function. Conduction system infiltration causes arrhythmias and heart block.

3. ECG Findings

- **Low QRS voltage despite echo LVH (voltage-amyloid paradox):** Pathognomonic; total 12-lead QRS amplitude <10 mm OR limb lead voltage <5 mm
- **Pseudoinfarction pattern:** Q waves in inferior and/or lateral leads without obstructive CAD
- **Conduction disease:** First-degree AV block, bundle branch block, complete heart block
- **Atrial fibrillation:** Very common, often the initial presentation
- **Nonspecific ST-T changes:** Diffuse ST depression, T wave flattening/inversion
- **Reduced limb lead voltages:** Progressive decline as disease advances

4. Required ECG Features

DIAGNOSTIC ECG CLUES

- Low voltage + increased LV wall thickness on echo = cardiac amyloidosis until proven otherwise
- Voltage-to-mass ratio (Cornell voltage / echo LV mass) < 0.5 suggests amyloidosis
- Pseudoinfarction Q waves without CAD history supports diagnosis
- Progressive conduction disease in the absence of ischemia or medications

5. Diagnostic Criteria

- Echocardiography: Increased LV wall thickness (>12 mm) with small LV cavity, biatrial enlargement, thickened valves/interatrial septum, sparkling/granular myocardial texture, diastolic dysfunction (restrictive pattern), preserved EF until late
- CMR (gold standard for detection): Diffuse subendocardial or transmural late gadolinium enhancement with characteristic difficulty nulling myocardium, elevated native T1, increased ECV
- Bone scintigraphy (99mTc-PYP/DPD/HMDP): Grade 2-3 myocardial uptake with slight or absent bone uptake = ATTR amyloidosis; negative scan effectively excludes ATTR (does not exclude AL)
- Serum free light chains + SPEP/UPEP with immunofixation: Screen for AL
- Genetic testing: TTR gene sequencing for hereditary ATTR
- Endomyocardial biopsy: Gold standard for AL confirmation; Congo red staining with apple-green birefringence under polarized light; mass spectrometry for amyloid type

6. Differential Diagnosis

Condition	Differentiating Features
Hypertrophic cardiomyopathy	High QRS voltage matching wall thickness; family history; sarcomere mutations; asymmetric septal hypertrophy; no LGE on CMR
Hypertensive heart disease	High QRS voltage; long-standing hypertension history; no conduction disease progression
Fabry disease	Low voltage may occur; alpha-galactosidase deficiency; early-onset; skin angiokeratomas; enzyme replacement available
Sarcoidosis	Non-caseating granulomas; extracardiac manifestations; different CMR LGE pattern (midwall, epicardial)
Hemochromatosis	Iron deposition; low voltage; transferrin saturation elevated; phlebotomy treatment

7. Symptoms

- **Dyspnea on exertion:** Restrictive physiology limits cardiac output
- **Peripheral edema, ascites:** Right heart failure predominates early
- **Orthopnea, PND:** Elevated filling pressures
- **Syncope:** Especially postural (autonomic neuropathy in AL and ATTR); arrhythmia; fixed stroke volume
- **Fatigue:** Low cardiac output
- **Extra-cardiac AL symptoms:** Macroglossia, periorbital purpura (raccoon eyes), carpal tunnel syndrome, nephrotic syndrome, hepatomegaly
- **Extra-cardiac ATTR symptoms:** Bilateral carpal tunnel syndrome, lumbar spinal stenosis, autonomic neuropathy (orthostatic hypotension, GI dysmotility), biceps tendon rupture (shoulder pad sign)

8. Risk Factors

- **Age:** ATTRwt typically >70 years; ATTRv age varies by mutation (Val122Ile: >65, African ancestry)
- **African ancestry:** Val122Ile mutation carrier rate 3-4% in African Americans
- **Family history:** ATTRv (autosomal dominant)
- **Plasma cell dyscrasia:** Multiple myeloma, MGUS for AL amyloidosis
- **Bilateral carpal tunnel syndrome:** Often precedes cardiac diagnosis by 5-10 years in ATTR
- **Lumbar spinal stenosis, biceps tendon rupture:** Red flags for ATTR

9. Immediate Actions

ACUTE MANAGEMENT

1. Diuretics for congestion (careful - preload dependent)
2. Treat AF with rate control; anticoagulation (regardless of CHA_2DS_2-VASc)
3. Avoid digoxin in AL amyloidosis (toxic at therapeutic levels due to binding amyloid)
4. Pacemaker if high-grade AV block
5. ICD generally not beneficial (primarily pulseless electrical activity as arrest rhythm, not shockable)
6. Low threshold for temporary pacing if conduction disease worsens

10. Treatment Overview

AL amyloidosis: Treat underlying plasma cell dyscrasia. Chemotherapy (bortezomib-based regimens preferred) with/without autologous stem cell transplant in selected patients. Daratumumab (anti-CD38) showing promise. Cardiac response to treatment improves survival. Avoid stem cell transplant if advanced cardiac involvement (NT-proBNP >5000 or troponin elevated).

ATTRwt and ATTRv: Tafamidis (transthyretin stabilizer): reduces mortality and cardiovascular hospitalizations; approved for both wild-type and variant ATTR. Diflunisal (NSAID TTR stabilizer - off-label). Patisiran and vutrisiran (gene silencers, siRNA) for hereditary ATTR with polyneuropathy. Revusiran and others in development.

Heart transplant: Selected AL patients after hematologic complete response; ATTR patients (disease limited to heart). Combined heart-liver transplant for some hereditary ATTR variants.

HF management: Diuretics for congestion. Avoid ACE-I/ARB if orthostatic hypotension. Avoid beta-blockers if not tolerated (fixed stroke volume, rate is needed to maintain output).

11. Follow-up Questions

- Duration of heart failure symptoms? Progressive course?
- Bilateral carpal tunnel syndrome? If so, when? (Often precedes cardiac involvement by years in ATTR)
- Any history of lumbar spinal stenosis or biceps tendon rupture?
- Family history of cardiomyopathy or neuropathy? (ATTR hereditary)
- African ancestry? (Val122Ile mutation)
- Plasma cell disorder, MGUS, or multiple myeloma? (AL amyloidosis)
- Symptoms of autonomic neuropathy: orthostatic dizziness, GI issues, erectile dysfunction?

- Macroglossia, periorbital purpura, easy bruising? (AL)

12. References

1. Garcia-Pavia P, et al. Diagnosis and treatment of cardiac amyloidosis: a position statement of the ESC. *Eur Heart J*. 2021;42(16):1554-1568.
2. Maurer MS, et al. Tafamidis Treatment for Patients With Transthyretin Amyloid Cardiomyopathy. *N Engl J Med*. 2018;379(11):1007-1016.
3. Gertz MA, et al. Pathophysiology and treatment of cardiac amyloidosis. *Nat Rev Cardiol*. 2015;12(2):91-102.
4. Falk RH, et al. AL (Light-Chain) Cardiac Amyloidosis: A Review of Diagnosis and Therapy. *J Am Coll Cardiol*. 2016;68(12):1323-1341.

Cardiac Sarcoidosis

Category: Infiltrative/Inflammatory **Priority: HIGH** **ICD-10:** D86.85

1. Definition

Cardiac sarcoidosis is a manifestation of systemic sarcoidosis characterized by the formation of non-caseating granulomas in the myocardium, leading to conduction disease, ventricular arrhythmias, heart failure, and sudden cardiac death. Sarcoidosis is a multisystem inflammatory disease of unknown etiology. Cardiac involvement is clinically evident in approximately 5% of sarcoidosis patients but detected in 20-30% at autopsy, making it the leading cause of sarcoidosis-related death in developed countries.

2. Mechanism

Unknown etiology (environmental triggers in genetically susceptible individuals). Activated CD4+ T cells drive granuloma formation with accumulation of epithelioid histiocytes and giant cells. In the heart, granulomas most commonly involve the left ventricular free wall and basal interventricular septum (explaining high predilection for conduction disease). Active inflammation causes myocardial edema; chronic phase leads to replacement fibrosis creating arrhythmogenic substrate. Progression: acute inflammation → granuloma formation → fibrosis → scar. The scar creates reentrant circuits for VT and replaces conduction tissue.

3. ECG Findings

- **Complete heart block:** Most common presenting feature (may be the first manifestation); especially in patients <60 years without CAD
- **Bundle branch blocks:** RBBB or LBBB due to septal involvement
- **PR interval prolongation:** First-degree AV block
- **VT from granulomatous scar:** May mimic idiopathic VT; often monomorphic; scar-based reentry
- **Frequent PVCs:** Due to irritable myocardium adjacent to granulomas
- **ST-T changes:** Nonspecific; pathological Q waves from scar in distribution not matching coronary anatomy
- **Q waves in leads not matching coronary distribution:** Highly suggestive (e.g., pseudo-infarct pattern in lateral leads without LCX disease)

4. Required ECG Features

RED FLAGS FOR CARDIAC SARCOIDOSIS

- Complete heart block in patient <60 years without CAD or other explanation = cardiac sarcoidosis until proven otherwise
- VT with non-coronary scar pattern (e.g., patchy mid-wall or epicardial distribution)
- Pathological Q waves in non-coronary distribution (e.g., lateral, anteroseptal without LAD disease)
- New bundle branch block or progressive conduction disease in patient with known sarcoidosis

5. Diagnostic Criteria

- Japanese Ministry of Health criteria (modified 2016): Major criteria (complete heart block, basal thinning of IVS, LVEF <40%, perfusion defect on scintigraphy, delayed myocardial enhancement on CMR, endomyocardial biopsy showing non-caseating granulomas) + Minor criteria (abnormal ECG, abnormal echocardiography, nuclear imaging, gallium uptake, indication of cardiac involvement on FDG-PET, extracardiac sarcoidosis)
- Heart Rhythm Society (HRS) 2014 criteria: Histological diagnosis from myocardial tissue OR clinical diagnosis from invasive and non-invasive findings in context of histologically confirmed extracardiac sarcoidosis
- CMR: T2 mapping (edema/inflammation), LGE (scar/fibrosis - typically mid-myocardial, epicardial, or patchy, NOT subendocardial)
- FDG-PET: Focal myocardial uptake indicating active inflammation
- Endomyocardial biopsy: Low sensitivity due to patchy involvement (20-30%); non-caseating granulomas = definitive
- Extrapulmonary sarcoidosis on biopsy (skin, lymph node, conjunctiva) supports diagnosis

6. Differential Diagnosis

Condition	Differentiating Features
ARVC	RV involvement, epsilon wave, desmosomal mutations; CMR shows RV findings; no extracardiac sarcoid
Viral myocarditis	Acute presentation, viral prodrome, diffuse myocardial involvement; endomyocardial biopsy differentiates
Idiopathic AV block	No granulomas on biopsy; no extracardiac manifestations; no CMR LGE; no PET uptake
Coronary artery disease	Risk factors, subendocardial LGE matching coronary territory, angiographic stenosis
Giant cell myocarditis	Fulminant course; multinucleated giant cells with necrosis; worse prognosis

7. Symptoms

- **Complete heart block in patient <60 years:** May be presenting feature; syncope, presyncope
- **VT with non-coronary scar pattern:** Palpitations, syncope, cardiac arrest
- **Palpitations, syncope:** From arrhythmias
- **Extra-cardiac sarcoid symptoms:** Skin lesions (erythema nodosum, lupus pernio), pulmonary (cough, dyspnea, CXR with bilateral hilar lymphadenopathy), ocular (uveitis), neurologic (cranial nerve palsies)
- **Heart failure symptoms:** Dyspnea, fatigue, edema if extensive myocardial involvement
- **Many asymptomatic:** Detected through screening in known sarcoidosis patients

8. Risk Factors

- Known extracardiac sarcoidosis (pulmonary, skin, ocular)
- Age 20-50 years (bimodal peaks: 20-30 and 45-65)
- Female predominance (sarcoidosis overall slightly female predominant)
- Black race (3x higher incidence than Whites; more severe disease)
- Japanese population (higher rate of cardiac involvement)
- Specific HLA types: HLA-DRB1*03, DQB1*0201 associated with risk

9. Immediate Actions

EMERGENCY MANAGEMENT

1. High-grade AV block or complete heart block: Temporary or permanent pacing
2. VT/VF: ACLS protocol; ICD often required for secondary prevention
3. Heart failure: Standard therapy; consider immunosuppression for active inflammation
4. Document rhythm with telemetry; prepare for emergent pacing

10. Treatment Overview

Immunosuppression: Indicated for active inflammation (CMR T2 elevation, FDG-PET positive). Corticosteroids (prednisone 20-40 mg/day) first-line; taper based on response. Methotrexate, mycophenolate, azathioprine, or infliximab as steroid-sparing agents. Evidence base limited but expert consensus supports treatment.

Conduction disease: Permanent pacemaker for complete heart block or symptomatic high-grade AV block. Consider EP study for risk stratification.

Arrhythmia management: ICD for secondary prevention (cardiac arrest, sustained VT with hemodynamic compromise). Antiarrhythmics (amiodarone, sotalol) for VT suppression. Catheter ablation for recurrent VT (often epicardial approach needed due to basal lateral scar).

Heart failure: Standard GDMT; cardiac transplantation for end-stage refractory HF.

Screening: All newly diagnosed sarcoidosis patients should undergo cardiac screening (ECG, echo, CMR if available).

11. Follow-up Questions

- Known sarcoidosis diagnosis? What organs are involved?
- Skin lesions, eye problems, lung symptoms?
- Syncope, palpitations, or documented arrhythmias?
- Race/ethnicity? (Higher rates in Black and Japanese populations)
- Age at presentation? (<60 years with heart block highly suggestive)
- Any prior ECG abnormalities?
- Family history of sarcoidosis or cardiomyopathy?

12. References

1. Birnie DH, et al. HRS expert consensus statement on the diagnosis and management of arrhythmias associated with cardiac sarcoidosis. *Heart Rhythm*. 2014;11(7):1305-1323.

2. Terasaki F, et al. Japanese Circulation Society Joint Working Group report on guidelines for diagnosis and treatment of cardiac sarcoidosis. *Circ J*. 2016;80(7):1617-1625.

3. Kandolin R, et al. Cardiac sarcoidosis: epidemiology, characteristics, and outcome over 25 years in a nationwide study. *Circulation*. 2015;131(7):624-632.

Cardiomyopathy (General)

Category: Cardiomyopathy **Priority: HIGH** **ICD-10:** I42.9

1. Definition

Cardiomyopathy is a heterogeneous group of diseases of the myocardium associated with mechanical and/or electrical dysfunction, usually exhibiting inappropriate ventricular hypertrophy or dilation, due to a variety of causes that frequently are genetic. The ESC classification recognizes five morphological and functional phenotypes: hypertrophic cardiomyopathy (HCM), dilated cardiomyopathy (DCM), arrhythmogenic cardiomyopathy (ACM), restrictive cardiomyopathy (RCM), and unclassified (including left ventricular non-compaction). Cardiomyopathies are a leading cause of heart failure and sudden cardiac death, particularly in the young.

2. Mechanism

Genetic: Sarcomere protein mutations (MYH7, MYBPC3) in HCM; cytoskeletal and sarcomere mutations (TTN truncating variants in ~20% of DCM) in DCM; desmosomal mutations in ACM. Mutations disrupt force generation, transmission, or myocyte structural integrity. **Acquired:** Myocarditis (viral, autoimmune), toxins (alcohol, anthracyclines, cocaine), endocrine (thyroid, pheochromocytoma), nutritional (thiamine deficiency), tachycardia-induced. **Infiltrative:** Amyloidosis, sarcoidosis, hemochromatosis, Fabry disease, glycogen storage diseases. Final common pathways involve myocyte death, fibrosis replacement, and progressive ventricular remodeling.

3. ECG Findings

- **Hypertrophic:** Massive LVH voltage (Sokolow-Lyon often >35 mm), repolarization strain pattern (deep ST depression, T wave inversion in lateral leads), pathological Q waves in inferior/lateral (narrow, without Q wave evolution)
- **Dilated:** LBBB, QRS widening, chamber enlargement (LAE, biatrial), nonspecific ST-T changes; AF common
- **Restrictive:** Low voltage (amyloid), LAE, pseudoinfarction patterns
- **Arrhythmogenic:** Epsilon wave, T wave inversion V1-V3, terminal QRS activation delay >55 ms
- **LV non-compaction:** LVH voltage, ST-T changes, pre-excitation, arrhythmias

4. Required ECG Features

ECG CLUES BY TYPE

- HCM: Voltage criteria for LVH + deep T wave inversion + pathological Q waves without CAD = classic triad
- DCM: LBBB in young patient suggests familial DCM or muscular dystrophy
- RCM: Low voltage with echo LVH = infiltrative (amyloid); high voltage = non-infiltrative
- ACM: Epsilon wave or T wave inversion V1-V3 beyond age 14 in RBBB-free patient

5. Diagnostic Criteria

- Echocardiography: Primary diagnostic tool for morphologic classification (wall thickness, cavity size, EF, strain analysis)
- CMR: Gold standard for tissue characterization (LGE, T1/T2 mapping, ECV); identifies fibrosis, edema, infiltrative patterns
- Genetic testing: Recommended for all HCM, ACM, unexplained DCM; cascade screening of family members
- Biomarkers: BNP/NT-proBNP elevated proportionally to severity; troponin may be mildly elevated
- Endomyocardial biopsy: When infiltrative, inflammatory, or storage disease suspected and non-invasive testing inconclusive
- Specific criteria exist for each subtype (see individual disease entries)

6. Differential Diagnosis

Condition	Differentiating Features
Athletic heart	Physiologic LVH with normal diastolic function; deconditioning reverses changes; no LGE; no family history
Hypertensive heart disease	Long-standing hypertension; concentric LVH without disproportionate septal thickening; no sarcomere mutations
Ischemic cardiomyopathy	Prior MI, regional wall motion abnormalities matching coronary territories; angiographic CAD
Pericardial constriction	Pericardial thickening/calcification; septal bounce; preserved LVEF; CT/MRI differentiates

7. Symptoms

- **Dyspnea on exertion:** Most common; due to elevated filling pressures and reduced cardiac output
- **Fatigue, exercise intolerance:** Reduced cardiac output
- **Chest pain:** Angina-like (HCM from supply-demand mismatch) or atypical
- **Palpitations:** AF, VT, PVCs
- **Syncope:** Arrhythmia or outflow obstruction (HCM)
- **Peripheral edema:** Right heart failure
- **Family history:** SCD, cardiomyopathy (strongly suggests genetic etiology)

8. Risk Factors

- **Genetic/family history:** Most common risk factor; autosomal dominant inheritance common
- **Alcohol:** Heavy, long-term use (toxic cardiomyopathy)
- **Chemotherapy:** Anthracyclines (doxorubicin), trastuzumab, immune checkpoint inhibitors
- **Viral myocarditis:** Enteroviruses, adenovirus, parvovirus B19, SARS-CoV-2
- **Pregnancy/peripartum:** Peripartum cardiomyopathy (last month to 5 months postpartum)
- **HIV:** HIV-associated cardiomyopathy
- **Autoimmune:** SLE, scleroderma, polymyositis
- **Nutritional:** Thiamine deficiency (wet beriberi), selenium deficiency
- **Endocrine:** Hypothyroidism, hyperthyroidism, acromegaly, pheochromocytoma

9. Immediate Actions

> **ACUTE MANAGEMENT**
>
> 1. Heart failure decompensation: Diuresis, vasodilators, inotropes if needed
> 2. Sustained VT or VF: Cardioversion/defibrillation; antiarrhythmics (amiodarone)
> 3. Complete heart block: Temporary pacing
> 4. Acute pulmonary edema: Non-invasive ventilation, nitrates, diuretics
> 5. Identify reversible cause: Acute ischemia, alcohol, thyrotoxicosis, infection

10. Treatment Overview

HCM: See dedicated entry. Beta-blockers, disopyramide, myectomy, alcohol septal ablation.

DCM: GDMT for HFrEF (ARNI/ACE-I, evidence-based beta-blockers, MRA, SGLT2i, diuretics). ICD if EF <35% despite optimal therapy. CRT if LBBB with QRS >150 ms. Identify and treat cause (alcohol cessation, immunosuppression for myocarditis).

RCM: Diuretics for congestion; treat underlying cause; cardiac transplant for end-stage. Avoid nitrates and vasodilators (preload dependent).

ACM: See dedicated entry. Exercise restriction, beta-blockers, antiarrhythmics, ICD, catheter ablation.

Genetic counseling: Recommended for all hereditary cardiomyopathies; cascade family screening.

11. Follow-up Questions

- Family history of cardiomyopathy, heart failure, or sudden death?
- Alcohol consumption? Chemotherapy history?
- Recent viral illness or COVID-19?
- Pregnancy or postpartum status?
- Symptoms of thyroid disease?
- Exercise history? (Distinguish athletic heart from HCM)
- Autoimmune disease symptoms?

12. References

1. Arbustini E, et al. The MOGE(S) classification of cardiomyopathy for clinicians. *J Am Coll Cardiol*. 2014;64(3):304-318.
2. Elliott P, et al. Classification of the cardiomyopathies: a position statement from the ESC Working Group on Myocardial and Pericardial Diseases. *Eur Heart J*. 2008;29(2):270-276.
3. Heidenreich PA, et al. 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure. *Circulation*. 2022;145(18):e895-e1032.

Congenital Heart Disease

Category: Congenital **Priority: HIGH** **ICD-10:** Q24.9

1. Definition

Congenital heart disease (CHD) comprises structural abnormalities of the heart and great vessels present at birth, resulting from abnormal embryological development during the first 8 weeks of gestation. It is the most common congenital anomaly, affecting approximately 8 per 1000 live births. Lesions range from simple defects (ASD, VSD, mild pulmonary stenosis) that may be asymptomatic, to complex cyanotic defects (single ventricle, hypoplastic left heart syndrome) requiring multiple staged surgeries in infancy. With advances in surgical and interventional techniques, >90% of children born with CHD now survive to adulthood.

2. Mechanism

Genetic factors (chromosomal abnormalities, single gene mutations, copy number variants) account for ~20% of cases; environmental factors (maternal diabetes, alcohol/fetal alcohol syndrome, rubella infection, teratogens like lithium, retinoic acid) for ~10%; the majority are multifactorial. Disruption of normal cardiac development at specific time points produces specific lesions: septation defects (ASD, VSD), conotruncal anomalies (tetralogy of Fallot, transposition of the great arteries), outflow obstruction (aortic stenosis, coarctation, pulmonary atresia), and complex univentricular hearts. The hemodynamic consequences depend on lesion type, severity, and presence of shunting.

3. ECG Findings

- **Chamber hypertrophy patterns:** Depend on specific lesion and hemodynamics - RVH with ASD, VSD, pulmonary stenosis; LVH with VSD, PDA, aortic stenosis; biventricular hypertrophy with large VSD
- **Conduction abnormalities:** RBBB after VSD or tetralogy repair; LBBB after LVOT procedures; complete heart block with AV septal defects or after surgery
- **Arrhythmias:** Atrial flutter/AF (very common in adult CHD), VT (especially repaired TOF with QRS >180 ms), complete heart block
- **Axis deviation:** Left axis deviation with AV septal defects (primum ASD); right axis deviation with ASD, TOF
- **P wave abnormalities:** Atrial enlargement from chronic volume/pressure overload
- **QRS duration:** Prolongation in repaired TOF (>180 ms predicts VT risk)

4. Required ECG Features

ECG BY LESION

- ASD: Right axis deviation, incomplete RBBB, RVH pattern
- VSD: BVH, LAE, LVH (large shunt); isolated LVH (moderate)
- TOF: RVH, RAE, right axis deviation (pre-repair); RBBB (post-repair)
- TGA: RVH pattern (pre-repair, RV is systemic ventricle); arrhythmias post-atrial switch
- AV septal defect: Left axis deviation, first-degree AV block
- Coarctation: LVH with strain pattern

5. Diagnostic Criteria

- Echocardiography: First-line diagnostic tool; defines anatomy, shunt direction/severity, chamber sizes, pressures
- Cardiac MRI/CT: Complex anatomy, quantification of ventricular volumes and function, vascular imaging
- Cardiac catheterization: Hemodynamic assessment, oximetry for shunt quantification, intervention
- Screening: Fetal echocardiography for high-risk pregnancies (family history, genetic syndrome, maternal diabetes)
- Classification: Simple (ASD, VSD, PDA, PS), moderate (TOF, coarctation, AVSD), complex (single ventricle, HLHS, TGA)

6. Differential Diagnosis

Condition	Differentiating Features
Normal newborn transition	RV dominance, RBBB pattern normal in neonates; resolves over first months
Pulmonary hypertension	RVH, RAE, right axis deviation; echo shows elevated PA pressures; no structural defect
Myocarditis	Acute presentation; normal cardiac anatomy; diffuse dysfunction
Cardiomyopathy	No structural congenital lesion; primary myocardial disease
Primary arrhythmia	Normal cardiac anatomy; rhythm disturbance without structural cause

7. Symptoms

- **Cyanosis:** Central (tongue/lips) in right-to-left shunts; clubbing in chronic cyanosis
- **Feeding difficulties, poor growth:** In infants with large left-to-right shunts
- **Tachypnea, recurrent respiratory infections:** Pulmonary overcirculation
- **Exercise intolerance:** Common in unrepaired or palliated lesions
- **Palpitations:** Arrhythmias very common in ACHD
- **Syncope:** Arrhythmia or hemodynamic compromise
- **Heart failure:** In unrepaired large shunts, valvular lesions, or failing systemic/pulmonary ventricles
- **Many asymptomatic:** Small ASD, mild PS may be incidentally detected

8. Risk Factors

- **Genetic syndromes:** Down syndrome (AVSD, VSD), Turner syndrome (coarctation, BAV), DiGeorge (conotruncal), Noonan (PS, HCM), Marfan (aortic root dilation), Holt-Oram (ASD, VSD)
- **Maternal factors:** Diabetes, phenylketonuria, obesity, advanced age, febrile illness in first trimester
- **Environmental:** Alcohol (fetal alcohol syndrome), smoking, lithium, retinoids, antiepileptics
- **Family history:** Increased risk (multifactorial inheritance); specific mutations in some families

9. Immediate Actions

EMERGENCIES IN CHD

1. Cyanotic spell (TOF): Knee-chest position, supplemental O2, IV fluids, morphine, beta-blocker (esmolol), phenylephrine
2. Supraventricular tachycardia: Adenosine or synchronized cardioversion
3. Heart failure: Diuretics, afterload reduction, inotropes if needed
4. Complete heart block: Temporary pacing
5. Infective endocarditis: Blood cultures, antibiotics

10. Treatment Overview

Interventional catheterization: ASD/PDA device closure, balloon valvuloplasty (pulmonary, aortic), coarctation stenting, pulmonary artery stenting.

Surgical: VSD closure, tetralogy repair, arterial switch for TGA, Fontan palliation for single ventricle, valve repair/replacement.

Arrhythmia management: Antiarrhythmics, catheter ablation, pacemaker/ICD implantation. ACHD patients at high risk for SVT and VT.

Heart failure: GDMT adapted for CHD (e.g., Fontan physiology requires different approach). Cardiac transplantation for end-stage.

Transition to ACHD care: Specialized adult CHD centers; lifelong follow-up; endocarditis prophylaxis for high-risk lesions.

11. Follow-up Questions

- Specific CHD diagnosis? Original anatomy and surgical repairs?
- Age at repair? Type of repair (patch, conduit, valve replacement)?
- Current functional status? NYHA class?
- Arrhythmia history?
- Cyanosis (oxygen saturation)?
- Need for re-intervention anticipated?
- Pregnancy counseling? Contraception?

12. References

1. Warnes CA, et al. ACC/AHA 2008 Guidelines for the Management of Adults with Congenital Heart Disease. *Circulation*. 2008;118(23):e714-e833.
2. Baumgartner H, et al. 2020 ESC Guidelines for the management of adult congenital heart disease. *Eur Heart J*. 2021;42(6):563-645.
3. Marelli AJ, et al. Lifetime prevalence of congenital heart disease in the general population from 2000 to 2010. *Circulation*. 2014;130(9):749-756.

Coronary Heart Disease

Category: Coronary Artery Disease **Priority: HIGH** **ICD-10:** I25.1

1. Definition

Coronary heart disease (CHD), also called coronary artery disease (CAD) or ischemic heart disease, is a condition in which atherosclerotic plaque buildup in the epicardial coronary arteries narrows the arterial lumen, reducing myocardial blood flow and oxygen supply. CHD is the most common cardiovascular disease globally and the leading cause of death worldwide. It encompasses a spectrum from stable angina through acute coronary syndromes to sudden cardiac death and ischemic cardiomyopathy.

2. Mechanism

Atherosclerosis begins with endothelial dysfunction and LDL cholesterol infiltration into the subendothelial space. Oxidized LDL triggers an inflammatory response with monocyte recruitment, foam cell formation, and fatty streak development. Progressive plaque growth causes luminal narrowing; plaque rupture with superimposed thrombosis causes acute coronary syndromes. Positive remodeling (outward expansion) initially preserves lumen; negative remodeling causes progressive stenosis. Endothelial shear stress patterns determine lesion location (bifurcations, inner curves). Vulnerable plaques have thin fibrous caps, large lipid cores, and increased inflammation - prone to rupture even when not hemodynamically significant.

3. ECG Findings

- **Normal between episodes:** Common in stable CAD; or nonspecific ST-T changes
- **Dynamic ST-T changes:** During acute episodes - ST depression, T wave inversion
- **Pathological Q waves:** With persistent ST-T changes indicating prior MI
- **ST depression >= 1 mm:** Horizontal or downsloping during stress testing or symptoms
- **T wave inversion:** In leads corresponding to ischemic territory
- **LVH:** From chronic pressure overload and compensatory mechanisms
- **Left anterior fascicular block:** May indicate proximal LAD disease
- **Resting abnormalities:** Unrelated to acute presentation but may indicate prior events

4. Required ECG Features

ECG IN STABLE CAD

No required ECG features for diagnosis of stable CAD. ECG serves to: (1) detect prior MI, (2) assess baseline rhythm, (3) identify LVH, (4) detect dynamic changes during stress testing. A normal resting ECG does not exclude significant CAD. Stress-induced ST depression >= 1 mm is the most commonly used diagnostic ECG criterion.

5. Diagnostic Criteria

- Clinical: Typical angina symptoms (chest pressure with exertion, relieved by rest/nitroglycerin)
- Non-invasive testing: Exercise ECG (treadmill), stress echocardiography, myocardial perfusion imaging (SPECT/PET), coronary CTA
- Invasive coronary angiography: Gold standard; defines stenosis severity, location, number of vessels involved; SYNTAX score for complexity
- FFR: Functional significance of intermediate lesions (40-70%); FFR < 0.80 indicates ischemia-producing lesion
- Intravascular imaging (IVUS, OCT): Characterizes plaque morphology and guides stenting
- Coronary CTA: High negative predictive value for excluding CAD; calcium scoring for risk assessment

6. Differential Diagnosis

Condition	Differentiating Features
Non-cardiac chest pain	Pleuritic, positional, reproducible; normal stress testing; no risk factors
GERD	Burning epigastric pain, relation to meals, relief with antacids/PPIs
Anxiety/panic	Pleuritic, sharp, associated with hyperventilation; normal cardiac workup
Myocarditis	Younger, viral prodrome, diffuse ST changes, elevated troponin, non-obstructive coronaries
Microvascular angina	Typical angina symptoms with non-obstructive coronaries (INOCA); diagnosis of exclusion
Vasospastic angina	Rest pain with transient ST elevation; responds to nitrates; acetylcholine challenge positive

7. Symptoms

- **Stable angina:** Substernal pressure/tightness with exertion, emotional stress, cold; relieved by rest or nitroglycerin within 5 minutes; duration 2-15 minutes
- **Atypical angina:** Dyspnea, fatigue, epigastric discomfort, jaw/arm pain without chest pain
- **Angina equivalents:** Especially in elderly, women, diabetics - exertional dyspnea without chest discomfort
- **Progressive symptoms:** Decreasing exercise tolerance, increasing frequency/severity (unstable angina)
- **Silent ischemia:** No symptoms despite objective evidence of ischemia (common in diabetics)

8. Risk Factors

- Major modifiable: Smoking, hypertension, dyslipidemia (elevated LDL, low HDL), diabetes, obesity, physical inactivity
- Major non-modifiable: Age (men >45, women >55), male sex, family history of premature CAD
- Emerging: Lp(a), hs-CRP, CKD, inflammatory diseases (RA, psoriasis, SLE), HIV, psychosocial factors (depression, stress), OSA

9. Immediate Actions

UNSTABLE SYMPTOMS = ACS PROTOCOL

1. If unstable symptoms: Aspirin, nitroglycerin, anticoagulation per ACS protocol
2. Troponins to exclude MI
3. Risk stratification: GRACE or TIMI score
4. Early invasive strategy if high-risk

10. Treatment Overview

Medical therapy: Anti-ischemic (beta-blockers first-line, CCBs, nitrates, ranolazine), antiplatelet (aspirin 75-100 mg), high-intensity statin (atorvastatin 40-80 mg or rosuvastatin 20-40 mg), ACE-I/ARB (especially diabetes, CKD, or prior MI), SGLT2i (empagliflozin, dapagliflozin for diabetes with CAD).

Revascularization: PCI with stenting or CABG. Indications: left main stenosis >50%, 3-vessel disease (especially with diabetes - CABG preferred), 2-vessel disease with proximal LAD, significant stenosis with LV dysfunction, refractory angina despite OMT. FFR-guided PCI for intermediate lesions.

Secondary prevention: Cardiac rehabilitation, smoking cessation, weight management, diabetes control, BP control (<130/80), LDL <70 for very high risk, regular exercise.

Risk factor modification: This is the cornerstone of long-term management. Aggressive lipid lowering, BP control, glycemic control, lifestyle changes.

11. Follow-up Questions

- Typical angina symptoms? CCS class?
- Risk factors: smoking, BP, cholesterol, diabetes, family history?
- Prior stress test, angiography, or revascularization?
- Current antianginal medications and response?
- Symptoms at rest or minimal exertion? (unstable)
- Exercise tolerance baseline?

12. References

1. Knuuti J, et al. 2019 ESC Guidelines for the diagnosis and management of chronic coronary syndromes. *Eur Heart J*. 2020;41(3):407-477.Fihn SD, et al. 2012 ACCF/AHA Guideline for the diagnosis and management of stable ischemic heart disease. *Circulation*. 2012;126(25):e354-e471.
2. Lawton JS, et al. 2021 ACC/AHA/SCAI Guideline for Coronary Artery Revascularization. *J Am Coll Cardiol*. 2022;79(2):e21-e129.

COVID-19 (Cardiac Manifestations)

Category: Infectious **Priority: HIGH** **ICD-10:** U07.1

1. Definition

COVID-19, caused by SARS-CoV-2, has significant cardiac manifestations ranging from acute myocardial injury (20-30% of hospitalized patients) to myocarditis, pericarditis, heart failure, arrhythmias, acute coronary syndrome, pulmonary embolism, and stress cardiomyopathy. Cardiovascular complications are major determinants of prognosis, with elevated troponin being an independent predictor of mortality. Post-acute sequelae ("long COVID") include persistent cardiac symptoms and exercise intolerance.

2. Mechanism

Multiple mechanisms: (1) Direct viral myocardial invasion via ACE2 receptor expressed on cardiomyocytes; (2) Cytokine storm and systemic inflammation causing myocardial injury; (3) Hypoxemia from pulmonary disease causing supply-demand mismatch (Type 2 MI); (4) Coronary plaque instability from inflammation and prothrombotic state (Type 1 MI); (5) Stress cardiomyopathy from catecholamine surge; (6) Prothrombotic state causing PE and microvascular thrombosis; (7) Drug toxicity (hydroxychloroquine-induced QT prolongation - now rarely used). Post-vaccine myocarditis (mRNA vaccines) predominantly affects young males, typically mild and self-limiting.

3. ECG Findings

- **Diffuse ST elevation with PR depression:** Myopericarditis pattern
- **ST elevation MI:** Type 1 (plaque rupture) or Type 2 (demand ischemia)
- **QT prolongation:** Drug-induced (if treated with HCQ, azithromycin, antipsychotics); monitor closely
- **Low voltage, electrical alternans:** Pericardial effusion
- **Any arrhythmia type:** AF, VT, bradyarrhythmias, heart block
- **Sinus tachycardia:** Most common rhythm disturbance; may persist as POTS-like syndrome
- **S1Q3T3 pattern:** If PE
- **T wave inversions:** Nonspecific or associated with myocardial injury

4. Required ECG Features

> **ECG MONITORING**
>
> No specific ECG finding is pathognomonic for COVID-19 cardiac involvement. Continuous telemetry is recommended for all hospitalized COVID-19 patients. Daily 12-lead ECGs if receiving QT-prolonging therapy. New ST changes, arrhythmias, or conduction abnormalities should prompt cardiac evaluation with troponins and echocardiography.

5. Diagnostic Criteria

- COVID-19 diagnosis: PCR or antigen testing for SARS-CoV-2
- Cardiac involvement: Elevated troponin (hs-cTn I or T), new ECG abnormalities, echo wall motion abnormalities, or CMR findings
- Myocarditis: CMR Lake Louise criteria (T2 mapping, LGE, early gadolinium enhancement) or endomyocardial biopsy
- PE: CTPA, elevated D-dimer, clinical probability assessment
- ACS: Per standard criteria with angiographic confirmation
- Post-vaccine myocarditis: Typically presents 2-5 days after mRNA vaccination; troponin elevation with diffuse ST elevation; CMR confirms

6. Differential Diagnosis

Condition	Differentiating Features
Acute MI	Regional wall motion abnormalities matching coronary territory; culprit lesion on angiography
Takotsubo	Apical ballooning; emotional/physical trigger; postmenopausal women; no viral symptoms
Myocarditis (non-COVID)	Viral prodrome (different virus), same diagnostic criteria; temporal relation to COVID determines etiology
PE	CTPA diagnostic; S1Q3T3 pattern; dyspnea out of proportion
Drug-induced cardiotoxicity	Temporal relation to specific medication (HCQ, azithromycin); QT prolongation

7. Symptoms

- **COVID-19 symptoms:** Fever, cough, dyspnea, fatigue, loss of taste/smell, myalgias

- **Chest pain:** Myopericarditic (sharp, pleuritic) or anginal (pressure, exertional)
- **Palpitations:** AF, sinus tachycardia, PVCs
- **Syncope:** Arrhythmia or hemodynamic compromise
- **Dyspnea disproportionate to respiratory symptoms:** Suggests cardiac involvement
- **Post-COVID (long COVID):** Exercise intolerance, chest discomfort, palpitations, POTS-like orthostatic intolerance

8. Risk Factors

- Pre-existing cardiovascular disease (CAD, HF, hypertension)
- Age >65 years
- Male sex
- Diabetes, obesity (BMI >30), CKD
- Severe COVID-19 requiring hospitalization, ICU admission, mechanical ventilation
- Elevated baseline troponin, BNP, D-dimer
- Post-vaccine myocarditis: Young males (12-30 years), mRNA vaccines, dose 2

9. Immediate Actions

ACUTE MANAGEMENT

1. Troponin and BNP on admission; serial troponins if elevated
2. ECG on admission and daily if hospitalized
3. Echocardiography if elevated troponin, new ECG changes, hemodynamic instability, or clinical concern
4. Continuous telemetry monitoring
5. Monitor QTc if receiving QT-prolonging medications; hold if QTc >500 ms
6. D-dimer and CTPA if PE suspected
7. Coronary angiography if STEMI presentation
8. Supportive care; treat specific cardiac complications per standard protocols

10. Treatment Overview

Myocardial injury/Type 2 MI: Treat underlying COVID-19; oxygen; avoid unnecessary anticoagulation if not indicated. Supportive care.

Myocarditis: Supportive; avoid NSAIDs in early viral phase (animal data - controversial); colchicine for pericarditis component; IVIG and corticosteroids for severe cases; mechanical circulatory support if cardiogenic shock.

ACS: Standard ACS management; primary PCI for STEMI; consider thrombosis risk with COVID prothrombotic state.

Arrhythmias: Standard management; AF rate/rhythm control; avoid QT-prolonging drugs; electrolyte repletion.

PE: Standard anticoagulation per PE guidelines; consider higher index of suspicion.

Post-vaccine myocarditis: Supportive; NSAIDs, colchicine; typically self-limited; cardiology follow-up; avoid strenuous exercise until recovered and CMR normalized.

Long COVID cardiac: Graduated return to exercise, cardiac rehabilitation, beta-blockers for POTS-like symptoms, treat residual symptoms.

11. Follow-up Questions

- COVID-19 symptoms and timeline?
- Hospitalization? ICU admission? Oxygen requirement?
- Prior cardiovascular disease?
- Chest pain - characterize (pleuritic vs. pressure)?
- Current medications, especially QT-prolonging drugs?
- Vaccination status? If post-vaccine symptoms, which vaccine and timing?
- Exercise tolerance post-recovery? Orthostatic symptoms?
- Ongoing dyspnea, fatigue, palpitations?

12. References

1. Driggin E, et al. Cardiovascular Considerations for Patients, Health Care Workers, and Health Systems During the COVID-19 Pandemic. *J Am Coll Cardiol*. 2020;75(18):2352-2371.
2. Clerkin KJ, et al. COVID-19 and Cardiovascular Disease. *Circulation*. 2020;141(20):1648-1655.
3. Ammirati E, et al. Prevalence, Characteristics, and Outcomes of COVID-19-Associated Acute Myocarditis. *Circulation*. 2022;145(15):1123-1139.
4. Oster ME, et al. Myocarditis Cases Reported After mRNA-Based COVID-19 Vaccination in the US From December 2020 to August 2021. *JAMA*. 2022;327(4):331-340.

Dilated Cardiomyopathy (DCM)

Category: Cardiomyopathy **Priority: HIGH** **ICD-10:** I42.0

1. Definition

Dilated cardiomyopathy (DCM) is a myocardial disorder characterized by left ventricular or biventricular dilation with systolic dysfunction (ejection fraction <40%) in the absence of abnormal loading conditions (hypertension, valvular disease) or coronary artery disease sufficient to cause the degree of systolic impairment. DCM is the most common cardiomyopathy and the most common indication for heart transplantation. It affects approximately 1 in 2500 individuals and represents the final common phenotype of diverse etiologies including genetic, infectious, toxic, and autoimmune processes.

2. Mechanism

Genetic causes (30-50% of cases): mutations in cytoskeletal (TTN truncating variants in ~20%), sarcomere, nuclear envelope, and desmosomal proteins leading to impaired force transmission, abnormal mechanosensing, and progressive myocyte death. Acquired causes: viral myocarditis (enteroviruses, adenovirus, parvovirus B19, HHV-6), autoimmune (anti-cardiac antibodies), toxins (alcohol, anthracyclines, cocaine), nutritional deficiencies (thiamine, selenium, carnitine), endocrine (thyroid disease, acromegaly, pheochromocytoma), peripartum, tachycardia-induced. Final common pathway: myocyte injury → necrosis/apoptosis → replacement fibrosis → progressive ventricular dilation and spherical remodeling → worsening systolic function.

3. ECG Findings

- **Atrial fibrillation:** Very common due to atrial stretch and elevated filling pressures
- **LBBB:** Very common; may be first finding to suggest DCM; wide QRS (>120 ms) with characteristic morphology
- **Biatrial enlargement:** Tall P waves in II (RAE), biphasic P in V1 (LAE)
- **QRS prolongation:** >120 ms with LBBB; QRS >150 ms indicates dyssynchrony and CRT candidacy
- **Low voltage:** If concurrent pericardial effusion
- **PVCs, NSVT:** Common; sustained VT = poor prognostic sign
- **Non-specific ST-T changes:** Diffuse ST depression, T wave inversion

4. Required ECG Features

ECG CLUES

- LBBB in young patient without CAD suggests familial DCM or myocarditis
- QRS >150 ms with LBBB = CRT indication (improves EF, symptoms, and survival)
- AF in a patient with new DCM suggests alcohol, thyroid disease, or familial DCM
- Frequent PVCs (>15% of total beats) can cause PVC-induced cardiomyopathy (reversible with ablation)

5. Diagnostic Criteria

- Echocardiography: LV dilation (LV end-diastolic dimension >55 mm or >117% predicted for body surface area) with EF <45%
- CMR: LGE pattern (mid-wall stripe) suggests non-ischemic etiology; T2 mapping for edema; ECV for diffuse fibrosis
- Genetic testing: TTN, LMNA, MYH7, MYBPC3, SCN5A, BAG3, RBM20; cascade screening of first-degree relatives
- Biomarkers: Elevated BNP/NT-proBNP; elevated troponin indicates worse prognosis
- Coronary angiography: To exclude ischemic cardiomyopathy in patients with risk factors
- Endomyocardial biopsy: When inflammatory or infiltrative etiology suspected and non-invasive testing inconclusive
- Family screening: First-degree relatives with echo regardless of genetic test result (30-50% familial)

6. Differential Diagnosis

Condition	Differentiating Features
Ischemic cardiomyopathy	Prior MI, regional WMA, angiographic CAD; LGE in subendocardial coronary distribution on CMR
Hypertensive heart disease	Long-standing uncontrolled hypertension; concentric LVH initially; EF improves with BP control
Valvular cardiomyopathy	Severe AR or MR causing volume overload; improves with valve surgery
Tachycardia-mediated CMP	Reversible with rate control; AF with uncontrolled RVR most common cause
Alcoholic cardiomyopathy	Heavy alcohol use >80 g/day for >5 years; improves with abstinence
Peripartum cardiomyopathy	Last month of pregnancy to 5 months postpartum; higher recovery rate

7. Symptoms

- **Progressive dyspnea:** Initially exertional, later at rest (backward failure)
- **Orthopnea, PND:** Elevated LV filling pressures
- **Peripheral edema:** Right heart failure
- **Fatigue, weakness:** Low forward cardiac output
- **Palpitations:** AF, PVCs, VT
- **Syncope:** VT or severe bradycardia
- **Family history of DCM or sudden death** in 30-50% of cases

8. Risk Factors

- **Genetic:** Family history of DCM or SCD; TTN truncating variants most common
- **Viral myocarditis:** Enteroviruses, adenovirus, parvovirus B19, HHV-6, SARS-CoV-2
- **Alcohol:** Heavy consumption (>80 g/day for >5 years)
- **Chemotherapy:** Anthracyclines (doxorubicin), trastuzumab
- **Autoimmune:** Anti-cardiac antibodies, SLE, scleroderma
- **Peripartum:** Late pregnancy and early postpartum
- **Nutritional:** Thiamine (beriberi), selenium (Keshan disease), carnitine deficiency
- **Tachycardia:** Chronic uncontrolled tachyarrhythmias

9. Immediate Actions

ACUTE DECOMPENSATION

1. Diuresis for congestion (IV loop diuretics)
2. Vasodilators (nitrates, hydralazine) if not hypotensive
3. Inotropes (dobutamine, milrinone) if cardiogenic shock
4. Non-invasive ventilation if pulmonary edema
5. Treat precipitant: infection, arrhythmia, medication non-adherence, ischemia
6. Temporary mechanical circulatory support (Impella, ECMO) if refractory

10. Treatment Overview

GDMT for HFrEF: Quadruple therapy: (1) ARNI (sacubitril/valsartan) preferred over ACE-I; (2) Evidence-based beta-blocker (metoprolol succinate, bisoprolol, carvedilol); (3) MRA (spironolactone, eplerenone); (4) SGLT2 inhibitor (dapagliflozin, empagliflozin). Diuretics for congestion.

Device therapy: ICD for primary prevention if EF <35% despite optimal medical therapy for >3 months. CRT (biventricular pacing) if LBBB with QRS >150 ms, EF <35%, NYHA II-IV despite GDMT.

Specific therapies: Treat underlying cause (alcohol cessation, immunosuppression for myocarditis/giant cell, thyroid replacement, removal of offending drugs). Genetic counseling for familial DCM.

Advanced therapies: Inotrope dependency, INTERMACS profile 2-4: durable LVAD (bridge to transplant or destination therapy). Heart transplantation for end-stage refractory to all therapies.

11. Follow-up Questions

- Symptom progression timeline?
- Family history of cardiomyopathy or sudden death?
- Alcohol consumption (quantify grams/day)?
- History of chemotherapy or radiation?
- Recent viral illness or COVID-19?
- Pregnancy/postpartum status?
- Current medications and adherence?
- Any history of arrhythmias?

12. References

1. Heidenreich PA, et al. 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure. *Circulation*. 2022;145(18):e895-e1032.
2. Pinto YM, et al. Proposal for a revised definition of dilated cardiomyopathy, hypokinetic non-dilated cardiomyopathy, and its implications for clinical practice. *Eur Heart J*. 2016;37(23):1850-1858.
3. Hershberger RE, et al. Genetic evaluation of cardiomyopathy - a Heart Failure Society of America practice guideline. *J Card Fail*. 2009;15(2):83-97.

Endocarditis

Category: Infectious Heart Disease **Priority: HIGH** **ICD-10:** I33.0

1. Definition

Infective endocarditis (IE) is an infection of the endocardial surface of the heart, most commonly affecting heart valves (native or prosthetic), but may also involve the mural endocardium, chordae tendineae, or intracardiac devices. It is characterized by vegetation formation (fibrin, platelets, microorganisms, inflammatory cells) on heart valves, leading to valvular destruction, embolic phenomena, and metastatic infection. IE has high morbidity and mortality (in-hospital mortality 15-30%) despite modern antimicrobial and surgical therapies.

2. Mechanism

Endothelial injury (turbulent flow across abnormal valves, jet lesions, indwelling catheters) exposes subendothelial collagen, promoting platelet and fibrin deposition forming sterile thrombotic vegetations. Transient bacteremia (dental procedures, IV drug use, indwelling catheters) seeds these vegetations. Organisms adhere to fibronectin and platelets, multiply, and form mature vegetations protected from host defenses and antibiotics. Valvular destruction causes regurgitation; septic emboli cause stroke, splenic abscesses, mycotic aneurysms, and metastatic infection; immune complex deposition causes glomerulonephritis and vasculitis.

3. ECG Findings

- **ECG often normal:** Changes indicate complications rather than primary disease
- **New PR prolongation (first-degree AV block):** Suggests periannular abscess (especially aortic valve IE extending into AV conduction system)
- **Worsening conduction or complete heart block:** Surgical emergency - indicates septal abscess extension
- **AF:** May develop; embolic risk high
- **New Q waves, ST changes:** If myocardial abscess or coronary embolism
- **Low voltage:** If pericardial effusion from valve ring abscess rupture

4. Required ECG Features

ECG WARNING SIGNS

- New AV block in a patient with aortic valve endocarditis = periannular abscess until proven otherwise
- Progressive PR prolongation to complete heart block indicates catastrophic extension requiring emergent surgery
- These ECG changes may be the first sign of complication even when patient appears stable

5. Diagnostic Criteria

- Modified Duke Criteria: Definite IE = 2 major, or 1 major + 3 minor, or 5 minor criteria
- Major criteria: (1) Blood cultures positive for typical organism (Strep viridans, Strep bovis, Staph aureus, Enterococcus) from 2 separate cultures, or persistently positive blood cultures; (2) Evidence of endocardial involvement on echo (oscillating vegetation, abscess, new valvular regurgitation, prosthetic valve dehiscence)
- Minor criteria: Predisposing heart condition or IV drug use, fever >38°C, vascular phenomena (arterial emboli, septic pulmonary infarcts, mycotic aneurysm, ICH, conjunctival hemorrhages, Janeway lesions), immunological phenomena (glomerulonephritis, Osler nodes, Roth spots, RF), microbiological evidence not meeting major criteria
- TTE first-line; TEE if prosthetic valve, poor TTE windows, or suspicion of complication (abscess, fistula)

6. Differential Diagnosis

Condition	Differentiating Features
Non-bacterial thrombotic endocarditis (NBTE/marantic)	Sterile vegetations in hypercoagulable states (cancer, APS); blood cultures negative; no fever
Libman-Sacks endocarditis	SLE-associated; small vegetations on both sides of valve; sterile; serologic markers
Myxoma	Mobile mass on echo; constitutional symptoms; negative blood cultures
Vasculitis	ANCA-associated; no vegetations; different organ involvement; biopsy diagnostic
Acute rheumatic fever	Children/young adults; migratory polyarthritis, chorea, erythema marginatum; ASO titers elevated

7. Symptoms

- **Fever:** Most common (>90%); may be low-grade or absent in elderly, immunocompromised, or CHF patients
- **New murmur or change in existing murmur:** Regurgitant murmur from valvular destruction
- **Embolic phenomena:** Stroke, splenic infarcts/abscess, peripheral arterial emboli
- **Immune phenomena:** Osler nodes (tender subcutaneous nodules on fingers/toes), Roth spots (retinal hemorrhages with pale centers), Janeway lesions (non-tender macules on palms/soles), splinter hemorrhages
- **Back pain:** Vertebral osteomyelitis, epidural abscess, or mycotic aneurysm
- **Glomerulonephritis:** Hematuria, proteinuria, elevated creatinine (immune complex deposition)
- **Night sweats, weight loss, fatigue:** Constitutional symptoms

8. Risk Factors

- **Structural heart disease:** Rheumatic valve disease, bicuspid aortic valve, degenerative valvular disease, MVP with thickening/regurgitation, HCM
- **Prosthetic valves:** Risk 1-6% per year; highest in first 6 months
- **Prior endocarditis:**
- **Intravenous drug use:** Right-sided IE (tricuspid valve); Staph aureus predominant
- **Intracardiac devices:** Pacemakers, ICDs, LVADs

- **Dental procedures:** With poor dentition or gingival disease
- **Indwelling catheters:** Hemodialysis, central lines
- **Immunosuppression:** HIV, transplant recipients, chronic steroids

9. Immediate Actions

EMERGENCY MANAGEMENT

1. Obtain 3 sets of blood cultures (aerobic and anaerobic) from different sites before antibiotics if patient stable
2. Empiric IV antibiotics immediately if patient unstable, septic, or after cultures obtained
3. Emergent TTE/TEE to confirm diagnosis and assess complications
4. Assess for complications: CNS emboli (CT head if neurologic symptoms), heart failure, abscess
5. **Surgical indications (urgent):** Acute heart failure from valve dysfunction, periannular abscess, fungal/ highly resistant organism, persistent bacteremia >7 days despite therapy, prosthetic valve dehiscence, recurrent emboli
6. Remove infected intracardiac devices

10. Treatment Overview

Antibiotics: Prolonged IV therapy (typically 4-6 weeks) based on organism and susceptibilities. Native valve: penicillin/ampicillin + gentamicin for streptococci; vancomycin for Staph aureus or MRSA; antistaphylococcal penicillin/ vancomycin + gentamicin for enterococci. Prosthetic valve: longer duration, broader coverage.

Surgery: Indicated for: heart failure due to valve dysfunction, uncontrolled infection (abscess, persistent bacteremia), prevention of embolism (large mobile vegetations >10 mm on aortic/mitral valve with embolic events), prosthetic valve endocarditis, fungal infection. Early surgery (within 48 hours) for heart failure or uncontrolled infection improves outcomes.

Monitoring: Serial blood cultures, TTE/TEE for vegetation changes, renal function, CNS imaging if embolic events. Anticoagulation controversial in native valve IE (generally avoid); continue if prosthetic valve and no contraindication.

Prophylaxis: Only for highest risk procedures (dental with manipulation of gingiva) in highest risk patients (prosthetic valves, prior IE, unrepaired cyanotic CHD, heart transplant with valvulopathy). Amoxicillin 2 g single dose 30-60 minutes before procedure.

11. Follow-up Questions

- Fever pattern? Duration? Highest temperature?

- Heart murmur history? Valve disease? Prosthetic valve?
- Intravenous drug use? (Assess non-judgmentally)
- Recent dental work, invasive procedures, or catheters?
- Embolic symptoms: stroke, back pain, abdominal pain, limb ischemia?
- Skin findings: Osler nodes, Janeway lesions, splinter hemorrhages?
- Urinary symptoms (glomerulonephritis)?
- Immunosuppression, HIV, transplant status?
- Prior endocarditis?

12. References

1. Habib G, et al. 2015 ESC Guidelines for the management of infective endocarditis. *Eur Heart J*. 2015;36(44):3075-3128.
2. Baddour LM, et al. Infective Endocarditis in Adults: Diagnosis, Antimicrobial Therapy, and Management of Complications. *Circulation*. 2015;132(15):1435-1486.
3. Wilson W, et al. Prevention of Infective Endocarditis: Guidelines From the AHA. *Circulation*. 2007;116(15):1736-1754.

Heart Failure

Category: Cardiac Function **Priority: HIGH** **ICD-10:** I50.9

1. Definition

Heart failure (HF) is a clinical syndrome characterized by structural or functional cardiac abnormalities resulting in impaired ventricular filling or ejection of blood, leading to cardinal symptoms (dyspnea, fatigue, exercise intolerance) and signs (edema, rales, elevated JVP) at rest or with stress. It affects approximately 64 million people worldwide with increasing prevalence due to aging populations. Classified by left ventricular ejection fraction: HFrEF (EF <=40%), HFmrEF (EF 41-49%), and HFpEF (EF >=50%).

2. Mechanism

HFrEF: Myocardial injury (MI, myocarditis, toxins) causes myocyte loss and replacement fibrosis, leading to chamber dilation and impaired contractility. Compensatory mechanisms (neurohormonal activation - RAAS, sympathetic nervous system, natriuretic peptides) initially maintain cardiac output but become maladaptive, causing vasoconstriction, volume overload, and progressive remodeling. **HFpEF:** Predominantly diastolic dysfunction from impaired ventricular relaxation and increased stiffness; associated with hypertension, diabetes, obesity, aging, and systemic inflammation. Increased filling pressures cause congestion despite preserved contractility. **HFmrEF:** Intermediate phenotype with features of both.

3. ECG Findings

- **AF:** Very common in both HFpEF and HFrEF; irregular rhythm predisposes to thromboembolism
- **LVH voltage criteria:** Suggests hypertensive heart disease or aortic stenosis as etiology
- **Left atrial enlargement:** P mitrale (broad, notched P in II); biphasic P in V1 with prominent terminal negative deflection
- **LBBB common in DCM:** QRS >150 ms indicates dyssynchrony and CRT candidacy
- **RBBB:** May indicate pulmonary hypertension or RV overload
- **ST-T changes:** Ischemic etiology if regional; nonspecific if diffuse
- **Q waves:** Prior MI (ischemic cardiomyopathy)
- **Low voltage:** Infiltrative (amyloid), pericardial effusion, COPD, obesity
- **Arrhythmias:** PVCs, VT (risk markers for SCD)

4. Required ECG Features

ECG CLUES TO ETIOLOGY

- Q waves in coronary distribution = ischemic cardiomyopathy (most common cause of HFrEF)
- LBBB in young patient = familial DCM or myocarditis
- LVH with strain + short PR + delta wave = WPW (tachycardia-mediated CMP)
- Low voltage with echo LVH = cardiac amyloidosis
- RVH, RAE = pulmonary hypertension, congenital heart disease

5. Diagnostic Criteria

- Framingham criteria (clinical): Major (paroxysmal nocturnal dyspnea, JVP >6 cm, rales, cardiomegaly, acute pulmonary edema, S3 gallop, CVP >16 cm H2O, hepatojugular reflux, weight loss >4.5 kg in 5 days on diuretics) + Minor (bilateral ankle edema, nocturnal cough, dyspnea on exertion, hepatomegaly, pleural effusion, vital capacity reduced by 1/3, tachycardia >120). Definite HF: 2 major or 1 major + 2 minor.
- Echocardiography: Defines EF, chamber sizes, wall thickness, valvular function, diastolic parameters, hemodynamics
- BNP/NT-proBNP: Elevated in HF (diagnostic and prognostic); NT-proBNP >900 pg/mL suggests HF in acute setting (age-adjusted cutoffs)
- CMR: Tissue characterization for etiology (LGE pattern in amyloid, myocarditis, sarcoidosis)
- 6-minute walk test: Functional capacity assessment
- NYHA functional class: I (no symptoms), II (symptoms with ordinary activity), III (symptoms with minimal activity), IV (symptoms at rest)

6. Differential Diagnosis

Condition	Differentiating Features
COPD/asthma	Wheezing, prolonged expiration, smoking history; echo normal; BNP not elevated
Pulmonary embolism	Acute dyspnea, pleuritic pain, risk factors; CTPA diagnostic; echo shows RV strain
Cirrhosis/ascites	Chronic liver disease, spider angiomas, caput medusae; normal cardiac function
Nephrotic syndrome	Heavy proteinuria, hypoalbuminemia; normal JVP; no cardiac abnormalities
Anemia	Pallor, fatigue out of proportion; normal JVP; no peripheral edema; elevated symptoms improve with transfusion
Obesity/deconditioning	Gradual dyspnea on exertion; normal echo; no congestion signs; improves with weight loss/exercise
Obstructive sleep apnea	Nocturnal symptoms, loud snoring, daytime somnolence; often coexists with HF

7. Symptoms

- **Left-sided (forward failure/low output):** Dyspnea on exertion (most common), fatigue, weakness, exercise intolerance, nocturnal cough (cardiac asthma)
- **Left-sided (backward failure/pulmonary congestion):** Orthopnea, paroxysmal nocturnal dyspnea, cough with pink frothy sputum (pulmonary edema)
- **Right-sided (backward failure/systemic congestion):** Peripheral edema (ankles, sacrum), abdominal bloating, early satiety, nausea (hepatic congestion), right upper quadrant pain, weight gain from fluid retention
- **Advanced:** Cachexia (cardiac wasting), Cheyne-Stokes respiration, renal dysfunction (cardiorenal syndrome)

8. Risk Factors

- **Coronary artery disease:** Most common cause of HFrEF (~60-70%)
- **Hypertension:** Most important cause of HFpEF; also contributes to HFrEF
- **Diabetes mellitus:** Increases risk 2-5 fold; independent risk factor
- **Valvular heart disease:** Aortic stenosis, mitral regurgitation

- **Obesity:** Strong risk factor for HFpEF
- **AF:** Causes and worsens HF; tachycardia-mediated cardiomyopathy
- **CKD:** Bidirectional relationship with HF (cardiorenal syndrome)
- **Sleep apnea:** Both central and obstructive contribute to HF progression
- **Cardiotoxic therapies:** Anthracyclines, trastuzumab, immune checkpoint inhibitors
- **Alcohol, cocaine:** Toxic cardiomyopathy
- **Family history:** Familial cardiomyopathies

9. Immediate Actions

ACUTE DECOMPENSATED HF

1. Sit upright, supplemental O2 (target SpO2 >92%), IV access
2. Loop diuretics: IV furosemide (1-2.5x home oral dose) or bumetanide; add thiazide (metolazone) if diuretic resistance
3. Vasodilators if SBP >110: IV nitroglycerin or nitroprusside (reduce preload and afterload)
4. Non-invasive ventilation (BiPAP) if respiratory distress or acidosis
5. Inotropes (dobutamine, milrinone) if cardiogenic shock or end-organ hypoperfusion
6. Treat trigger: infection, arrhythmia (rate control/chemical cardioversion for AF), ischemia, medication non-adherence
7. Renal function and electrolyte monitoring (K, Mg) with diuresis
8. ICU admission if: SBP <85, altered mental status, cold extremities, anuria, elevated lactate, escalating oxygen requirements

10. Treatment Overview

HFrEF - Guideline-Directed Medical Therapy (GDMT):

- Quadruple therapy: (1) ARNI (sacubitril/valsartan) preferred over ACE-I/ARB; (2) Evidence-based beta-blocker (metoprolol succinate, bisoprolol, carvedilol); (3) MRA (spironolactone, eplerenone); (4) SGLT2 inhibitor (dapagliflozin 10 mg, empagliflozin 10 mg)
- Diuretics for congestion (loop diuretics, add thiazide if resistant)
- Hydralazine/isosorbide dinitrate: Add if African American (A-HeFT trial) or if ACE-I/ARB/ARNI not tolerated
- Ivabradine: If sinus rhythm, EF <35%, HR >70 despite maximally tolerated beta-blocker
- Device therapy: ICD (EF <35% despite optimal therapy >3 months); CRT (LBBB + QRS >150 ms + EF <35%)

HFpEF: No therapy has shown mortality benefit. Treat comorbidities aggressively (hypertension, diabetes, obesity, AF). SGLT2i (dapagliflozin/empagliflozin) approved to reduce CV death and HF hospitalization. Diuretics for congestion. Exercise training improves symptoms and functional capacity.

Advanced HF: Inotrope-dependent, INTERMACS profile 2-4: durable LVAD (bridge to transplant or destination therapy). Heart transplantation for selected end-stage patients.

11. Follow-up Questions

- NYHA class? How far can you walk? How many pillows to sleep?
- Recent weight change? Daily weights trend?
- Current medications and adherence? (SGLT2i, ARNI, beta-blocker, MRA, diuretic doses)
- Any medication side effects? (Hypotension, hyperkalemia, worsening renal function)
- Triggers: missed doses, dietary indiscretion (salt), infection, arrhythmia?
- Baseline EF? Date of last echo? Trend?
- Device therapy? ICD/CRT? Last interrogation?
- Renal function baseline? Recent creatinine?
- Advanced care planning? Goals of care?

12. References

1. Heidenreich PA, et al. 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure. *Circulation*. 2022;145(18):e895-e1032.
2. McDonagh TA, et al. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*. 2021;42(36):3599-3726.
3. Packer M, et al. Effect of catheter-based renal denervation on mortality, cardiovascular events, and blood pressure. *J Am Coll Cardiol*. 2023;81(21):2024-2033.

Heart Valve Disease

Category: Valvular Heart Disease **Priority: HIGH** **ICD-10:** I38

1. Definition

Heart valve disease encompasses structural and functional abnormalities of any of the four cardiac valves (aortic, mitral, tricuspid, pulmonary) causing stenosis (restricted opening), regurgitation (incomplete closure), or both. Valvular heart disease affects approximately 2.5% of the population, with prevalence increasing significantly with age. Aortic stenosis and mitral regurgitation are the most common clinically significant lesions. Untreated severe valve disease causes progressive heart failure, arrhythmias, and death.

2. Mechanism

Aortic stenosis: Degenerative calcification (most common in elderly), bicuspid aortic valve (congenital, presents earlier), or rheumatic heart disease. **Aortic regurgitation:** Aortic root dilation (Marfan, hypertension, age-related), bicuspid valve, endocarditis, rheumatic disease. **Mitral stenosis:** Almost exclusively rheumatic heart disease (developing countries) or calcific (elderly in developed countries). **Mitral regurgitation:** Primary (organic: prolapse, endocarditis, rheumatic, chordal rupture) or secondary (functional: LV dilation causing annular dilation and leaflet tethering from ischemic or dilated cardiomyopathy). **Tricuspid regurgitation:** Functional (secondary to RV dilation from pulmonary hypertension or left-sided disease) or organic (endocarditis, carcinoid, rheumatic, Ebstein anomaly). **Pulmonary stenosis:** Congenital (most common), carcinoid, rheumatic (rare).

3. ECG Findings

- **Aortic stenosis:** LVH with strain pattern, LBBB, left atrial enlargement, AF in severe cases
- **Aortic regurgitation:** LVH with volume overload pattern (tall R waves, prominent Q waves, steep T waves), left axis deviation
- **Mitral stenosis:** LAE (P mitrale, biphasic P V1), AF (very common), RVH if pulmonary hypertension develops, right axis deviation
- **Mitral regurgitation:** LAE, LVH (volume overload), AF
- **Tricuspid regurgitation:** RAE (tall peaked P in II), RVH, AF
- **Pulmonary stenosis:** RVH (tall R V1, RAD), RAE

4. Required ECG Features

ECG CLUES

- New AF in a patient with known valvular disease often indicates hemodynamic worsening
- RVH pattern in mitral stenosis indicates advanced disease with pulmonary hypertension
- LAE is the earliest ECG finding in mitral valve disease
- AF + mitral stenosis = high thromboembolic risk (anticoagulation mandatory regardless of CHA_2DS_2-VASc)

5. Diagnostic Criteria

- Echocardiography: First-line; defines valve anatomy, severity of stenosis/regurgitation, chamber sizes, ventricular function, pulmonary pressures
- Stenosis severity: Valve area, mean gradient, jet velocity (aortic); valve area and pressure half-time (mitral)
- Regurgitation severity: Vena contracta width, PISA method, regurgitant volume/fraction, Doppler jet area, holodiastolic flow reversal in descending aorta (AR)
- CMR: Quantifies regurgitant fraction and ventricular volumes when echo is inconclusive
- Cardiac CT: Aortic valve calcium scoring; coronary CTA before TAVR; annular sizing
- Exercise testing: Objective assessment of functional capacity and symptom threshold in apparently asymptomatic severe valve disease

6. Differential Diagnosis

Condition	Differentiating Features
Physiologic valvular regurgitation	Trivial MR/TR on echo; normal valve structure; no chamber enlargement; no symptoms
Functional MR vs. organic MR	Functional: normal leaflets, annular dilation, tethering; organic: prolapse/flail, thickening, vegetations
Aortic sclerosis without stenosis	Thickening/calcification without significant gradient; Vmax <2.5 m/s
Hypertrophic obstructive cardiomyopathy	Dynamic LVOT obstruction; systolic anterior motion of MV; no valvular abnormality

7. Symptoms

- **Exertional dyspnea:** Universal across all severe valve lesions
- **Reduced exercise tolerance, fatigue:** Reduced cardiac output
- **Chest pain:** AS (supply-demand mismatch), AR (less common)
- **Syncope:** AS (exertional), any severe lesion with low output
- **Palpitations:** AF (common in MS, MR)
- **Peripheral edema:** Right-sided failure (TR, pulmonary hypertension from MS)

8. Risk Factors

- Age (degenerative calcification)
- Rheumatic fever history (MS, MR, AS, TR in developing countries)
- Bicuspid aortic valve (AS, AR)
- Connective tissue disorders (Marfan - AR, MVP)
- Infective endocarditis (any valve)
- Radiation therapy (aortic and mitral)
- Intravenous drug use (tricuspid endocarditis)
- Carcinoid syndrome (tricuspid and pulmonary, left-sided protected by pulmonary inactivation)
- Anorexigen drugs (fenfluramine/phentermine - now withdrawn)

9. Immediate Actions

ACUTE VALVE EMERGENCIES

1. Acute MR (papillary muscle/ chordal rupture, endocarditis): Afterload reduction (nitroprusside), IABP, emergent surgery
2. Acute AR (endocarditis, aortic dissection): Aortic dissection protocol if suspected; emergent surgery
3. AF with rapid response and MS: Rate control (avoid beta-blockers if possible), anticoagulation; urgent cardioversion if unstable
4. Pulmonary edema from any severe valve lesion: Diuretics, nitrates (avoid in AS), non-invasive ventilation

10. Treatment Overview

Severe symptomatic valve disease: Intervention indicated regardless of EF (symptom onset without intervention carries poor prognosis).

Aortic stenosis: TAVR (preferred for elderly/high-risk) or SAVR (younger/low-risk). See Aortic Stenosis entry.

Aortic regurgitation: Aortic valve replacement (SAVR). Root replacement if aortic root >5.0 cm (or >4.5 cm in Marfan). Vasodilators (nifedipine, ACE-I) may delay surgery in chronic severe AR.

Mitral stenosis: Percutaneous balloon commissurotomy if suitable anatomy (pliable, non-calcified valve, no LA thrombus); valve replacement if not. Anticoagulation mandatory if AF.

Mitral regurgitation: Primary MR: Mitral repair preferred over replacement (lower mortality, preserves LV function); transcatheter edge-to-edge repair (MitraClip) for high surgical risk. Secondary MR: GDMT for underlying cardiomyopathy; MitraClip if persistent symptomatic severe MR despite GDMT (COAPT criteria).

Tricuspid regurgitation: Annuloplasty (ring) at time of left-sided surgery if moderate+; isolated surgery if symptomatic severe TR despite medical therapy.

11. Follow-up Questions

- Known valve lesion? Severity? When last assessed?
- Symptom progression? Exercise tolerance?
- AF history? Anticoagulation status?
- Prior endocarditis? Prophylaxis indicated?
- Prior interventions (repair, replacement, valvuloplasty)?
- Symptoms of right heart failure? (edema, ascites)
- Pregnancy considerations if applicable?

12. References

1. Otto CM, et al. 2020 ACC/AHA Guideline for the Management of Patients With Valvular Heart Disease. *Circulation*. 2021;143(5):e72-e227.
2. Vahanian A, et al. 2021 ESC/EACTS Guidelines for the management of valvular heart disease. *Eur Heart J*. 2022;43(7):561-632.
3. Stone GW, et al. Transcatheter Mitral-Valve Repair in Patients with Heart Failure. *N Engl J Med*. 2018;379(24):2307-2318.

Hypertrophic Cardiomyopathy (HCM)

Category: Cardiomyopathy **Priority: HIGH** **ICD-10:** I42.1

1. Definition

Hypertrophic cardiomyopathy (HCM) is an inherited cardiomyopathy characterized by unexplained left ventricular hypertrophy (wall thickness >=15 mm in any segment, or >=13 mm with family history of HCM) in the absence of other causes of increased afterload (hypertension, aortic stenosis, athlete's heart). It is the most common inherited cardiomyopathy, affecting approximately 1 in 500 individuals (over 700,000 in the US). The hallmark is asymmetric septal hypertrophy with myofibrillar disarray and dynamic LV outflow tract obstruction in 70% of patients.

2. Mechanism

Autosomal dominant inheritance of mutations in sarcomere protein genes (MYBPC3 ~40%, MYH7 ~30%, TNNI3, TNNT2, TPM1, ACTC1, MYL2, MYL3). Penetrance incomplete and age-related. Mutations cause increased myofilament calcium sensitivity, abnormal force generation, and myocardial disarray (chaotic arrangement of myocytes with interstitial fibrosis). The hypertrophy is a compensatory response to impaired contractile function. Dynamic LVOT obstruction results from SAM (systolic anterior motion) of the mitral valve due to Venturi effect from accelerated flow in the outflow tract and papillary muscle displacement. Diastolic dysfunction from reduced compliance and ischemia from supply-demand mismatch contribute to symptoms.

3. ECG Findings

- **Massive QRS voltages:** Sokolow-Lyon: S V1 + R V5/V6 >= 35 mm (massive voltages possible >50 mm); Cornell: R aVL + S V3 >28 mm (M) or >20 mm (F)
- **Deep ST depression with T wave inversion:** In lateral leads (I, aVL, V5-V6) with downsloping ST segment - "strain pattern"
- **Asymmetric T wave inversion with downsloping ST:** Apical variant shows deep T wave inversion in V4-V6
- **Pathological Q waves:** In inferior (II, III, aVF) and/or lateral (I, aVL, V5-V6) leads - narrow (<40 ms), deep, without Q wave evolution (different from MI); due to initial septal depolarization vectors from hypertrophied septum
- **Left atrial enlargement:** From elevated LVEDP and MR
- **NSVT on monitoring:** Risk marker for SCD
- **WPW pattern:** Pre-excitation in PRKAG2 syndrome (glycogen storage, mimics HCM)

4. Required ECG Features

DIAGNOSTIC ECG CLUES

- Massive QRS voltage (>35 mm) + deep T wave inversion in lateral leads = classic HCM pattern
- "Giant" negative T waves in V4-V6 = apical HCM (Japanese variant, Yamaguchi syndrome)
- Q waves in II, III, aVF, I, aVL WITHOUT history of MI or CAD = septal Q waves from HCM
- LVH criteria + strain pattern in young patient without hypertension = HCM until proven otherwise

5. Diagnostic Criteria

- Echocardiography: Wall thickness >=15 mm (or >=13 mm with family history) in any myocardial segment, not explained by other causes
- CMR: Gold standard for quantification of hypertrophy pattern (asymmetric, concentric, apical, mid-ventricular); LGE (patchy mid-wall, often at RV insertion points) identifies fibrosis and SCD risk; can detect apical variant missed by echo
- Genetic testing: Identifies pathogenic sarcomere mutation in 60-70%; cascade screening of first-degree relatives recommended regardless of echo findings
- Exercise testing: Assess for exertional obstruction, arrhythmias, BP response (abnormal if SBP falls or fails to rise >20 mmHg)
- Holter monitoring: Detect NSVT (risk stratification)
- SCD risk stratification: 5 major markers (massive LVH >30 mm, unexplained syncope, NSVT, abnormal BP response to exercise, family history of SCD from HCM)

6. Differential Diagnosis

Condition	Differentiating Features
Athlete's heart	LVH <15 mm; symmetric; normal diastolic function; deconditioning reverses in 3 months; no LGE; no family history
Hypertensive heart disease	Long-standing HTN history; concentric LVH usually <15 mm; no SAM; no outflow gradient; responds to BP control
Aortic stenosis	Valvular calcification; high transvalvular gradients; no SAM; surgery indicated
Fabry disease	Alpha-galactosidase deficiency; concentric LVH; low voltage may be present; enzyme replacement; genetic testing
Amyloidosis	Low QRS voltage despite LVH; restrictive physiology; Congo red positive; different LGE pattern on CMR
PRKAG2 syndrome	WPW + conduction disease + glycogen storage mimicking HCM; PRKAG2 mutation

7. Symptoms

- **Dyspnea on exertion:** Most common (90%); due to elevated filling pressures and outflow obstruction
- **Chest pain:** Angina-like (from supply-demand mismatch despite normal coronaries) or atypical
- **Syncope or presyncope:** Exertional syncope is RED FLAG for SCD risk (due to arrhythmia or outflow obstruction with vasodilation)
- **Palpitations:** AF (20-25% lifetime risk), VT
- **Family history of sudden cardiac death:** Key component of risk stratification
- **Asymptomatic:** Many detected through family screening or incidental finding

8. Risk Factors

- **Genetic:** Autosomal dominant; family history of HCM or SCD
- **Double mutations:** Worse prognosis
- **Massive LVH:** >30 mm wall thickness (strongest SCD risk factor)
- **NSVT:** On ambulatory monitoring
- **Abnormal BP response to exercise:** SBP falls or fails to rise >20 mmHg
- **Unexplained syncope:** Especially exertional

- **Apical aneurysm:** Increasingly recognized high-risk subgroup

9. Immediate Actions

ACUTE MANAGEMENT

1. If AF with RVR: Urgent rate/rhythm control; anticoagulation (very high stroke risk in HCM + AF); avoid class IC antiarrhythmics
2. Sustained VT or VF: Defibrillation; ICD if not present
3. Exertional syncope: Avoid exercise until evaluated; comprehensive risk stratification
4. Acute decompensated HF: Diuretics, avoid vasodilators and inotropes if obstructive (worsen gradient)

10. Treatment Overview

Medical: Beta-blockers (first-line) or non-DHP CCBs (verapamil, diltiazem) for symptom control and outflow gradient reduction. Disopyramide (Class IA antiarrhythmic with negative inotropic effect) for refractory symptoms. Avoid vasodilators, diuretics, and digoxin in obstructive HCM (reduce preload, worsen obstruction). Diuretics cautiously for volume overload. Avoid strenuous exercise, competitive sports.

Septal reduction therapy: For symptomatic patients (NYHA III-IV) despite medical therapy with resting/provocable gradient >50 mmHg. Surgical myectomy (Morrow procedure): gold standard, most effective, lowest recurrence. Alcohol septal ablation: catheter-based, less invasive, suitable for older patients or comorbidities.

SCD prevention: Shared decision-making for ICD. Primary prevention if >=1 major risk factor (massive LVH, unexplained syncope, NSVT, abnormal BP response, high-risk genetic mutation). Secondary prevention for survivors of cardiac arrest or sustained VT.

AF management: Anticoagulation mandatory regardless of CHA_2DS_2-VASc. Rate/rhythm control. Disopyramide has both antiarrhythmic and outflow gradient benefits.

Screening: First-degree relatives: clinical screening (ECG, echo) at diagnosis, every 12-18 months ages 12-18, every 5 years adults; genetic testing if mutation

known.

11. Follow-up Questions

- Any syncope, especially during or after exercise? (RED FLAG)
- Family history of sudden death or HCM?
- Current symptoms: dyspnea, chest pain, palpitations?
- Exercise tolerance? Any restriction?
- Any prior genetic testing results?
- Any history of AF?
- Last echo results? Maximum wall thickness? Gradient?
- Holter monitoring results? Any NSVT?

12. References

1. Ommen SR, et al. 2020 AHA/ACC Guideline for the Diagnosis and Treatment of Patients With Hypertrophic Cardiomyopathy. *Circulation*. 2020;142(25):e558-e631.

2. Maron BJ, et al. Contemporary Natural History and Management of Nonobstructive Hypertrophic Cardiomyopathy. *J Am Coll Cardiol*. 2016;67(12):1399-1409.

3. Maron MS, et al. Hypertrophic Cardiomyopathy Is Predominantly a Disease of Left Ventricular Outflow Tract Obstruction. *Circulation*. 2006;114(21):2232-2239.

4. Elliott PM, et al. 2014 ESC Guidelines on diagnosis and management of hypertrophic cardiomyopathy. *Eur Heart J*. 2014;35(39):2733-2779.

Ischaemic Heart Disease

Category: Coronary Artery Disease **Priority: HIGH** **ICD-10:** I25.1

1. Definition

Ischaemic heart disease (IHD), synonymous with coronary artery disease (CAD), is a condition in which reduced blood supply to the myocardium causes myocardial ischemia, leading to angina pectoris, myocardial infarction, heart failure, arrhythmias, and sudden cardiac death. IHD is the leading cause of death globally, responsible for approximately 9 million deaths annually. Chronic IHD refers to stable coronary disease with symptoms of angina or ischemia without acute plaque rupture.

2. Mechanism

Atherosclerosis of the epicardial coronary arteries is the fundamental process. Endothelial dysfunction, lipid accumulation, inflammation, and smooth muscle proliferation cause progressive luminal narrowing. Plaque rupture or erosion with superimposed thrombosis causes ACS. In chronic stable IHD, fixed stenosis (>70%) limits coronary flow reserve. During exertion or stress, increased myocardial oxygen demand cannot be met by flow increase through the stenotic vessel, causing ischemia. Endothelial dysfunction also causes abnormal vasomotor tone (paradoxical vasoconstriction). Microvascular dysfunction contributes to ischemia in non-obstructive coronary arteries (INOCA).

3. ECG Findings

- **ST depression:** Horizontal or downsloping ST depression during symptoms or stress testing (reversible ischemia)
- **T wave inversion:** In leads corresponding to ischemic territory; may be persistent or dynamic
- **Prior Q waves:** From old infarction; persistent ST-T changes in territory of prior MI
- **LVH:** From chronic pressure overload and compensatory mechanisms
- **Dynamic changes:** ST-T changes with exertion or symptoms that normalize with rest
- **Resting ECG often normal:** Especially in single-vessel disease; does not exclude significant CAD

4. Required ECG Features

ECG IN CHRONIC IHD

- Exercise-induced ST depression >= 1 mm in 2+ contiguous leads = positive stress test, indicates ischemia
- ST depression at rest that worsens with pain suggests high-risk unstable angina (transitioning to ACS)
- Pathological Q waves with persistent ST-T changes indicate prior MI
- Normal resting ECG does NOT exclude significant CAD

5. Diagnostic Criteria

- Clinical: Typical angina (substernal pressure with exertion, relieved by rest/nitroglycerin)
- Non-invasive testing: Exercise ECG (treadmill), stress echocardiography (wall motion abnormalities with stress), myocardial perfusion imaging (SPECT or PET - reversible defects indicate ischemia)
- Coronary CTA: Anatomic assessment; high negative predictive value; calcium scoring for risk assessment
- Invasive coronary angiography: Gold standard for anatomy; FFR for functional significance of intermediate lesions
- ISCHEMIA trial: Moderate-severe ischemia on stress testing does not necessarily mandate invasive strategy if medically stable - OMT first

6. Differential Diagnosis

Condition	Differentiating Features
ACS/NSTEMI	Rest pain, dynamic ECG changes, elevated troponins; requires urgent evaluation
Microvascular angina (INOCA)	Typical angina with non-obstructive coronaries; abnormal coronary flow reserve; acetylcholine testing
Vasospastic angina	Rest pain with transient ST elevation; responds to nitrates; acetylcholine challenge positive
Hypertrophic cardiomyopathy	Younger, murmur, LVH on echo, normal coronaries
Anemia	Pallor, fatigue, high-output state; anemia confirmed on CBC; improves with transfusion

7. Symptoms

- **Stable angina:** Substernal pressure, tightness, burning with exertion; relieved by rest/nitroglycerin within 5 minutes; duration 2-15 minutes
- **Angina equivalents:** Dyspnea, fatigue, epigastric discomfort (elderly, women, diabetics)
- **Silent ischemia:** No symptoms despite objective evidence of ischemia
- **Progressive symptoms:** Decreasing exercise tolerance, increasing frequency (warrants evaluation for ACS)

8. Risk Factors

- Major: Age, male sex, smoking, hypertension, diabetes, dyslipidemia, obesity, family history
- Emerging: Lp(a), CKD, inflammatory diseases, OSA, depression

9. Immediate Actions

> **ACUTE SYMPTOMS**
>
> 1. If rest pain or crescendo symptoms: evaluate as ACS (troponins, serial ECGs)
> 2. Otherwise: optimize medical therapy; non-invasive testing for diagnosis/risk stratification

10. Treatment Overview

Medical: Anti-ischemic (beta-blockers, CCBs, nitrates, ranolazine), antiplatelet (aspirin), high-intensity statin, ACE-I/ARB. SGLT2i for diabetic patients.

Revascularization: PCI or CABG for left main disease, 3-vessel disease, proximal LAD, or refractory symptoms despite OMT. FFR-guided PCI for intermediate lesions.

Lifestyle: Smoking cessation, cardiac rehabilitation, weight management, BP control, diabetes management.

11. Follow-up Questions

- CCS angina class? Trend over time?
- Risk factor assessment and management?
- Current medications and response?
- Prior stress test, angiography, revascularization?

12. References

1. Knuuti J, et al. 2019 ESC Guidelines for chronic coronary syndromes. *Eur Heart J*. 2020;41(3):407-477.

2. Maron DJ, et al. Initial Invasive or Conservative Strategy for Stable Coronary Disease. *N Engl J Med*. 2020;382(15):1395-1407.

Kawasaki Disease

Category: Vasculitis (Pediatric) **Priority: HIGH** **ICD-10:** M30.3

1. Definition

Kawasaki disease (KD) is an acute, self-limited vasculitis of childhood predominantly affecting medium-sized arteries, especially the coronary arteries. It is the leading cause of acquired heart disease in children in developed countries. Without treatment, 20-25% of patients develop coronary artery aneurysms; with IVIG treatment within 10 days of fever onset, this risk is reduced to <5%. KD most commonly affects children under 5 years (80% of cases) and has a male predominance (1.5:1).

2. Mechanism

Unknown etiology, though infectious trigger in genetically susceptible individuals is suspected. Aberrant immune activation causes pan-vasculitis with endothelial injury and destruction of the arterial media. In the acute phase, neutrophilic infiltration damages coronary arterial walls. Subacute phase (7-21 days): pro-inflammatory cytokine storm with coronary arteritis; highest risk of aneurysm formation. Convalescent phase: granulation tissue formation, myofibroblast proliferation, potential aneurysm formation, thrombosis, stenosis. Chronic phase: vascular remodeling, calcification, progressive stenosis of giant aneurysms. Pathophysiology involves T-cell activation, B-cell polyclonal expansion, and cytokine-mediated vascular injury.

3. ECG Findings

- **Acute phase:** Sinus tachycardia (fever, inflammation), ST-T changes (myocarditis, pericarditis component)
- **Prolonged PR interval:** First-degree AV block (myocarditis, edema)
- **Prolonged QT interval:** Due to inflammation or electrolyte disturbances
- **Arrhythmias:** Supraventricular and ventricular arrhythmias during acute myocarditis
- **Low voltage QRS:** Pericardial effusion
- **Ischemic changes:** If coronary artery thrombosis or giant aneurysm with stenosis
- **Q waves:** If prior myocardial infarction from aneurysm thrombosis
- **Diffuse ST-T changes:** During acute myocarditis phase

4. Required ECG Features

> **ECG IN KD**
>
> No specific ECG finding is diagnostic of Kawasaki disease. ECG serves to assess myocarditis, pericarditis, arrhythmias, and ischemia. Serial ECGs during acute illness and long-term follow-up for patients with coronary aneurysms. New ischemic changes in a child with known KD and coronary aneurysms indicate thrombosis or progressive stenosis.

5. Diagnostic Criteria

- Fever >5 days plus >=4 of 5 principal features: (1) Bilateral conjunctival injection (non-exudative); (2) Mucosal changes (strawberry tongue, fissured lips, injected pharynx); (3) Cervical lymphadenopathy (>1.5 cm, usually unilateral); (4) Polymorphous rash (truncal, extremities); (5) Extremity changes (erythema/edema of palms/soles, periungual desquamation in convalescent phase)
- Incomplete (atypical) KD: Fever with <4 criteria plus supportive features (elevated CRP/ESR, anemia, elevated LFTs, thrombocytosis after day 7, pyuria, aseptic meningitis, ultrasound evidence of gallbladder hydrops)
- Echocardiography: Assess coronary arteries; aneurysm defined as internal diameter >3 mm in children <5 years or >4 mm in older children, or segment diameter >1.5x adjacent segment
- Z-score: Coronary artery dimension adjusted for body surface area; Z >2.5 = dilation, Z >5 = large/giant aneurysm

6. Differential Diagnosis

Condition	Differentiating Features
Scarlet fever	Streptococcal pharyngitis, sandpaper rash, desquamation, ASO positive; no conjunctivitis or extremity changes
Measles	Cough, coryza, conjunctivitis, Koplik spots; rash cephalocaudal spread; different timeline
Stevens-Johnson syndrome	Mucosal erosions, target lesions, medication trigger; no coronary involvement
Toxic shock syndrome	Hypotension, multi-organ failure, desquamation; Staph/Strep toxin-mediated
Juvenile idiopathic arthritis	Arthritis, rash, fever; no mucosal or conjunctival changes; no coronary involvement
Leptospirosis	Exposure history, renal involvement, conjunctival suffusion; serology diagnostic

7. Symptoms

- **Fever >5 days:** High-spiking, often >39°C, unresponsive to antibiotics
- **Irritability:** Especially in infants (very prominent feature)
- **Bilateral conjunctival injection:** Non-exudative, limbal sparing
- **Mucosal changes:** Strawberry tongue (prominent papillae), fissured/cracked lips, injected pharynx
- **Cervical lymphadenopathy:** Often unilateral, >1.5 cm, non-fluctuant
- **Polymorphous rash:** Trunk and extremities; various morphologies
- **Extremity changes:** Erythema and edema of palms/soles (acute); periungual desquamation (subacute, week 2-3)
- **Arthralgia/arthritis:** Especially knees and ankles
- **Abdominal pain, vomiting, diarrhea:** Gastrointestinal involvement

8. Risk Factors

- Age <5 years (peak incidence 6 months - 2 years)
- Male sex (1.5:1)
- Asian or Pacific Islander ancestry (highest incidence in Japan)
- Genetic polymorphisms: ITPKC, CASP3, HLA-B
- Seasonal: Winter and spring peaks; geographic outbreaks suggest infectious trigger

- Delay in treatment >10 days: Higher risk of coronary aneurysm
- IVIG resistance: Persistent or recrudescent fever after initial IVIG

9. Immediate Actions

ACUTE MANAGEMENT

1. **IVIG 2 g/kg** single infusion over 10-12 hours within 10 days of fever onset (reduces aneurysm risk from 25% to <5%)
2. **High-dose aspirin** (80-100 mg/kg/day divided QID) until afebrile 48-72 hours, then low-dose (3-5 mg/kg/day) for antiplatelet effect (duration 6-8 weeks or longer if aneurysms present)
3. Echocardiogram at diagnosis, 2 weeks, 6-8 weeks; more frequently if aneurysms identified
4. **Rescue therapy for IVIG resistance:** Second IVIG dose, IV methylprednisolone, infliximab, or cyclosporine
5. Anticoagulation (warfarin or LMWH) for giant aneurysms (>8 mm)

10. Treatment Overview

Acute: IVIG + high-dose aspirin. Monitor for IVIG resistance (fever persists or returns 36 hours after IVIG completion). Second IVIG dose if resistant. Add corticosteroids or infliximab for persistent resistance.

Subacute/convalescent: Low-dose aspirin antiplatelet therapy. Serial echo surveillance. Risk stratification based on coronary artery status.

Long-term management: No aneurysms: aspirin 6-8 weeks; normal echo at 6-8 weeks = low risk, no restrictions. Small aneurysms (<5 mm): long-term aspirin until regression. Medium aneurysms (5-8 mm): aspirin indefinitely + additional antiplatelet; annual stress testing after age 10. Giant aneurysms (>8 mm): dual antiplatelet or warfarin (target INR 2-2.5); cardiology follow-up every 6-12 months with stress testing; angiography if ischemia detected.

Intervention for coronary complications: PCI or CABG for significant stenosis; cardiac transplantation for end-stage ischemic cardiomyopathy.

Live vaccines: Delay MMR and varicella for 11 months after IVIG (passive antibody interference).

11. Follow-up Questions

- Age at diagnosis? Date of fever onset?
- Was IVIG given? On what day of illness? (Earlier = better outcomes)

- Any fever recurrence after IVIG? (IVIG resistance)
- Coronary artery status: normal, dilated, small/medium/giant aneurysm?
- Current medications (aspirin, anticoagulation)?
- Any chest pain, palpitations, or exercise intolerance? (ischemia)
- Last echocardiogram results?
- Immunization status (live vaccines delayed post-IVIG)?

12. References

1. McCrindle BW, et al. Diagnosis, Treatment, and Long-Term Management of Kawasaki Disease: A Scientific Statement for Health Professionals From the AHA. *Circulation*. 2017;135(17):e927-e999.
2. Newburger JW, et al. A Single Intravenous Infusion of Gamma Globulin as Compared with Four Infusions in the Treatment of Acute Kawasaki Syndrome. *N Engl J Med*. 1991;324(23):1633-1639.
3. Suda K, et al. Coronary artery lesions of Kawasaki disease: cardiac catheterization findings of 1100 cases. *Pediatr Cardiol*. 1989;10(4):161-165.

Marfan Syndrome

Category: Connective Tissue Disorder **Priority: HIGH** **ICD-10:** Q87.4

1. Definition

Marfan syndrome is an autosomal dominant connective tissue disorder caused by mutations in the FBN1 gene encoding fibrillin-1, a glycoprotein essential for elastic fiber formation and extracellular matrix integrity. It affects the skeletal, ocular, cardiovascular, and pulmonary systems. Cardiovascular manifestations are the most life-threatening, including aortic root dilation, aortic dissection, and mitral valve prolapse. The prevalence is approximately 1 in 5,000 individuals with no ethnic or geographic predilection.

2. Mechanism

FBN1 mutations cause abnormal fibrillin-1 microfibrils, leading to: (1) weakened connective tissue with loss of elastic fiber integrity in the aortic wall media (cystic medial necrosis), causing progressive aortic root dilation and dissection risk; (2) myxomatous degeneration of mitral valve leaflets causing prolapse and regurgitation; (3) abnormal skeletal growth (excessive longitudinal bone overgrowth). TGF-beta signaling dysregulation contributes to pathogenesis - losartan (ARB) may reduce TGF-beta signaling and slow aortic growth. The aortic root dilation begins at the sinuses of Valsalva and progresses to involve the ascending aorta.

3. ECG Findings

- **ECG often normal:** Changes relate to cardiac complications
- **Repolarization changes:** Nonspecific ST-T changes from mitral valve prolapse
- **Atrial/ventricular arrhythmias:** From MVP with significant MR
- **AF, VT:** May occur with severe MVP or aortic regurgitation
- **Left axis deviation:** May be present with MVP
- **Q waves:** May mimic inferior MI due to septal changes (uncommon)
- **Prolonged QT interval:** In some patients
- **Bradycardia:** From increased vagal tone or conduction abnormalities

4. Required ECG Features

ECG IN MARFAN SYNDROME

ECG is not diagnostic for Marfan syndrome. The primary cardiac assessment is echocardiographic measurement of aortic root dimensions. ECG serves to detect arrhythmias from MVP, monitor for ischemia if aortic dissection suspected, and establish baseline for perioperative monitoring.

5. Diagnostic Criteria

- Revised Ghent Criteria (2010): Score-based system incorporating: aortic root Z-score (>=2 = 1 point, >3 = 2 points, >4 or dissection = 3 points), FBN1 mutation (with known aortic disease = 2 points, without known aortic disease = 1 point), ectopia lentis (major criterion = 2 points), systemic features (wrist and thumb sign = 3 points, either alone = 1 point; pectus carinatum = 2 points, excavatum or chest asymmetry = 1 point; hindfoot deformity = 2 points, plain pes planus = 1 point; pneumothorax = 2 points; dural ectasia = 2 points; protusio acetabuli = 2 points; reduced US/LS ratio or increased arm/height and no scoliosis = 1 point; scoliosis or thoracolumbar kyphosis = 1 point; reduced elbow extension = 1 point; facial features = 1 point; skin striae = 1 point; myopia >3 diopters = 1 point; mitral valve prolapse = 1 point)
- Aortic root Z-score on echo: >=2 in adults (adjusted for BSA and age)
- Genetic testing: FBN1 pathogenic mutation confirms
- Exclusion of alternative diagnoses: Loeys-Dietz (TGFBR1/2), Ehlers-Danlos vascular type (COL3A1), familial aortic aneurysm

6. Differential Diagnosis

Condition	Differentiating Features
Loeys-Dietz syndrome	Widely spaced eyes, bifid uvula/cleft palate, arterial tortuosity; more aggressive vascular disease; TGFBR1/2 mutation
Ehlers-Danlos vascular type	Thin translucent skin, easy bruising, arterial/uterine rupture at young age; COL3A1 mutation
Familial thoracic aortic aneurysm	Isolated aortic disease without systemic features; ACTA2, MYH11, or FBN1 mutation with incomplete expression
MASS phenotype	Mitral valve, Aorta, Skin, Skeletal features but NO ectopia lentis and NO aortic root >Z=2; FBN1 variant
Homocystinuria	Intellectual disability, ectopia lentis (downward), thromboembolism; elevated homocysteine

7. Symptoms

- **Tall stature:** Often >95th percentile; arm span/height ratio >1.05
- **Chest pain:** Aortic dissection must always be excluded (sudden, severe, tearing)
- **Palpitations:** From MVP, arrhythmias
- **Dyspnea:** From MR, AR, or heart failure
- **Visual problems:** Ectopia lentis (upward lens dislocation), myopia, retinal detachment
- **Skeletal:** Pectus excavatum/carinatum, scoliosis, arachnodactyly, joint hypermobility
- **Spontaneous pneumothorax:** Apical lung bullae
- **Back pain:** Dural ectasia (enlargement of thecal sac at lumbosacral level)

8. Risk Factors

- **Genetic:** Autosomal dominant FBN1 mutation; 25% de novo mutations
- **Family history:** First-degree relative with Marfan or aortic dissection
- **Aortic root diameter:** >5.0 cm is high risk for dissection
- **Rapid aortic growth:** >0.5 cm/year
- **Prior aortic dissection:** High risk for recurrence
- **Pregnancy:** Increased hemodynamic stress; risk of dissection, especially if aortic root >4.0 cm
- **Competitive sports, isometric exercise:** Increase aortic wall stress
- **Hypertension:** Accelerates aortic dilation

9. Immediate Actions

AORTIC DISSECTION EMERGENCY

1. If acute chest/back pain with tearing quality: Immediate beta-blockade (esmolol or labetalol IV)
2. Emergent CT angiography or TEE to confirm/exclude dissection
3. Avoid isolated vasodilators (increase dP/dT)
4. Type A dissection: Emergent cardiothoracic surgery
5. Type B dissection: Medical management unless complications

10. Treatment Overview

Aortic protection: Beta-blockers (propranolol, atenolol) first-line to reduce aortic wall stress (dP/dT); add ARB (losartan) to reduce TGF-beta signaling (showed benefit in pediatric trials). Target BP <130/80 in adults, <95th percentile in children.

Aortic root surveillance: Annual echocardiography (every 6 months if rapidly growing or near threshold). MRI/CT for poor echo windows or beyond ascending aorta.

Prophylactic surgery: Aortic root replacement when diameter reaches 5.0 cm in adults (4.5 cm if Loeys-Dietz features, family history of dissection at <5.0 cm, or rapid growth >0.5 cm/year); 4.2-4.6 cm in children (based on BSA); before pregnancy if >4.0-4.5 cm.

Valve-sparing root replacement (David or Yacoub procedure): Preferred in experienced centers; preserves native aortic valve, avoiding prosthetic valve and anticoagulation.

Activity restrictions: Avoid competitive/isometric sports (weightlifting, wrestling), contact sports; aerobic exercise encouraged at moderate intensity.

MVP management: Surgery if severe MR; endocarditis prophylaxis only if prior IE.

Pregnancy: Preconception counseling; echo assessment; beta-blockers continued; vaginal delivery acceptable if aortic root <4.0 cm; cesarean if >4.5 cm.

11. Follow-up Questions

- Height, arm span, arm span/height ratio?
- Any chest pain? Characterize - tearing? sudden onset? radiation?
- Family history of Marfan, tall stature, sudden death, aortic dissection?
- Eye problems? Lens dislocation? High myopia? Retinal detachment?
- Skeletal features: pectus, scoliosis, joint laxity, long fingers?

- Prior aortic root measurements? Trend?
- Current medications (beta-blocker, ARB)?
- Blood pressure control?
- Pregnancy considerations? Planning pregnancy?

12. References

1. Loeys BL, et al. The revised Ghent nosology for the Marfan syndrome. *J Med Genet*. 2010;47(7):476-485.
2. Judge DP, Dietz HC. Marfan's syndrome. *Lancet*. 2005;366(9501):1965-1976.
3. Pyeritz RE. Marfan Syndrome and Related Disorders. In: *Emery and Rimoin's Principles and Practice of Medical Genetics*. 2013.
4. Baumgartner H, et al. 2021 ESC/EACTS Guidelines for the management of valvular heart disease. *Eur Heart J*. 2022;43(7):561-632.

Myocarditis

Category: Inflammatory Heart Disease **Priority: HIGH** **ICD-10:** I51.4

1. Definition

Myocarditis is an inflammatory disease of the myocardium characterized by myocyte injury in the absence of ischemia, most commonly due to viral infection but also caused by autoimmune conditions, toxins, drugs, and hypersensitivity reactions. It can present with a wide clinical spectrum from asymptomatic ECG changes to fulminant heart failure, cardiogenic shock, arrhythmias, and sudden death. Myocarditis accounts for approximately 10-20% of sudden cardiac death in young adults and is a leading cause of dilated cardiomyopathy.

2. Mechanism

Three phases: (1) Acute phase (days 1-3): Viral entry via Coxsackie-adenovirus receptor (CAR) and decay-accelerating factor (DAF); direct viral cytotoxicity causes myocyte necrosis. (2) Subacute phase (days 4-14): Innate and adaptive immune response; macrophage and T-cell infiltration clears virus but may cause collateral myocardial damage. (3) Chronic phase (>14 days): Either resolution with healing (fibrosis), persistent inflammation causing DCM, or transition to autoimmune myocarditis with anti-heart antibodies. In hypersensitivity/eosinophilic myocarditis, drug-induced immune reaction causes eosinophilic infiltration. Giant cell myocarditis involves aggressive T-cell mediated destruction with multinucleated giant cells.

3. ECG Findings

- **Diffuse ST elevation:** Most common finding; pericarditis pattern if epicardial involvement (concave, widespread)
- **ST depression:** May mimic ischemia; regional or diffuse
- **T wave inversion:** Diffuse or regional; often deep and widespread
- **Any type of arrhythmia:** Sinus tachycardia (most common), VT, complete heart block, AF, frequent PVCs
- **AV block:** First-degree, Mobitz, or complete heart block (especially Lyme disease, sarcoidosis, giant cell)
- **Bundle branch blocks:** Due to septal inflammation
- **Q waves:** May develop if necrosis is significant (pseudo-infarct pattern)
- **Low voltage QRS:** In severe cases with extensive myocardial edema
- **QT prolongation:** Due to repolarization abnormalities

4. Required ECG Features

ECG CLUES

- Diffuse ST elevation + PR depression + pericarditic chest pain = myopericarditis
- Young patient + ST elevation/Q waves + normal coronaries on angiography = myocarditis
- Any arrhythmia (including complete heart block) in context of recent viral illness suggests myocarditis
- Pseudo-infarct pattern (Q waves without CAD) with recent viral prodrome is highly suggestive

5. Diagnostic Criteria

- Dallas criteria (endomyocardial biopsy): Inflammatory infiltrate with myocyte necrosis in the absence of ischemia
- CMR (Lake Louise criteria, 2018 update): At least one T2-based criterion (T2-weighted imaging or elevated T2 mapping indicating edema/inflammation) AND at least one T1-based criterion (LGE, elevated T1 mapping, or elevated ECV indicating myocardial injury/necrosis/fibrosis)
- Elevated troponins (hs-cTnI or hs-cTnT): Sensitive but nonspecific; higher levels indicate more extensive injury
- Elevated CRP, ESR: Inflammatory markers
- NT-proBNP: Elevated with ventricular dysfunction
- PCR/viral studies: Blood, nasopharyngeal, or myocardial tissue for viral genome detection
- Autoimmune markers: ANA, anti-dsDNA, anti-Ro/La if autoimmune suspected
- Exclude MI: Coronary angiography or CTA if risk factors or atypical presentation

6. Differential Diagnosis

Condition	Differentiating Features
Acute MI (STEMI/NSTEMI)	Risk factors, regional ECG changes matching coronary territory, culprit lesion on angiography
Takotsubo cardiomyopathy	Postmenopausal woman, emotional trigger, apical ballooning, no diffuse ST elevation/PR depression
Pericarditis	Diffuse ST elevation + PR depression without myocardial injury markers; preserved LV function
Drug-induced cardiomyopathy	Temporal relation to offending drug (anthracyclines, immune checkpoint inhibitors); different histology
Giant cell myocarditis	Fulminant course, complete heart block, refractory VT; multinucleated giant cells on biopsy
Sarcoidosis	Non-caseating granulomas; extracardiac manifestations; different CMR LGE pattern (mid-wall/epicardial)
Sepsis-induced cardiac dysfunction	Systemic infection; shock; reversible with source control; no specific cardiac inflammation

7. Symptoms

- **Viral prodrome:** Recent fever, myalgias, URI symptoms, GI illness (1-2 weeks prior)
- **Chest pain:** Pleuritic, sharp (pericarditic component) or anginal (myocardial involvement)
- **Dyspnea, fatigue:** Heart failure symptoms from LV dysfunction
- **Palpitations:** Arrhythmias
- **Syncope:** High-grade AV block or VT
- **Recent COVID-19 or vaccination:** mRNA vaccines (rare, typically mild, self-limited in young males)
- **Fulminant presentation:** Cardiogenic shock requiring mechanical circulatory support

8. Risk Factors

- **Viral infections:** Enteroviruses (Coxsackie B), adenoviruses, parvovirus B19, HHV-6, influenza, SARS-CoV-2
- **Autoimmune diseases:** SLE, sarcoidosis, systemic sclerosis, inflammatory bowel disease
- **Hypersensitivity reactions:** Antibiotics (penicillins, sulfonamides), anticonvulsants, diuretics, NSAIDs

- **Toxins:** Alcohol, cocaine, amphetamines, heavy metals (lead, mercury)
- **Drug reactions:** Clozapine, immune checkpoint inhibitors (pembrolizumab, nivolumab)
- **Vaccines:** mRNA COVID-19 vaccines (rare, typically mild)
- **Young age and male sex:** Higher incidence
- **Giant cell myocarditis:** Thymoma, autoimmune disorders

9. Immediate Actions

EMERGENCY MANAGEMENT

1. Hospital admission with telemetry monitoring
2. Troponins, BNP, ESR/CRP, CBC (eosinophil count)
3. Echocardiography: LV function, wall motion abnormalities, pericardial effusion
4. CMR if hemodynamically stable: Tissue characterization (Lake Louise criteria)
5. Cardiogenic shock: Inotropes, mechanical circulatory support (Impella, ECMO) if needed
6. VT/VF storm: Antiarrhythmics, temporary pacing for complete heart block
7. Exclude MI with coronary angiography if risk factors or atypical presentation
8. Endomyocardial biopsy: If giant cell myocarditis suspected, refractory HF, or diagnosis uncertain with treatment implications

10. Treatment Overview

Supportive care: Most cases self-limited. Activity restriction (no competitive sports for 3-6 months). Treat HF with standard GDMT if LV dysfunction present.

Antiviral therapy: Generally not beneficial; exception: IVIG for proven enteroviral myocarditis in children.

Immunosuppression: Indicated for: eosinophilic myocarditis (high-dose corticosteroids), giant cell myocarditis (combination immunosuppression - steroids + cyclosporine + OKT3 or ATG), autoimmune myocarditis (SLE, sarcoidosis), and proven virus-negative chronic myocarditis (tailored immunosuppression per TIMIC trial). Not indicated for acute viral myocarditis (may worsen).

Giant cell myocarditis: Aggressive immunosuppression; rapid progression; mechanical circulatory support; cardiac transplantation often required.

Arrhythmias: Temporary pacing for complete heart block; antiarrhythmics for VT; ICD if persistent high-risk arrhythmias.

Mechanical circulatory support: Impella, ECMO for cardiogenic shock as bridge to recovery or transplant.

Prognosis: Fulminant myocarditis has paradoxically better prognosis if patient survives acute phase (complete recovery common). Chronic persistent myocarditis may progress to DCM.

11. Follow-up Questions

- Recent viral illness? Fever, URI, GI symptoms? When?
- Chest pain - pleuritic or pressure-like?
- Shortness of breath? Exercise tolerance?
- Palpitations or syncope?
- Recent COVID-19 infection or vaccination? (Type and timing)
- Current medications? Any recent new prescriptions? (Hypersensitivity)
- Autoimmune disease history?
- Substance use: alcohol, cocaine, amphetamines?
- Family history of cardiomyopathy?
- Any history of cancer immunotherapy (checkpoint inhibitors)?

12. References

1. Caforio ALP, et al. Current state of knowledge on aetiology, diagnosis, management, and therapy of myocarditis: a position statement of the ESC Working Group on Myocardial and Pericardial Diseases. *Eur Heart J*. 2013;34(33):2636-2648.
2. Ferreira VM, et al. Cardiovascular Magnetic Resonance in Nonischemic Myocardial Inflammation: Expert Recommendations. *J Am Coll Cardiol*. 2018;72(24):3158-3176.
3. Ammirati E, et al. Survival and Left Ventricular Function Changes in Fulminant Versus Nonfulminant Acute Myocarditis. *Circulation*. 2017;136(6):529-545.
4. Frustaci A, et al. Randomized study on the efficacy of immunosuppressive therapy in patients with virus-negative inflammatory cardiomyopathy. *Eur Heart J*. 2009;30(16):1995-2002.

Pericardial Effusion

Category: Pericardial Disease **Priority: HIGH** **ICD-10:** I31.3

1. Definition

Pericardial effusion is the accumulation of fluid in the pericardial space between the visceral and parietal layers of the pericardium. Small effusions are common and often asymptomatic. Large or rapidly accumulating effusions can compress the heart, impair ventricular filling, and progress to life-threatening cardiac tamponade. The pericardium normally contains 15-50 mL of serous fluid; effusions can range from trivial (<10 mm echo-free space) to massive (>20 mm), containing up to several liters.

2. Mechanism

Fluid accumulation results from: (1) Increased capillary permeability (inflammation, infection, malignancy, autoimmune); (2) Decreased lymphatic drainage (malignant obstruction); (3) Transudation from elevated venous pressure (heart failure, constrictive pericarditis, hypoalbuminemia); (4) Hemorrhage (trauma, aortic dissection rupture, post-cardiac surgery, anticoagulation, coagulopathy). The clinical significance depends on the volume and rate of accumulation; rapid accumulation of even small volumes can cause tamponade (non-compliant pericardium cannot stretch), while slow accumulation of large volumes may be well-tolerated (pericardial stretch adaptation).

3. ECG Findings

- **Low QRS amplitude:** <5 mm in limb leads, <10 mm in precordial leads (due to insulating effect of pericardial fluid)
- **Electrical alternans:** Beat-to-beat alternation of QRS amplitude (highly specific for large effusion with tamponade physiology; due to swinging heart in fluid)
- **Diffuse ST elevation:** If concurrent pericarditis (concave, without reciprocal changes)
- **PR depression:** If concurrent pericarditis
- **T wave inversion:** After ST elevation normalizes (pericarditis evolution)
- **Tachycardia:** Compensatory response to reduced stroke volume
- **AF:** May occur due to atrial irritation

4. Required ECG Features

ECG CLUES

- Low voltage + electrical alternans = large pericardial effusion with tamponade risk
- Electrical alternans is highly specific but insensitive (present in ~20-30% of tamponade cases)
- Diffuse ST elevation + PR depression indicates concomitant pericarditis
- Progression from low voltage to electrical alternans suggests increasing effusion size

5. Diagnostic Criteria

- Echocardiography (gold standard): Echo-free space around the heart; size grading (small: <10 mm diastolic, moderate: 10-20 mm, large: >20 mm); signs of tamponade (RA/RV collapse, IVC plethora, respiratory variation in transvalvular flows)
- CT/MRI: Superior for loculated effusions, pericardial thickening, mass lesions
- Pericardiocentesis: Diagnostic and therapeutic; fluid analysis (cell count, cytology, cultures, TB PCR, triglycerides for chylopericardium, hematocrit for hemorrhagic)
- Hemodynamic assessment: If tamponade physiology suspected - equalization of diastolic pressures (RA = RV = PA diastolic = PCWP), pulsus paradoxus >10 mmHg
- Pericardial biopsy: If tuberculosis or malignancy suspected and fluid analysis nondiagnostic

6. Differential Diagnosis

Condition	Differentiating Features
Cardiac tamponade	Clinical syndrome of hemodynamic compromise from effusion; echo signs of chamber collapse; equalized pressures
Dilated cardiomyopathy	Large cardiac silhouette on CXR but echo shows chamber dilation, not pericardial fluid
Pleural effusion	Fluid outside pericardium; echo may show fluid adjacent but not surrounding heart; chest imaging differentiates
Pericardial cyst	Congenital, localized, usually right cardiophrenic angle; stable over time; no tamponade risk

7. Symptoms

- **Small effusion:** Often asymptomatic
- **Dyspnea:** Most common symptom; compression of adjacent lung, reduced cardiac output

- **Chest fullness or discomfort:** Pericardial distension
- **Palpitations:** Tachycardia or arrhythmias
- **Cough:** Bronchial or phrenic nerve irritation
- **Hiccups:** Phrenic nerve irritation
- **Symptoms of underlying cause:** Fever (infection), weight loss (malignancy, TB), symptoms of autoimmune disease
- **Tamponade symptoms:** Dyspnea, hypotension, syncope, cold clammy extremities (see Cardiac Tamponade entry)

8. Risk Factors / Etiologies

- **Malignancy:** Lung, breast, lymphoma, leukemia (metastatic); most common cause of large symptomatic effusion in developed countries
- **Viral pericarditis:** Most common overall; usually small-moderate, self-limited
- **Uremia:** Chronic kidney disease; responds to dialysis
- **Autoimmune:** SLE, rheumatoid arthritis, scleroderma, Dressler syndrome
- **Post-cardiac injury syndrome:** Post-MI (Dressler), post-surgery, post-radiation
- **Tuberculosis:** Most common cause worldwide in endemic areas
- **Bacterial:** Purulent pericarditis (rare, life-threatening)
- **Hypothyroidism:** Myxedema; slow accumulation, often massive without tamponade
- **Trauma:** Penetrating chest injury, cardiac surgery, catheter-related perforation
- **Aortic dissection:** Rupture into pericardium - rapidly fatal
- **Drugs:** Hydralazine, procainamide, isoniazid, phenytoin, anticoagulants

9. Immediate Actions

ACUTE MANAGEMENT

1. Echocardiography to confirm diagnosis, quantify size, assess for tamponade physiology
2. Assess hemodynamic stability; if tamponade signs present, urgent pericardiocentesis
3. IV volume expansion (500-1000 mL) if hypotensive (temporary measure only)
4. Drain effusion if: large (>20 mm), symptomatic, tamponade physiology, or uncertain etiology requiring diagnostic analysis
5. If due to aortic dissection: emergent surgery, NOT pericardiocentesis
6. Treat underlying cause

10. Treatment Overview

Conservative: Small asymptomatic effusions: observe with serial echo. Treat underlying cause.

Pericardiocentesis: Indicated for tamponade, large symptomatic effusion, or diagnostic uncertainty. Echo-guided preferred. Leave drainage catheter for 24-48 hours.

Surgical: Pericardial window (subxiphoid or thoracoscopic) for recurrent effusions, loculated effusions, thick/clot-filled pericardium, or malignancy expected to recur. Pericardiectomy for chronic recurrent pericardial effusion or constriction.

Specific therapy: Malignant: intrapericardial chemotherapy, systemic therapy, radiation. TB: anti-TB therapy. Uremic: dialysis. Autoimmune: treat underlying disease, NSAIDs, colchicine, corticosteroids. Hypothyroid: thyroid replacement.

11. Follow-up Questions

- When did symptoms start? Rate of progression?
- Any recent cardiac procedure, trauma, or surgery?
- History of cancer, especially lung, breast, lymphoma?
- Fever, weight loss, night sweats? (TB, malignancy)
- Autoimmune disease symptoms?
- Kidney function? (Uremic effusion)
- Thyroid history? (Hypothyroidism)
- Anticoagulant or antiplatelet use?
- Known aortic aneurysm or dissection risk factors?

12. References

1. Adler Y, et al. 2015 ESC Guidelines for the diagnosis and management of pericardial diseases. *Eur Heart J*. 2015;36(42):2921-2964.
2. LeWinter MM. Clinical Practice. Acute Pericarditis. *N Engl J Med*. 2014;371(25):2410-2416.
3. Spodick DH. Acute Cardiac Tamponade. *N Engl J Med*. 2003;349(7):684-690.

Adult Congenital Heart Disease (ACHD)

Category: Congenital Heart Disease **Priority: HIGH** **ICD-10:** Q24.9

1. Definition

Adult congenital heart disease (ACHD) refers to structural heart defects present at birth that continue into adulthood. With advances in pediatric cardiac surgery and interventional catheterization, >90% of children born with CHD now survive to adulthood, resulting in a rapidly growing ACHD population (estimated >1.4 million adults in the US, exceeding the pediatric CHD population). ACHD encompasses a broad spectrum from simple defects requiring occasional follow-up to complex cyanotic lesions with multiple prior surgeries requiring specialized, lifelong multidisciplinary care.

2. Mechanism

Same embryological mechanisms as pediatric CHD. The focus in ACHD shifts to long-term consequences: (1) Residual hemodynamic lesions causing progressive chamber enlargement, valvular dysfunction, and heart failure; (2) Surgical sequelae including conduit obstruction, baffle leaks, residual shunts, and arrhythmia substrates from surgical scars; (3) Eisenmenger syndrome in unrepaired large shunts with irreversible pulmonary vascular disease; (4) Fontan-associated liver disease, protein-losing enteropathy, and plastic bronchitis in single ventricle patients; (5) Myocardial dysfunction from chronic volume/pressure overload. The combination of native anatomy, surgical palliation, and acquired cardiovascular disease creates complex hemodynamic substrates unique to each patient.

3. ECG Findings

- **Depends on specific lesion:** Often shows sequelae of repair
- **RBBB:** Very common after VSD repair or tetralogy of Fallot repair (surgical injury to right bundle)
- **LBBB:** After LVOT or arch procedures, DCM
- **Chamber hypertrophy patterns:** RVH in unrepairedTOF or severe PS; LVH in AS, CoA; biventricular in large VSD
- **SVT:** Atrial flutter most common (due to surgical atrial scars, especially after Mustard/Senning or Fontan); reentrant circuits around surgical incisions
- **VT:** From ventricular scars (especially repaired TOF with QRS >180 ms)
- **Complete heart block:** After VSD repair, AV canal repair, or progressive conduction disease
- **QT prolongation:** Especially in Fontan patients

4. Required ECG Features

ECG CLUES

- RBBB in a young adult with prior sternotomy = repaired VSD or TOF
- Atrial flutter in ACHD is often intra-atrial reentrant tachycardia (IART) rather than typical cavotricuspid isthmus-dependent flutter
- QRS >180 ms in repaired TOF indicates VT risk and consideration for ICD
- Complete heart block after VSD repair may require permanent pacing

5. Diagnostic Criteria

- Confirmation of CHD diagnosis from childhood records, surgical reports, or current imaging
- Echocardiography: Primary follow-up tool; assesses anatomy, function, hemodynamics
- CMR: Gold standard for quantification of ventricular volumes, function, flow quantification; essential for TOF, systemic RV, Fontan, coarctation
- CT: Coronary evaluation, airway anatomy, aortic arch, prior conduit assessment
- Cardiac catheterization: Hemodynamic assessment, reactivity testing for pulmonary hypertension, intervention
- Functional assessment: 6-minute walk test, cardiopulmonary exercise testing
- ACHD classification (AHA): Simple, moderate, severe (complex)

6. Differential Diagnosis

Condition	Differentiating Features
Acquired heart disease	No childhood cardiac history; normal anatomy except for acquired lesions; different risk factors
Primary arrhythmia	No structural heart disease; normal cardiac anatomy
Pulmonary hypertension	May be secondary to CHD (Eisenmenger) or primary; different management
Deconditioning	Reduced exercise capacity without hemodynamic abnormalities; improves with rehabilitation

7. Symptoms

- **Dyspnea on exertion:** Most common; indicates worsening hemodynamics or deconditioning
- **Palpitations:** Very common; SVT most frequent (IART/flutter > AF)

- **Syncope:** Arrhythmia or hemodynamic compromise
- **Cyanosis:** If unrepaired/palliated defects with right-to-left shunting; clubbing in chronic cyanosis
- **Exercise intolerance:** Reduced functional capacity
- **Heart failure symptoms:** Orthopnea, edema (especially Fontan, systemic RV)
- **Hepatic dysfunction:** Fontan-associated liver disease
- **Hemoptysis:** Eisenmenger syndrome (pulmonary hemorrhage, bronchial collaterals)
- **Paradoxical embolism:** Stroke from venous thrombus crossing intracardiac shunt

8. Risk Factors

- Specific CHD lesion and complexity
- Number and type of prior surgical/interventional procedures
- Residual hemodynamic lesions
- Chamber enlargement and dysfunction
- Arrhythmia burden
- Pulmonary hypertension
- Cyanosis (hematocrit >50% increases thrombosis risk)
- Pregnancy (hemodynamic stress)

9. Immediate Actions

EMERGENCIES IN ACHD

1. Atrial flutter/IART with rapid ventricular response: Rate control, anticoagulation, cardioversion; often requires antiarrhythmic therapy or ablation
2. VT/VF: ACLS; ICD if sustained VT or cardiac arrest
3. Heart failure decompensation: Tailored therapy (Fontan patients especially challenging)
4. Hemoptysis (Eisenmenger): Conservative management; avoid anticoagulation if possible
5. Paradoxical embolism/stroke: Anticoagulation; closure of residual shunt if feasible
6. Endocarditis: Blood cultures, antibiotics per guidelines

10. Treatment Overview

Arrhythmia management: Antiarrhythmics; catheter ablation for IART/flutter (often complex, requiring 3D mapping); AF management (rate/rhythm control, anticoagulation); device therapy (pacemaker/ICD).

Heart failure: Tailored to specific physiology. Systemic RV (TGA, ccTGA): ACE-I may help. Fontan: preload-dependent, afterload reduction, rhythm control, Fontan conversion or cardiac transplant for failing Fontan.

Re-intervention: Pulmonary valve replacement for repaired TOF (when RV dilation or dysfunction develops). Conduit replacement. Residual lesion closure. Coarctation re-intervention.

Eisenmenger syndrome: PAH-targeted therapy (bosentan, sildenafil); avoid phlebotomy unless symptomatic hyperviscosity; heart-lung transplant for end-stage.

Pregnancy: Pre-conception risk stratification (WHO risk classes); multidisciplinary care; hemodynamic monitoring.

Endocarditis prophylaxis: For highest risk lesions (prior IE, prosthetic valve, unrepaired cyanotic CHD).

11. Follow-up Questions

- Original CHD diagnosis? Detailed anatomy?
- All prior surgeries and catheter interventions with dates?
- Current functional status? NYHA class? Exercise capacity?
- Arrhythmia history? Current antiarrhythmic medications?
- Device therapy (pacemaker/ICD)? Last interrogation?
- Cyanosis? Oxygen saturation?
- Current medications? Adherence?
- Pregnancy considerations?
- Last echo/CMR results? RV/LV function? Any residual lesions?
- Need for re-intervention discussed?

12. References

1. Warnes CA, et al. ACC/AHA 2008 Guidelines for the Management of Adults with Congenital Heart Disease. *Circulation*. 2008;118(23):e714-e833.
2. Baumgartner H, et al. 2020 ESC Guidelines for the management of adult congenital heart disease. *Eur Heart J*. 2021;42(6):563-645.
3. Marelli AJ, et al. Lifetime Prevalence of Congenital Heart Disease in the General Population From 2000 to 2010. *Circulation*. 2014;130(9):749-756.

5. Moderate Priority Conditions

Important conditions requiring structured evaluation and management

Pericarditis

Category: Pericardial Disease **Priority: MODERATE** **ICD-10:** I31.9

1. Definition

Pericarditis is inflammation of the pericardium, the fibroserous sac surrounding the heart, typically presenting with sharp pleuritic chest pain, pericardial friction rub, and characteristic ECG changes. It is the most common pericardial disease and accounts for approximately 5% of emergency department chest pain presentations. Most cases are idiopathic (presumed viral) and self-limited, but recurrence occurs in 15-30% of patients after initial episode.

2. Mechanism

Inflammation of the visceral and parietal pericardium causes pain from stretching of the adjacent parietal pleura and release of inflammatory mediators. The inflammatory process typically involves the epicardium as well (myopericarditis). Causes: idiopathic/viral (most common), post-MI (Dressler syndrome - autoimmune, now rare with reperfusion therapy), post-cardiac surgery, autoimmune (SLE, RA, scleroderma), uremia, malignancy (metastatic lung, breast, lymphoma), tuberculosis (endemic areas), bacterial (rare, life-threatening), radiation, drugs (procainamide, hydralazine, isoniazid). Recurrent pericarditis occurs due to inadequate initial treatment or autoimmune mechanisms.

3. ECG Findings

- **Stage 1 (acute):** Diffuse concave ST elevation in I, II, III, aVF, V2-V6; PR depression (especially in II, aVF, V4-V6); reciprocal ST depression and PR elevation in aVR (and sometimes V1)
- **Stage 2:** Normalization of ST segments; may occur within days
- **Stage 3:** Diffuse T wave inversion (after ST segments normalize)
- **Stage 4:** Normalization of all findings (may take weeks to months)
- **Differentiation from MI:** Diffuse ST elevation (vs. regional in STEMI); concave morphology (vs. convex); PR depression present; no reciprocal changes except aVR; no Q waves
- **Electrical alternans:** If large effusion develops
- **Low voltage:** If effusion present

4. Required ECG Features

DIAGNOSTIC ECG CRITERIA

- Diffuse concave ST elevation with PR depression = acute pericarditis
- ST elevation in aVR with PR elevation = reciprocal changes supporting pericarditis diagnosis
- Absence of regional reciprocal ST depression (except aVR) excludes STEMI
- Note: Only ~60% of pericarditis patients have classic Stage 1 ECG; atypical presentations common

5. Diagnostic Criteria

- 2015 ESC guidelines: At least 2 of 4 criteria: (1) Pericarditic chest pain (sharp, pleuritic, worse with inspiration and supine, better sitting forward); (2) Pericardial friction rub (scratchy, triphasic - atrial, ventricular, diastolic); (3) New widespread ST elevation or PR depression on ECG; (4) Pericardial effusion (new or worsening)
- Supporting findings: Elevated inflammatory markers (CRP, ESR, WBC), pericardial effusion on imaging, fever
- Echocardiography: Assess for effusion, tamponade, ventricular function
- CMR: Late gadolinium enhancement of pericardium confirms active inflammation; assess for myocarditis (epicardial LGE)
- CT: Pericardial thickening/calcification in constrictive pericarditis
- Exclude MI (troponins, coronary angiography if risk factors)

6. Differential Diagnosis

Condition	Differentiating Features
STEMI	Regional ST elevation, reciprocal changes, elevated troponin, coronary territory distribution
Early repolarization	Stable ECG over time, no symptoms, no PR depression, no inflammation markers
Myocarditis	Greater myocardial involvement (elevated troponin, ventricular dysfunction); CMR shows myocardial LGE
Pulmonary embolism	Pleuritic pain, dyspnea, tachycardia; S1Q3T3; CT pulmonary angiography diagnostic
Pneumonia/pleurisy	Fever, cough, infiltrate on CXR; friction rub absent; cardiac enzymes normal

7. Symptoms

- **Chest pain:** Sharp, pleuritic, retrosternal; worsens with inspiration and lying supine; improves sitting forward
- **Pericardial friction rub:** Pathognomonic but transient; best heard at left lower sternal border with patient leaning forward
- **Fever:** Low-grade common; high fever suggests bacterial or TB
- **Dyspnea:** From pain-limited inspiration or effusion
- **Symptoms of underlying cause:** Weight loss/malaise (malignancy, TB), joint pain (autoimmune)

8. Risk Factors

- Recent viral illness (most common trigger)
- Prior pericarditis (recurrence risk 15-30%)
- Post-MI (Dressler syndrome), post-cardiac surgery, post-radiation
- Autoimmune disease (SLE, RA, scleroderma, vasculitis)
- Renal failure (uremic pericarditis)
- Malignancy (metastatic, especially lung, breast, lymphoma, melanoma)
- Tuberculosis (endemic areas)
- Hypothyroidism
- Medications: procainamide, hydralazine, isoniazid, phenytoin, doxorubicin

9. Immediate Actions

ACUTE MANAGEMENT

1. Exclude STEMI (troponin, ECG pattern)
2. NSAIDs: Ibuprofen 600-800 mg q8h or aspirin 750-1000 mg q8h (first-line; aspirin preferred if concomitant myocarditis)
3. Colchicine 0.5 mg BID (or 0.6 mg BID) for 3 months; reduces recurrence by 50%
4. Gastric protection with PPI
5. Rest and activity restriction until symptoms resolve and CRP normalizes
6. Monitor for tamponade (large effusion or clinical deterioration)
7. Avoid anticoagulation if large effusion present (hemorrhagic conversion risk)

10. Treatment Overview

First episode: NSAIDs + colchicine for 1-2 weeks until symptom resolution and CRP normalization, then taper. Typical total colchicine course 3 months.

Recurrent pericarditis: Same as first episode; taper NSAIDs slowly over weeks. Colchicine maintenance for 6 months. Corticosteroids (prednisone 0.2-0.5 mg/kg/day) only if NSAIDs and colchicine contraindicated or refractory; slow taper essential to avoid recurrence.

Refractory/recurrent: Azathioprine, IVIG, anakinra (IL-1 receptor antagonist), pericardiectomy (last resort).

Tamponade: Urgent pericardiocentesis.

Constrictive pericarditis: Diuretics; pericardiectomy if symptomatic despite medical therapy.

Specific etiologies: Treat underlying cause (autoimmune, TB, malignancy, renal failure).

11. Follow-up Questions

- Chest pain - worse with inspiration? Better sitting forward?
- Duration of symptoms? Prior episodes? (recurrent)
- Recent viral illness?
- Fever pattern?
- Recent MI, cardiac surgery, or radiation?
- Autoimmune disease symptoms?
- Renal function history?
- Malignancy history?
- Current medications?
- TB exposure or travel to endemic areas?

12. References

1. Adler Y, et al. 2015 ESC Guidelines for the diagnosis and management of pericardial diseases. *Eur Heart J*. 2015;36(42):2921-2964.
2. Imazio M, et al. A randomized trial of colchicine for acute pericarditis. *N Engl J Med*. 2013;369(16):1522-1528.
3. Imazio M, et al. Evaluation and Treatment of Pericarditis: A Systematic Review. *JAMA*. 2015;314(14):1498-1506.

Pulmonary Hypertension

Category: Pulmonary Vascular **Priority: MODERATE** **ICD-10:** I27.0

1. Definition

Pulmonary hypertension (PH) is a hemodynamic and pathophysiological condition defined as an increase in mean pulmonary arterial pressure (mPAP) >20 mmHg at rest as assessed by right heart catheterization. It is classified into five groups based on etiology: Group 1 (pulmonary arterial hypertension - PAH), Group 2 (PH due to left heart disease - most common), Group 3 (PH due to lung disease/hypoxia), Group 4 (chronic thromboembolic PH - CTEPH), and Group 5 (PH with unclear/multifactorial mechanisms). PAH is a rare, progressive disease affecting the small pulmonary arterioles.

2. Mechanism

Group 1 PAH: Pathological remodeling of small pulmonary arterioles with intimal hyperplasia, medial hypertrophy, adventitial proliferation, and plexiform lesion formation. Vasoconstriction, thrombosis in situ, and inflammation contribute. Mutations in BMPR2 (bone morphogenetic protein receptor type 2) account for 70-80% of heritable PAH. Other causes: connective tissue disease (SSc-PAH most common), HIV, portal hypertension, congenital heart disease (Eisenmenger), drugs/toxins (fenfluramine). Increased pulmonary vascular resistance (PVR) causes RV pressure overload, hypertrophy, and eventual failure.

3. ECG Findings

- **RVH:** R wave V1 >7 mm, R/S ratio in V1 >1, RAD (right axis deviation >90°)
- **Right atrial enlargement:** P pulmonale (tall peaked P waves >2.5 mm in II)
- **RV strain:** ST depression and T wave inversion in V1-V4 (anterior ischemia from RV dilation)
- **Supraventricular arrhythmias:** AF, atrial flutter common in advanced disease
- **Prolonged QRS:** RV dilation correlates with QRS duration
- **Right bundle branch block:** From RV dilation
- **S1Q3T3 pattern:** If acute PE is cause (Group 4)

4. Required ECG Features

ECG CLUES

- RVH + RAD + RAE in patient with dyspnea = PH until proven otherwise
- RV strain (T wave inversions V1-V4) indicates advanced disease with RV dilation/dysfunction
- Normal ECG does not exclude PH (especially early disease)
- Severity of ECG abnormalities correlates with PH severity

5. Diagnostic Criteria

- Right heart catheterization (gold standard): mPAP >20 mmHg, PAWP <=15 mmHg (precapillary), PVR >2 Wood units
- Echocardiography: Estimates PA pressure from TR jet velocity (RVSP); assesses RV size/function, left heart disease
- CT pulmonary angiography: For CTEPH evaluation (Group 4)
- PFTs with DLCO: Low DLCO suggests PAH or PVOD
- V/Q scan: High sensitivity for CTEPH; normal scan virtually excludes CTEPH
- Connective tissue disease screening: ANA, anti-Scl-70, anti-centromere, SSA/SSB
- Genetic testing: BMPR2, EIF2AK4 for PVOD
- 6-minute walk test: Functional assessment and prognosis

6. Differential Diagnosis

Condition	Differentiating Features
Left heart failure (Group 2 PH)	Elevated PAWP (>15 mmHg), echo shows left heart disease, elevated BNP
Lung disease (Group 3 PH)	COPD, ILD, OSA on PFTs/imaging; hypoxemia; disproportionate PH
CTEPH (Group 4)	Prior PE history; V/Q scan defects; webs/pouches on CTA; surgical curable
Acute PE	Acute presentation, risk factors, S1Q3T3, RV strain on echo, CTPA diagnostic
RV infarction	Inferior STEMI, right-sided leads, elevated troponin, coronary angiography

7. Symptoms

- **Dyspnea on exertion:** Most common initial symptom; progressive
- **Fatigue:** Reduced cardiac output
- **Syncope:** Especially with exertion - poor prognostic sign (inability to increase cardiac output)
- **Chest discomfort:** RV angina (from RV wall stress)
- **Peripheral edema:** Right heart failure
- **Symptoms of underlying cause:** Cyanosis (Eisenmenger), Raynaud's (SSc), liver disease (portopulmonary)

8. Risk Factors

- Connective tissue disease (systemic sclerosis most common)
- Congenital heart disease with systemic-to-pulmonary shunts (Eisenmenger)
- Portal hypertension (portopulmonary hypertension)
- HIV infection
- BMPR2 mutation (heritable PAH)
- Anorexigen drugs (fenfluramine - now withdrawn)
- Methamphetamine, cocaine use
- Schistosomiasis (endemic regions)

9. Immediate Actions

ACUTE MANAGEMENT

1. Supplemental O2 to maintain SpO2 >90%
2. Diuretics for RV failure signs
3. Avoid systemic vasodilators (cause hypotension)
4. Anticoagulation: Consider for idiopathic/heritable PAH, CTEPH
5. Refer to PH specialist center
6. Right heart catheterization for definitive diagnosis and vasoreactivity testing

10. Treatment Overview

Group 1 PAH: Vasoreactivity testing (acute vasodilator challenge during RHC). Responders: high-dose CCBs (amlodipine, diltiazem, nifedipine). Non-responders: targeted PAH therapies - endothelin receptor antagonists (bosentan, ambrisentan, macitentan), PDE5 inhibitors (sildenafil, tadalafil), prostacyclin analogs (epoprostenol IV, treprostinil SC/IV/inhaled, iloprost inhaled), soluble guanylate cyclase stimulator (riociguat), prostacyclin receptor agonist (selexipag). Combination therapy preferred (initial dual oral, escalate to triple including prostacyclin if needed).

Group 2 (left heart disease): Treat left heart disease (GDMT for HFpEF/HFrEF); PH-targeted therapies NOT beneficial and potentially harmful.

Group 3 (lung disease): Optimize lung disease treatment, supplemental O2, consider PAH therapies if severe.

Group 4 (CTEPH: Surgical pulmonary endarterectomy (curative if proximal disease); riociguat and balloon pulmonary angioplasty for inoperable disease.

Transplantation: Heart-lung or lung transplant for refractory PAH.

11. Follow-up Questions

- Progressive dyspnea? Duration? Functional limitation?
- Any syncope, especially with exertion?
- Raynaud's phenomenon? Sclerodactyly? (SSc)
- Prior PE or DVT?
- Liver disease? Portal hypertension?
- HIV status?
- Family history of PAH or BMPR2 mutation?
- Congenital heart disease history?

12. References

1. Humbert M, et al. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. *Eur Heart J*. 2022;43(38):3618-3731.
2. Galie N, et al. 2015 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. *Eur Heart J*. 2016;37(1):67-119.

Supraventricular Tachycardia (SVT)

Category: Arrhythmia **Priority: MODERATE** **ICD-10:** I47.1

1. Definition

Supraventricular tachycardia (SVT) is a broad term encompassing regular tachycardias originating above the ventricles, involving the atria, AV node, or accessory pathways, at a rate typically 150-250 bpm with a narrow QRS complex (<120 ms). Common types include AV nodal reentrant tachycardia (AVNRT, 60% of cases), AV reentrant tachycardia (AVRT, 30% - includes orthodromic and antidromic), and atrial tachycardia (10%). SVT is the most common symptomatic arrhythmia in clinical practice and often presents with abrupt onset and offset.

2. Mechanism

AVNRT: Reentrant circuit within the AV node using dual pathways (fast and slow). Typical (slow-fast) AVNRT: antegrade conduction down the slow pathway and retrograde up the fast pathway; P waves buried in QRS or immediately after (pseudo-R' in V1, pseudo-S in inferior leads). Atypical (fast-slow) AVNRT: opposite direction; long RP interval. **AVRT:** Requires an accessory pathway (bypass tract) connecting atria and ventricles. Orthodromic AVRT: antegrade via AV node, retrograde via accessory pathway; narrow QRS; P wave after QRS (short RP). Antidromic AVRT: antegrade via accessory pathway, retrograde via AV node; wide QRS (mimics VT). **Atrial tachycardia:** Enhanced automaticity or micro-reentry in atrial tissue; distinct P wave morphology different from sinus P waves.

3. ECG Findings

- **Rate:** 150-250 bpm with sudden onset/offset
- **Regular, narrow complex tachycardia:** QRS <120 ms unless aberrant conduction or antidromic AVRT
- **AVNRT:** P waves buried in QRS or immediately after; pseudo-R' in V1; pseudo-S waves in II, III, aVF
- **AVRT (orthodromic):** P wave after QRS with short RP interval (<70 ms); ST/T wave changes may reveal retrograde P
- **Atrial tachycardia:** Abnormal P wave morphology preceding QRS; may have AV block (variable conduction)
- **Short RP:** RP < PR (AVNRT, orthodromic AVRT)
- **Long RP:** RP > PR (atypical AVNRT, atrial tachycardia, PJRT)
- **Pre-excitation in sinus rhythm (WPW):** Short PR, delta wave, wide QRS - indicates accessory pathway

4. Required ECG Features

DIAGNOSTIC FEATURES

- Regular narrow complex tachycardia 150-250 bpm = SVT
- Visible retrograde P waves after QRS (short RP) = AVRT or typical AVNRT
- Pseudo-R' in V1 or pseudo-S in inferior leads = typical AVNRT
- Short PR + delta wave in sinus rhythm = WPW syndrome (accessory pathway)
- Irregular narrow complex with varying R-R = not SVT; consider AF or atrial flutter

5. Diagnostic Criteria

- ECG: 12-lead during tachycardia establishes diagnosis
- Adenosine response: Transient AV block reveals underlying atrial activity; AVNRT and AVRT terminate; atrial tachycardia may continue with AV block
- Carotid sinus massage or Valsalva: May terminate AVNRT/AVRT or slow rate to reveal P waves
- EP study: Definitive diagnosis; maps reentrant circuit; guides ablation
- Exercise testing: If exertion-triggered
- Monitoring: Event recorder or implantable loop recorder if infrequent episodes

6. Differential Diagnosis

Condition	Differentiating Features
Sinus tachycardia	Gradual onset/offset; P waves normal morphology; rate usually <160 in adults
Atrial flutter	Sawtooth flutter waves; atrial rate ~300; regular ventricular response with AV block
Atrial fibrillation	Irregularly irregular; no distinct P waves
VT	Wide QRS (>120 ms); AV dissociation; fusion/capture beats; extreme axis
Antidromic AVRT	Wide QRS due to accessory pathway conduction; may mimic VT; pre-excitation in sinus rhythm

7. Symptoms

- **Palpitations:** Sudden start/stop; regular, rapid heartbeat

- **Dyspnea, chest tightness:** Rate-related
- **Dizziness, near-syncope:** From reduced cardiac output
- **Polyuria:** ANP release during tachycardia causing diuresis
- **Anxiety:** Sensation of rapid heartbeat
- **Neck pounding:** Cannon A waves from simultaneous atrial/ventricular contraction (AVNRT)

8. Risk Factors

- Younger age (AVNRT typically 20-40; AVRT younger)
- Female sex (AVNRT 2:1 female predominance)
- Anxiety, stress
- Caffeine, alcohol, stimulants
- Exercise (catecholamine-mediated)
- Pregnancy (increased episodes due to hormonal/volume changes)
- Structural heart disease (atrial tachycardia more common)
- Prior cardiac surgery (scar-related atrial tachycardia)

9. Immediate Actions

ACUTE MANAGEMENT

1. If unstable (hypotension, altered mental status, ischemic chest pain, acute HF): Synchronized cardioversion (50-100 J)
2. If stable: Vagal maneuvers first (Valsalva 15 seconds, carotid sinus massage, ice water to face)
3. Adenosine 6 mg rapid IV push followed by 20 mL saline flush (if no response, 12 mg, then 12 mg); short-acting, terminates AVNRT/AVRT by transient AV block; may help diagnose AT
4. Calcium channel blockers (verapamil 2.5-5 mg IV or diltiazem 15-20 mg IV) if adenosine contraindicated or ineffective
5. Beta-blockers (metoprolol 25-50 mg PO or 5 mg IV) as alternative
6. Avoid AV nodal blockers if pre-excited AF (WPW with AF) - risk of accelerating conduction down accessory pathway

10. Treatment Overview

Acute termination: Vagal maneuvers, adenosine, CCBs, beta-blockers, cardioversion if unstable.

Chronic prevention: Catheter ablation is curative and first-line for symptomatic patients (AVNRT success >95%, AVRT accessory pathway >95%, atrial tachycardia 80-90%).

Medical therapy (if ablation declined): Beta-blockers, non-DHP CCBs, flecainide, sotalol, amiodarone (last resort due to toxicity).

WPW: Risk stratification (symptomatic = ablation; asymptomatic with high-risk features on EP study = ablation).

Lifestyle: Avoid triggers (caffeine, alcohol, stimulants); stress management.

11. Follow-up Questions

- How do episodes start and stop? Sudden or gradual?
- Rate? Regular or irregular?
- Any syncope or near-syncope?
- Frequency and duration of episodes?
- Triggers: exercise, stress, caffeine, alcohol?
- Pregnancy status or plans?
- Any ECG during tachycardia?
- Prior adenosine or vagal maneuver response?

12. References

1. Page RL, et al. 2015 ACC/AHA/HRS Guideline for the Management of Adult Patients With Supraventricular Tachycardia. *Circulation*. 2016;133(14):e506-e574.
2. Blomstrom-Lundqvist C, et al. ACC/AHA/ESC guidelines for the management of patients with supraventricular arrhythmias. *Circulation*. 2003;108(15):1871-1909.

Tetralogy of Fallot (TOF)

Category: Congenital Heart Disease **Priority: MODERATE** **ICD-10:** Q21.3

1. Definition

Tetralogy of Fallot (TOF) is the most common cyanotic congenital heart defect, characterized by four anatomical abnormalities: (1) ventricular septal defect (VSD), (2) right ventricular outflow tract obstruction (RVOTO - infundibular or valvular pulmonary stenosis), (3) overriding aorta (straddling the VSD), and (4) right ventricular hypertrophy (secondary to RVOTO). It accounts for 10% of all congenital heart defects. Complete surgical repair is typically performed in infancy, and adults with repaired TOF now represent the largest group of cyanotic ACHD patients.

2. Mechanism

Abnormal anterior and cephalad deviation of the infundibular (conal) septum during embryological development causes: VSD (typically large, perimembranous), narrowing of the RVOT, and aortic override. The severity of RVOTO determines the degree of right-to-left shunting and cyanosis. Pathophysiology: RVOTO increases RV pressure; blood shunts right-to-left across the VSD into the overriding aorta, causing systemic cyanosis. RVH develops from pressure overload. Post-repair: VSD patch closure and RVOT obstruction relief (transannular patch or infundibular resection) leave patients with chronic pulmonary regurgitation, which causes progressive RV dilation and dysfunction over decades - the major late complication.

3. ECG Findings

- **Pre-repair:** RVH (tall R wave V1, R/S >1 after age 1), RAD, RAE
- **Post-repair RBBB:** rsR' pattern in V1 (surgical injury to right bundle during VSD closure); present in nearly all repaired TOF patients
- **QRS duration:** Prolongation >180 ms = increased VT risk; correlate with RV size and arrhythmia substrate
- **RVH persistence post-repair:** May persist if residual RVOT obstruction or severe pulmonary regurgitation
- **Q waves in left precordial leads:** From surgical incision; may mimic anterior MI
- **Atrial arrhythmias:** AF, atrial flutter (due to RA dilation from chronic volume overload)
- **VT:** Monomorphic VT from RV surgical scar; risk increases with QRS >180 ms
- **Complete heart block:** Uncommon post-repair (surgical injury)

4. Required ECG Features

ECG CLUES

- RBBB + history of sternotomy in young adult = repaired TOF until proven otherwise
- QRS >180 ms in repaired TOF indicates high VT risk and need for further evaluation (CMR, EP study)
- Progressive QRS prolongation over serial ECGs indicates progressive RV dilation from pulmonary regurgitation
- New atrial arrhythmias may indicate hemodynamic deterioration requiring re-intervention

5. Diagnostic Criteria

- Echocardiography: Defines VSD, RVOTO severity, aortic override, RV pressure, post-repair pulmonary regurgitation severity, RV function
- CMR: Gold standard for post-repair surveillance; quantifies RV volume, function, pulmonary regurgitation fraction; identifies RV scar and fibrosis; QRS >180 ms correlates with RV dilation
- Cardiac CT: Coronary anatomy (left anterior descending from right coronary artery - crosses RVOT - surgical importance); conduit evaluation
- Cardiac catheterization: Hemodynamic assessment; intervention (pulmonary valve implantation)
- Exercise testing: Functional capacity; arrhythmia induction

6. Differential Diagnosis

Condition	Differentiating Features
Double outlet right ventricle	Both great arteries arise from RV; more complex anatomy; echo/CMR differentiates
Pulmonary atresia with VSD	No connection between RV and PA; more severe cyanosis; collateral-dependent pulmonary blood flow
Transposition with VSD and PS	Aorta arises from RV, PA from LV; different surgical approach; anatomy defines
Eisenmenger VSD	Large VSD with pulmonary vascular disease causing shunt reversal; no RVOTO

7. Symptoms

- **Cyanosis (unrepaired or palliated):** Central cyanosis, clubbing, polycythemia
- **Squatting behavior:** Children squat to increase SVR and reduce right-to-left shunt
- **Tet spells (hypercyanotic spells):** Paroxysmal dyspnea, cyanosis, irritability in infants; triggered by crying, feeding, defecation; treated with knee-chest position, morphine, beta-blockers, fluids
- **Post-repair dyspnea on exertion:** From pulmonary regurgitation causing RV dilation and dysfunction
- **Palpitations:** Atrial arrhythmias or VT
- **Syncope:** VT or hemodynamic compromise
- **Right heart failure:** Peripheral edema, ascites (late sign of severe pulmonary regurgitation)

8. Risk Factors

- Genetic: 22q11.2 deletion (DiGeorge syndrome) in ~15%; trisomy 21, trisomy 18
- Maternal: Diabetes, phenylketonuria, advanced maternal age
- Post-repair complications: Transannular patch (free pulmonary regurgitation), residual VSD, residual RVOTO, conduit stenosis
- Late deterioration: Chronic pulmonary regurgitation (most common), arrhythmias, RV dysfunction, sudden death

9. Immediate Actions

EMERGENCIES

1. Tet spell (infant): Knee-chest position, supplemental O2, IV fluids, morphine, beta-blocker (esmolol), phenylephrine
2. VT/VF: ACLS; ICD if sustained VT or cardiac arrest
3. Atrial flutter: Rate control, anticoagulation, cardioversion; often requires antiarrhythmic or ablation
4. Decompensated HF: Diuretics; evaluate for pulmonary valve replacement

10. Treatment Overview

Primary repair: VSD closure + RVOT reconstruction (typically in infancy, 3-6 months). Transannular patch relieves RVOTO but causes free pulmonary regurgitation.

Pulmonary valve replacement (PVR): Indicated when CMR shows severe pulmonary regurgitation with RV dilation (RVEDVI >150-160 mL/m^2), declining RV function, increasing QRS duration, symptomatic arrhythmias, or declining exercise capacity. Percutaneous pulmonary valve implantation (Melody or Sapien valve) now available for conduit or native RVOT.

Arrhythmia management: Antiarrhythmics; catheter ablation for IART/flutter (often complex due to surgical scars); ICD for sustained VT or cardiac arrest (secondary prevention) or high-risk primary prevention (QRS >180 ms + additional risk factors).

Heart failure: Diuretics; ACE-I/ARB may help RV function; cardiac resynchronization may be considered; heart transplant for end-stage.

Surveillance: Annual ACHD clinic visit with echo; CMR every 2-3 years; exercise testing periodically.

11. Follow-up Questions

- Age at repair? Type of repair? (transannular patch vs. infundibular resection)
- Any prior pulmonary valve replacements?
- Last CMR results? RV volumes? QRS duration?
- Exercise tolerance? NYHA class?
- Any arrhythmias? Palpitations? Syncope?
- Current medications?
- Endocarditis prophylaxis needed?

12. References

1. Knauth AL, et al. Ventricular size and function assessed by cardiac MRI predict major adverse clinical outcomes late after tetralogy of Fallot repair. *Heart*. 2008;94(2):211-216.
2. Bhatt AB, et al. Congenital Heart Disease in the Older Adult. *Circulation*. 2007;115(14):1934-1948.
3. Warnes CA. Transposition of the Great Arteries. *Circulation*. 2006;114(24):2699-2709.

Transposition of the Great Arteries (TGA/D-TGA)

Category: Congenital Heart Disease **Priority: MODERATE** **ICD-10:** Q20.3

1. Definition

Transposition of the great arteries (TGA), also called dextro-transposition or complete TGA (D-TGA), is a cyanotic congenital heart defect in which the aorta arises from the right ventricle and the pulmonary artery arises from the left ventricle, creating two separate parallel circuits (systemic and pulmonary) rather than the normal series circulation. Without mixing of blood between the circuits (via ASD, VSD, or PDA), it is incompatible with life. It accounts for 5-7% of all congenital heart defects and is the most common cyanotic lesion in newborns.

2. Mechanism

Failure of normal rotation and septation of the truncus arteriosus during embryological development results in ventriculoarterial discordance: RV connects to aorta, LV connects to PA. Deoxygenated blood recirculates to the body, and oxygenated blood recirculates to the lungs - incompatible with survival without intercirculatory mixing. Atrial switch (Mustard/Senning, historical): Baffles redirect venous return at atrial level (systemic venous to LV/PA; pulmonary venous to RV/aorta). Arterial switch (Jatene, current standard): Switches the great arteries and coronary arteries, establishing normal ventriculoarterial connections. Post-atrial switch complications: systemic RV failure, baffle obstruction/leak, SVT, sinus node dysfunction. Post-arterial switch: coronary ostial stenosis, supravalvular PS, neoaortic regurgitation.

3. ECG Findings

- **Pre-repair:** RVH pattern, RAD (RV is systemic ventricle)
- **Post-atrial switch (Mustard/Senning):** RVH persists (RV remains systemic ventricle); SVT very common (IART/flutter from atrial suture lines); sick sinus syndrome from sinus node injury
- **Post-arterial switch (Jatene):** May be near-normal or show LVH if supravalvular PS; coronary ischemia patterns if ostial stenosis
- **Post-Rastelli (VSD baffled to aorta + conduit from RV to PA):** Conduit dysfunction may cause arrhythmias; LV-to-aorta pathway obstruction
- **Sick sinus syndrome post-atrial switch:** Sinus bradycardia, sinus arrest, junctional rhythm; may require pacing
- **Atrial flutter/IART:** Very common post-atrial switch (due to extensive atrial surgery)

4. Required ECG Features

ECG CLUES

- RVH pattern in young adult with cyanotic CHD history = unrepaired or atrial-switch TGA
- SVT in patient with prior atrial switch operation = intra-atrial reentrant tachycardia until proven otherwise
- Sick sinus syndrome (bradycardia, junctional escape) common after atrial switch
- Post-arterial switch: ECG may be normal; ischemic changes suggest coronary ostial stenosis

5. Diagnostic Criteria

- Echocardiography: Demonstrates ventriculoarterial discordance (aorta from RV, PA from LV); assesses repair type and complications
- CMR: Gold standard for post-atrial switch (systemic RV function, baffle assessment, PA/branch stenosis); post-arterial switch (coronary patency, neo-aortic root, branch PA stenosis)
- CT coronary angiography: Coronary ostial assessment post-arterial switch
- Cardiac catheterization: Hemodynamics, baffle leak assessment, intervention
- Exercise testing: Functional capacity; arrhythmia provocation

6. Differential Diagnosis

Condition	Differentiating Features
Congenitally corrected TGA (ccTGA)	Atrioventricular and ventriculoarterial discordance; systemic LV; different natural history; AV block common
Double outlet right ventricle (DORV)	Both great vessels from RV; variable relationship; different surgical approach
Truncus arteriosus	Single great vessel overriding VSD; different physiology; single semilunar valve
Tetralogy of Fallot	PA from LV, aorta overrides VSD; subpulmonic stenosis; different repair

7. Symptoms

- **Type of repair:** Determines late presentation

- **Post-atrial switch:** Dyspnea on exertion (systemic RV failure), palpitations (SVT), syncope (arrhythmia or baffle obstruction), peripheral edema
- **Post-arterial switch:** Usually asymptomatic; exercise intolerance if coronary or supravalvular PS; arrhythmias less common
- **Pre-Rastelli:** Conduit dysfunction (dyspnea), arrhythmias
- **Cyanosis:** If baffle leak with right-to-left shunt (atrial switch) or PA obstruction

8. Risk Factors

- Maternal diabetes (strong association)
- Genetic: Less commonly associated with chromosomal abnormalities than other CHD
- Type of repair: Atrial switch (Mustard/Senning) has worse long-term outcomes than arterial switch
- Post-atrial switch complications: Systemic RV failure, baffle leak/obstruction, arrhythmias, sudden death
- Post-arterial switch: Coronary ostial stenosis (risk of ischemia and sudden death)

9. Immediate Actions

EMERGENCIES

1. SVT post-atrial switch: Rate control, anticoagulation, cardioversion; antiarrhythmic or ablation
2. Baffle obstruction: Diuretics; intervention (balloon dilation, stenting, or surgical revision)
3. Systemic RV failure: Diuretics, afterload reduction (ACE-I/ARB), consider cardiac transplant
4. Coronary ischemia post-arterial switch: Standard ACS management; urgent angiography

10. Treatment Overview

Atrial switch patients: ACE-I/ARB for systemic RV (limited evidence but commonly used); diuretics for congestion; rhythm control for SVT; anticoagulation for AF/flutter; cardiac transplantation for refractory systemic RV failure.

Arterial switch patients: Surveillance for coronary ostial stenosis (CT coronary angiography), neo-aortic root dilation, branch PA stenosis, neoaortic regurgitation; intervention as needed.

Arrhythmia management: Antiarrhythmics; catheter ablation for IART (complex, 3D mapping); pacemaker for sick sinus syndrome; ICD for sustained VT or cardiac arrest.

Cardiac transplantation: For end-stage systemic RV failure (atrial switch) or inoperable coronary/hemodynamic issues (arterial switch).

Surveillance: Annual ACHD clinic; CMR every 2-3 years; cardiopulmonary exercise testing.

11. Follow-up Questions

- Type of repair? (Mustard/Senning atrial switch vs. Jatene arterial switch vs. Rastelli)
- Age at repair?
- Exercise tolerance? NYHA class?
- Any arrhythmias? Palpitations? Syncope?
- Pacemaker or ICD?
- Last CMR/CT results?
- Any baffle obstruction or leak? (atrial switch)
- Any coronary assessment? (arterial switch)

12. References

1. Warnes CA. Transposition of the Great Arteries. *Circulation*. 2006;114(24):2699-2709.
2. Khairy P, et al. PACES/HRS Expert Consensus Statement on the Recognition and Management of Arrhythmias in Adult Congenital Heart Disease. *Heart Rhythm*. 2014;11(10):e53-e96.

Aortic Aneurysm

Category: Aortic Disease **Priority: MODERATE** **ICD-10:** I71.9

1. Definition

An aortic aneurysm is a localized or diffuse dilation of the aorta to greater than 1.5 times its normal diameter (typically >3.0 cm for thoracic aorta, >3.5 cm for abdominal aorta). Thoracic aortic aneurysm (TAA) most commonly involves the aortic root and ascending aorta; abdominal aortic aneurysm (AAA) occurs below the renal arteries. Aneurysms are true (involving all three vessel wall layers) or false (pseudoaneurysm, usually post-traumatic). Aortic aneurysm is the 19th leading cause of death overall; rupture carries 85-90% mortality.

2. Mechanism

TAAs: Cystic medial necrosis (degeneration of the aortic media with smooth muscle cell loss, elastic fiber fragmentation, and mucoid accumulation) is the primary mechanism. Genetic conditions (Marfan, Loeys-Dietz, vascular Ehlers-Danlos, bicuspid aortic valve) accelerate this process through defective connective tissue proteins. AAAs: Atherosclerotic degeneration with transmural inflammation, proteolytic enzyme activation (MMPs), and wall remodeling. Risk factors (smoking, hypertension, age) drive this inflammatory-degenerative process. Aneurysm expansion follows the Law of Laplace (wall tension proportional to radius); as diameter increases, wall stress increases, creating a vicious cycle of progressive dilation.

3. ECG Findings

- **ECG often normal:** Changes reflect associated conditions or complications
- **LVH voltage:** If associated aortic regurgitation or hypertension
- **Left atrial enlargement:** From aortic regurgitation or chronic hypertension
- **AF:** If aortic root dilation causes atrial stretch
- **ST-T changes:** Nonspecific or from associated cardiac conditions
- **Low voltage:** Rare, if very large thoracic aneurysm with pericardial effusion
- **Acute changes (dissection/rupture):** May show ischemia from coronary involvement, tamponade, or simply sinus tachycardia from pain/hemorrhage

4. Required ECG Features

ECG IN AORTIC ANEURYSM

ECG is not diagnostic for aortic aneurysm. The primary diagnostic tools are imaging (TTE, CT angiography, MRI). ECG serves to: (1) establish baseline for surgical planning, (2) detect associated conditions (AR, hypertension), (3) exclude concurrent ischemic heart disease, (4) detect complications (dissection extending to coronaries, tamponade from rupture).

5. Diagnostic Criteria

- Imaging: TTE (aortic root, ascending aorta), CT angiography (entire aorta, gold standard), MRI (no radiation, serial monitoring)
- TAA: Ascending aorta >3.5-4.0 cm; aortic root >3.5 cm; arch >3.5 cm; descending >3.0 cm
- AAA: Infrarenal aorta >3.0 cm diameter
- Connective tissue disorder screening: FBN1 (Marfan), TGFBR1/2 (Loeys-Dietz), COL3A1 (vEDS), ACTA2
- Family screening: First-degree relatives of TAA patients should undergo aortic imaging

6. Differential Diagnosis

Condition	Differentiating Features
Aortic dissection	Intimal flap on imaging; acute symptoms; different management urgency
Aortic ectasia	Mild uniform dilation <1.5x normal; often age-related; slower progression
Postsurgical pseudoaneurysm	Prior cardiac or aortic surgery; contained rupture; saccular appearance
Mycotic aneurysm	Infectious cause; rapid growth; associated with bacteremia/endocarditis; saccular

7. Symptoms

- **Most are asymptomatic:** Detected incidentally on imaging for other indications
- **Thoracic:** Substernal chest pain (deep, aching), back pain, dyspnea (airway/bronchial compression), hoarseness (recurrent laryngeal nerve), dysphagia (esophageal compression), stridor (tracheal compression)

- **Abdominal:** Pulsatile abdominal mass (palpable if >5 cm), abdominal/back pain, flank pain
- **Rupture symptoms:** Sudden severe pain, hypotension, syncope, shock - surgical emergency

8. Risk Factors

- Age >65 (AAA), >60 (TAA)
- Male sex (AAA 4-6:1)
- Smoking (strongest modifiable risk for AAA)
- Hypertension
- Family history (especially TAA - autosomal dominant inheritance in 20%)
- Connective tissue disorders (Marfan, Loeys-Dietz, vEDS)
- Bicuspid aortic valve (ascending aorta)
- Prior aortic dissection
- Atherosclerosis

9. Immediate Actions

ACUTE MANAGEMENT (RUPTURE/DISSECTION)

1. Immediate beta-blockade (reduce dP/dT)
2. Blood pressure control
3. Emergent surgery for rupture or symptomatic aneurysm
4. Type A dissection: emergent surgery
5. Contained rupture: emergent repair

10. Treatment Overview

Surveillance: Asymptomatic aneurysms below surgical threshold require periodic imaging (every 6-12 months depending on size/growth rate). BP control (target <130/80), beta-blockers, smoking cessation.

Surgical thresholds: Ascending TAA >5.5 cm (4.5-5.0 cm in Marfan/Loeys-Dietz); descending TAA >6.0 cm; AAA >5.5 cm (men) or >5.0 cm (women); rapid growth >0.5 cm/year; symptomatic regardless of size.

Endovascular: TEVAR for descending thoracic; EVAR for infrarenal AAA; increasingly preferred over open repair in suitable anatomy.

Open repair: Ascending aorta and arch replacement with hypothermic circulatory arrest.

11. Follow-up Questions

- Size and location of aneurysm?
- Rate of growth on serial imaging?
- Symptoms - chest pain, back pain, compression symptoms?
- Family history of aneurysm or sudden death?
- Connective tissue disorder features?
- Bicuspid aortic valve?
- Smoking history? BP control?

12. References

1. Hiratzka LF, et al. 2010 ACCF/AHA/AATS/ACR/ASA/SCA/SCAI/SIR/STS/SVM Guidelines for thoracic aortic disease. *Circulation*. 2010;121(13):e266-e369.
2. Erbel R, et al. 2014 ESC Guidelines on the diagnosis and treatment of aortic diseases. *Eur Heart J*. 2014;35(41):2873-2926.

Atherosclerosis

Category: Vascular Disease **Priority: MODERATE** **ICD-10:** I70.9

1. Definition

Atherosclerosis is a chronic inflammatory disease of the arterial wall characterized by the formation of lipid-rich plaques (atheromas) within the intima of large and medium-sized arteries. It is the underlying pathological process responsible for coronary artery disease, cerebrovascular disease, peripheral artery disease, and aortic aneurysm. Atherosclerosis is the single leading cause of morbidity and mortality worldwide, responsible for approximately 50% of all deaths in developed countries.

2. Mechanism

The response-to-injury hypothesis: Endothelial dysfunction (from hypertension, smoking, hyperlipidemia, diabetes, inflammation) increases permeability to LDL cholesterol, which enters the subendothelial space and becomes oxidized. Oxidized LDL recruits monocytes which differentiate into macrophages, engulf oxLDL to become foam cells, and form fatty streaks. Smooth muscle cells migrate from media to intima, proliferate, and secrete extracellular matrix forming a fibrous cap over the lipid-rich necrotic core. Plaques with thick fibrous caps are stable; those with thin caps, large lipid cores, and high inflammatory cell content are vulnerable to rupture. Plaque rupture exposes thrombogenic material, triggering platelet aggregation and thrombosis - the acute event (MI, stroke, sudden death).

3. ECG Findings

- **No specific ECG findings for atherosclerosis itself:** ECG changes reflect complications (ischemia, infarction, arrhythmias)
- **Prior MI changes:** Pathological Q waves, persistent ST-T changes
- **Ischemic changes:** ST depression, T wave inversion during angina
- **LVH:** From chronic hypertension or prior MI remodeling
- **Arrhythmias:** AF, PVCs (associated with CAD)

4. Required ECG Features

> **ECG ROLE**
>
> ECG does not diagnose atherosclerosis. It serves to detect established CAD complications (prior MI, ischemia, arrhythmias) as markers of disease presence. Resting ECG is normal in ~50% of patients with significant CAD.

5. Diagnostic Criteria

- Clinical: Risk factor assessment, symptoms of ischemia
- Non-invasive: Coronary CTA (anatomical), stress testing (functional), CAC scoring
- Invasive: Coronary angiography (gold standard for luminal stenosis)
- Intravascular imaging (IVUS, OCT): Characterizes plaque morphology
- Carotid ultrasound: IMT and plaque assessment
- Ankle-brachial index: Peripheral atherosclerosis

6. Differential Diagnosis

Condition	Differentiating Features
Vasospastic angina	Normal coronaries or non-obstructive disease; transient ST elevation; responds to nitrates/CCBs
Myocarditis	Younger, viral prodrome, elevated troponin, non-obstructive coronaries
Coronary artery spasm	Rest pain, transient ST elevation, acetylcholine positive
Coronary artery dissection	SCAD: young women, non-atherosclerotic, intimal flap on OCT

7. Symptoms

- **Coronary:** Angina, MI, dyspnea
- **Cerebrovascular:** TIA, stroke
- **Peripheral:** Claudication, limb ischemia
- **Often asymptomatic:** Until acute event occurs

8. Risk Factors

- Non-modifiable: Age, male sex, family history, genetic polymorphisms

- Major modifiable: Smoking, hypertension, diabetes, dyslipidemia (LDL, TG, low HDL), obesity
- Emerging: Lp(a), hs-CRP, CKD, OSA, psychosocial stress, sedentary lifestyle, poor diet

9. Immediate Actions

ACS PRESENTATION

1. If ACS: standard ACS protocol (aspirin, anticoagulation, revascularization as indicated)
2. Risk stratification
3. Secondary prevention initiation

10. Treatment Overview

Primary prevention: Risk factor modification - smoking cessation, BP control, statins for elevated LDL, diabetes management, lifestyle (diet, exercise, weight).

Secondary prevention (established ASCVD): High-intensity statin (LDL <70 if very high risk), aspirin 75-100 mg, ACE-I/ARB, beta-blocker post-MI, SGLT2i if diabetes. Aggressive risk factor targets.

Revascularization: As indicated for symptomatic CAD (PCI/CABG).

11. Follow-up Questions

- Risk factor assessment: smoking, BP, lipids, diabetes?
- Prior ASCVD events (MI, stroke, PAD)?
- Family history of premature ASCVD?
- Current medications for prevention?
- Lipid profile and targets?

12. References

1. Libby P. The vascular biology of atherosclerosis. In: *Braunwald's Heart Disease*. 12th ed. 2018.
2. 2019 ESC/EAS Guidelines for the management of dyslipidaemias. *Eur Heart J*. 2020;41(1):111-188.

Atrial Septal Defect (ASD)

Category: Congenital Heart Disease **Priority: MODERATE** **ICD-10:** Q21.1

1. Definition

Atrial septal defect (ASD) is a congenital opening in the interatrial septum allowing abnormal communication between the atria. It is one of the most common congenital heart defects diagnosed in adulthood (most common adult CHD after bicuspid aortic valve). There are four types: secundum ASD (ostium secundum defect in the region of the fossa ovalis - 75%), primum ASD (lower septum - part of AV septal defects - 15%), sinus venosus ASD (superior or inferior to fossa ovalis near vena cavae - 10%), and coronary sinus ASD (rare). Secundum ASDs may be amenable to device closure.

2. Mechanism

Secundum ASD: Deficiency or excessive resorption of the septum primum during septation. The direction of shunt depends on relative ventricular compliance; left-to-right shunt occurs because the LV is thicker and less compliant than the RV, so LA blood follows the pressure gradient to the RA. This causes chronic volume overload of the RA, RV, and pulmonary circulation. Increased pulmonary blood flow (>1.5:1 ratio = significant) leads to RA/RV dilation, pulmonary artery dilation, and eventually pulmonary hypertension if long-standing and uncorrected. Paradoxical embolism (venous thrombus crossing to systemic circulation) can cause cryptogenic stroke.

3. ECG Findings

- **Secundum ASD:** Right axis deviation, incomplete RBBB (rsR' in V1 - from RV volume overload), RVH pattern, RAE (tall peaked P waves in II)
- **Primum ASD (AV septal defect):** Left axis deviation (superior axis) - pathognomonic; first-degree AV block; may have RBBB
- **Sinus venosus ASD:** Similar to secundum; may be associated with partial anomalous pulmonary venous return (PAPVR)
- **AF or atrial flutter:** Common in adults with ASD (>40 years), due to RA dilation
- **First-degree AV block:** Especially with primum ASD
- **Arrhythmias from surgical patch:** If surgically closed

4. Required ECG Features

DIAGNOSTIC ECG CLUES

- Incomplete RBBB + RAD + RAE in adult with unexplained right heart enlargement = ASD
- Left axis deviation in a patient with known ASD = primum ASD (AV septal defect)
- New AF in a young/middle-aged adult with no other cause: consider ASD
- Secundum ASD is the most common cause of incomplete RBBB with RAD

5. Diagnostic Criteria

- Echocardiography: Visualizes ASD, shunt direction, Qp/Qs ratio, RA/RV size, estimated PA pressure
- Significant ASD: Qp/Qs >1.5; RA/RV enlargement
- CMR/CT: ASD size and anatomy; associated PAPVR with sinus venosus ASD; Qp/Qs quantification
- Cardiac catheterization: Confirm hemodynamics; exclude pulmonary vascular disease (PVR) before closure; device closure procedure
- Agitated saline contrast echo (bubble study): Opacified RA first, then LA after 3-5 cardiac cycles (delayed contrast appearance in LA)
- TEE: Definitive assessment of ASD anatomy and suitability for device closure

6. Differential Diagnosis

Condition	Differentiating Features
Patent foramen ovale (PFO)	Small tunnel-like opening; no RA/RV enlargement; normal ECG; associated with cryptogenic stroke
Partial anomalous pulmonary venous return	May coexist with sinus venosus ASD; one or more pulmonary veins drain to RA/SVC; similar physiology
Primary pulmonary hypertension	No intracardiac shunt; isolated PAH; no ASD on echo
Ebstein anomaly	Tricuspid valve apically displaced; massive RA; may have ASD/PFO; different anatomy

7. Symptoms

- **Many adults asymptomatic:** Discovered incidentally on echo or by murmur

- **Exertional dyspnea:** Most common symptom; progressive
- **Fatigue, exercise intolerance:** Reduced forward cardiac output
- **Palpitations:** AF or atrial flutter from RA dilation
- **Cryptogenic stroke:** Paradoxical embolism
- **Right heart failure:** Edema, ascites (if Eisenmenger develops from long-standing large shunt)

8. Risk Factors

- Genetic: Secundum ASD has familial recurrence in ~10%; GATA4, NKX2-5, TBX5 mutations
- Down syndrome (primum ASD/AV septal defect)
- Holt-Oram syndrome (TBX5 - ASD with radial limb abnormalities)
- Secundum ASD: 2:1 female predominance
- Late presentation: Many diagnosed in adulthood when symptoms or AF develop

9. Immediate Actions

MANAGEMENT CONSIDERATIONS

1. Echocardiographic assessment of shunt size, direction, RA/RV size, PA pressure
2. If AF: Rate/rhythm control; anticoagulation for stroke prevention; consider ASD closure
3. Exclude Eisenmenger physiology (irreversible PAH) before closure
4. Closure is NOT recommended if Eisenmenger syndrome (fixed PAH with right-to-left shunt)

10. Treatment Overview

Closure indications: Significant ASD (Qp/Qs >1.5) with RA/RV enlargement, paradoxical embolism (especially cryptogenic stroke), or significant symptoms. Closure can be performed regardless of age (previously thought to be contraindicated after age 40, but evidence shows benefit at any age if PA pressure not severely elevated).

Device closure: Percutaneous closure with septal occluder device (Amplatzer, Gore Helex) for secundum ASD with adequate rims. Most common approach. Contraindicated for primum and sinus venosus ASDs.

Surgical closure: For primum ASD, sinus venosus ASD, or secundum ASD unsuitable for device (inadequate rims, large defect). Surgical patch closure

with concurrent PAPVR repair if sinus venosus type.

Contraindications to closure: Eisenmenger syndrome (fixed PAH), severe LV dysfunction (needs ASD as pop-off), reversible PAH (test with vasodilators).

AF management: AF may persist after closure if atrial remodeling is advanced; anticoagulation if CHA_2DS_2-VASc indicates; consider catheter ablation.

Endocarditis prophylaxis: Not routinely recommended for ASD (unrepaired or closed) unless prior endocarditis.

11. Follow-up Questions

- Known ASD? Type? Size?
- Any prior closure (device or surgical)? When?
- Qp/Qs ratio?
- RA/RV size? PA pressure?
- Any history of AF?
- Any cryptogenic stroke or TIA?
- Exercise tolerance? Symptoms?
- Any paradoxical embolism symptoms (migraine with aura)?

12. References

1. Warnes CA, et al. ACC/AHA 2008 Guidelines for the Management of Adults with Congenital Heart Disease. *Circulation*. 2008;118(23):e714-e833.
2. Stout KK, et al. 2018 AHA/ACC Guideline for the Management of Adults With Congenital Heart Disease. *Circulation*. 2019;139(14):e698-e800.

Diabetes (Cardiac Manifestations)

Category: Metabolic Cardiovascular Disease **Priority: MODERATE** **ICD-10:** E11.59

1. Definition

Diabetes mellitus is a major risk factor for cardiovascular disease, with cardiovascular complications being the leading cause of morbidity and mortality in diabetic patients. Diabetes increases the risk of coronary artery disease 2-4 fold, heart failure 2-5 fold, and stroke 1.5-3 fold. Diabetic cardiomyopathy is a distinct entity characterized by structural and functional changes in the myocardium independent of hypertension or CAD, leading to diastolic and eventually systolic dysfunction. The cardiac manifestations of diabetes encompass accelerated atherosclerosis, microvascular disease, autonomic neuropathy, and direct myocardial metabolic dysfunction.

2. Mechanism

Multiple mechanisms contribute to diabetic cardiac disease: (1) **Hyperglycemia:** Advanced glycation end-products (AGEs) cross-link collagen causing myocardial stiffness; protein kinase C activation causes vascular dysfunction; oxidative stress from increased mitochondrial superoxide production. (2) **Insulin resistance:** Impaired myocardial glucose uptake shifts metabolism toward free fatty acids, increasing oxygen consumption and reducing efficiency. (3) **Lipotoxicity:** Accumulation of ceramides and other lipid intermediates in cardiomyocytes causes apoptosis and fibrosis. (4) **Microvascular dysfunction:** Capillary basement membrane thickening, endothelial dysfunction, impaired vasodilation. (5) **Autonomic neuropathy:** Sympathetic/vagal imbalance causing resting tachycardia, orthostatic hypotension, silent ischemia, QT prolongation. (6) **Accelerated atherosclerosis:** Endothelial dysfunction, platelet hyperaggregability, pro-inflammatory state, dyslipidemia (high TG, low HDL, small dense LDL).

3. ECG Findings

- **Silent ischemia:** ST depression during stress testing without symptoms (due to autonomic neuropathy)
- **Resting ECG may be normal:** Despite significant CAD
- **Prior MI changes:** Q waves, persistent ST-T changes (diabetics more likely to have had unrecognized MI)
- **LVH:** From hypertension (common comorbidity) or diabetic cardiomyopathy
- **QT prolongation:** From autonomic neuropathy, electrolyte disturbances, or medications
- **Resting tachycardia:** From autonomic neuropathy (loss of vagal tone)
- **Orthostatic changes:** Heart rate and BP fail to compensate with position change

- **AF:** Increased risk (diabetes is part of CHA_2DS_2-VASc)

4. Required ECG Features

> **ECG CONSIDERATIONS IN DIABETES**
>
> - Silent ischemia is common; normal ECG does not exclude significant CAD
> - Q waves without known MI history suggest prior silent infarction
> - QT prolongation increases arrhythmic risk; monitor with QT-prolonging drugs
> - Resting HR >100 with no other cause suggests autonomic neuropathy

5. Diagnostic Criteria

- Diabetes: Fasting glucose >=126 mg/dL, HbA1c >=6.5%, 2-hour OGTT >=200 mg/dL
- Diabetic cardiomyopathy: LV diastolic dysfunction (echo: E/e' >15, LA enlargement) with preserved EF in the absence of hypertension, CAD, or valvular disease; or HFrEF with these exclusions
- Stress testing: Lower threshold for screening due to high pre-test probability and silent ischemia
- Coronary CTA or angiography: For symptomatic patients or high-risk features
- Biomarkers: BNP/NT-proBNP elevated with cardiac involvement; troponin if MI
- Autonomic testing: Heart rate variability, Valsalva maneuver, tilt table if indicated

6. Differential Diagnosis

Condition	Differentiating Features
Ischemic cardiomyopathy	Prior MI, regional WMA, obstructive CAD on angiography
Hypertensive heart disease	Long-standing uncontrolled hypertension; may coexist with diabetes
Valvular heart disease	Valvular abnormalities on echo; different etiology
Alcoholic cardiomyopathy	Heavy alcohol use; improves with abstinence

7. Symptoms

- **Silent ischemia:** No chest pain despite objective ischemia (autonomic neuropathy)
- **Dyspnea on exertion:** HFpEF from diabetic cardiomyopathy or ischemic heart disease
- **Fatigue, reduced exercise tolerance:** Diastolic dysfunction, reduced cardiac output
- **Orthostatic dizziness:** Autonomic neuropathy
- **Peripheral edema:** Heart failure or nephrotic syndrome from diabetic nephropathy

- **Nocturnal dyspnea:** Elevated filling pressures

8. Risk Factors

- Type 1 or Type 2 diabetes mellitus
- Duration of diabetes (>10 years higher risk)
- Poor glycemic control (HbA1c >7%)
  Concomitant hypertension, dyslipidemia, obesity
- Diabetic nephropathy (albuminuria, reduced eGFR)
- Smoking
- Family history of CAD
- Insulin resistance/metabolic syndrome

9. Immediate Actions

ACUTE CARDIAC EVENTS IN DIABETES

1. ACS: Same management as non-diabetic; glycemic control (target 140-180 mg/dL; avoid hypoglycemia)
2. Acute HF: Diuretics; SGLT2i continuation if hemodynamically stable
3. Check HbA1c if not known; tight glycemic control reduces microvascular but not necessarily macrovascular events in acute setting

10. Treatment Overview

Glycemic control: HbA1c target <7% for most (individualize). Metformin first-line. SGLT2 inhibitors (empagliflozin, dapagliflozin, canagliflozin) and GLP-1 receptor agonists (liraglutide, semaglutide) have proven cardiovascular benefit and reduce MACE, HF hospitalization, and mortality independent of glycemic control.

Cardiovascular risk reduction: High-intensity statin (atorvastatin 40-80 mg or rosuvastatin 20-40 mg) regardless of baseline LDL. Aspirin 75-100 mg for secondary prevention or primary prevention if high ASCVD risk (>10% 10-year risk). BP target <130/80 (ACE-I or ARB preferred). Smoking cessation.

Heart failure: SGLT2 inhibitors are now foundational therapy for both HFrEF and HFpEF in diabetic patients. Standard GDMT for HF.

Screening: Stress testing for symptomatic patients or those with high-risk features; not routine in asymptomatic patients due to poor specificity.

11. Follow-up Questions

- Type and duration of diabetes?
- Most recent HbA1c? Glycemic trend?
- Current antihyperglycemic medications? (SGLT2i, GLP-1 RA?)
- Any symptoms of ischemia or heart failure?
- Cardiovascular risk factors: lipids, BP, smoking, family history?
- Renal function? Albuminuria?
- Any prior cardiovascular events?
- Signs of autonomic neuropathy (orthostasis, resting tachycardia, gastroparesis)?

12. References

1. American Diabetes Association. Standards of Medical Care in Diabetes - 2024. *Diabetes Care*. 2024;47(Suppl 1):S179-S218.
2. Zinman B, et al. Empagliflozin, Cardiovascular Outcomes, and Mortality in Type 2 Diabetes. *N Engl J Med*. 2015;373(22):2117-2128.
3. Marso SP, et al. Liraglutide and Cardiovascular Outcomes in Type 2 Diabetes. *N Engl J Med*. 2016;375(4):311-322.

Ehlers-Danlos Syndrome

Category: Connective Tissue Disorder **Priority: MODERATE** **ICD-10:** Q79.6

1. Definition

Ehlers-Danlos syndrome (EDS) is a heterogeneous group of inherited connective tissue disorders characterized by joint hypermobility, skin hyperextensibility, and tissue fragility. The vascular type (vEDS, formerly type IV) is the most clinically significant for cardiac care, caused by mutations in COL3A1 encoding type III collagen. vEDS is associated with arterial fragility, spontaneous arterial dissection and rupture (most commonly mesenteric, splenic, and renal arteries; less frequently aortic), uterine rupture in pregnancy, and colonic perforation. Vascular EDS has the most severe prognosis among EDS subtypes, with median survival ~51 years.

2. Mechanism

Type III collagen is a major component of the walls of hollow organs and blood vessels, providing tensile strength and elasticity. In vEDS, mutations in COL3A1 lead to deficient or abnormal type III collagen, causing thinning of the arterial media, loss of elastic fibers, and fragility of the vessel wall. Arterial dissection and rupture occur without preceding aneurysm formation, distinguishing vEDS from Marfan syndrome. The absence of type III collagen in vessel walls means standard surgical repair is technically challenging and prone to failure.

3. ECG Findings

- **ECG usually normal:** Unless cardiac complication occurs
- **Acute changes with dissection:** ST elevation or depression if coronary ostia involved (rare in aortic dissection with vEDS)
- **Changes related to associated conditions:** Mitral valve prolapse may show repolarization abnormalities

4. Required ECG Features

ECG IN EDS

No specific ECG findings. The role of ECG is primarily to evaluate acute complications (arterial dissection, MI) and establish baseline for perioperative monitoring. The diagnosis is clinical and genetic.

5. Diagnostic Criteria

- Villefranche criteria (1998): Major criteria (thin translucent skin, arterial/uterine/intestinal fragility or rupture, easy bruising, characteristic facial appearance), minor criteria (acrogeria, hypermobility of small joints, tendon/muscle rupture, talipes equinovarus, varicose veins, pneumothorax, gingival recession, positive family history)
- Genetic testing: COL3A1 pathogenic mutation confirms vEDS
- Skin biopsy: Reduced or absent type III collagen on immunofluorescence; abnormal collagen fibrils on electron microscopy
- Imaging: Vascular imaging (CTA/MRA) for surveillance; no routine screening due to fragility risk

6. Differential Diagnosis

Condition	Differentiating Features
Marfan syndrome	Aortic root dilation, ectopia lentis, tall stature; FBN1 mutation; aortic aneurysm precedes dissection
Loeys-Dietz syndrome	Widely spaced eyes, bifid uvula, arterial tortuosity; TGFBR1/2 mutation
Other EDS types	Hypermobile type (joint laxity, pain, no vascular risk); classical type (skin hyperextensibility, atrophic scarring)
Familial arterial aneurysm	Isolated arterial disease without skin/joint findings; different genetic basis

7. Symptoms

- **Skin:** Thin, translucent, easily bruised; poor wound healing; atrophic scars
- **Arterial rupture/dissection:** Sudden severe abdominal, chest, or limb pain; pulsatile masses
- **Joint hypermobility:** Milder than other EDS types; mainly small joints
- **Characteristic facies:** Thin lips, small chin, prominent eyes, thin nose
- **Pneumothorax:** Spontaneous from lung fragility
- **GI perforation:** Colonic or sigmoid perforation

8. Risk Factors

- Genetic: Autosomal dominant COL3A1 mutation; 50% de novo
- Family history of vEDS, sudden death, arterial rupture, or uterine rupture
- Pregnancy: High risk of uterine and arterial rupture; 5% maternal mortality per pregnancy

- Trauma: Even minor trauma can cause arterial injury

9. Immediate Actions

ARTERIAL EMERGENCY

1. Suspected arterial rupture: Immediate imaging (CTA); permissive hypotension
2. Avoid invasive procedures unless absolutely necessary (arterial puncture carries high risk)
3. Surgical or endovascular repair only if life-threatening; tissue fragility complicates procedures
4. Genetic counseling for family

10. Treatment Overview

Monitoring: No consensus on routine vascular imaging due to procedure risks. Clinical vigilance for symptoms. Some centers use non-invasive imaging (MRA) cautiously.

Medical: Celiprolol (beta-blocker with alpha-2 agonist activity) shown to reduce arterial events in vEDS (BBEST trial). Standard beta-blockers may be used. Avoid anticoagulation unless clearly indicated.

Surgery: High perioperative mortality; endovascular approaches increasingly preferred. Only for life-threatening emergencies.

Lifestyle: Avoid contact sports, isometric exercise, heavy lifting; injury prevention; med-alert bracelet.

Pregnancy: High-risk; multidisciplinary care; consider avoiding; celiprolol continued; planned cesarean delivery if pregnancy pursued.

11. Follow-up Questions

- Known vEDS diagnosis? Genetic confirmation?
- Family history of arterial rupture or sudden death?
- Any prior arterial events?
- Skin characteristics: thin, translucent, easy bruising?
- Joint hypermobility?
- Pregnancy considerations?

12. References

1. Byers PH. Vascular Ehlers-Danlos Syndrome. In: *GeneReviews*. University of Washington, Seattle; 1999-2024.
2. Frank M, et al. Vascular Ehlers-Danlos Syndrome: Long-Term Observational Study. *J Am Coll Cardiol*. 2019;73(15):1948-1957.
3. Ong KT, et al. Effect of celiprolol on prevention of cardiovascular events in vascular Ehlers-Danlos syndrome: a prospective randomised, open, blinded-endpoints trial. *Lancet*. 2010;376(9751):1476-1484.

Familial Hypercholesterolemia

Category: Genetic Dyslipidemia **Priority: MODERATE** **ICD-10:** E78.0

1. Definition

Familial hypercholesterolemia (FH) is an autosomal dominant genetic disorder of lipid metabolism characterized by severely elevated LDL cholesterol from birth, leading to premature atherosclerotic cardiovascular disease. It is caused by loss-of-function mutations in the LDL receptor (LDLR - 90%), apolipoprotein B (APOB - 5-10%), or gain-of-function mutations in PCSK9 (<1%). Homozygous FH is rare (1 in 300,000) but causes extreme LDL levels (>500 mg/dL) and childhood CAD. Heterozygous FH affects approximately 1 in 250 individuals worldwide but is underdiagnosed (<10% identified).

2. Mechanism

LDL receptor (LDLR) on hepatocytes normally binds LDL particles and mediates their clearance from circulation. In FH, mutations in LDLR reduce the number or function of receptors, impairing hepatic LDL clearance and causing markedly elevated circulating LDL. The excess LDL penetrates the arterial endothelium, becomes oxidized, and drives accelerated atherosclerosis from early childhood. Tendon xanthomas develop from cholesterol deposition in tendons. APOB mutations impair the ligand (ApoB100 on LDL) that binds to LDLR. PCSK9 gain-of-function increases degradation of LDLR, reducing LDL clearance.

3. ECG Findings

- **ECG is normal in uncomplicated FH:** Changes reflect atherosclerotic complications
- **Prior MI changes:** Q waves, persistent ST-T changes (premature CAD)
- **Ischemic changes:** ST depression during stress testing or ACS
- **LVH:** From chronic hypertension
- **Arrhythmias:** AF, VT if significant CAD develops

4. Required ECG Features

ECG IN FH

No specific ECG findings for FH itself. ECG assesses for established ASCVD complications. All FH patients require cardiovascular risk assessment and monitoring for ischemic changes given their very high lifetime ASCVD risk.

5. Diagnostic Criteria

- Dutch Lipid Clinic Network criteria: Points for LDL level (>330 = 8 pts, 250-329 = 5, 190-249 = 3, 155-189 = 1), tendon xanthomas (6), corneal arcus <45 years (4), family history of FH (1) or premature CAD/stroke (2), personal premature CAD (2) or stroke (1), genetic mutation (8). Score >8 = definite, 6-8 = probable, 3-5 = possible, <3 = unlikely.
- LDL levels: Heterozygous FH typically >190 mg/dL in adults; homozygous FH >500 mg/dL
- Genetic testing: LDLR, APOB, PCSK9 mutations
- Clinical: Tendon xanthomas (Achilles, extensor tendons of hands), corneal arcus (<45 years suggests FH), xanthelasma
- Screen first-degree relatives (cascade screening) if FH mutation identified

6. Differential Diagnosis

Condition	Differentiating Features
Polygenic hypercholesterolemia	Less severe LDL elevation; no tendon xanthomas; no FH mutation; family history less dramatic
Familial combined hyperlipidemia	Elevated TG and LDL; ApoB elevated; different genetic basis; no xanthomas
Secondary hyperlipidemia	Hypothyroidism, nephrotic syndrome, drugs (steroids, antiretrovirals); corrects with treatment of cause
Sitosterolemia	Elevated plant sterols; tendon xanthomas; homozygous; ABCG5/G8 mutations

7. Symptoms

- **Usually asymptomatic until ASCVD develops:**
- **Tendon xanthomas:** Firm nodular swellings over Achilles, extensor tendons (patience, elbows)
- **Xanthelasma:** Yellow cholesterol deposits on eyelids
- **Corneal arcus:** Cholesterol deposit in cornea; <45 years suggests FH
- **Angina, MI, stroke:** Premature ASCVD (men <55, women <60)
- **Homozygous FH:** Childhood angina, MI, aortic stenosis from supravalvular deposit, visible xanthomas in first decade

8. Risk Factors

- Family history of FH or premature CAD

- LDLR, APOB, or PCSK9 mutation
- Homozygous or compound heterozygous state (severe disease)
- Additional risk factors compound FH risk: smoking, hypertension, diabetes, obesity, elevated Lp(a)
- Untreated FH: ~50% risk of fatal or non-fatal MI by age 50 in men, age 60 in women

9. Immediate Actions

ASCVD EVENT

1. If ACS: Standard ACS management
2. Intensify lipid-lowering therapy to maximum tolerated dose
3. Cascade screening of first-degree relatives

10. Treatment Overview

Lifestyle: Heart-healthy diet (reduce saturated fat, eliminate trans fats), exercise, weight management, smoking cessation. Important but insufficient alone due to genetic defect.

Statins: High-intensity (atorvastatin 40-80 mg or rosuvastatin 20-40 mg) first-line; often need maximum dose. Target LDL <100 in adults, <70 if ASCVD present, <70 in very high risk.

Ezetimibe: Add if LDL goal not achieved on statin alone; inhibits intestinal cholesterol absorption.

PCSK9 inhibitors: Evolocumab or alirocumab for statin-refractory or statin-intolerant patients; can achieve LDL <40 mg/dL; very effective in FH.

Inclisiran: siRNA against PCSK9; twice-yearly dosing; effective LDL reduction.

Lipoprotein apheresis: For homozygous FH or severe heterozygous FH not controlled with drugs; every 1-2 weeks.

Homozygous FH: Lomitapide (microsomal triglyceride transfer protein inhibitor), evinacumab (ANGPTL3 monoclonal antibody), liver transplantation.

Cascade screening: Test all first-degree relatives; genetic testing preferred.

11. Follow-up Questions

- LDL cholesterol level? At diagnosis? Current? Trend?
- Tendon xanthomas? Corneal arcus? Xanthelasma?
- Family history of high cholesterol or premature CAD/stroke?
- Genetic testing results?

- Current lipid-lowering therapy and response?
- Any prior cardiovascular events?
- Cascade screening completed for relatives?
- Additional risk factors: smoking, BP, diabetes, Lp(a)?

12. References

1. Defesche JC, et al. Familial hypercholesterolaemia. *Nat Rev Dis Primers*. 2017;3:17093.
2. 2019 ESC/EAS Guidelines for the management of dyslipidaemias. *Eur Heart J*. 2020;41(1):111-188.
3. Raal FJ, et al. Evinacumab for Homozygous Familial Hypercholesterolemia. *N Engl J Med*. 2020;383(8):711-720.

Fibromuscular Dysplasia (FMD)

Category: Vascular Disease **Priority: MODERATE** **ICD-10:** I77.3

1. Definition

Fibromuscular dysplasia (FMD) is a non-atherosclerotic, non-inflammatory vascular disease affecting medium-sized arteries, most commonly the renal arteries (60-75%) and extracranial carotid/vertebral arteries (25-30%), causing stenosis, occlusion, aneurysm, or dissection. It predominantly affects women (90% of cases), typically aged 20-60 years. Medial fibroplasia ("string of beads" appearance) is the most common histological subtype. FMD is a significant cause of secondary hypertension, stroke, and spontaneous coronary artery dissection (SCAD).

2. Mechanism

The etiology remains unclear but likely involves hormonal factors (estrogen, given female predominance), genetic predisposition, and mechanical factors (stretching of vessel wall). Pathologically, FMD involves abnormal fibrous tissue and smooth muscle hyperplasia within the vessel wall (media most commonly), causing alternating areas of stenosis and dilation ("string of beads"). Renal artery FMD causes secondary hypertension through renin-angiotensin activation. Cervicocranial FMD can cause stenosis, aneurysm, or dissection leading to TIA/stroke. The association with SCAD is increasingly recognized (up to 50-80% of SCAD patients have concomitant extracoronary vascular abnormalities consistent with FMD).

3. ECG Findings

- **ECG usually normal:** Changes reflect complications
- **LVH:** From long-standing hypertension (renal artery FMD)
- **Ischemic changes:** If associated SCAD occurs
- **Arrhythmias:** If stroke or TIA affects autonomic centers

4. Required ECG Features

ECG IN FMD

No specific ECG findings. ECG may show LVH from secondary hypertension. In patients presenting with SCAD and FMD, ECG shows typical ACS patterns.

5. Diagnostic Criteria

- Imaging (gold standard): CT angiography, MR angiography, or conventional angiography showing characteristic "string of beads" (multifocal type) or smooth tubular stenosis (focal type)
- Duplex ultrasound: Screening tool; elevated velocities in affected segments
- Diagnosis of exclusion: No atherosclerosis, no inflammatory markers, no connective tissue disorder
- Screening for SCAD: Coronary angiography if cardiac symptoms
- Imaging from head to pelvis to assess all vascular beds (CTA or MRA)

6. Differential Diagnosis

Condition	Differentiating Features
Atherosclerotic renal artery stenosis	Older, male, smokers, ostial location, atherosclerotic disease elsewhere
Vasculitis (Takayasu, PAN)	Inflammatory markers elevated, systemic symptoms, different angiographic pattern
Connective tissue disorders	Marfan (root dilation), vascular EDS (arterial rupture); genetic testing differentiates
Renal parenchymal hypertension	Kidney disease on labs/urine; no vascular lesion on imaging

7. Symptoms

- **Renal FMD:** Hypertension (often severe, resistant, onset <35 years in women); renal bruit (lateralizing)
- **Cervicocranial FMD:** TIA, stroke, pulsatile tinnitus, headache, neck pain (dissection), Horner syndrome
- **Mesenteric FMD:** Postprandial abdominal pain, weight loss, mesenteric ischemia
- **Coronary (SCAD):** ACS symptoms
- **Iliac/Extremity FMD:** Claudication, limb ischemia
- **Many asymptomatic:** Incidentally detected on imaging

8. Risk Factors

- Female sex (90%)
- Age 20-60 (mean ~50)

- Smoking (associated but not causative)
- Hormonal factors (estrogen)
- Possible genetic predisposition (familial clustering reported)
- Prior SCAD (screen for FMD)

9. Immediate Actions

ACUTE COMPLICATIONS

1. SCAD: See SCAD entry for management
2. Stroke/TIA from cervical FMD: Standard stroke management
3. Dissection: BP control (beta-blockers); anticoagulation or antiplatelet if no hemorrhage
4. Hypertensive crisis: IV antihypertensives

10. Treatment Overview

Renal artery FMD: Balloon angioplasty (without stent) is treatment of choice for hemodynamically significant stenosis causing hypertension or renal impairment; cure or improve hypertension in ~60-70%. Medical therapy: ACE-I/ARB (bilateral disease may worsen renal function - monitor), beta-blockers, CCBs.

Cervicocranial FMD: Antiplatelet therapy (aspirin 81 mg) for all patients unless contraindicated; no proven benefit of anticoagulation. Revascularization (angioplasty/stenting or surgery) for symptomatic stenosis or aneurysm causing symptoms.

SCAD: Conservative management preferred; see SCAD entry.

Screening: All SCAD patients should be screened for extracoronary vascular abnormalities (CTA/MRA head to pelvis). First-degree relative screening controversial but may be considered.

Lifestyle: Smoking cessation; BP control; avoid estrogen-containing contraceptives/HRT.

11. Follow-up Questions

- Hypertension onset age? Severity? Response to medications?
- Renal bruit?

- Any TIA/stroke symptoms?
- Any SCAD history?
- Headaches? Pulsatile tinnitus?
- Smoking history?
- Estrogen use (OCP, HRT)?
- Family history of early hypertension, stroke, vascular disease?

12. References

1. Olin JW, et al. Diagnosis, management, and future developments of fibromuscular dysplasia. *J Vasc Surg*. 2021;73(2 Suppl):S31-S43.
2. Saw J, et al. Spontaneous coronary artery dissection: association with predisposing arteriopathies and precipitating stressors. *J Am Coll Cardiol*. 2015;67(14):1713-1722.

High Blood Pressure (Hypertension)

Category: Risk Factor/Vascular **Priority: MODERATE** **ICD-10:** I10

1. Definition

Hypertension is a sustained elevation of systemic arterial blood pressure, defined as systolic BP >=130 mmHg or diastolic BP >=80 mmHg based on the average of >=2 properly measured readings on >=2 occasions (2017 ACC/AHA guidelines). Hypertension is the most common modifiable risk factor for cardiovascular disease, affecting approximately 1.3 billion adults worldwide. It is the leading attributable risk factor for global death, responsible for approximately 10.4 million deaths annually through stroke, ischemic heart disease, heart failure, and chronic kidney disease.

2. Mechanism

Essential (primary) hypertension: Accounts for 90-95% of cases. Complex interplay of genetic, environmental, and behavioral factors: increased sympathetic nervous system activity, activation of RAAS, endothelial dysfunction, sodium sensitivity, insulin resistance, arterial stiffness, and abnormal renal sodium handling. Over time, sustained elevated BP causes arteriolar hyaline degeneration, medial hypertrophy, and reduced arterial compliance, creating a vicious cycle of increasing systolic pressure and widened pulse pressure. **Secondary hypertension:** Renal parenchymal disease, renal artery stenosis, primary aldosteronism, pheochromocytoma, Cushing syndrome, coarctation of aorta, OSA, drug-induced (NSAIDs, OCPs, steroids, stimulants).

3. ECG Findings

- **LVH voltage criteria:** Sokolow-Lyon: S V1 + R V5/V6 >35 mm; Cornell: R aVL + S V3 >28 mm (M), >20 mm (F)
- **Strain pattern:** ST depression with T wave inversion in lateral leads (I, aVL, V5-V6) with downsloping ST segment - indicates severe, long-standing hypertension with repolarization abnormality
- **Left atrial enlargement:** P mitrale (broad, notched P in II); biphasic P in V1 with prominent negative terminal deflection
- **Left axis deviation:** May be present
- **Prolonged QT interval:** From LVH, medications, or electrolyte disturbances
- **AF:** Associated with hypertension; elevated filling pressures and atrial stretch
- **Left bundle branch block:** In severe, long-standing cases

4. Required ECG Features

ECG CLUES

- LVH with strain pattern = severe, long-standing hypertension with target organ damage
- LA enlargement often precedes LVH and indicates elevated filling pressures
- AF in hypertensive patient suggests structural remodeling from chronic pressure overload
- Strain pattern is associated with increased cardiovascular risk independent of LVH voltage

5. Diagnostic Criteria

- Office BP: Average of >=2 readings on >=2 occasions: SBP >=130 mmHg or DBP >=80 mmHg (ACC/AHA 2017); or SBP >=140 or DBP >=90 (ESC/ESH 2018)
- Home BP monitoring: Average >=135/85 mmHg
- Ambulatory BP monitoring (ABPM): 24-hour average >=130/80; daytime >=135/85; nighttime >=120/70
- Classification: Normal (<120/<80), Elevated (120-129/<80), Stage 1 (130-139/80-89), Stage 2 (>=140/>=90) per ACC/AHA
- Target organ damage assessment: LVH on ECG/echo, proteinuria, eGFR, retinopathy
- Secondary hypertension workup: If young onset (<30), resistant HTN, or suggestive features

6. Differential Diagnosis

Condition	Differentiating Features
White coat hypertension	Elevated office BP but normal home/ABPM; cardiovascular risk intermediate between normotensive and sustained hypertensive
Masked hypertension	Normal office BP but elevated home/ABPM; higher cardiovascular risk than white coat; common in smokers, diabetics, OSA
Secondary hypertension	Young age, severe/resistant HTN, abrupt onset, hypokalemia (primary aldosteronism), abdominal bruit (renal artery stenosis)
Pseudohypertension	Elderly with calcified arteries; cuff pressure exceeds intra-arterial pressure; Osler maneuver may be positive

7. Symptoms

- **Usually asymptomatic:** "Silent killer" - most patients have no symptoms
- **Headache:** Occipital, morning; not consistently correlated with BP level
- **Dizziness:** Often medication-related rather than from hypertension itself
- **Symptoms of complications:** Dyspnea (HF), chest pain (CAD), vision changes (hypertensive retinopathy), neurological symptoms (stroke/TIA)
- **Hypertensive emergency:** BP >180/120 with acute target organ damage (encephalopathy, papilledema, acute HF, renal failure, aortic dissection)

8. Risk Factors

- Non-modifiable: Age, family history, genetic polymorphisms, Black race (higher incidence, severity, target organ damage)
- Modifiable: High sodium intake, obesity, physical inactivity, excessive alcohol, smoking, stress, low potassium intake
- Associated conditions: Diabetes, CKD, OSA, dyslipidemia
- Medications: NSAIDs, OCPs, corticosteroids, decongestants, stimulants, calcineurin inhibitors, erythropoietin

9. Immediate Actions

HYPERTENSIVE EMERGENCY VS. URGENCY

1. **Emergency (BP >180/120 with acute target organ damage):** Immediate controlled BP reduction with IV agents (nicardipine, clevidipine, labetalol); reduce MAP by no more than 20-25% in first hour to avoid hypoperfusion
2. **Urgency (very high BP without target organ damage):** Oral medications; gradual reduction over 24-48 hours; do NOT acutely normalize BP (risk of ischemia)
3. Specific scenarios: Aortic dissection (target SBP <120 within minutes); Acute ischemic stroke (permissive hypertension unless thrombolysis candidate); Acute HF (IV vasodilators + diuretics); Acute coronary syndrome (oral beta-blockers + nitrates)

10. Treatment Overview

Lifestyle: Weight reduction (5-10 kg loss = 5-20 mmHg SBP reduction), DASH diet (rich in fruits, vegetables, low-fat dairy, reduced saturated fat), sodium restriction (<1.5-2.3 g/day), potassium supplementation (if no CKD), regular aerobic exercise (150 min/week), limit alcohol (<2 drinks/day men, <1 women), smoking cessation. Can achieve 10-20 mmHg reduction alone.

Pharmacotherapy: Thiazide-like diuretics (chlorthalidone, indapamide), ACE-I/ARB (especially diabetes, CKD, CAD, HF), calcium channel blockers (amlodipine - particularly effective in Black patients), beta-blockers (not first-line unless CAD, HF, or pregnancy). Target: <130/80 (ACC/AHA) or <140/90 (ESC/ESH); individualized.

Resistant hypertension: Failure of 3 drugs including diuretic at maximum tolerated doses; evaluate for secondary causes, medication adherence, volume overload; consider spironolactone (PATHWAY-2 showed superior effect), renal denervation (in selected patients).

11. Follow-up Questions

- Home BP readings? Trend? Frequency?
- Current antihypertensive medications and doses? Adherence?
- Any side effects? (Cough with ACE-I, edema with CCBs)
- Dietary sodium intake? Alcohol? Exercise?
- Weight history? Sleep apnea symptoms?
- Current medications that may raise BP (NSAIDs, decongestants, stimulants)?
- Target organ damage history (stroke, MI, HF, CKD, retinopathy)?
- Family history of hypertension or complications?
- Home BP monitoring technique? Cuff size? Arm position?

12. References

1. Whelton PK, et al. 2017 ACC/AHA Guideline for the Prevention, Detection, Evaluation, and Management of High Blood Pressure in Adults. *Hypertension*. 2018;71(6):e13-e115.
2. Williams B, et al. 2018 ESC/ESH Guidelines for the management of arterial hypertension. *Eur Heart J*. 2018;39(33):3021-3104.
3. Williams B, et al. Spironolactone versus placebo, bisoprolol, and doxazosin to determine the optimal treatment for drug-resistant hypertension (PATHWAY-2). *Lancet*. 2015;386(10008):2059-2068.

Mitral Stenosis

Category: Valvular Heart Disease **Priority: MODERATE** **ICD-10:** I05.0

1. Definition

Mitral stenosis (MS) is narrowing of the mitral valve orifice causing obstruction of left ventricular inflow, elevated left atrial pressure, and pulmonary venous hypertension. It is almost exclusively caused by rheumatic heart disease (historically) or age-related degenerative calcification (in developed countries). The normal mitral valve area is 4-6 cm^2; mild MS is >1.5 cm^2, moderate 1.0-1.5 cm^2, severe <1.0 cm^2. MS is associated with a high thromboembolic risk due to left atrial enlargement, stasis, and atrial fibrillation.

2. Mechanism

Rheumatic MS: Autoimmune cross-reactivity following group A streptococcal pharyngitis causes progressive commissural fusion, leaflet thickening, chordal fusion, and calcification over decades. The valve leaflets become rigid with a "fish-mouth" or "buttonhole" appearance. Degenerative (calcific) MS: Age-related calcium deposition on the mitral annulus and leaflets; different pathology with preserved commissures; less likely to respond to balloon valvotomy. Elevated LA pressure transmits backward to pulmonary veins and pulmonary capillaries, causing exertional dyspnea and eventually pulmonary edema. Chronic LA dilation causes AF; blood stasis in the dilated LA (especially the left atrial appendage) promotes thrombus formation.

3. ECG Findings

- **Left atrial enlargement (LAE):** P mitrale (broad, notched P wave >0.12s in lead II); biphasic P in V1 with prominent terminal negative deflection >0.04s and >1 mm deep
- **Atrial fibrillation:** Very common once LA significantly enlarged; irregularly irregular rhythm
- **Right ventricular hypertrophy (RVH):** Tall R wave in V1 (R/S >1), RAD, RAE if pulmonary hypertension develops
- **Right axis deviation:** Indication of pulmonary hypertension
- **RV strain pattern:** ST depression and T wave inversion in V1-V4 if RV pressure overload
- **Normal ECG:** May be normal in mild MS before AF develops

4. Required ECG Features

ECG CLUES

- LAE is the earliest and most consistent ECG finding in MS
- AF + LAE in a patient with no other obvious cause = consider MS
- RVH/RAD in MS indicates advanced disease with pulmonary hypertension
- MS is the classic valvular lesion where AF = high stroke risk regardless of CHA_2DS_2-VASc

5. Diagnostic Criteria

- Echocardiography: Valve area by planimetry or pressure half-time (PHT); mean transmitral gradient; assessment of valve morphology (commissural fusion, leaflet mobility, calcification, subvalvular fusion)
- Severe MS: Valve area <1.0 cm^2 (or <0.6 cm^2/m^2), mean gradient >5-10 mmHg
- Wilkins score (echo): Assesses suitability for balloon valvotomy (leaflet mobility, thickening, calcification, subvalvular fusion; score <=8 favorable)
- TEE: Exclude LA thrombus before valvotomy or cardioversion; assess LAA for thrombus
- Exercise testing: If symptoms discordant with resting hemodynamics

6. Differential Diagnosis

Condition	Differentiating Features
Left atrial myxoma	Mobile mass on echo; tumor plop sound; constitutional symptoms; no commissural fusion
Cor triatriatum	Fibromuscular membrane dividing LA; no valve abnormality; echo differentiates
Pulmonary vein stenosis	Post-ablation; echo shows normal MV; CT/MRI of pulmonary veins
Other causes of pulmonary hypertension	Normal mitral valve; other etiology for PAH identified

7. Symptoms

- **Exertional dyspnea:** Earliest symptom; due to elevated LA pressure during exertion
- **Orthopnea, PND:** Pulmonary venous congestion when supine

- **Hemoptysis:** From ruptured bronchial veins (pink frothy sputum from pulmonary edema or frank blood from bronchial varices)
- **Palpitations:** AF (once LA enlarged)
- **Fatigue, reduced exercise tolerance:** Reduced cardiac output (LV underfilled due to inflow obstruction)
- **Right heart failure symptoms:** Edema, ascites, hepatomegaly if pulmonary hypertension develops

8. Risk Factors

- Rheumatic fever history (most common worldwide)
- Age (degenerative calcific MS in elderly women in developed countries)
- Female predominance (2:1)
- Developing country residence (rheumatic MS)
- Genetic susceptibility to rheumatic fever
- Recurrent streptococcal infections

9. Immediate Actions

ACUTE DECOMPENSATION

1. AF with RVR: Rate control (beta-blocker, digoxin, non-DHP CCB); urgent cardioversion if hemodynamically unstable but need TEE first to exclude LA thrombus
2. Pulmonary edema: Diuretics, nitrates, sit upright; avoid afterload reduction (stenosis is pre-load dependent, not afterload dependent)
3. Anticoagulation mandatory if AF (warfarin or DOACs)

10. Treatment Overview

Medical: Diuretics for congestion; beta-blockers or heart rate-limiting CCBs to prolong diastolic filling time. Anticoagulation: Mandatory for AF; consider if prior embolism, LA thrombus, or spontaneous echo contrast.

Percutaneous balloon mitral valvotomy (BMV): First-line for symptomatic severe MS with favorable valve morphology (Wilkins score <=8, no significant MR, no LA thrombus); avoids prosthetic valve.

Surgical: Commissurotomy (if valve can be preserved but BMV not feasible) or mitral valve replacement (if heavily calcified, significant MR, or failed BMV).

Bioprosthetic valve preferred for women of childbearing age (warfarin teratogenicity).

AF management: Rate control or rhythm control; anticoagulation mandatory regardless of CHA_2DS_2-VASc.

Pregnancy: Tolerated well if mild-moderate; severe MS may decompensate (physiologic volume expansion increases transmitral gradient); BMV during pregnancy if necessary.

11. Follow-up Questions

- History of rheumatic fever? Age at onset?
- Symptom progression? Exercise tolerance?
- Any AF episodes?
- Any thromboembolic events (stroke, TIA, peripheral)?
- Current anticoagulation status?
- Echo results? Valve area? Mean gradient? Pulmonary pressure?
- Wilkins score if valvotomy considered?
- Pregnancy considerations?

12. References

1. Otto CM, et al. 2020 ACC/AHA Guideline for the Management of Patients With Valvular Heart Disease. *Circulation*. 2021;143(5):e72-e227.
2. Nishimura RA, et al. 2014 AHA/ACC Guideline for the Management of Patients With Valvular Heart Disease. *J Am Coll Cardiol*. 2014;63(22):e57-e185.

Mitral Valve Prolapse

Category: Valvular Heart Disease **Priority: MODERATE** **ICD-10:** I34.1

1. Definition

Mitral valve prolapse (MVP) is a valvular abnormality characterized by systolic displacement of one or both mitral valve leaflets into the left atrium >=2 mm beyond the annular plane on echocardiography in the parasternal long-axis view, typically with leaflet thickening >5 mm (classic MVP) or without thickening (non-classic). It affects 2-3% of the population, with a slight female predominance. Most cases are benign, but complications include significant mitral regurgitation, infective endocarditis, arrhythmias (ventricular tachycardia, sudden death), and cerebral ischemic events.

2. Mechanism

Myxomatous degeneration of the mitral valve leaflets and chordae due to disruption of the structural extracellular matrix (increased proteoglycans, fragmentation of collagen and elastin). This causes leaflet thickening, redundancy, elongation, and increased area. Genetic basis: autosomal dominant with variable penetrance; associated with Marfan syndrome (FBN1 mutations), Loeys-Dietz, Ehlers-Danlos. In MVP, the leaflets billow back into the LA during systole, which may progress to MR if chordal elongation or rupture occurs. The arrhythmic substrate arises from mechanical stretch of the papillary muscles and LV myocardium, creating foci for ventricular arrhythmias.

3. ECG Findings

- **ECG usually normal:** In uncomplicated MVP
- **Inferolateral T wave inversions:** Nonspecific but common; may reflect papillary muscle tension
- Nonspecific, often in lateral leads
- **Arrhythmias:** PVCs (especially from LV outflow tract/fascicular origin), NSVT; rare sustained VT
- **Prolonged QT interval:** May be present
- **BBB:** May occur with severe MR and chamber enlargement

4. Required ECG Features

ECG IN MVP

No diagnostic ECG criteria. ECG may show nonspecific repolarization changes (inferolateral ST-T changes) and arrhythmias. Complex ventricular arrhythmias in a patient with known MVP warrant risk stratification. The primary diagnostic tool is echocardiography.

5. Diagnostic Criteria

- Echocardiography: Systolic leaflet displacement >=2 mm into LA in parasternal long-axis; leaflet thickness >5 mm = classic MVP; <5 mm = non-classic
- Assess MR severity, leaflet morphology (flail, prolapsing segments), LA size, LV function
- Risk stratification: Severe MR, flail leaflet, LVEF <60%, LA >40 mm, age >50, male sex, bileaflet prolapse, complex ventricular arrhythmias

6. Differential Diagnosis

Condition	Differentiating Features
Physiologic billowing	Mild displacement; no leaflet thickening; no MR; no arrhythmias
Rheumatic MR	Thickened restricted leaflets; commissural fusion; no prolapse
Functional MR	Normal leaflets; annular dilation from LV remodeling; no prolapse
Papillary muscle dysfunction	Ischemic; regional wall motion abnormality; different etiology

7. Symptoms

- **Asymptomatic:** Most common
- **Palpitations:** PVCs from mechanical irritation
- **Atypical chest pain:** Sharp, non-exertional; mechanism unclear
- **Symptoms of MR:** Dyspnea, fatigue if regurgitation severe
- **Syncope/rare sudden death:** In high-risk subset (bileaflet prolapse, severe MR, complex VT)

8. Risk Factors

- Genetic/family history; connective tissue disorders
- Female sex (slight predominance)

- Younger age at diagnosis (higher progression risk)
- Bileaflet prolapse (higher complication risk)
- Leaflet thickness >5 mm

9. Immediate Actions

> **MANAGEMENT**
>
> 1. Complex arrhythmias: Cardiology referral; risk stratification
> 2. Severe acute MR from chordal rupture: Afterload reduction, emergent surgery

10. Treatment Overview

Asymptomatic: Reassurance; serial echo every 3-5 years.

Severe MR: Mitral valve repair (preferred over replacement) when symptomatic or LV dysfunction develops.

Arrhythmias: Beta-blockers for symptomatic PVCs; ablation if refractory. ICD only for sustained VT/VF survivors.

Endocarditis prophylaxis: Only if prior endocarditis, prosthetic valve, or unrepaired cyanotic CHD.

11. Follow-up Questions

- Any symptoms? Palpitations? Atypical chest pain?
- Any syncope?
- Severity of MR on echo?
- Leaflet morphology (bileaflet? flail? thickness?)
- Arrhythmia history?
- Family history of MVP, sudden death, connective tissue disorder?

12. References

1. Otto CM, et al. 2020 ACC/AHA Guideline for the Management of Patients With Valvular Heart Disease. *Circulation*. 2021;143(5):e72-e227.

Obesity (Cardiac Manifestations)

Category: Metabolic Cardiovascular Disease **Priority: MODERATE** **ICD-10:** E66.9

1. Definition

Obesity (BMI >=30 kg/m^2) is a major modifiable risk factor for cardiovascular disease and an independent contributor to heart failure (especially HFpEF), coronary artery disease, atrial fibrillation, sudden cardiac death, and thromboembolic disease. The obesity paradox (improved survival in overweight/obese patients with established HF compared to normal weight) is observed but does not negate the adverse cardiovascular effects of excess adiposity. Central obesity (waist circumference >102 cm men, >88 cm women) carries greater cardiovascular risk than peripheral obesity.

2. Mechanism

Multiple mechanisms: (1) Hemodynamic: Increased blood volume and cardiac output to perfuse adipose tissue; LV dilation and eccentric hypertrophy; eventually systolic and diastolic dysfunction. (2) Metabolic: Insulin resistance, type 2 diabetes, dyslipidemia, systemic inflammation (elevated CRP, IL-6, TNF-alpha from adipose tissue). (3) Neurohormonal: Activation of RAAS and sympathetic nervous system. (4) Structural: Pericardial fat deposition (paracrine pro-inflammatory effects on adjacent myocardium), epicardial adipose tissue thickness correlates with AF risk. (5) OSA: Highly prevalent in obesity; causes nocturnal hypoxia, sympathetic activation, pulmonary hypertension. (6) Thrombotic: Prothrombotic state from elevated fibrinogen, PAI-1, platelet activation. (7) Adipokine dysregulation: Leptin resistance, reduced adiponectin (protective).

3. ECG Findings

- **Low QRS voltage:** Increased distance between heart and electrodes due to adipose tissue; especially limb leads
- **Leftward shift of QRS axis:** Due to diaphragmatic elevation and horizontal heart position
- **Flat T waves in inferior leads:** Nonspecific; may relate to diaphragmatic position
- **LA enlargement:** From elevated filling pressures and diastolic dysfunction
- **LVH:** Eccentric hypertrophy pattern
- **AF:** Increased risk (5% increase per BMI unit); pericardial fat-mediated
- **QT prolongation:** Risk of Torsades de Pointes
- **OSA-related:** Bradycardia during apneas, tachycardia on arousal, arrhythmias

4. Required ECG Features

> **ECG IN OBESITY**
>
> No specific diagnostic ECG criteria. Low voltage in limb leads is common but nonspecific. Clinical correlation required - do not attribute all cardiac symptoms to obesity. ECG changes should prompt appropriate cardiac evaluation rather than dismissal due to obesity.

5. Diagnostic Criteria

- BMI >=30 kg/m² (overweight 25-29.9; class I obesity 30-34.9; class II 35-39.9; class III/morbid >=40)
- Waist circumference: >102 cm men, >88 cm women (metabolic syndrome component)
- Obesity cardiomyopathy: HF (especially HFpEF) in obese patient without other clear cause; echo shows LV dilation, hypertrophy, diastolic dysfunction
- BNP/NT-proBNP may be paradoxically lower in obese patients despite HF (natriuretic peptide clearance receptors in adipose tissue)

6. Differential Diagnosis

Condition	Differentiating Features
Other causes of HFpEF	Hypertension, diabetes, ischemia may coexist; comprehensive workup needed
Sleep apnea alone	OSA can cause PH and R-sided HF without direct obesity cardiomyopathy
Hypothyroidism	Can cause both obesity and HF; check TSH

7. Symptoms

- **Dyspnea on exertion:** From HFpEF or obesity itself (mechanical restriction)
- **Orthopnea:** From HF, OSA, or both
- **Excessive daytime somnolence:** OSA
- **Peripheral edema:** Right HF, venous insufficiency, or both
- **Palpitations:** AF, PACs from atrial stretch
- **Chest discomfort:** CAD risk elevated; also non-cardiac causes common

8. Risk Factors

- BMI >=30; central obesity worse than peripheral
- Weight gain trajectory
- Obesity duration
- Metabolic syndrome components (hypertension, diabetes, dyslipidemia)
- OSA
- Physical inactivity
- Poor diet quality
- Genetic/family history

9. Immediate Actions

MANAGEMENT

1. Assess for acute HF if dyspnea: echo, BNP, CXR
2. Screen for OSA if symptoms suggest
3. Evaluate for CAD if chest pain
4. Do not dismiss symptoms as "just obesity" without appropriate evaluation

10. Treatment Overview

Weight loss: 5-10% weight loss produces clinically meaningful improvement in BP, lipids, glycemic control, and HF symptoms. GLP-1 receptor agonists (semaglutide, liraglutide) and SGLT2 inhibitors produce significant weight loss with proven cardiovascular benefit. Bariatric surgery (Roux-en-Y, sleeve gastrectomy) produces durable weight loss and reduces mortality in severe obesity.

HF management: SGLT2 inhibitors now approved for HFpEF and reduce weight concurrently. Diuretics for congestion. Exercise training improves functional capacity independent of weight loss.

OSA treatment: CPAP reduces nocturnal desaturation, improves sleep quality, may reduce BP and arrhythmia burden.

AF management: Weight reduction (LEGACY study showed 10% weight loss reduces AF burden and recurrence post-ablation). Rate/rhythm control; anticoagulation per CHA_2DS_2-VASc.

Do not withhold cardiac therapies due to obesity: ICD, CRT, transplantation eligibility should be assessed using adjusted criteria.

11. Follow-up Questions

- BMI? Weight trajectory? Duration of obesity?
- Central obesity (waist circumference)?
- Comorbidities: diabetes, hypertension, dyslipidemia, OSA?
- Current weight loss efforts? Prior bariatric surgery?
- Symptoms: dyspnea, orthopnea, edema, palpitations, snoring?
- Exercise tolerance? Functional limitations?
- Current medications including GLP-1 RA or SGLT2i?

12. References

1. Powell-Wiley TM, et al. Obesity and Cardiovascular Disease: A Scientific Statement From the AHA. *Circulation*. 2021;143(21):e984-e1010.
2. Pathak RK, et al. Long-term effect of goal-directed weight management in an atrial fibrillation cohort: A long-term follow-up study (LEGACY). *J Am Coll Cardiol*. 2015;65(20):2159-2169.
3. Kosiborod M, et al. Dapagliflozin in Patients with Cardiometabolic Risk Factors Hospitalised with Heart Failure. *Eur J Heart Fail*. 2023;25(5):627-637.

Peripheral Artery Disease (PAD)

Category: Vascular Disease **Priority: MODERATE** **ICD-10:** I73.9

1. Definition

Peripheral artery disease (PAD) is atherosclerotic narrowing of non-coronary, non-cerebral arteries, most commonly affecting the lower extremities (femoropopliteal, tibial, aortoiliac). It affects approximately 230 million adults worldwide, but remains underdiagnosed because many patients are asymptomatic or have atypical symptoms. PAD confers equivalent cardiovascular risk to coronary artery disease - a PAD patient has the same 5-year risk of MI, stroke, or cardiovascular death as a patient with CAD. Ankle-brachial index (ABI) <0.90 is diagnostic.

2. Mechanism

Atherosclerosis of the lower extremity arteries, sharing the same risk factors and pathophysiology as coronary atherosclerosis. Plaque formation causes progressive luminal narrowing, reducing distal perfusion. Claudication occurs when exercise-induced metabolic demand exceeds blood supply. Critical limb ischemia (CLI) develops when resting perfusion is insufficient to maintain tissue viability. In diabetes, additional microvascular disease and neuropathy contribute to tissue loss. PAD is a systemic marker of atherosclerosis - patients with PAD have high burden of concomitant CAD and cerebrovascular disease.

3. ECG Findings

- **ECG not diagnostic for PAD:** Used to screen for concomitant CAD
- **Prior MI changes:** Common due to high CAD prevalence in PAD patients
- **Ischemic changes:** May be present from concurrent CAD
- **Arrhythmias:** AF (increases stroke and embolic risk to lower limbs)

4. Required ECG Features

ECG ROLE

ECG screens for coexisting CAD (high prevalence in PAD patients). A normal ECG does not exclude CAD. Resting ECG should be obtained in all PAD patients before vascular surgery.

5. Diagnostic Criteria

- ABI <0.90 (sensitivity 95%, specificity 100% for >50% stenosis)

- ABI >1.40 suggests noncompressible vessels (diabetes, elderly) - use toe-brachial index instead
- Segmental pressure measurement, pulse volume recording
- Duplex ultrasound: Anatomic localization and stenosis severity
- CTA/MRA: Anatomic mapping for intervention
- Contrast angiography: Gold standard; pre-intervention

6. Differential Diagnosis

Condition	Differentiating Features
Neurogenic claudication (spinal stenosis)	Bilateral; worse with standing/walking downhill; improves with sitting or leaning forward; normal pulses
Venous claudication	Heavy, aching pain after prolonged standing; improves with leg elevation; venous disease history
Musculoskeletal pain	Localized, reproducible tenderness; worsens with specific movements; not exertion-related
Popliteal artery entrapment	Young, athletic; symptoms with specific foot positions; MRI/CT angiography with foot positions
Cystic adventitial disease	Young, no risk factors; gelatinous cyst in arterial wall; MRI characteristic

7. Symptoms

- **Intermittent claudication:** Exercise-induced aching, cramping, fatigue in calf, thigh, or buttock; relieved by rest within 10 minutes
- **Atypical leg pain:** Present in more patients than classic claudication
- **Critical limb ischemia:** Rest pain (worse with leg elevation, improves with dependency), non-healing wounds, gangrene
- **Acute limb ischemia (6 Ps):** Pain, Pallor, Pulselessness, Paralysis, Paresthesia, Poikilothermia (cold)
- **Many asymptomatic:** Up to 40-50% of patients with ABI <0.90 have no symptoms

8. Risk Factors

- Smoking (strongest modifiable risk factor; 2-3x increased risk)
- Diabetes (2-4x increased risk; more distal disease)
- Age >65 (or >50 with risk factors)
- Hypertension

- Dyslipidemia
- Homocysteinemia
- Chronic kidney disease
- Family history of PAD, CAD, or stroke

9. Immediate Actions

ACUTE LIMB ISCHEMIA

1. Immediate anticoagulation (heparin) to prevent thrombus propagation
2. Emergent vascular surgery consultation
3. Imaging to define level of occlusion
4. Revascularization within 6 hours (catheter-directed thrombolysis, surgical embolectomy, or bypass)
5. Monitor for compartment syndrome (fasciotomy if needed)

10. Treatment Overview

Risk factor modification: Antiplatelet (aspirin 75-100 mg or clopidogrel 75 mg); high-intensity statin; smoking cessation (most effective intervention); diabetes control; BP control; supervised exercise therapy (SET) - 30-60 min sessions, 3x/week for 12 weeks improves walking distance by 50-200%.

Claudication: Cilostazol (PDE3 inhibitor) 100 mg BID (avoid in HF); revascularization if lifestyle-limiting despite medical therapy and SET.

Critical limb ischemia: Revascularization (endovascular first-line for femoropopliteal disease; bypass for long-segment or failed endovascular). Amputation only if non-salvageable limb.

Vigilant foot care: Especially in diabetic patients; regular podiatry; protective footwear; prompt treatment of wounds.

11. Follow-up Questions

- Walking distance before pain onset? Rest time for relief?
- Any rest pain? Non-healing wounds? Gangrene?
- Smoking history? (most important modifiable risk)
- Diabetes, hypertension, dyslipidemia control?
- ABI result?

- Current medications: antiplatelet, statin?
- Prior revascularization procedures?

12. References

1. Aboyans V, et al. 2017 ESC Guidelines on the Diagnosis and Treatment of Peripheral Arterial Diseases. *Eur Heart J*. 2018;39(9):763-816.
2. Gerhard-Herman MD, et al. 2016 AHA/ACC Guideline on the Management of Patients With Peripheral Artery Disease. *Circulation*. 2017;135(12):e726-e779.

Pulmonary Stenosis

Category: Congenital/Valvular **Priority: MODERATE** **ICD-10:** Q22.1

1. Definition

Pulmonary stenosis (PS) is narrowing of the pulmonary valve or right ventricular outflow tract (RVOT) causing obstruction of blood flow from the right ventricle to the pulmonary artery. It is one of the most common congenital heart defects (8-10% of all CHD), usually isolated but can be associated with other defects (most notably tetralogy of Fallot). Valvular PS accounts for 80% of cases. Severity is graded by peak instantaneous gradient: mild (<40 mmHg), moderate (40-60 mmHg), severe (>60 mmHg or >50 mmHg with symptoms).

2. Mechanism

Congenital valvular PS results from abnormal fusion or thickening of the pulmonary valve leaflets, creating a narrowed orifice (dome-shaped valve in 80%). The RV responds to pressure overload with concentric hypertrophy. Post-stenotic dilation of the main pulmonary artery occurs from the high-velocity jet. Subvalvular PS (infundibular stenosis) can be fixed (muscle bundle) or dynamic (hypertrophic response to valvular stenosis). Supravalvular PS is rare, associated with Williams syndrome (elastin gene mutation). PS can also occur as part of more complex congenital heart disease (TOF, DORV) or be acquired (carcinoid syndrome, rheumatic fever).

3. ECG Findings

- **RVH:** Tall R wave in V1 (R/S ratio >1 in adults), RAD (>90 degrees), RV strain pattern (ST depression, T wave inversion V1-V3)
- **RAE:** Tall peaked P waves in II (>2.5 mm) - P pulmonale
- **RBBB:** May develop as part of RV volume overload if associated ASD
- **Normal ECG:** Mild PS may have normal ECG

4. Required ECG Features

ECG CLUES

- RVH + RAD + RAE = significant PS
- RV strain pattern indicates severe PS with RV pressure overload
- Degree of RVH correlates with severity of stenosis
- PS with normal ECG suggests mild, hemodynamically insignificant stenosis

5. Diagnostic Criteria

- Echocardiography: Defines valve anatomy, peak gradient, RV pressure, RV hypertrophy, RV function; associated lesions
- Cardiac MRI/CT: Supplementary for complex anatomy, PA branch stenosis, RV quantification
- Cardiac catheterization: Direct gradient measurement; balloon valvuloplasty
- Exercise testing: Assess functional capacity and exercise-induced symptoms

6. Differential Diagnosis

Condition	Differentiating Features
TOF with PS	VSD, overriding aorta, RVH; more complex hemodynamics
Peripheral PA stenosis	Normal valve; stenosis in PA branches; different murmur location
RV cardiomyopathy	No valvular obstruction; different etiology
Carcinoid syndrome	Acquired; thickened immobile leaflets; hepatic metastases; elevated 5-HIAA

7. Symptoms

- **Asymptomatic:** Mild to moderate PS often discovered incidentally
- **Exertional dyspnea:** Reduced cardiac output from RV obstruction
- **Chest pain:** RV angina from hypertrophy
- **Syncope:** With severe PS, especially exertion
- **Peripheral edema:** Right heart failure if severe/long-standing

8. Risk Factors

- Congenital (most common)
- Noonan syndrome (dysplastic pulmonary valve)
- Carcinoid syndrome (acquired)
- Rheumatic heart disease (rare)

9. Immediate Actions

> **MANAGEMENT**
>
> 1. RV failure: Diuretics
> 2. Balloon valvuloplasty for severe/symptomatic PS
> 3. Surgery if balloon valvuloplasty fails or dysplastic valve

10. Treatment Overview

Mild: No intervention; periodic monitoring.

Severe/symptomatic: Balloon valvuloplasty (first-line); very effective with low recurrence. Surgical valvotomy if dysplastic valve or associated lesions.

Re-stenosis: Rare after successful valvuloplasty.

11. Follow-up Questions

- Known since childhood? Interventions?
- Current symptoms? Exercise tolerance?
- Last echo: gradient? RV function?
- Any associated CHD?

12. References

1. Warnes CA, et al. ACC/AHA 2008 Guidelines for the Management of Adults with Congenital Heart Disease. *Circulation*. 2008;118(23):e714-e833.

Rheumatic Heart Disease

Category: Valvular/Inflammatory **Priority: MODERATE** **ICD-10:** I09.9

1. Definition

Rheumatic heart disease (RHD) is chronic valvular damage resulting from one or more episodes of acute rheumatic fever (ARF), an autoimmune inflammatory disease triggered by group A Streptococcus (GAS) pharyngeal infection. RHD remains a major cause of cardiovascular morbidity and mortality in low- and middle-income countries, affecting approximately 40 million people worldwide. Mitral valve is most commonly affected (involved in >90% of cases), followed by aortic valve. Tricuspid and pulmonary valves are less frequently involved.

2. Mechanism

Molecular mimicry between GAS antigens (M protein, N-acetyl-glucosamine) and human cardiac proteins (myosin, laminin, vimentin) causes cross-reactive immune-mediated damage. The valvular endothelium is activated, allowing T-cell infiltration and autoimmune attack on valvular tissue. Acute phase: valvulitis with leaflet edema, erosion, and vegetation formation. Chronic phase: leaflet fibrosis, thickening, calcification, commissural fusion, chordal fusion, and shortening. Progressive valvular stenosis and/or regurgitation develops over years to decades after the initial insult.

3. ECG Findings

- **MS pattern:** LAE (P mitrale, biphasic P V1), AF, RVH if pulmonary hypertension
- **AR pattern:** LVH with volume overload pattern
- **MR pattern:** LAE, LVH, AF
- **Combined lesions:** Complex ECG patterns reflecting multiple valve involvement
- **Acute rheumatic fever:** First-degree AV block (PR prolongation), myocarditis changes, tachycardia
- **Pancarditis:** Diffuse ST-T changes if active carditis

4. Required ECG Features

ECG CLUES

- LAE + AF + RVH in patient from endemic region = severe MS from RHD
- PR prolongation during acute illness suggests acute rheumatic carditis
- Coarse AF in MS indicates severe LA enlargement
- Multiple valve involvement patterns on ECG suggest RHD over degenerative valve disease

5. Diagnostic Criteria

- Modified Jones criteria (ARF diagnosis): Requires evidence of preceding GAS infection plus 2 major OR 1 major + 2 minor criteria. Major: carditis, polyarthritis, chorea, erythema marginatum, subcutaneous nodules. Minor: arthralgia, fever, elevated ESR/CRP, prolonged PR interval.
- Echocardiography: Valve morphology (thickened, restricted leaflets, commissural fusion); severity of stenosis/regurgitation
- WHF criteria: Echo-based criteria for RHD diagnosis in endemic settings
- Streptococcal serology: ASO titers, anti-DNase B

6. Differential Diagnosis

Condition	Differentiating Features
Degenerative calcific MS	Elderly, developed country; calcified annulus; preserved commissures; no RHD history
Myxomatous MVP	Redundant leaflets; prolapse; no commissural fusion; different pathology
Congenital MS	Childhood presentation; parachute mitral valve; no ARF history
Endocarditis	Vegetations, fever, bacteremia; acute presentation

7. Symptoms

- **Exertional dyspnea:** Most common; from MS, MR, or AR
- **Palpitations:** AF from LA enlargement
- **Peripheral edema, ascites:** Right heart failure from pulmonary hypertension
- **Systemic embolism:** From AF and LA thrombus
- **Acute rheumatic fever symptoms:** Arthritis, fever, carditis, chorea, rash

8. Risk Factors

- Prior acute rheumatic fever (single or recurrent episodes)
- Untreated or undertreated streptococcal pharyngitis
- Overcrowded living conditions, poverty (endemic regions)
- Age 5-15 years (ARF peak incidence)
- Family history (genetic susceptibility)
- Residence in endemic area (sub-Saharan Africa, South Asia, Pacific Islands, Indigenous Australians)

9. Immediate Actions

> **ACUTE RHEUMATIC FEVER**
>
> 1. Eradicate GAS: Benzathine penicillin G IM single dose or oral penicillin V 10 days
> 2. Anti-inflammatory: Aspirin (first-line for arthritis/carditis) or NSAIDs; corticosteroids for severe carditis
> 3. Bed rest during acute phase
> 4. Secondary prophylaxis: Benzathine penicillin IM every 3-4 weeks until age 21 or 10 years after last ARF

10. Treatment Overview

Secondary prophylaxis: Benzathine penicillin G 1.2 million units IM every 3-4 weeks until age 21 (minimum 10 years after last ARF). Erythromycin if penicillin-allergic.

Valvular management: Percutaneous balloon mitral valvotomy for pure MS with favorable valve; surgical repair/replacement for MR or complex disease; aortic valve replacement for severe AR/AS.

AF management: Rate/rhythm control; anticoagulation (vitamin K antagonists preferred over DOACs in moderate-severe MS due to limited data).

Heart failure: Standard therapy; diuretics for congestion.

11. Follow-up Questions

- History of acute rheumatic fever? How many episodes? Age at onset?
- Secondary prophylaxis adherence?

- Which valves affected? Severity?
- Symptoms? Exercise tolerance?
- Any AF? Anticoagulation?
- Endemic area residence?

12. References

1. Watkins DA, et al. Global, Regional, and National Burden of Rheumatic Heart Disease, 1990-2015. *N Engl J Med*. 2017;377(8):713-722.
2. Gewitz MH, et al. Revision of the Jones Criteria for the Diagnosis of Acute Rheumatic Fever. *Circulation*. 2015;131(20):1806-1818.

Ventricular Septal Defect (VSD)

Category: Congenital Heart Disease **Priority: MODERATE** **ICD-10:** Q21.0

1. Definition

Ventricular septal defect (VSD) is a hole in the interventricular septum allowing communication between the left and right ventricles. It is the most common congenital heart defect (30-40% of all CHD, excluding bicuspid aortic valve). VSDs are classified by location: perimembranous (70-80%), muscular (5-20%), supracristal/outlet/subarterial (5-10%, associated with aortic regurgitation), and inlet (AV canal type). Small restrictive VSDs may be hemodynamically insignificant; large non-restrictive VSDs cause significant left-to-right shunting leading to pulmonary overcirculation, heart failure, and Eisenmenger syndrome if unrepaired.

2. Mechanism

Failure of complete fusion of the interventricular septum during embryological development. Perimembranous VSD: defect adjacent to the membranous septum near the tricuspid and aortic valves. Muscular VSD: defect within the muscular septum; may be single or multiple (Swiss cheese septum). Supracristal VSD: defect in the outlet septum below the aortic and pulmonary valves; the aortic valve may prolapse into the defect causing AR. Hemodynamics: Left-to-right shunt occurs due to higher LV pressure; magnitude depends on defect size and PVR. Large defects cause pulmonary overcirculation, elevated PA pressures, LV volume overload, and eventually pulmonary vascular remodeling (Eisenmenger if unrepaired).

3. ECG Findings

- **Small VSD:** Normal ECG or may show mild LVH
- **Moderate VSD:** LVH (tall R waves V5-V6, deep S V1), LAE (biphasic P V1), deep Q waves in lateral leads (volume overload pattern)
- **Large VSD:** Biventricular hypertrophy (tall R/S in V1 and tall R in V5-V6), biventricular strain pattern, LAE, RAE
- **Eisenmenger VSD:** RVH (tall R V1, RAD), RAE (P pulmonale) - shunt reversal
- **Supracristal VSD:** May have LAE if AR develops from aortic valve prolapse
- **Post-repair:** RBBB common after surgical closure; complete heart block if surgical injury to conduction system

4. Required ECG Features

ECG CLUES

- LVH in a child without hypertension = significant L-to-R shunt (VSD or PDA)
- BVH in a child = large VSD
- RVH + RAD in a cyanotic patient with VSD = Eisenmenger syndrome
- Post-surgical RBBB = prior VSD repair
- Complete heart block after VSD repair indicates surgical injury to conduction system

5. Diagnostic Criteria

- Echocardiography: Defines VSD location, size, direction and velocity of shunt, Qp/Qs ratio, chamber sizes, PA pressure, associated lesions
- Significant VSD: Qp/Qs >1.5; evidence of LV volume overload
- Non-restrictive VSD: Equal RV and LV pressures; >75% of aortic root diameter
- Restrictive VSD: High-velocity jet across small defect with normal RV pressure
- CMR: Qp/Qs quantification if echo inconclusive
- Cardiac catheterization: Hemodynamics, PVR assessment, intervention

6. Differential Diagnosis

Condition	Differentiating Features
PDA	Continuous murmur at left upper sternal border; diastolic flow in PA on echo
ASD	Fixed split S2; RBBB with RAD; no LV volume overload
AV septal defect	Primum ASD + inlet VSD; common AV valve; LAD on ECG; Down syndrome
DORV	Both great vessels from RV; more complex; echo defines

7. Symptoms

- **Small VSD:** Usually asymptomatic; harsh holosystolic murmur
- **Moderate VSD:** Mild dyspnea, poor growth (infants), recurrent respiratory infections
- **Large VSD (infant):** Tachypnea, feeding difficulties, failure to thrive, diaphoresis, recurrent respiratory infections, heart failure
- **Eisenmenger VSD:** Cyanosis, clubbing, dyspnea, hemoptysis, syncope
- **Adults with unrepaired VSD:** Dyspnea, palpitations, exercise intolerance

8. Risk Factors

- Genetic: Isolated VSD multifactorial; associated with trisomy 21, trisomy 18, trisomy 13, DiGeorge syndrome
- Maternal: Diabetes, phenylketonuria, alcohol, rubella
- Family history of CHD
- Most VSDs are sporadic

9. Immediate Actions

MANAGEMENT

1. Infant with large VSD and HF: Diuretics, afterload reduction (ACE-I), calorie-dense feeds, surgical referral
2. Endocarditis: If suspected, blood cultures and antibiotics
3. Eisenmenger crisis: Avoid IV fluids (worsens shunt), supplemental O2, pulmonary vasodilators

10. Treatment Overview

Natural history: 30-50% of small VSDs close spontaneously (mostly muscular); perimembranous less likely; supracristal never close. Large defects require intervention.

Indications for closure: Qp/Qs >1.5 with LV volume overload or PAH, recurrent endocarditis, progressive AR (supracristal), aortic valve prolapse.

Surgical: Patch closure (pericardial or synthetic); gold standard; low mortality. May require pulmonary artery banding in infancy if definitive repair not feasible.

Device closure: Percutaneous device for selected muscular VSDs; less commonly for perimembranous (risk of conduction block, AR).

Eisenmenger VSD: Closure contraindicated (fixed PAH); PAH-targeted therapies; heart-lung transplant for end-stage.

Post-closure: Most patients do well; RBBB common after surgery; residual VSD rare; endocarditis prophylaxis only if residual VSD or prosthetic material within 6 months.

11. Follow-up Questions

- VSD type? Size? Location?
- Qp/Qs ratio? PA pressure?
- Prior closure? When? Surgical or device?
- Current symptoms? Exercise tolerance?
- Any residual shunt? AR? (especially supracristal)
- Any arrhythmias?
- Need for endocarditis prophylaxis?

12. References

1. Penny DJ, Vick GW 3rd. Ventricular Septal Defect. *Lancet*. 2011;377(9771):1103-1112.
2. Warnes CA, et al. ACC/AHA 2008 Guidelines for the Management of Adults with Congenital Heart Disease. *Circulation*. 2008;118(23):e714-e833.

6. References

This knowledge base references major international cardiovascular guidelines and consensus statements. Below is a consolidated list of key references organized by clinical domain.

General Cardiology Guidelines

1. 2022 AHA/ACC/HFSA Guideline for the Management of Heart Failure. *Circulation*. 2022;145(18):e895-e1032.
2. 2021 ESC Guidelines for the diagnosis and treatment of acute and chronic heart failure. *Eur Heart J*. 2021;42(36):3599-3726.
3. 2019 ESC Guidelines for the diagnosis and management of chronic coronary syndromes. *Eur Heart J*. 2020;41(3):407-477.
4. 2020 ACC/AHA Guideline for the Management of Patients With Valvular Heart Disease. *Circulation*. 2021;143(5):e72-e227.
5. 2021 ESC/EACTS Guidelines for the management of valvular heart disease. *Eur Heart J*. 2022;43(7):561-632.

Arrhythmia Guidelines

1. 2020 ESC Guidelines for the diagnosis and management of atrial fibrillation. *Eur Heart J*. 2021;42(5):373-498.
2. 2019 AHA/ACC/HRS Focused Update of the 2014 Guideline for the Management of Patients With Atrial Fibrillation. *Circulation*. 2019;140(2):e125-e151.
3. 2017 AHA/ACC/HRS Guideline for Management of Patients With Ventricular Arrhythmias. *Circulation*. 2018;138(13):e210-e271.
4. 2015 ESC Guidelines for the management of patients with ventricular arrhythmias. *Eur Heart J*. 2015;36(41):2793-2867.
5. 2015 ACC/AHA/HRS Guideline for the Management of Adult Patients With Supraventricular Tachycardia. *Circulation*. 2016;133(14):e506-e574.

Congenital Heart Disease Guidelines

1. 2018 AHA/ACC Guideline for the Management of Adults With Congenital Heart Disease. *Circulation*. 2019;139(14):e698-e800.
2. 2020 ESC Guidelines for the management of adult congenital heart disease. *Eur Heart J*. 2021;42(6):563-645.

Cardiomyopathy Guidelines

1. 2020 AHA/ACC Guideline for the Diagnosis and Treatment of Patients With Hypertrophic Cardiomyopathy. *Circulation*. 2020;142(25):e558-e631.
2. 2017 AHA/ACC/HRS Guideline for Management of Patients With Ventricular Arrhythmias. *Circulation*. 2018;138(13):e210-e271.

Acute Cardiovascular Care

1. 2018 ESC/EACTS Guidelines on myocardial revascularization. *Eur Heart J*. 2019;40(2):87-165.
2. 2014 AHA/ACC Guideline for the Management of Patients With NSTE-ACS. *Circulation*. 2014;130(25):e344-e426.
3. 2013 ACCF/AHA Guideline for the Management of STEMI. *Circulation*. 2013;127(4):e362-e425.
4. 2019 ESC Guidelines for acute pulmonary embolism. *Eur Heart J*. 2020;41(4):543-603.
5. 2015 ESC Guidelines for the diagnosis and management of pericardial diseases. *Eur Heart J*. 2015;36(42):2921-2964.

Knowledge Base Architecture References

1. Johnson AEW, et al. MIMIC-III, a freely accessible critical care database. *Sci Data*. 2016;3:160035.
2. Rajpurkar P, et al. AI in health and medicine. *Nat Med*. 2022;28(1):31-38.
3. Wang LL, et al. Retrieval-Augmented Generation for Medical Question Answering. *npj Digital Medicine*. 2024;7:23.